

Engineering Model Go Repository Design

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1. Introduction

This architecture models the engineering-model-go repository as a model-driven toolchain. It captures authored functional intent, export surfaces, and verification-oriented traceability. The model is intended to guide implementation work in this repository using stable IDs and support paths.

1.1. Scope and Assumptions

- Functional Groups and Functional Units are the stable authored design backbone.
- Runtime elements and code elements are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- Requirements and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (VIEW-ARCHITECTURE-INTENT)	in-review	Initial repository self-model with stable top-level domains; implementation details will be expanded iteratively.	18	18	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Security View (VIEW-SECURITY)	in-review	Security scope starts from MCP boundary and export-validation supply chain paths.	12	12	9	high	Counts a unit as covered when explicit authored attack-vector mapping exists.
Traceability View (VIEW-TRACEABILITY)	in-review	Traceability scope is seeded and will be tightened as requirements are added.	9	9	9	high	Counts a unit as covered when direct inferred runtime or code evidence exists.
State Lifecycle View (VIEW-STATE-LIFECYCLE)	draft	Captures model validity and artifact freshness lifecycle behavior.	0	0	0	n/a	Lifecycle heuristic is currently basic; treat authoredStatus as primary signal.
Interaction Flow View (VIEW-INTERACTION-FLOW)	draft	Core implementation workflow is captured to guide model-driven repository changes.	3	3	0	high	Flow heuristic is step/branch-oriented; treat authoredStatus plus projection completeness as the primary signal.

4. Terms and Definitions

Term	Definition
a model workspace replacement is declared (FEAT-MODEL-WORKSPACE-REPLACEMENT)	Feature enabling a CUE module identity to resolve from an explicit sibling repository path during local development. Aliases:workspace replacement
actor (TERM-ACTOR)	A person or operational role that interacts with functional units.
ai agent (ACT-AI-AGENT)	Consumes MCP outputs to plan and execute model-guided edits. Aliases:coding agent
allocation trace (FU-ALLOCATION-TRACE)	Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems. Aliases:allocation

Term	Definition
architecture author (ACT-ARCHITECTURE-AUTHOR)	Maintains model structure, mappings, and architecture intent. Aliases:model author
architecture decision (TERM-DECISION)	Authored architecture decision record with status, context, decision text, and consequences.
architecture decision records are authored (FEAT-ARCHITECTURE-DECISION-RECORDS)	Feature indicating the architecture model includes first-class ADR entries. Aliases:adrs authored
architecture decision records document (DATA-ARCHITECTURE-DECISION-RECORDS)	Generated document containing authored architecture decision records. Aliases:adr document, decisions document
architecture model (DATA-ARCHITECTURE-MODEL)	Authored architecture source describing functional intent and mappings. Aliases:architecture yaml, candidate architecture model
artifact generated event is raised (EVT-ARTIFACT-GENERATED)	Event indicating one or more generated outputs were produced successfully. Aliases:artifact generated
artifact generation (FG-ARTIFACT-GENERATION)	Generation of AsciiDoc and exchange artifacts from model inputs. Aliases:generation
AsciiDoc diagram (TERM-ASCIIDOC-DIAGRAM)	Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.
AsciiDoc document (TERM-ASCIIDOC-DOCUMENT)	Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.
asciidoc generator (FU-ASCIIDOC-GENERATOR)	Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts. Aliases:engdoc
AsciiDoc section (TERM-ASCIIDOC-SECTION)	Structured generated document section or row model used to render authored and inferred architecture content.
attack vector (TERM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
authored mapping (TERM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
canonical semantic model (DATA-CANONICAL-SEMANTIC-MODEL)	SysML v2 and KerML aligned semantic representation used by validation and every output projection. Aliases:canonical model, semantic model
catalog (TERM-CATALOG)	Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.
catalog document (DATA-CATALOG-DOCUMENT)	Canonical authored vocabulary YAML document used by strict requirement and model validation. Aliases:catalog yaml

Term	Definition
catalog entry (TERM-CATALOG-ENTRY)	A stable catalog term with a definition and aliases for linting, linking, and generated references.
catalog references resolve successfully (COND-CATALOG-REFERENCES-RESOLVE)	Condition requiring all authored references to match known catalog entries. Aliases:catalog resolved
ci pipeline (ACT-CI-PIPELINE)	Runs automated validation and export checks for pull requests. Aliases:ci
ci validation mode is active (MODE-CI-VALIDATION)	Automated pipeline mode focused on deterministic checks and artifacts. Aliases:ci mode
CLI command (TERM-CLI-COMMAND)	Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.
cli orchestration (FU-CLI-ORCHESTRATION)	Command entrypoints that orchestrate loading, validation, generation, and export flows. Aliases:cmd layer
code element (TERM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
codemap inference (FU-CODEMAP-INFERENCE)	Infers runtime, code, and verification evidence from repository sources and test markers. Aliases:inference
compliance engineer (ACT-COMPLIANCE-ENGINEER)	Reviews controls, risks, and traceability outputs for compliance needs. Aliases:assurance
compliance mapping (TERM-COMPLIANCE-MAPPING)	Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.
control (TERM-CONTROL)	An authored security, policy, or operational control used to constrain behavior and reduce risk.
control verification (TERM-CONTROL-VERIFICATION)	Authored control verification record with method, status, threat/risk scope, findings, and evidence.
cue oci module registry (REF-CUE-OCI-MODULE-REGISTRY)	OCI-compliant registry resolved through CUE_REGISTRY and used to distribute immutable versioned Engineering Model modules. Aliases:cue registry, model registry, oci registry
data object (TERM-DATA-OBJECT)	An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.
decisions document (DATA-DECISIONS-DOCUMENT)	Canonical authored architecture decision YAML document. Aliases:decisions yaml
deployment target (TERM-DEPLOYMENT-TARGET)	An authored deployment destination such as environment, cluster, namespace, or registry scope.
design document (DATA-DESIGN-DOCUMENT)	Canonical authored design-view YAML document. Aliases:design yaml

Term	Definition
design document (TERM-DESIGN)	Authored design narrative organized by model entities and architecture views.
design view (TERM-DESIGN-VIEW)	View-scoped design narrative attached to an authored functional group or functional unit.
do-178c dal c readiness output is enabled (FEAT-DO178C-DAL-C-READINESS)	Feature enabling evidence-readiness validation and draft lifecycle-data outputs for a DO-178C or ED-12C DAL C project profile. Aliases:dal c readiness
dry run mode is active (MODE-DRY-RUN)	Change-planning mode that computes and validates a candidate without writing canonical files. Aliases:preview mode
engineering model go (SYS-ENGINEERING-MODEL-GO)	Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts. Aliases:engineering model toolchain, engmodel
event (TERM-EVENT)	An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.
evidence (TERM-EVIDENCE)	A file, artifact, or observation used to support control, risk, threat, or verification claims.
example validation is requested (EVT-EXAMPLE-VALIDATION-REQUESTED)	Event indicating maintained reference workspaces must be loaded, composed, generated, and checked. Aliases:example check
external schema dependencies are available (COND-EXTERNAL-SCHEMA-AVAILABLE)	Condition requiring external schema artifacts to be present for validation. Aliases:schema available
flow (TERM-FLOW)	An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.
flow step (TERM-FLOW-STEP)	A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.
functional group (TERM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
functional unit (TERM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
gemara export is enabled (FEAT-GEMARA-EXPORT)	Feature enabling OpenSSF Gemara catalog and evaluation log generation. Aliases:gemara generation
gemara exporter (FU-GEMARA-EXPORTER)	Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK. Aliases:enggemara
generated artifacts are fresh (STATE-ARTIFACTS-FRESH)	State where generated outputs reflect the latest authored model inputs. Aliases:up to date
generation run is requested (EVT-GENERATION-RUN-REQUESTED)	Event indicating documentation or export generation execution starts. Aliases:generation requested

Term	Definition
generation workflow (TERM-GENERATION-WORKFLOW)	Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.
go toolchain (REF-GO-TOOLCHAIN)	Go compiler, test runner, and module tooling used by the repository. Aliases:go
hardware interfaces are authored (FEAT-HARDWARE-INTERFACES)	Feature enabling hardware items and hardware/software interface modeling. Aliases:hw interfaces
implementation engineer (ACT-IMPLEMENTATION-ENGINEER)	Implements code changes guided by model requirements and support paths. Aliases:engineer
inference hint (TERM-INFERENCE-HINT)	Authored source and ownership configuration used to discover runtime, code, and verification evidence.
inferred mapping (TERM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.
interface (TERM-INTERFACE)	An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.
lint run (TERM-LINT-RUN)	Requirement lint configuration that controls parsing and quality checks for requirement text.
LOBSTER activity trace (TERM-LOBSTER-ACTIVITY-TRACE)	Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.
lobster exporter (FU-LOBSTER-EXPORTER)	Generates LOBSTER traceability inputs and reports from model and verification links. Aliases:englobster
lobster toolchain (REF-LOBSTER-TOOLCHAIN)	LOBSTER core and TRLC integration tools used for traceability reporting. Aliases:lobster
lobster trace report (DATA-LOBSTER-TRACE-REPORT)	Traceability report output generated from TRLC and test links. Aliases:lobster report
local development mode is active (MODE-LOCAL-DEVELOPMENT)	Local developer mode for iterative model and code updates. Aliases:local mode
malformed model input (AV-MALFORMED-MODEL-INPUT)	Invalid or ambiguous model content that causes incorrect graph construction. Aliases:invalid model
mcp integration (FG-MCP-INTEGRATION)	MCP server interfaces that expose model-aware tooling for AI agents. Aliases:mcp
mcp server (FU-MCP-SERVER)	Serves model-backed MCP tools with validated inputs and structured responses. Aliases:engmcp
MCP server (TERM-MCP-SERVER)	Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.

Term	Definition
mcp server is enabled (FEAT-MCP-SERVER)	Feature enabling MCP runtime APIs for model-aware tooling. Aliases:mcp enabled
MCP stdio transport (TERM-MCP-STDIO-TRANSPORT)	Content-Length framed standard input/output transport used by the MCP command-line server.
MCP tool (TERM-MCP-TOOL)	Named MCP operation with a constrained input schema and structured response payload.
mcp tool call is received (EVT-MCP-TOOL-CALL-RECEIVED)	Event indicating the MCP server received a tool invocation request. Aliases:tool call
MCP tool response (TERM-MCP-TOOL-RESPONSE)	Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.
mcp tool result (DATA-MCP-TOOL-RESULT)	Structured MCP tool response payload carrying model-aware query results. Aliases:tool result
model (TERM-MODEL)	The authored architecture model root that composes functional structure, relationships, inference hints, and views.
model authoring (FG-MODEL-AUTHORING)	Authoring and loading of architecture, catalog, requirements, and design inputs. Aliases:authoring
model authoring contract (DATA-MODEL-AUTHORING-CONTRACT)	Machine-readable description of canonical YAML documents, schema versions, explicit references, top-level sections, identifier conventions, and authoring order. Aliases:authoring manifest, input contract
model bundle (TERM-BUNDLE)	Loaded architecture, catalog, and companion documents resolved as one working model context.
model change (FU-MODEL-CHANGE)	Plans, validates, summarizes, and atomically applies typed changes to authored model documents. Aliases:delta merger, requirements merger
model is invalid (STATE-MODEL-INVALID)	State where validation diagnostics include one or more errors. Aliases:invalid
model is valid (STATE-MODEL-VALID)	State where authored model passes validation checks. Aliases:valid
model loader (FU-MODEL-LOADER)	Validates canonical YAML documents against embedded CUE schemas, strictly decodes them into Go runtime representations, and normalizes them for downstream use. Aliases:bundle loader
model updated event is received (EVT-MODEL-UPDATED)	Event indicating architecture or requirement model source changed. Aliases:model change
naf export is enabled (FEAT-NAF-EXPORT)	Feature enabling NATO Architecture Framework architecture document generation. Aliases:naf generation
naf exporter (FU-NAF-EXPORTER)	Validates NATO Architecture Framework 4.1 profile metadata and generates NAF architecture products from canonical model views. Aliases:engnaf, naf

Term	Definition
naf profile authoring is enabled (FEAT-NAF-PROFILE)	Feature enabling NATO Architecture Framework stakeholder, concern, and viewpoint metadata in the architecture model. Aliases:naf authoring
naf v4.1 architecture document (DATA-NAF-V4-ARCHITECTURE)	Generated NATO Architecture Framework v4.1 architecture document assembled from profile metadata and canonical model views. Aliases:naf architecture, naf document
nato architecture framework v4.1 (REF-NAF-V4-1-SPECIFICATION)	NATO Architecture Framework version 4.1 viewpoint taxonomy and architecture-description guidance. Aliases:naf, naf v4.1
Open OTM document (TERM-OPEN-OTM-DOCUMENT)	Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.
open otm schema (REF-OPEN-OTM-SCHEMA)	Open Threat Model schema used to validate OTM output artifacts. Aliases:otm schema
OSCAL assessment results (TERM-OSCAL-ASSESSMENT-RESULTS)	Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.
oscal exporter (FU-OSCAL-EXPORTER)	Exports the OSCAL system security plan, assessment results, and plan of action and milestones. Aliases:engoscal
OSCAL POA&M (TERM-OSCAL-POAM)	Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.
OSCAL SSP (TERM-OSCAL-SSP)	Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.
oscal ssp artifact (DATA-OSCAL-SSP)	Generated OSCAL system security plan JSON output. Aliases:ssp json
path traversal in mcp calls (AV-PATH-TRAVERSAL-IN-MCP)	Host file access attempts through untrusted path arguments in MCP tools. Aliases:path traversal
POA&M item (TERM-POAM-ITEM)	Plan of action and milestones item linked to a modeled risk and supporting artifacts.
reference index (TERM-REFERENCE-INDEX)	Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.
referenced element (TERM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.
repository index (TERM-REPO-INDEX)	Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.
requirement (TERM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
requirement trace is complete (COND-REQUIREMENT-TRACE-COMPLETE)	Condition requiring requirements to map to implementation and verification evidence. Aliases:trace complete

Term	Definition
requirements change summary (DATA-REQUIREMENTS-DIFF)	Deterministic human-readable summary of added, updated, and removed requirements. Aliases:requirements diff
requirements delta (DATA-REQUIREMENTS-DELTA)	Versioned add, whole-entity update, and remove operations keyed by requirement stable ID. Aliases:requirements change
requirements delta apply is requested (EVT-REQUIREMENTS-DELTA-APPLY-REQUESTED)	Event indicating a validated requirements change should replace the canonical document. Aliases:requirements apply requested
requirements delta is received (EVT-REQUIREMENTS-DELTA-RECEIVED)	Event indicating a typed requirements change is ready for planning and validation. Aliases:requirements change received
requirements document (DATA-REQUIREMENTS-DOCUMENT)	Canonical authored requirements YAML document. Aliases:requirements yml
risk (TERM-RISK)	Authored risk record with likelihood, impact, response, scope, controls, and evidence.
runtime element (TERM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
schema supply chain tamper (AV-SCHEMA-SUPPLY-CHAIN-TAMPER)	Tampering or drift in external schemas used for export validation. Aliases:schema tamper
security context (TERM-SECURITY-CONTEXT)	Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.
six-view example output is enabled (FEAT-SIX-VIEW-OUTPUT)	Feature enabling use-case, logical, process, physical, implementation, and deployment view generation from one model bundle. Aliases:six-view output
source block (TERM-SOURCE-BLOCK)	Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.
state (TERM-STATE)	An authored lifecycle state used to model transition behavior, guards, and event-driven progression.
Structurizr DSL (TERM-STRUCTURIZR-DSL)	Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.
structurizr dsl artifact (DATA-STRUCTURIZR-DSL)	Generated Structurizr workspace DSL document. Aliases:dsl
structurizr export is enabled (FEAT-STRUCTURIZR-EXPORT)	Feature enabling Structurizr DSL generation and validation. Aliases:dsl export
structurizr exporter (FU-STRUCTURIZR-EXPORTER)	Exports Structurizr DSL and supports validation/static-site workflows. Aliases:engstruct
structurizr validator (REF-STRUCTURIZR-VALIDATOR)	Structurizr DSL validation runtime used to verify generated DSL. Aliases:structurizr

Term	Definition
sysml export is enabled (FEAT-SYSML-EXPORT)	Feature enabling SysML v2 textual and project interchange generation. Aliases:sysml generation
sysml exporter (FU-SYSML-EXPORTER)	Generates SysML v2 model artifacts from the canonical semantic model and reports conformance coverage. Aliases:engsysml
sysml v2 metamodel (DATA-SYSML-V2-METAMODEL)	Pinned formal SysML 2.0 and KerML 1.0 abstract syntax used to derive semantic coverage and schema constraints. Aliases:kerml metamodel, sysml metamodel
sysml v2 model artifact (DATA-SYSML-V2-MODEL)	Generated SysML v2 textual or project interchange representation of the canonical semantic model. Aliases:sysml artifact, sysml model
sysml v2 toolchain (REF-SYSML-V2-TOOLCHAIN)	SysML 2.0 and KerML 1.0 with Pilot Implementation 2026-04 artifact 0.59.0, the pinned validator wrapper, and Sysand 0.2.1 for normative KPAR packaging. Aliases:kerml, sysml
system composition (FU-SYSTEM-COMPOSITION)	Resolves downward subsystem references from local paths, external git repositories, or versioned CUE modules in OCI registries and composes a federated system-of-systems model. Aliases:composition, model dependencies
system composition is enabled (FEAT-SYSTEM-COMPOSITION)	Feature enabling recursive composition of subsystem models from local paths, external git repositories, or CUE modules in OCI registries. Aliases:system of systems
threat assumption (TERM-THREAT-ASSUMPTION)	Authored security assumption that records scope, rationale, owner, status, and evidence.
Threat Dragon document (TERM-THREAT-DRAGON-DOCUMENT)	Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.
threat dragon json artifact (DATA-THREAT-DRAGON-JSON)	Generated threat model artifact for OWASP Threat Dragon. Aliases:threat dragon output
threat dragon schemas (REF-THREAT-DRAGON-SCHEMAS)	External JSON schemas used to validate Threat Dragon exports. Aliases:threat dragon
threat export is enabled (FEAT-THREAT-EXPORT)	Feature enabling Threat Dragon and Open OTM exports. Aliases:threat modeling export
threat exporter (FU-THREAT-EXPORTER)	Exports threat model views to Threat Dragon and Open OTM representations. Aliases:engdragon
threat mitigation (TERM-THREAT-MITIGATION)	Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.
threat model export (TERM-THREAT-MODEL-EXPORT)	Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.
threat out-of-scope decision (TERM-THREAT-OUT-OF-SCOPE)	Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.
threat scenario (TERM-THREAT-SCENARIO)	Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.

Term	Definition
trace marker (TERM-TRACE-MARKER)	Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.
traceability and compliance (FG-TRACEABILITY-COMPLIANCE)	Requirement trace exports and compliance-oriented output generation. Aliases:compliance
traceability export is enabled (FEAT-TRACEABILITY-EXPORT)	Feature enabling TRLC and LOBSTER generation flows. Aliases:trace export
traceability gap drift (AV-TRACEABILITY-GAP-DRIFT)	Requirement links or tests drifting from implementation over time. Aliases:trace drift
trlc exporter (FU-TRLC-EXPORTER)	Exports requirements and model structures to TRLC artifacts. Aliases:engtrlc
trlc package (DATA-TRLC-PACKAGE)	Generated TRLC requirements package and model metadata. Aliases:trlc output
TRLC package (TERM-TRLC-PACKAGE)	Generated TRLC model and requirements package used for formal requirement traceability processing.
trlc toolchain (REF-TRLC-TOOLCHAIN)	TRLC parser and runtime used for requirements language validation. Aliases:trlc
trust boundary (TERM-TRUST-BOUNDARY)	An authored trust separation zone that marks policy, identity, or security control boundaries.
upward linking (TERM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.
validation (TERM-VALIDATION)	Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.
validation and analysis (FG-VALIDATION-ANALYSIS)	Validation, inference, and evidence analysis over authored and inferred model content. Aliases:analysis
validation diagnostic (TERM-VALIDATION-DIAGNOSTIC)	Structured validation finding with code, severity, message, and optional source path.
validation engine (FU-VALIDATION-ENGINE)	Validates IDs, references, and model consistency across authored entities and mappings. Aliases:validator
validation failed event is raised (EVT-VALIDATION-FAILED)	Event indicating model or export validation failed. Aliases:validation failure
verification check (TERM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.
view (TERM-VIEW)	Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.
view projection (FU-VIEW-PROJECTION)	Builds view-specific projections for architecture communication and traceability outputs. Aliases:projection

5. Architecture Decision Records

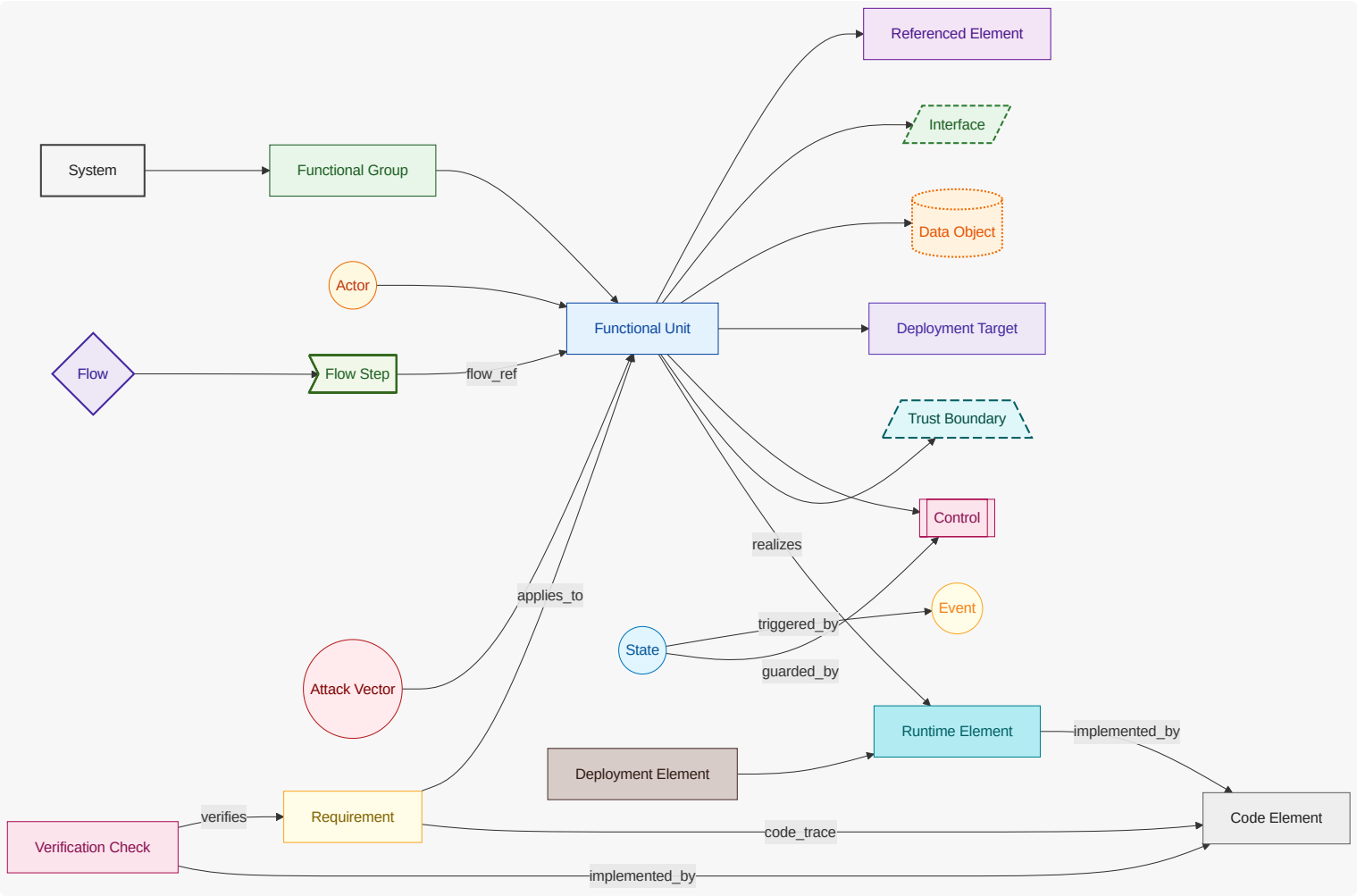
Architecture decision records are generated separately and linked here as a compact index.

ID	Title	Status	Date
<u>ADR-EMG-001</u>	Generate architecture decision records separately from the main architecture document	accepted	2026-04-29
<u>ADR-EMG-002</u>	Keep architecture decision records model-authored and separate from code links	accepted	2026-04-29
<u>ADR-EMG-003</u>	Render Gemara as a first-class output using the official go-gemara SDK types	accepted	2026-06-27
<u>ADR-EMG-004</u>	Source the Gemara evaluation log from authored control verifications, with pass and fail outcomes	accepted	2026-06-27
<u>ADR-EMG-005</u>	Keep hand-written OSCAL SSP/AR/POA&M; add Gemara-to-OSCAL only as an additive bridge	accepted	2026-06-27
<u>ADR-EMG-006</u>	Synthesize Gemara structure that the model does not author, with explicit fallbacks	accepted	2026-06-27
<u>ADR-EMG-007</u>	Attach trace and model-link markers to package-level var and const declarations	accepted	2026-06-27
<u>ADR-EMG-008</u>	Model multi-repository work as a recursive system-of-systems	accepted	2026-06-27
<u>ADR-EMG-009</u>	Subsystem references are downward-only with parent-owned allocation	accepted	2026-06-27
<u>ADR-EMG-010</u>	Resolve subsystem references from local subdirectories first	accepted	2026-06-27
<u>ADR-EMG-011</u>	Requirements carry no tiers; delegation is the link	accepted	2026-06-27
<u>ADR-EMG-012</u>	Trace links are existence-checked, scoped to the owning model, and exported as a matrix	accepted	2026-06-27
<u>ADR-EMG-013</u>	CI enforces the generation gates and artifact freshness	accepted	2026-06-28
<u>ADR-EMG-014</u>	Apply model deltas through YAML nodes and typed candidate validation	accepted	2026-09-15
<u>ADR-EMG-015</u>	Make SysML and KerML semantics canonical without duplicating Engineering Model concepts	accepted	2026-09-18
<u>ADR-EMG-016</u>	Use CUE as the schema contract for canonical YAML	accepted	2026-09-18
<u>ADR-EMG-017</u>	Pin the official SysML 2.0 parser baseline and delegate KPAR packaging to Sysand	accepted	2026-09-18
<u>ADR-EMG-018</u>	Implement NAF 4.1 as a profile over canonical model views	accepted	2026-09-19

ID	Title	Status	Date
<u>ADR-EMG-019</u>	Make the YAML document set self-describing without breaking legacy inputs	accepted	2026-09-19
<u>ADR-EMG-020</u>	Distribute reusable model systems as CUE modules in OCI registries	accepted	2026-09-19

5.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



6. Architecture Intent View

Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

Authored status: in-review

Authored scope and limitations: Initial repository self-model with stable top-level domains; implementation details will be expanded iteratively.

6.1. What This View Answers

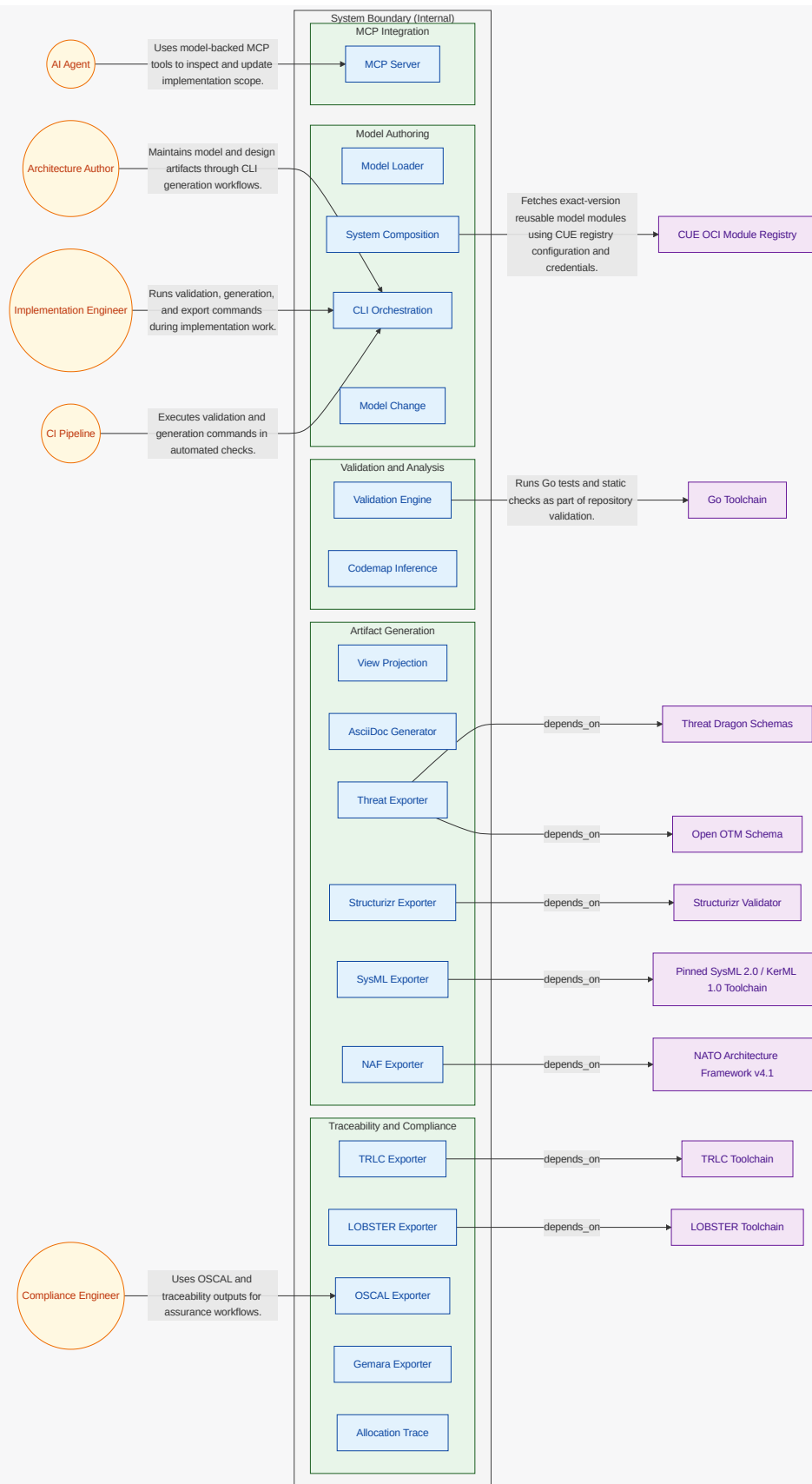
- What are the main capability areas?

- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

6.2. Authored View Diagram

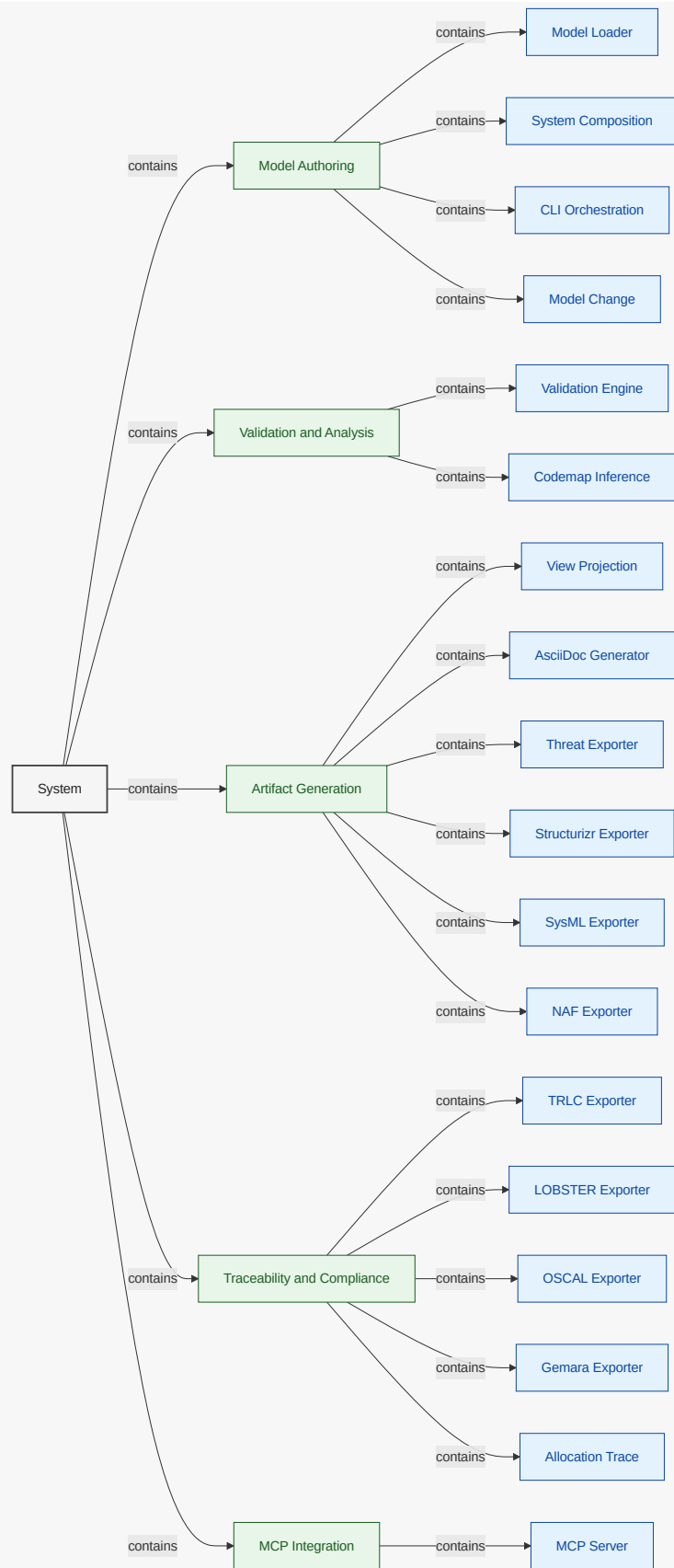
This diagram follows the selected roots, depth, kinds and relationships of `VIEW-ARCHITECTURE-INTENT` .

What this diagram shows: explicit internal vs external scope. Functional units are inside the **System Boundary (Internal)** container, while actors and referenced dependencies are shown outside the boundary.



6.4. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.



6.5. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

			AsciiDoc Generator	
			NAF Exporter	Allocation Trace
			Structurizr Exporter	Gemara Exporter
			SysML Exporter	LOBSTER Exporter
CLI Orchestration			Threat Exporter	OSCAL Exporter
Model Change	Codemap Inference		View Projection	TRLC Exporter
Model Loader	Validation Engine			MCP Server
System Composition				
Model Authoring	Validation and Analysis	Artifact Generation	Traceability and Compliance	MCP Integration

6.6. Functional Groups and Units (View Scoped)

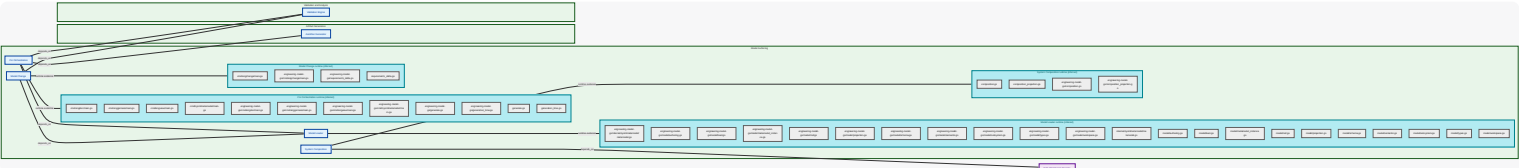
6.6.1. Model Authoring (FG-MODEL-AUTHORING)

Inputs and model loading responsibilities.

Maintains authored architecture, <<REF_IDX-ENGMODEL_TERM_REQUIREMENT,requirements>>, design, and <<REF_IDX-ENGMODEL_TERM_CATALOG,catalog>> handling.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



CLI Orchestration (FU-CLI-ORCHESTRATION)

Command entryptoints orchestrating model load, validation , generation , and exports.

Owned Responsibility

functional responsibility for cli orchestration ; includes decision and orchestration flow to AsciiDoc Generator and decision and orchestration flow to Model Loader

Key Collaborators

AsciiDoc Generator , Model Loader , Validation Engine

Linked Requirements

REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-004 , REQ-EMG-005 , REQ-EMG-006 , REQ-EMG-009 , REQ-EMG-013 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-033 , REQ-EMG-034 , REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-039 , REQ-EMG-040 , REQ-EMG-042 , REQ-EMG-043 , REQ-EMG-044 , REQ-EMG-046 , REQ-EMG-049 , REQ-EMG-051 , REQ-EMG-052 , REQ-EMG-055

Model Change (FU-MODEL-CHANGE)

Plans, validates, summarizes, and atomically applies typed stable-ID changes to authored model documents.

Owned Responsibility

functional responsibility for model change ; includes decision and orchestration flow to Model Loader and decision and orchestration flow to Validation Engine

Key Collaborators

Model Loader , Validation Engine

Linked Requirements

REQ-EMG-031 , REQ-EMG-032 , REQ-EMG-033 , REQ-EMG-034

Model Loader (FU-MODEL-LOADER)

Validates canonical YAML against the authoritative CUE contract before strict Go decoding and normalization.

Owned Responsibility

functional responsibility for model loader

Key Collaborators

none

Linked Requirements

REQ-EMG-001 , REQ-EMG-016 , REQ-EMG-027 , REQ-EMG-031 , REQ-EMG-035 , REQ-EMG-038 , REQ-EMG-041 , REQ-EMG-044 , REQ-EMG-045 , REQ-EMG-046 , REQ-EMG-047 , REQ-EMG-048 , REQ-EMG-051 , REQ-EMG-052 , REQ-EMG-053

System Composition (FU-SYSTEM-COMPOSITION)

Resolves downward subsystem references from local paths, external git repositories, or exact-version CUE modules fetched from OCI registries into the .engmod cache, and composes a federated system-of-systems model.

Owned Responsibility

functional responsibility for system composition ; includes decision and orchestration flow to CUE OCI Module Registry

Key Collaborators

CUE OCI Module Registry

Linked Requirements

REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-027 , REQ-EMG-047 , REQ-EMG-048 , REQ-EMG-049 , REQ-EMG-050 , REQ-EMG-051

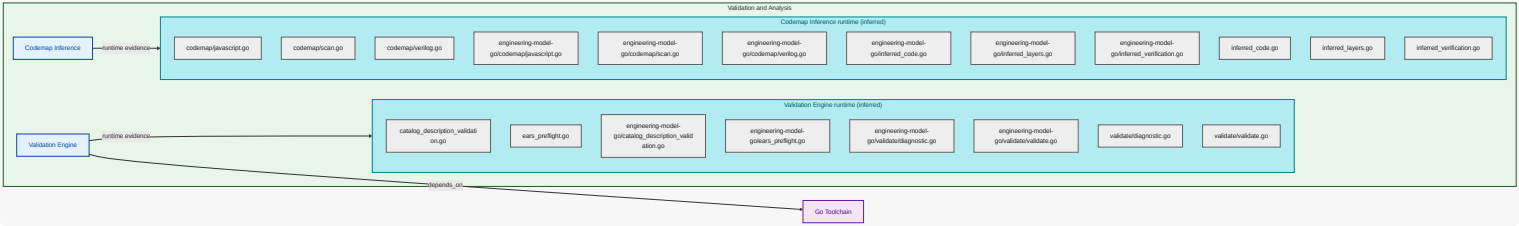
6.6.2. Validation and Analysis (FG-VALIDATION-ANALYSIS)

Validation and inference responsibilities.

Performs consistency checks and derives runtime, code, and verification <<REF_IDX-ENGMODEL_TERM_EVIDENCE,evidence>>.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Codemap Inference (FU-CODEMAP-INFERENCE)

Infers code/runtime ownership and verification links from source and tests.

Owned Responsibility

functional responsibility for codemap inference

Key Collaborators

none

Linked Requirements

REQ-EMG-010, REQ-EMG-012, REQ-EMG-028, REQ-EMG-030

Validation Engine (FU-VALIDATION-ENGINE)

Validates authored entities, IDs, references, and mapping consistency.

Owned Responsibility

functional responsibility for validation engine ; includes decision and orchestration flow to Go Toolchain

Key Collaborators

Go Toolchain

Linked Requirements

REQ-EMG-001, REQ-EMG-009, REQ-EMG-011, REQ-EMG-012, REQ-EMG-018, REQ-EMG-021, REQ-EMG-022, REQ-EMG-023, REQ-EMG-026, REQ-EMG-028, REQ-EMG-029, REQ-EMG-032, REQ-EMG-033, REQ-EMG-035, REQ-EMG-036, REQ-EMG-037, REQ-EMG-038, REQ-EMG-039, REQ-EMG-040, REQ-EMG-041, REQ-EMG-043, REQ-EMG-044, REQ-EMG-046, REQ-EMG-049, REQ-EMG-050, REQ-EMG-051, REQ-EMG-053, REQ-EMG-054, REQ-EMG-056

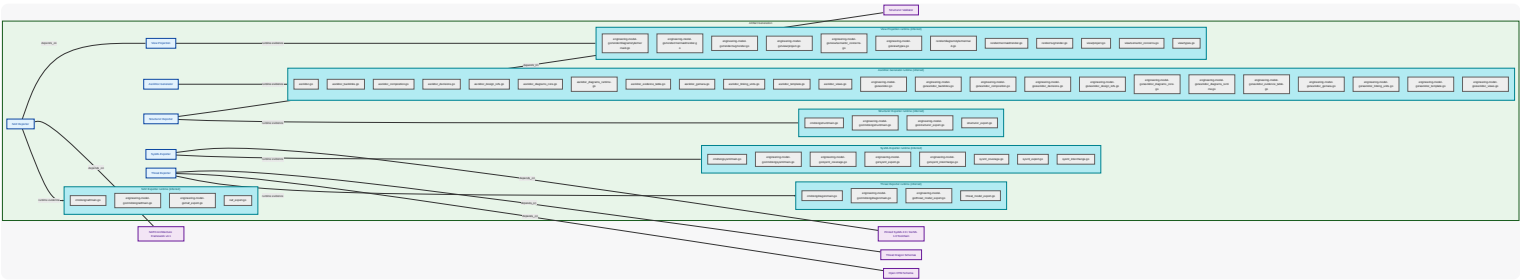
6.6.3. Artifact Generation (FG-ARTIFACT-GENERATION)

Publication and exchange artifact generation .

Produces AsciiDoc publication docs and external modeling outputs.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



AsciiDoc Generator (FU-ASCIIDOC-GENERATOR)

Renders architecture publication docs and view narratives for human consumption.

Owned Responsibility

functional responsibility for asciidoc generator

Key Collaborators

none

Linked Requirements

REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-012 , REQ-EMG-014 , REQ-EMG-024 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-050 , REQ-EMG-056

NAF Exporter (FU-NAF-EXPORTER)

Validates NAF 4.1 stakeholders, concerns, and viewpoint selections and renders architecture products from existing canonical views.

Owned Responsibility

functional responsibility for naf exporter ; includes decision and orchestration flow to NATO Architecture Framework v4.1 and decision and orchestration flow to View Projection

Key Collaborators

NATO Architecture Framework v4.1 , View Projection

Linked Requirements

REQ-EMG-041 , REQ-EMG-042 , REQ-EMG-043

Structurizr Exporter (FU-STRUCTURIZR-EXPORTER)

Emits Structurizr DSL and deployment-aware model views.

Owned Responsibility

functional responsibility for structurizr exporter ; includes decision and orchestration flow to Structurizr Validator

Key Collaborators

Structurizr Validator

Linked Requirements

REQ-EMG-001 , REQ-EMG-005 , REQ-EMG-011 , REQ-EMG-012

SysML Exporter (FU-SYSML-EXPORTER)

Generates SysML v2 textual and project interchange artifacts from the canonical semantic model while reporting conformance coverage.

Owned Responsibility

functional responsibility for sysml exporter ; includes decision and orchestration flow to Pinned SysML 2.0 / KerML 1.0 Toolchain

Key Collaborators

Pinned SysML 2.0 / KerML 1.0 Toolchain

Linked Requirements

REQ-EMG-012 , REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-038 , REQ-EMG-039 , REQ-EMG-040 , REQ-EMG-050 , REQ-EMG-052

Threat Exporter (FU-THREAT-EXPORTER)

Exports Threat Dragon and Open OTM model artifacts.

Owned Responsibility

functional responsibility for threat exporter ; includes decision and orchestration flow to Open OTM Schema and decision and orchestration flow to Threat Dragon Schemas

Key Collaborators

Open OTM Schema , Threat Dragon Schemas

Linked Requirements

REQ-EMG-001, REQ-EMG-004, REQ-EMG-011, REQ-EMG-012

View Projection (FU-VIEW-PROJECTION)

Builds projection graphs for architecture, traceability, security, and flow views.

Owned Responsibility

functional responsibility for view projection

Key Collaborators

none

Linked Requirements

REQ-EMG-003, REQ-EMG-024, REQ-EMG-042, REQ-EMG-052

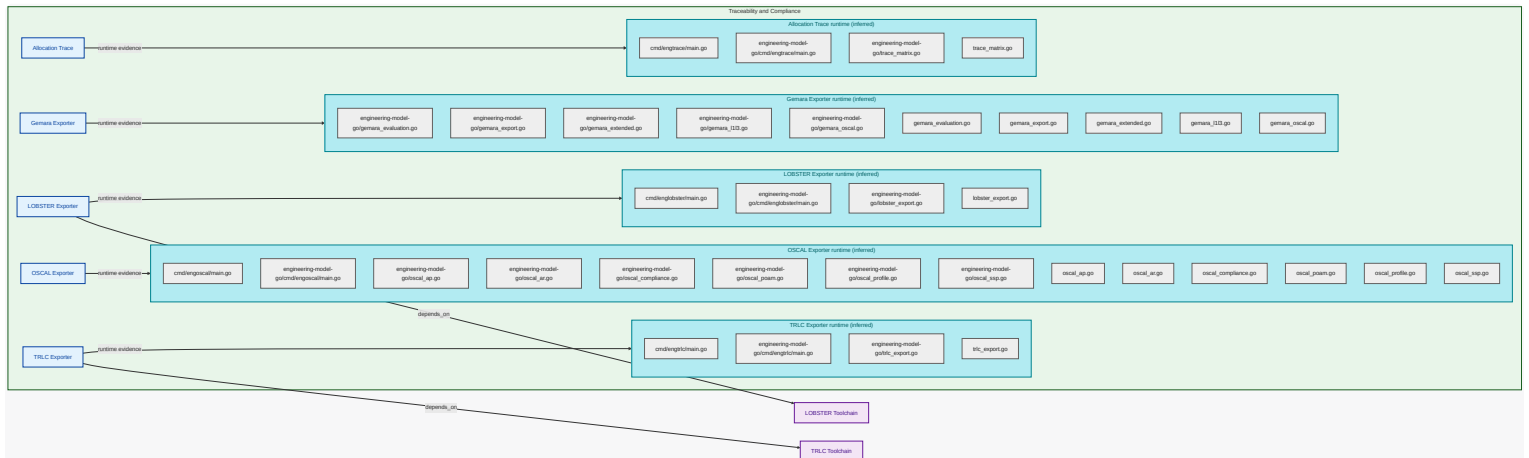
6.6.4. Traceability and Compliance (FG-TRACEABILITY-COMPLIANCE)

Trace and compliance export responsibilities.

Produces <<REF_IDX-CATALOG_TRLC,TRLC>>, <<REF_IDX-CATALOG_LOBSTER,LOBSTER>>, and OSCAL outputs from modeled content.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



Allocation Trace (FU-ALLOCATION-TRACE)

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems, without modifying subsystem models.

Owned Responsibility

functional responsibility for allocation trace

Key Collaborators

none

Linked Requirements

REQ-EMG-012, REQ-EMG-019, REQ-EMG-020, REQ-EMG-021, REQ-EMG-022, REQ-EMG-023, REQ-EMG-025, REQ-EMG-026, REQ-EMG-028, REQ-EMG-029, REQ-EMG-030

Gemara Exporter (FU-GEMARA-EXPORTER)

Renders OpenSSF Gemara L1-L7 catalogs and logs (vector, principle, guidance, control , capability, threat, risk , evaluation, enforcement, audit) with an optional OSCAL catalog and assessment-results bridge, validated by the Gemara SDK.

Owned Responsibility

functional responsibility for gemara exporter

Key Collaborators

none

Linked Requirements

REQ-EMG-012 , REQ-EMG-015 , REQ-EMG-050

LOBSTER Exporter (FU-LOBSTER-EXPORTER)

Generates LOBSTER traceability inputs and reports.

Owned Responsibility

functional responsibility for lobster exporter ; includes decision and orchestration flow to LOBSTER Toolchain

Key Collaborators

LOBSTER Toolchain

Linked Requirements

REQ-EMG-006 , REQ-EMG-012

OSCAL Exporter (FU-OSCAL-EXPORTER)

Exports OSCAL System Security Plan (SSP), Assessment Results (AR), and Plan of Action and Milestones (POA&M).

Owned Responsibility

functional responsibility for oscal exporter

Key Collaborators

none

Linked Requirements

REQ-EMG-001 , REQ-EMG-012 , REQ-EMG-013 , REQ-EMG-050

TRLC Exporter (FU-TRLC-EXPORTER)

Emits TRLC model and requirement artifacts.

Owned Responsibility

functional responsibility for trlc exporter ; includes decision and orchestration flow to TRLC Toolchain

Key Collaborators

TRLC Toolchain

Linked Requirements

REQ-EMG-006 , REQ-EMG-012

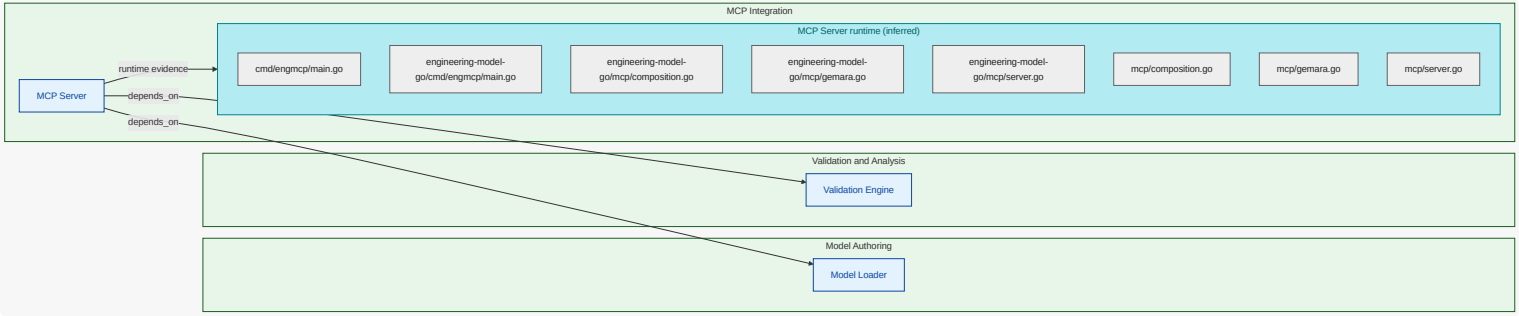
6.6.5. MCP Integration (FG-MCP-INTEGRATION)

AI-agent integration and runtime API responsibilities.

Exposes model-aware tools over MCP with strict input and error contracts.

Functional Unit Dependency Diagram

What this diagram shows: dependencies involving functional units in this functional group. Units in this group are rendered inside the group container; connected external units and referenced elements are shown outside it.



MCP Server (FU-MCP-SERVER)

Serves model-backed tools over MCP with path safety and structured errors.

Owned Responsibility

functional responsibility for `mcp server` ; includes decision and orchestration `flow` to `Model Loader` and decision and orchestration `flow` to `Validation Engine`

Key Collaborators

`Model Loader` , `Validation Engine`

Linked Requirements

`REQ-EMG-007` , `REQ-EMG-008` , `REQ-EMG-015` , `REQ-EMG-045`

7. Security View

Show the selected security relationships and boundary context. Interpret this projection within its authored scope; membership alone does not establish authentication, enforcement or complete threat coverage.

Authored status: in-review

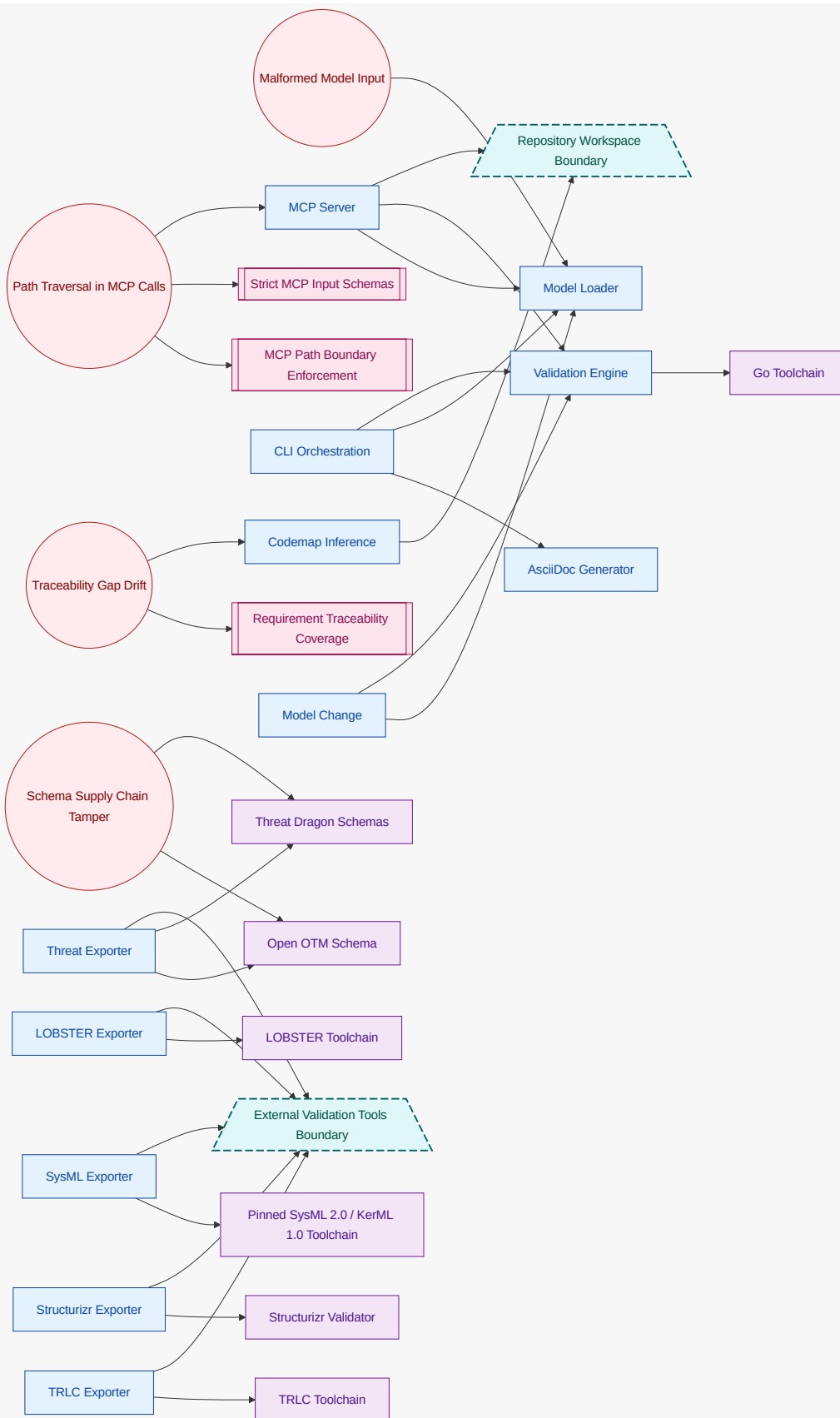
Authored scope and limitations: Security scope starts from MCP boundary and export-validation supply chain paths.

7.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

7.2. Authored View Diagram

This diagram follows the selected roots, depth, kinds and relationships of VIEW-SECURITY .



7.3. Coverage Gaps

Coverage summary: 3/12 units have direct evidence coverage in this view .

- AsciiDoc Generator has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- CLI Orchestration has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- LOBSTER Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Model Change has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Structurizr Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- SysML Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- TRLC Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Threat Exporter has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- Validation Engine has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

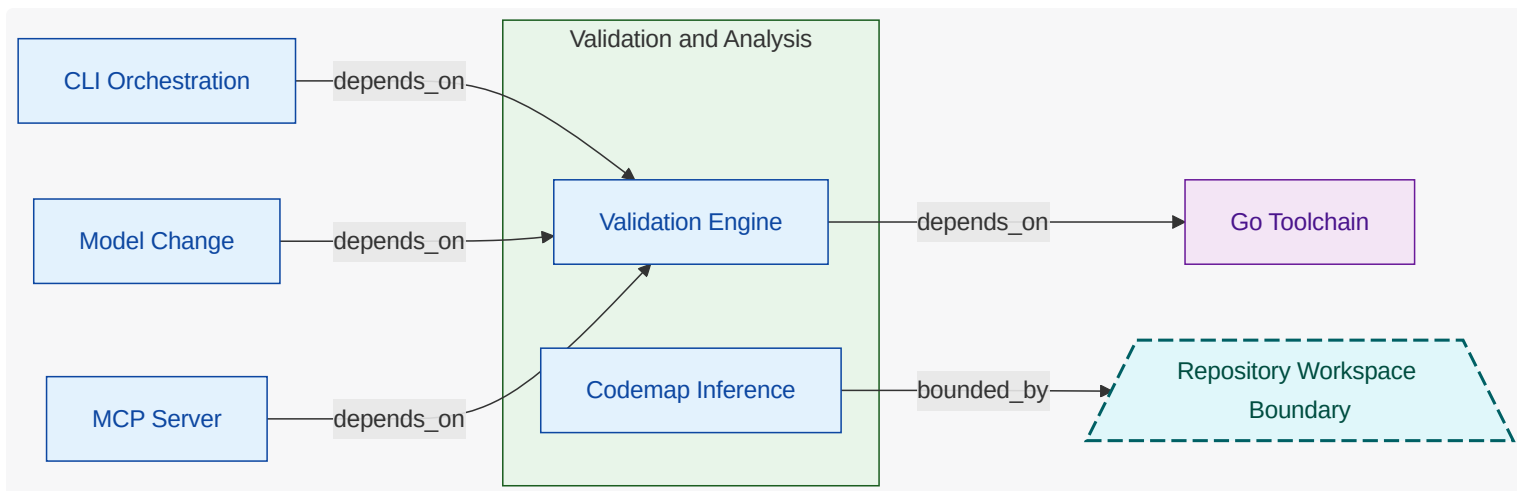
7.4. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/ events) for unresolved observability evidence .
- Map additional security-relevant dependencies and exposures into authored mappings for traceable coverage.

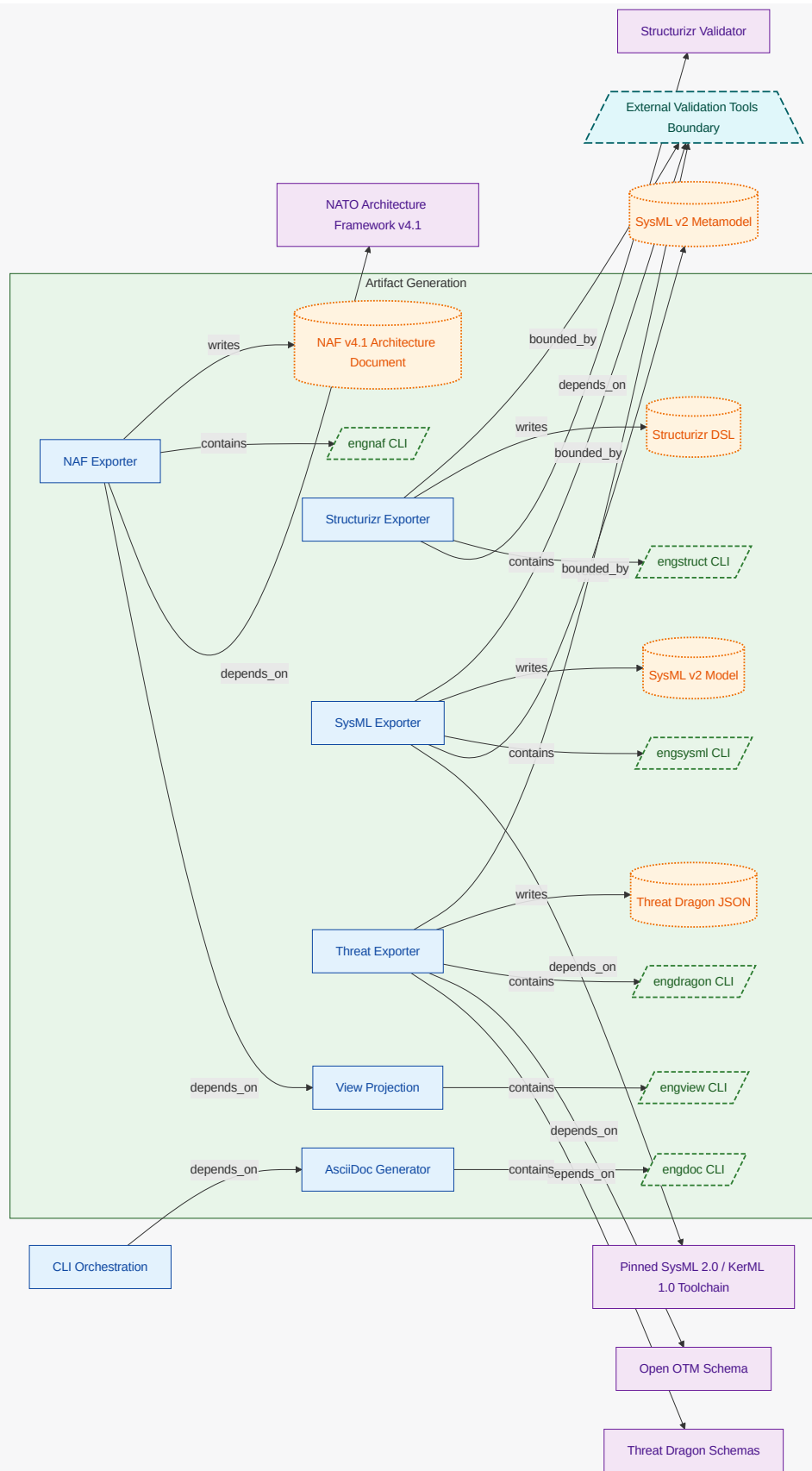
7.5. Threat Model Context Diagram

What these diagrams show: OWASP-style context views split by functional group. Each group diagram keeps owned functional units and owned security-relevant entities inside the functional-group container; connected actors, referenced elements, external functional units, and other external entities are rendered outside it.

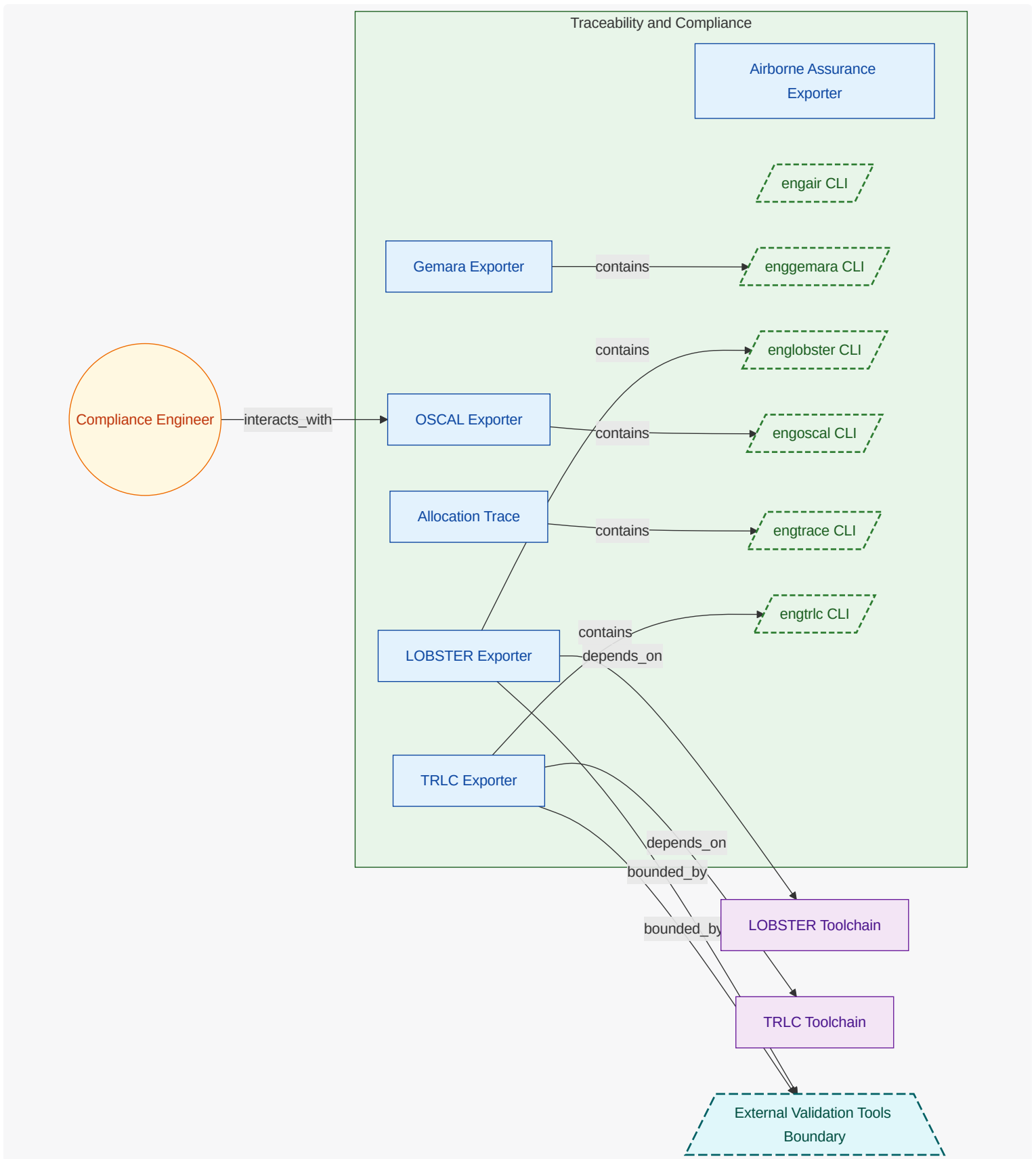
7.5.1. Model Authoring (FG-MODEL-AUTHORING)



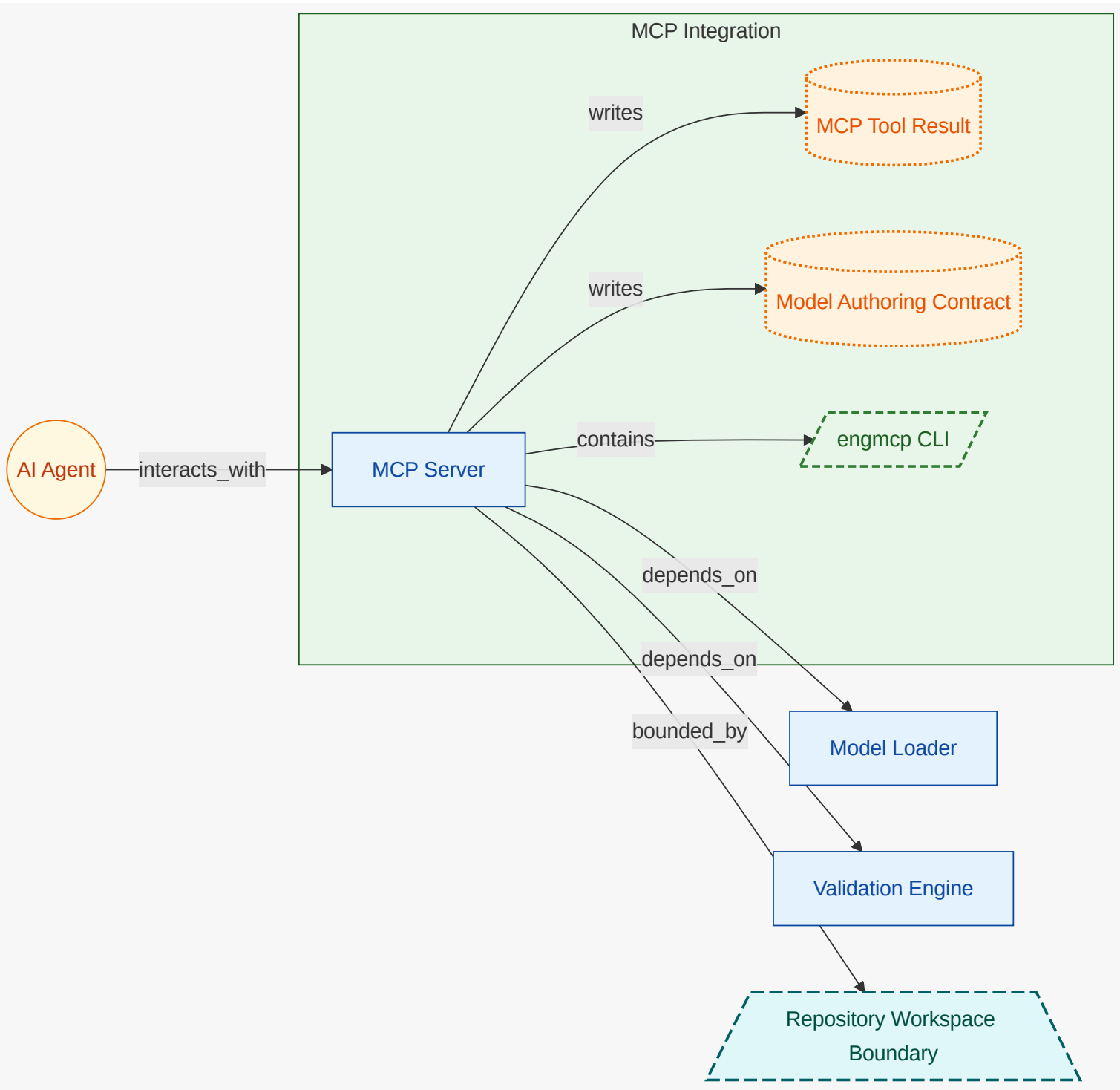
7.5.3. Artifact Generation (FG-ARTIFACT-GENERATION)



7.5.4. Traceability and Compliance (FG-TRACEABILITY-COMPLIANCE)



7.5.5. MCP Integration (FG-MCP-INTEGRATION)



7.6. Threat Scenario Register

7.6.1. Requirement traceability drifts from implementation (TS-TRACEABILITY-DRIFT)

Code and tests change without corresponding requirement trace updates.

Field	Value
Status	mitigating
Owner	Compliance Engineer

Field	Value
Attack Vector	Traceability Gap Drift
Scope	Codemap Inference , LOBSTER Exporter , TRLC Exporter
Flows	none
Likelihood	medium
Impact	medium
Severity	medium
Risk	none
Controls	Requirement Traceability Coverage
Mitigations	none
Verifications	none
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

VER-TRACEABILITY-COVERAGE

Field	Value
Control	Requirement Traceability Coverage
Method	analysis
Status	partial
Owner	Compliance Engineer
Last Tested	unknown
Risks	none
Findings	Coverage depends on TRLC-LINKS completeness in source and tests.

7.6.2. MCP path traversal accesses non-repo files (TS-MCP-PATH-TRAVERSAL)

Untrusted MCP input attempts to escape repo root for sensitive file reads.

Field	Value
Status	mitigating
Owner	Architecture Author
Attack Vector	Path Traversal in MCP Calls

Field	Value
Scope	MCP Server
Flows	Model Change to Verified Artifacts Flow
Likelihood	medium
Impact	high
Severity	high
Risk	none
Controls	MCP Path Boundary Enforcement , Strict MCP Input Schemas
Mitigations	none
Verifications	none
Operational Notes	Track closure through mitigation and verification linkage.

Control Verification Status

VER-MCP-PATH-BOUNDARY

Field	Value
Control	MCP Path Boundary Enforcement
Method	test
Status	pass
Owner	Architecture Author
Last Tested	unknown
Risks	none
Findings	Path traversal rejected and error payload returned in MCP tests.

7.7. Threat Mitigation Coverage

ID	Threat Scenario	Control	Status	Effectiveness	Owner	Verification Links	Notes
TM-MCP-PATH-BOUNDARY	TS-MCP-PATH-TRAVERSAL	MCP Path Boundary Enforcement	implemented	high	Architecture Author	VER-MCP-PATH-BOUNDARY	
TM-TRACE-COVERAGE	TS-TRACEABILITY-DRIFT	Requirement Traceability Coverage	implemented	medium	Compliance Engineer	none	

7.8. Security Flow Semantics

ID	Title	Kind	Direction	Frequency	Source	Destination	Protocol
FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS	Model Change to Verified Artifacts Flow	control	internal	on-demand	Implementation Engineer	CLI Orchestration	cli
FLOW-REQUIREMENTS-DELTA-TO-CANONICAL	Requirements Delta to Canonical Requirements Flow	control	internal	on-demand	Implementation Engineer	Model Change	cli

ID	Auth	Encryption	Integrity	Boundary Crossing	Trust Boundary	Threats	Data
FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS	repository-access	none	git-history	false	none	none	Architecture Model , MCP Tool Result , Structurizr DSL , Threat Dragon JSON
FLOW-REQUIREMENTS-DELTA-TO-CANONICAL	n/a	n/a	n/a	false	none	none	Architecture Model , Requirements Change Summary , Requirements Delta , Requirements Document

7.9. Security Logging and Observability

Signal	Layer	Owner	Evidence
application security telemetry hooks	code	FU-ALLOCATION-TRACE	code cmd/engtrace/main.go
application security telemetry hooks	code	FU-ALLOCATION-TRACE	code trace_matrix.go
application security telemetry hooks	code	FU-CODEMAP-INFERENCE	code codemap/verilog.go

Signal	Layer	Owner	Evidence
application security telemetry hooks	code	FU-RISK-SCORING	code examples/payments-engineering-sample/src/support/audit_envelope.ts
application security telemetry hooks	code	FU-VALIDATION-ENGINE	code catalog_description_validation.go
authorization and credential handling checks	code	FU-MODEL-LOADER	code model/authoring.go
fraud and abuse detection code paths	code	FU-RISK-SCORING	code examples/payments-engineering-sample/src/risk_scorer.ts

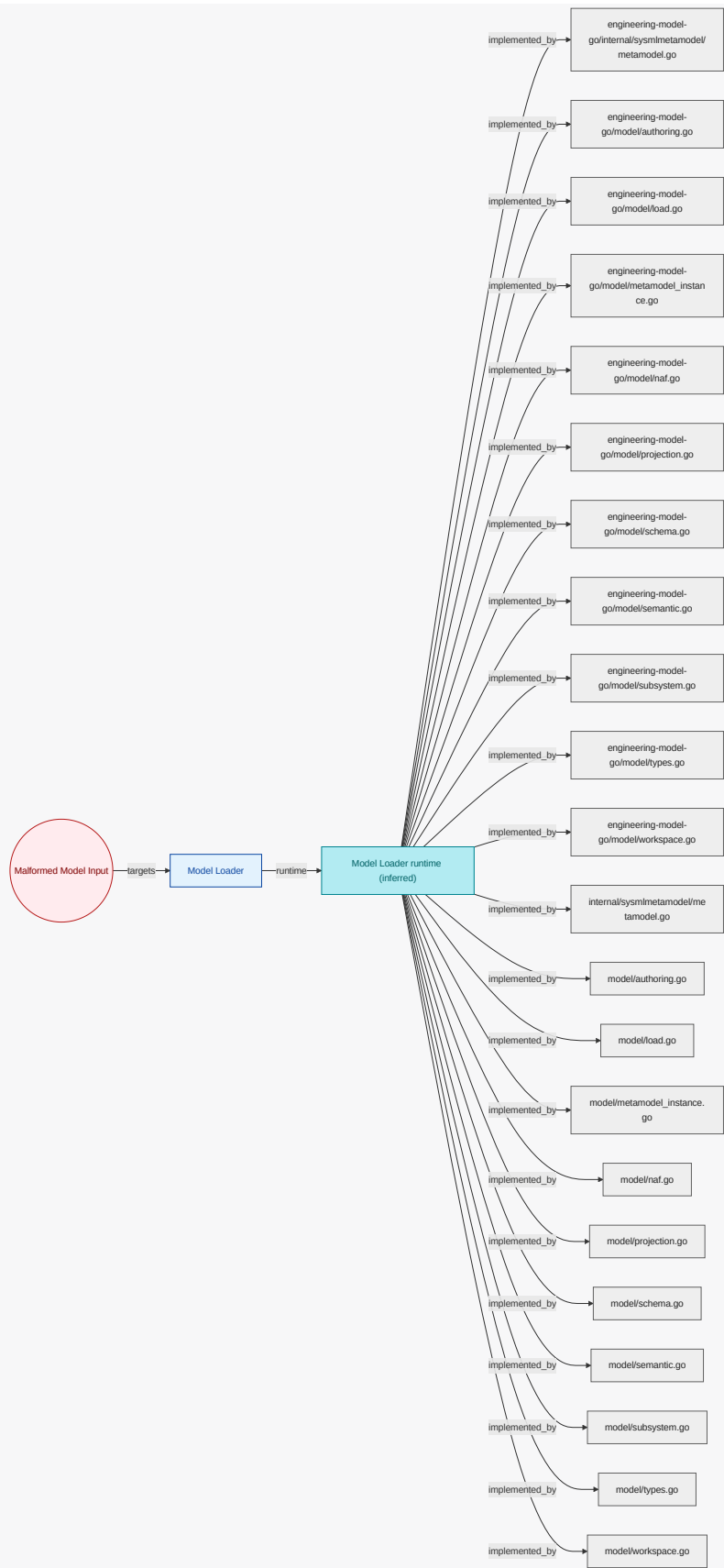
7.10. Attack Vector Chapters

7.10.1. Malformed Model Input (AV-MALFORMED-MODEL-INPUT)

Invalid or ambiguous model content causing incorrect parsing or graph interpretation.

Mitigated by: none

Trust boundaries: none



Model Loader (FU-MODEL-LOADER)

Rejects unknown fields, invalid cross-field constraints, invalid parsing states, and unresolved core model references with path-aware diagnostics.

Attack Surface

none

Security Evidence Present

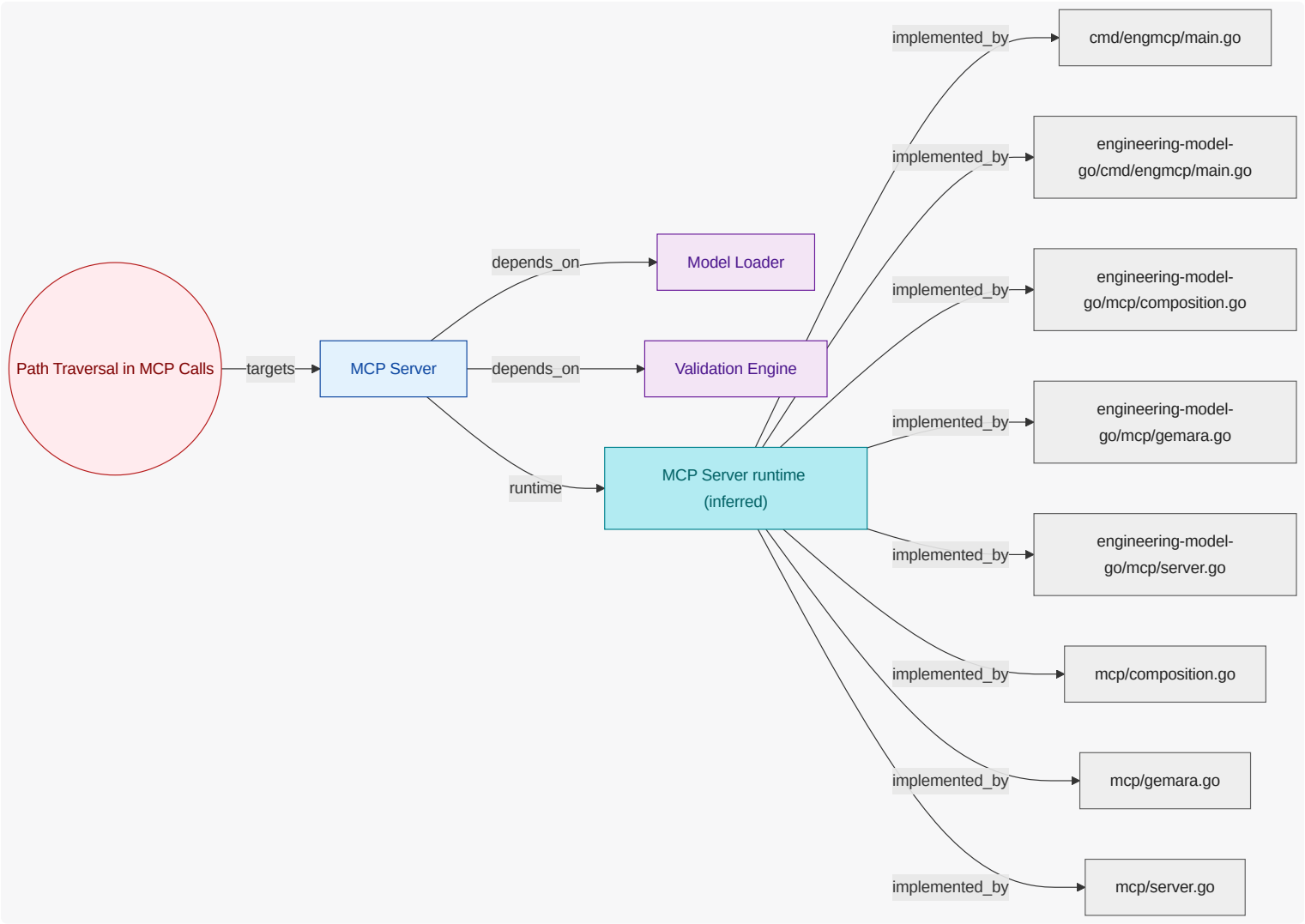
code modules: internal/sysmlmetamodel, model

7.10.2. Path Traversal in MCP Calls (AV-PATH-TRAVERSAL-IN-MCP)

Attempts to read files outside repository root via MCP path arguments.

Mitigated by: MCP Path Boundary Enforcement, Strict MCP Input Schemas

Trust boundaries: Repository Workspace Boundary



MCP Server (FU-MCP-SERVER)

Enforces strict schemas, panic containment, and repo-root path constraints.

Attack Surface

Model Loader, Validation Engine

Security Evidence Present

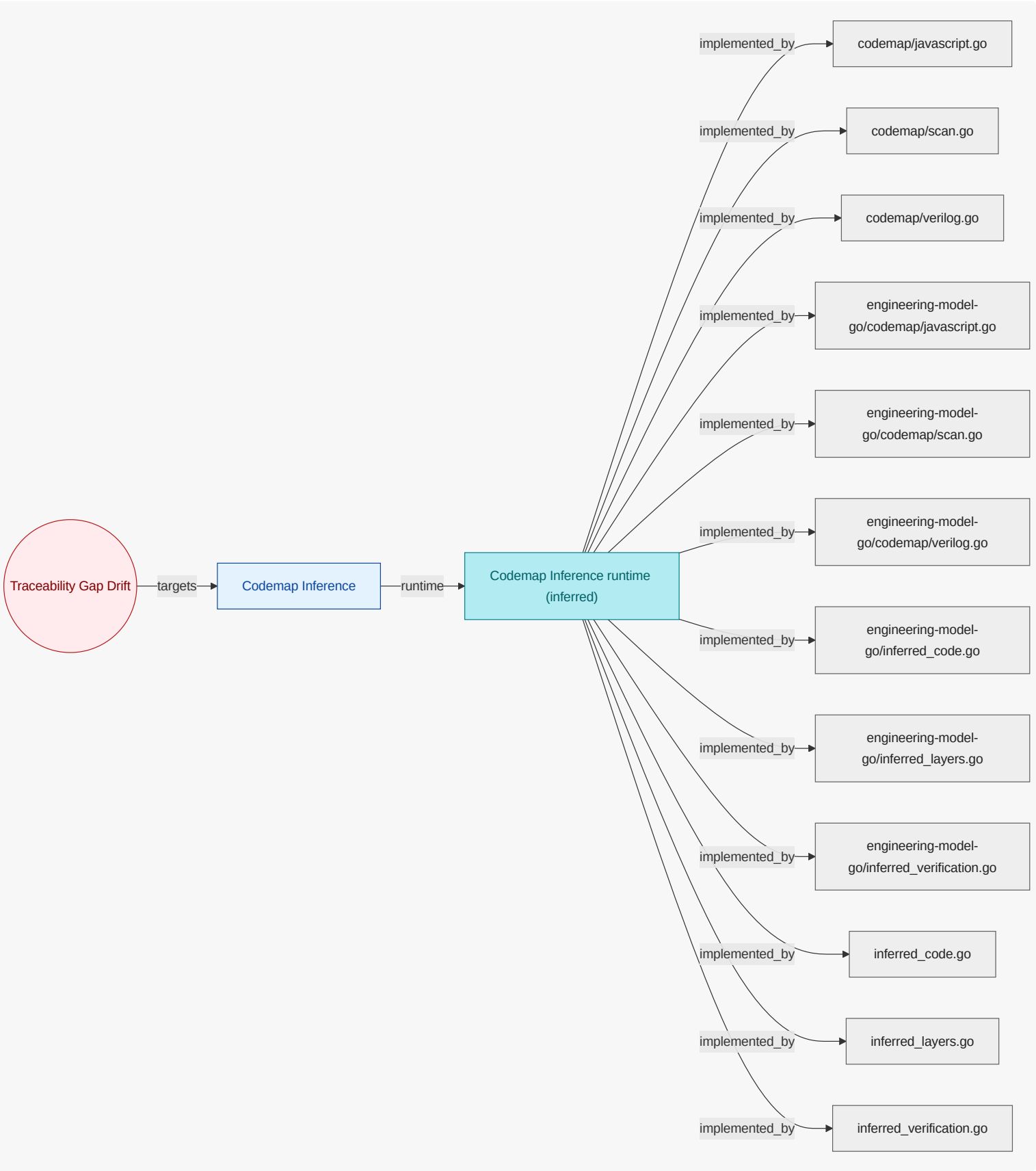
code modules: cmd/engmcp, mcp

7.10.3. Traceability Gap Drift (AV-TRACEABILITY-GAP-DRIFT)

Requirement and verification links diverge from implementation over time.

Mitigated by: Requirement Traceability Coverage

Trust boundaries: Repository Workspace Boundary



Codemap Inference (FU-CODEMAP-INFERENCE)

Limits `inference` scope to trusted repo paths and `evidence` formats.

Attack Surface

`none`

Security Evidence Present

code modules: `codemap`, `inferred_code`, `inferred_layers`, `inferred_verification`

8. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

Authored status: in-review

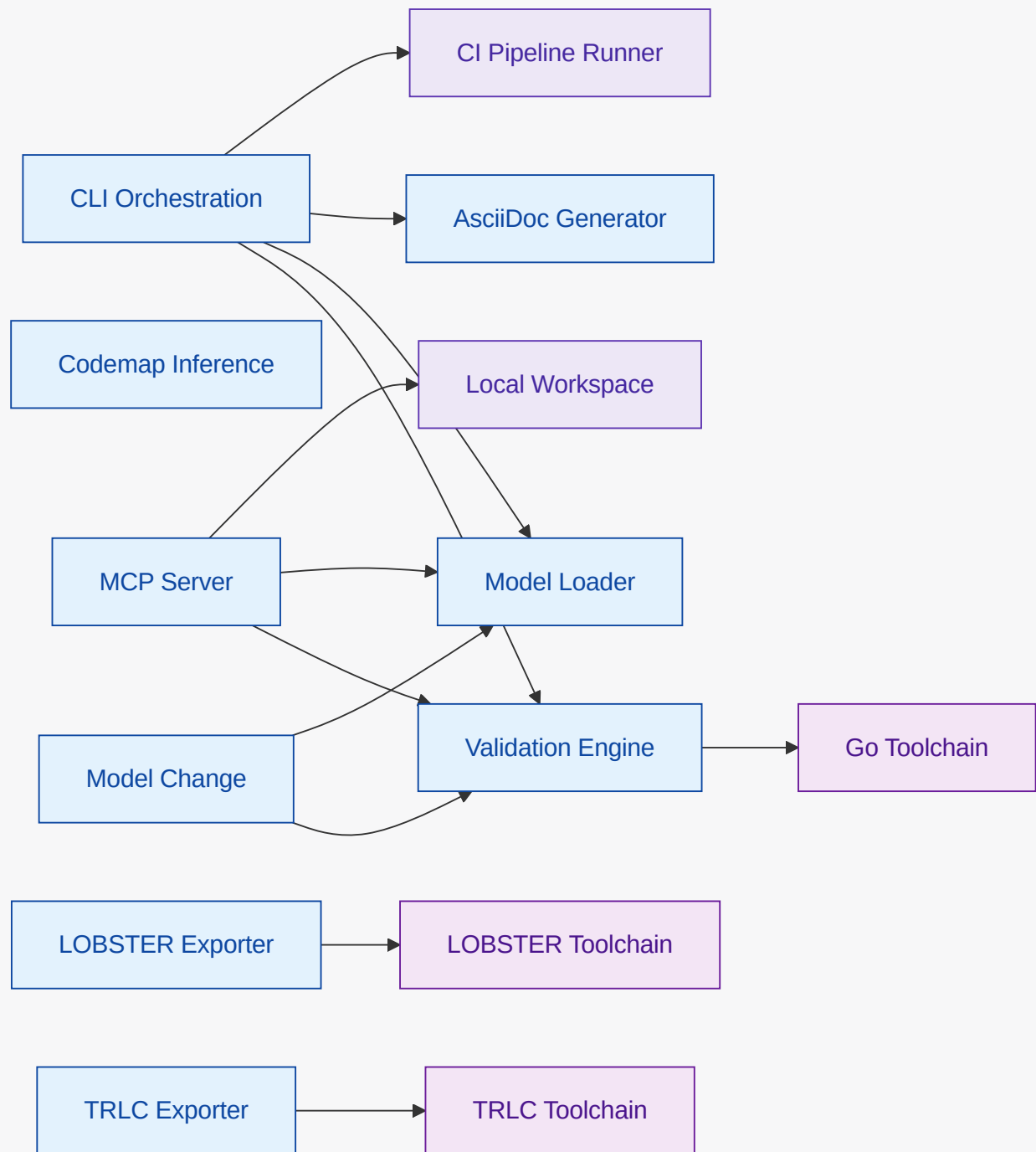
Authored scope and limitations: Traceability scope is seeded and will be tightened as requirements are added.

8.1. What This View Answers

- Which requirements map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

8.2. Authored View Diagram

This diagram follows the selected roots, depth, kinds and relationships of VIEW-TRACEABILITY .



8.3. Coverage Gaps

Coverage summary: 9/9 units have direct evidence coverage in this view.

- **AsciiDoc Generator** has no direct inferred runtime evidence yet.
- **CLI Orchestration** has no direct inferred runtime evidence yet.
- **Codemap Inference** has no direct inferred runtime evidence yet.
- **LOBSTER Exporter** has no direct inferred runtime evidence yet.
- **MCP Server** has no direct inferred runtime evidence yet.
- **Model Change** has no direct inferred runtime evidence yet.
- **Model Loader** has no direct inferred runtime evidence yet.

- `TRLCExporter` has no direct inferred runtime `evidence` yet.
- `ValidationEngine` has no direct inferred runtime `evidence` yet.

8.4. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code `evidence` .

9. State Lifecycle View

Show workflow state transitions, ownership of transitions, and key exceptional paths.

Authored status: draft

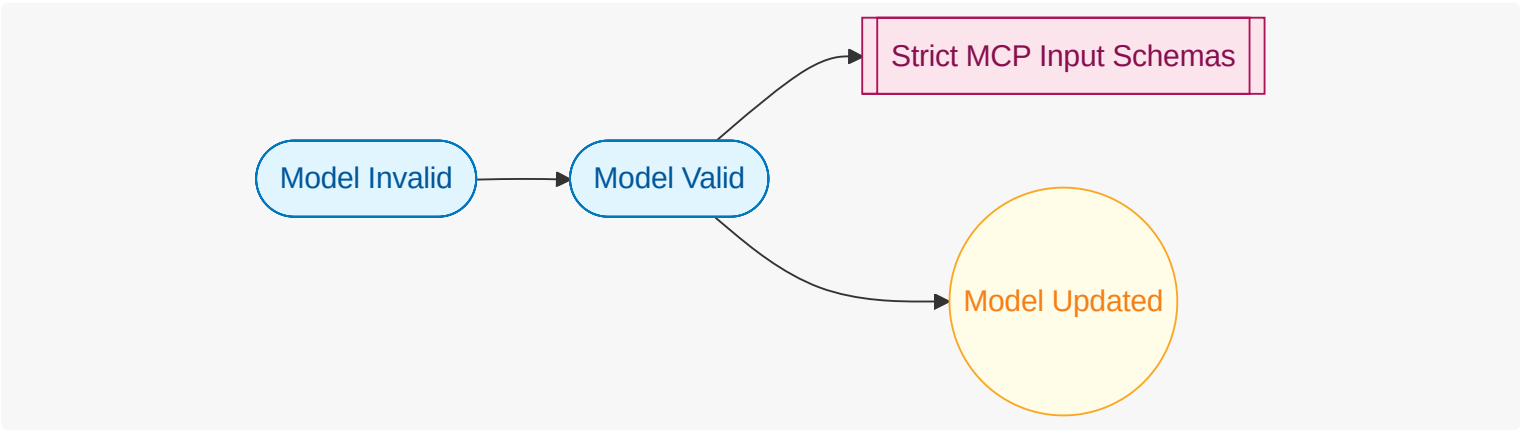
Authored scope and limitations: Captures model validity and artifact freshness lifecycle behavior.

9.1. What This View Answers

- What states exist and which events trigger transitions?
- Which unit owns each state transition?
- Where do automatic and manual flows diverge?

9.2. State Lifecycle Diagram

What this diagram shows: authored lifecycle states, triggering events, guard controls, trust boundaries, and lifecycle transitions for this view.



9.3. Coverage Gaps

Coverage summary: No functional units are in scope for this view .

- No major coverage gaps detected from current authored and inferred inputs.

9.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view .

9.5. Functional Groups and Units (View Scoped)

10. Interaction Flow View

Show user-intent-to-outcome causal flow , including ordered steps, data movement, and error/async branches.

Authored status: draft

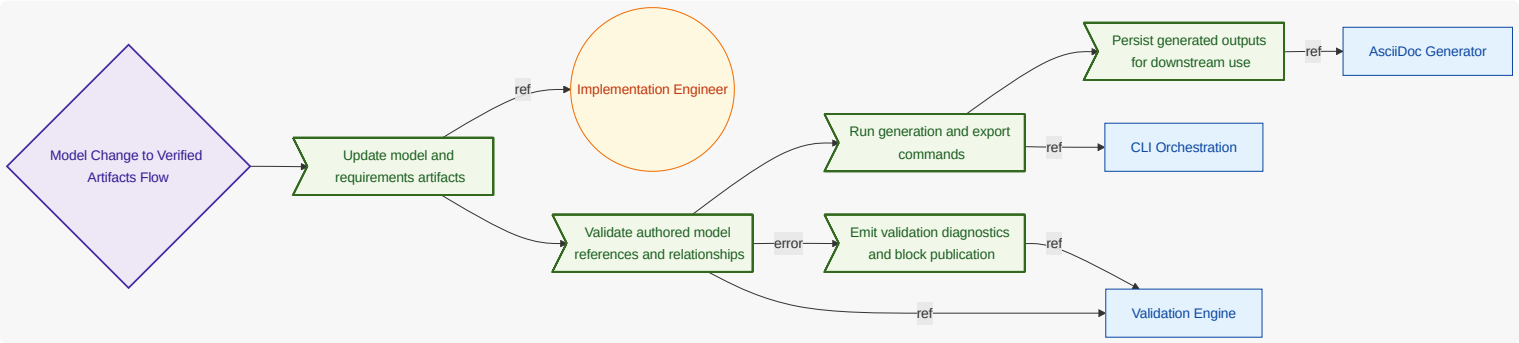
Authored scope and limitations: Core implementation workflow is captured to guide model-driven repository changes.

10.1. What This View Answers

- What happens from user intent to system outcome, step by step?
- What data enters/exits each step and where are decisions made?
- Where are async and error paths and what completes each path?

10.2. Interaction Flow Diagram

What this diagram shows: authored interaction flow steps, step ordering, async markers, and referenced actors/interfaces/units for this view.



10.3. Coverage Gaps

Coverage summary: 3/3 units have direct evidence coverage in this view .

- No major coverage gaps detected from current authored and inferred inputs.

10.4. Recommended Next Evidence Additions

- No immediate evidence additions are required for this view .

10.5. Functional Groups and Units (View Scoped)

10.6. Requirement-to-Unit Mapping (Compact)

What this matrix shows: authored requirement-to-unit mappings in a compact square layout (requirements as rows, functional units as columns).

Functional Unit Columns 1-7

Requirement	FU-AIRBORNE-ASSURANCE-EXPORTER	FU-ALLOCATION-TRACE	FU-ASCIIDOC-GENERATOR	FU-CLI-ORCHESTRATION	FU-CODEMAP-INFERENCE	FU-GEMARA-EXPORTER	FU-LOBSTER-EXPORTER
REQ-EMG-001			X	X			
REQ-EMG-003			X	X			
REQ-EMG-004				X			
REQ-EMG-005				X			
REQ-EMG-006				X			X
REQ-EMG-007							
REQ-EMG-008							
REQ-EMG-009				X			
REQ-EMG-010					X		
REQ-EMG-011							
REQ-EMG-012		X	X		X	X	X
REQ-EMG-013				X			
REQ-EMG-014			X	X			
REQ-EMG-015				X		X	
REQ-EMG-016							
REQ-EMG-017							
REQ-EMG-018							
REQ-EMG-019		X					
REQ-EMG-020		X					
REQ-EMG-021		X					
REQ-EMG-022		X					
REQ-EMG-023		X					
REQ-EMG-024			X				
REQ-EMG-025		X	X				
REQ-EMG-026		X	X				

Requirement	FU-AIRBORNE-ASSURANCE-EXPORTER	FU-ALLOCATION-TRACE	FU-ASCIIDOC-GENERATOR	FU-CLI-ORCHESTRATION	FU-CODEMAP-INFERENCE	FU-GEMARA-EXPORTER	FU-LOBSTER-EXPORTER
REQ-EMG-027							
REQ-EMG-028		X			X		
REQ-EMG-029		X					
REQ-EMG-030		X			X		
REQ-EMG-031							
REQ-EMG-032							
REQ-EMG-033				X			
REQ-EMG-034				X			
REQ-EMG-035							
REQ-EMG-036				X			
REQ-EMG-037				X			
REQ-EMG-038							
REQ-EMG-039				X			
REQ-EMG-040				X			
REQ-EMG-041							
REQ-EMG-042				X			
REQ-EMG-043				X			
REQ-EMG-044				X			
REQ-EMG-045							
REQ-EMG-046				X			
REQ-EMG-047							
REQ-EMG-048							
REQ-EMG-049				X			
REQ-EMG-050			X			X	
REQ-EMG-051				X			

Requirement	FU-AIRBORNE-ASSURANCE-EXPORTER	FU-ALLOCATION-TRACE	FU-ASCIIDOC-GENERATOR	FU-CLI-ORCHESTRATION	FU-CODEMAP-INFERENCE	FU-GEMARA-EXPORTER	FU-LOBSTER-EXPORTER
REQ-EMG-052				X			
REQ-EMG-053	X						
REQ-EMG-054	X						
REQ-EMG-055	X			X			
REQ-EMG-056	X		X				

Functional Unit Columns 8-14

Requirement	FU-MCP-SERVER	FU-MODEL-CHANGE	FU-MODEL-LOADER	FU-NAF-EXPORTER	FU-OSCAL-EXPORTER	FU-STRUCTURIZR-EXPORTER	FU-SYSML-EXPORTER
REQ-EMG-001			X		X	X	
REQ-EMG-003							
REQ-EMG-004							
REQ-EMG-005						X	
REQ-EMG-006							
REQ-EMG-007	X						
REQ-EMG-008	X						
REQ-EMG-009							
REQ-EMG-010							
REQ-EMG-011						X	
REQ-EMG-012					X	X	X
REQ-EMG-013					X		
REQ-EMG-014							
REQ-EMG-015	X						
REQ-EMG-016			X				
REQ-EMG-017							
REQ-EMG-018							

Requirement	FU-MCP-SERVER	FU-MODEL-CHANGE	FU-MODEL-LOADER	FU-NAF-EXPORTER	FU-OSCAL-EXPORTER	FU-STRUCTURIZR-EXPORTER	FU-SYSML-EXPORTER
REQ-EMG-019							
REQ-EMG-020							
REQ-EMG-021							
REQ-EMG-022							
REQ-EMG-023							
REQ-EMG-024							
REQ-EMG-025							
REQ-EMG-026							
REQ-EMG-027			X				
REQ-EMG-028							
REQ-EMG-029							
REQ-EMG-030							
REQ-EMG-031		X	X				
REQ-EMG-032		X					
REQ-EMG-033		X					
REQ-EMG-034		X					
REQ-EMG-035			X				X
REQ-EMG-036							X
REQ-EMG-037							X
REQ-EMG-038			X				X
REQ-EMG-039							X
REQ-EMG-040							X
REQ-EMG-041			X	X			
REQ-EMG-042				X			
REQ-EMG-043				X			
REQ-EMG-044			X				

Requirement	FU-MCP-SERVER	FU-MODEL-CHANGE	FU-MODEL-LOADER	FU-NAF-EXPORTER	FU-OSCAL-EXPORTER	FU-STRUCTURIZR-EXPORTER	FU-SYSML-EXPORTER
REQ-EMG-045	X		X				
REQ-EMG-046			X				
REQ-EMG-047			X				
REQ-EMG-048			X				
REQ-EMG-049							
REQ-EMG-050					X		X
REQ-EMG-051			X				
REQ-EMG-052			X				X
REQ-EMG-053			X				
REQ-EMG-054							
REQ-EMG-055							
REQ-EMG-056							

Functional Unit Columns 15-19

Requirement	FU-SYSTEM-COMPOSITION	FU-THREAT-EXPORTER	FU-TRLC-EXPORTER	FU-VALIDATION-ENGINE	FU-VIEW-PROJECTION
REQ-EMG-001		X		X	
REQ-EMG-003					X
REQ-EMG-004		X			
REQ-EMG-005					
REQ-EMG-006			X		
REQ-EMG-007					
REQ-EMG-008					
REQ-EMG-009				X	
REQ-EMG-010					
REQ-EMG-011		X		X	

Requirement	FU-SYSTEM-COMPOSITION	FU-THREAT-EXPORTER	FU-TRLC-EXPORTER	FU-VALIDATION-ENGINE	FU-VIEW-PROJECTION
REQ-EMG-012		X	X	X	
REQ-EMG-013					
REQ-EMG-014					
REQ-EMG-015					
REQ-EMG-016	X				
REQ-EMG-017	X				
REQ-EMG-018	X			X	
REQ-EMG-019	X				
REQ-EMG-020					
REQ-EMG-021				X	
REQ-EMG-022				X	
REQ-EMG-023				X	
REQ-EMG-024					X
REQ-EMG-025					
REQ-EMG-026				X	
REQ-EMG-027	X				
REQ-EMG-028				X	
REQ-EMG-029				X	
REQ-EMG-030					
REQ-EMG-031					
REQ-EMG-032				X	
REQ-EMG-033				X	
REQ-EMG-034					
REQ-EMG-035				X	
REQ-EMG-036				X	
REQ-EMG-037				X	

Requirement	FU-SYSTEM-COMPOSITION	FU-THREAT-EXPORTER	FU-TRLC-EXPORTER	FU-VALIDATION-ENGINE	FU-VIEW-PROJECTION
REQ-EMG-038				X	
REQ-EMG-039				X	
REQ-EMG-040				X	
REQ-EMG-041				X	
REQ-EMG-042					X
REQ-EMG-043				X	
REQ-EMG-044				X	
REQ-EMG-045					
REQ-EMG-046				X	
REQ-EMG-047	X				
REQ-EMG-048	X				
REQ-EMG-049	X			X	
REQ-EMG-050	X			X	
REQ-EMG-051	X			X	
REQ-EMG-052					X
REQ-EMG-053				X	
REQ-EMG-054				X	
REQ-EMG-055					
REQ-EMG-056				X	

10.7. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2	REQ-COF-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8	REQ-COF-002	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC	REQ-COF-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB	REQ-COF-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB	REQ-COF-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC	REQ-COF-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8	REQ-COF-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-BOUNDARY_TEST-2A7639	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-E2E_TEST-BE56F8	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-HARDWARE_TEST-445BFD	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_COMPLETE_DIAGRAM_TEST-8C18F7	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_EVIDENCE_TABLE_TEST-963640	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST- ASCIIDOC_LINKING_UNITS_TEST- 22FDE2	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- ASCIIDOC_PUBLICATION_SCOPE_TEST- 1BE464	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_TEST-F60765	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- ASCIIDOC_TEST_EVIDENCE_TEST- C6896C	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_VIEWS_TEST- BFC7F2	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-BOUNDARY_TEST- 2F6002	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-HARDWARE_TEST- 8CB069	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-PROJECT_TEST-2ECF72	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-RENDER_TEST-EECC23	REQ-EMG-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- THREAT_MODEL_EXPORT_TEST- 721A8D	REQ-EMG-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- STRUCTURIZR_EXPORT_TEST-886374	REQ-EMG-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOBSTER_EXPORT_TEST- 1C1C7E	REQ-EMG-006	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-TRLC_EXPORT_TEST- DD1345	REQ-EMG-006	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST- AB66B8	REQ-EMG-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-HARDWARE_TEST- F0D9BD	REQ-EMG-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-6B61AE	REQ-EMG-007	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-SERVER_TEST-389957	REQ-EMG-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-6B61AE	REQ-EMG-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SERVER_TEST-389957	REQ-EMG-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-BOUNDARY_TEST-2A7639	REQ-EMG-009	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26	REQ-EMG-009	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-HARDWARE_TEST-445BFD	REQ-EMG-009	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-009	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_TEST_EVIDENCE_TEST-C6896C	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_CODE_TEST-3CA82E	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_JAVASCRIPT_TEST-52EB3E	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-INFERRED_VERILOG_TEST-32D5FA	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-JAVASCRIPT_TEST-4E9F26	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCAN_TEST-B04F81	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VERILOG_TEST-44F499	REQ-EMG-010	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D	REQ-EMG-011	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-011	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COFFEE_SIX_VIEW_TEST-4B177C	REQ-EMG-012	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GEMARA_EXPORT_TEST-45C982	REQ-EMG-012	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E	REQ-EMG-012	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9	REQ-EMG-012	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_SSP_TEST-4B08EF	REQ-EMG-012	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-RENDER_TEST-5A1C59	REQ-EMG-012	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D	REQ-EMG-012	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9	REQ-EMG-013	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_PROFILE_TEST-9234C6	REQ-EMG-013	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_SSP_TEST-4B08EF	REQ-EMG-013	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-ASCIIDOC_TEST-F60765	REQ-EMG-014	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GEMARA_EXPORT_TEST-45C982	REQ-EMG-015	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA	REQ-EMG-015	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-016	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST- ASCIIDOC_HARDWARE_LABELS_TEST- AFE4FA	REQ-EMG-024	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-TRACE_MATRIX_TEST- 9FB80C	REQ-EMG-028	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-TRACE_MATRIX_TEST- 9FB80C	REQ-EMG-029	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- ASCIIDOC_TEST_EVIDENCE_TEST- C6896C	REQ-EMG-030	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST- AB66B8	REQ-EMG-030	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- INFERRED_JAVASCRIPT_TEST-52EB3E	REQ-EMG-030	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- INFERRED_VERILOG_TEST-32D5FA	REQ-EMG-030	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-TRACE_MATRIX_TEST- 9FB80C	REQ-EMG-030	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- REQUIREMENTS_DELTA_TEST-64E119	REQ-EMG-031	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- REQUIREMENTS_DELTA_TEST-64E119	REQ-EMG-032	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- REQUIREMENTS_DELTA_TEST-64E119	REQ-EMG-033	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-035	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-PROJECTION_TEST- D8D752	REQ-EMG-035	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_TEST-AB1606	REQ-EMG-035	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST- SCHEMA_V2_PROJECTION_TEST- 81EEBD	REQ-EMG-035	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SEMANTIC_TEST-CB8EF3	REQ-EMG-035	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E	REQ-EMG-035	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-035	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-AAEB3C	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-PROJECTION_TEST-D8D752	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_V2_PROJECTION_TEST-81EEBD	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SEMANTIC_TEST-CB8EF3	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_INTERCHANGE_TEST-CF2F5A	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-036	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-AAEB3C	REQ-EMG-037	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_V2_PROJECTION_TEST-81EEBD	REQ-EMG-037	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SEMANTIC_TEST-CB8EF3	REQ-EMG-037	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_INTERCHANGE_TEST-CF2F5A	REQ-EMG-037	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-037	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-METAMODEL_INSTANCE_TEST-33196D	REQ-EMG-038	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-METAMODEL_TEST-81AF9A	REQ-EMG-038	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-AAEB3C	REQ-EMG-039	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-METAMODEL_INSTANCE_TEST-33196D	REQ-EMG-039	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-METAMODEL_TEST-81AF9A	REQ-EMG-039	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_COVERAGE_TEST-9C7166	REQ-EMG-039	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E	REQ-EMG-039	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-AAEB3C	REQ-EMG-040	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-METAMODEL_INSTANCE_TEST-33196D	REQ-EMG-040	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_COVERAGE_TEST-9C7166	REQ-EMG-040	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E	REQ-EMG-040	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_INTERCHANGE_TEST-CF2F5A	REQ-EMG-040	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-NAF_EXPORT_TEST-26D0A4	REQ-EMG-041	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_TEST-AB1606	REQ-EMG-041	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-C71B6B	REQ-EMG-042	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-NAF_EXPORT_TEST-26D0A4	REQ-EMG-042	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-MAIN_TEST-C71B6B	REQ-EMG-043	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-NAF_EXPORT_TEST-26D0A4	REQ-EMG-043	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-SCHEMA_TEST-AB1606	REQ-EMG-043	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-044	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_TEST-AB1606	REQ-EMG-044	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-044	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-045	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SERVER_TEST-389957	REQ-EMG-045	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-046	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_TEST-AB1606	REQ-EMG-046	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_V2_FIXTURE_TEST-1B8AD1	REQ-EMG-046	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-VALIDATE_TEST-498C59	REQ-EMG-046	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_OCI_TEST-D86C11	REQ-EMG-047	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-047	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-047	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SERVER_TEST-389957	REQ-EMG-047	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E	REQ-EMG-047	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_OCI_TEST-D86C11	REQ-EMG-048	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-048	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-048	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SERVER_TEST-389957	REQ-EMG-048	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_OCI_TEST-D86C11	REQ-EMG-049	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-049	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-049	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-BOUNDARY_TEST-2A7639	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-BOUNDARY_TEST-2F6002	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_PROJECTION_TEST-71E4C3	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-EXAMPLES_COMPANY_TEST-A50783	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-HARDWARE_TEST-445BFD	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-HARDWARE_TEST-8CB069	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-HARDWARE_TEST-F0D9BD	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-OSCAL_SSP_TEST-4B08EF	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-RENDER_TEST-EECC23	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E	REQ-EMG-050	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-TEST-COMPOSITION_PROJECTION_TEST-71E4C3	REQ-EMG-051	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-8FE8F1	REQ-EMG-051	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COMPOSITION_TEST-AB66B8	REQ-EMG-051	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA	REQ-EMG-051	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-051	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-COFFEE_SIX_VIEW_TEST-4B177C	REQ-EMG-052	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-052	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-RENDER_TEST-5A1C59	REQ-EMG-052	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCHEMA_TEST-AB1606	REQ-EMG-052	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-AIRBORNE_ASSURANCE_TEST-E46ECA	REQ-EMG-053	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-AVIATION_TEST-42E2CB	REQ-EMG-053	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-053	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-AVIATION_TEST-42E2CB	REQ-EMG-054	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-AIRBORNE_ASSURANCE_TEST-E46ECA	REQ-EMG-055	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-AIRBORNE_ASSURANCE_TEST-E46ECA	REQ-EMG-056	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-LOAD_TEST-A58EB3	REQ-EMG-056	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367	REQ-PAY-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163	REQ-PAY-002	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671	REQ-PAY-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD	REQ-PAY-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163	REQ-PAY-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4	REQ-PAY-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671	REQ-PAY-006	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD	REQ-PAY-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-002	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-003	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003	REQ-PRR-004	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6	REQ-PRR-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F	REQ-PRR-005	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003	REQ-PRR-006	not-run	none; No parsed test-result artifact for this requirement yet.

Check ID	Requirement	Result	Evidence / Notes
VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4	REQ-PRR-007	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE	REQ-PRR-008	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCAN_TEST-B04F81	REQ-SAMPLE-001	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-SCAN_TEST-B04F81	REQ-SAMPLE-002	not-run	none; No parsed test-result artifact for this requirement yet.
VER-INF-TEST-JAVASCRIPT_TEST-4E9F26	REQ-WEB-001	not-run	none; No parsed test-result artifact for this requirement yet.

11. Traceability Appendix

11.1. Requirement Details

11.1.1. REQ-EMG-001

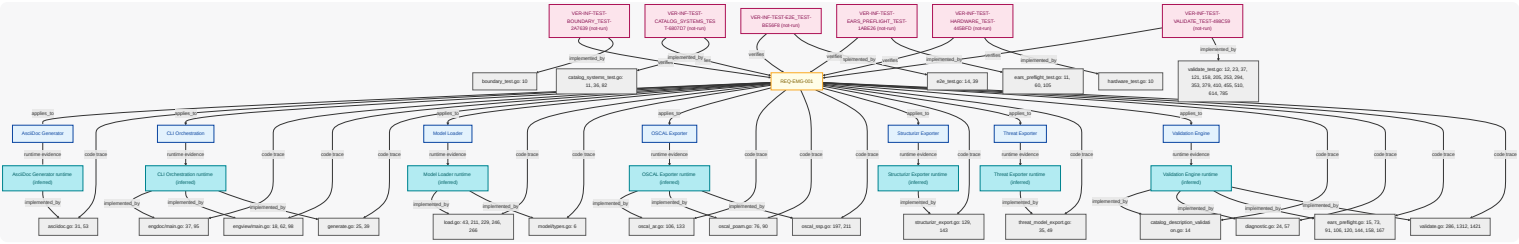
While local development mode is active , when model updated event is received , the engineering model go shall validate architecture model before generation run is requested .

NOTE

Validation must gate all generation flows.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.2. REQ-EMG-003

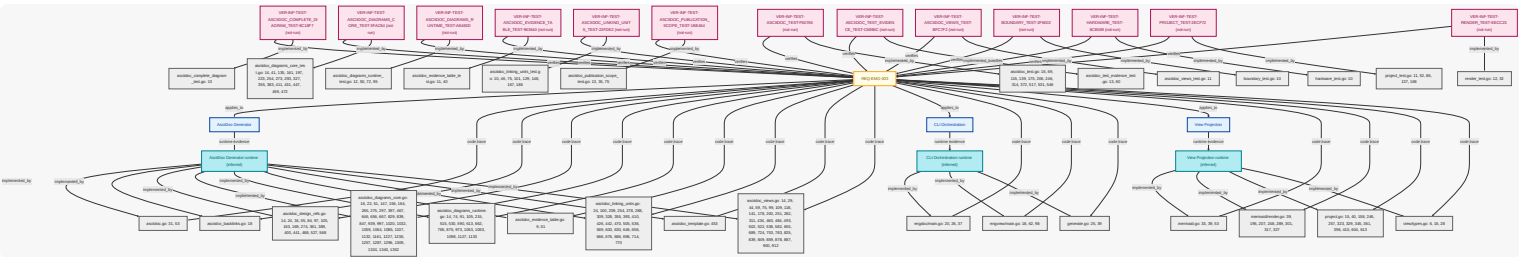
While model is valid , when generation run is requested , the engineering model go shall produce AsciiDoc architecture publication output without whitespace-only lines.

NOTE

Human-readable architecture output remains required and shall be clean for version control and downstream processing.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.3. REQ-EMG-004

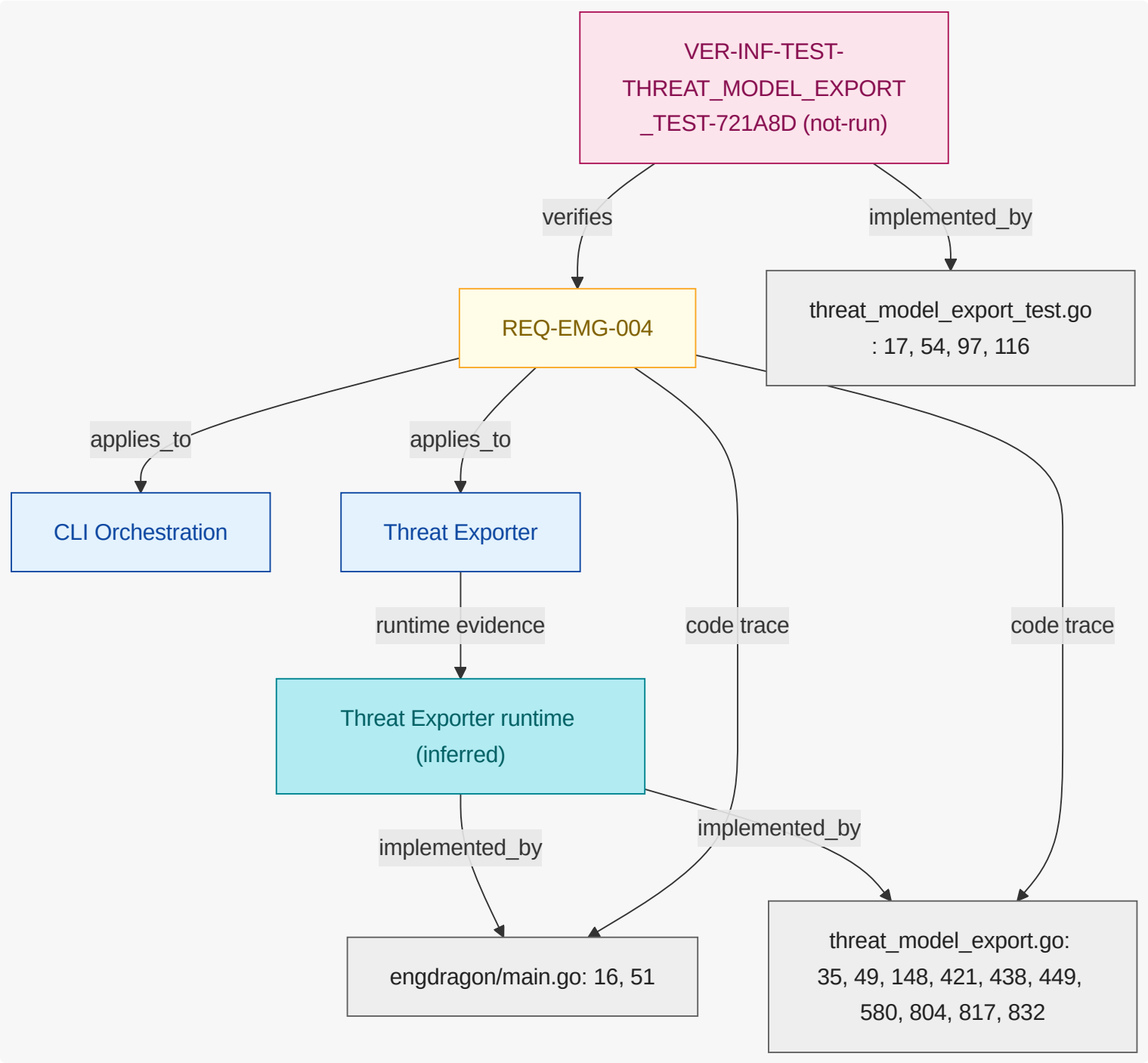
Where threat export is enabled , when generation run is requested , the engineering model go shall export threat dragon json artifact plus open threat model artifact.

NOTE

Threat model interchange output contract.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



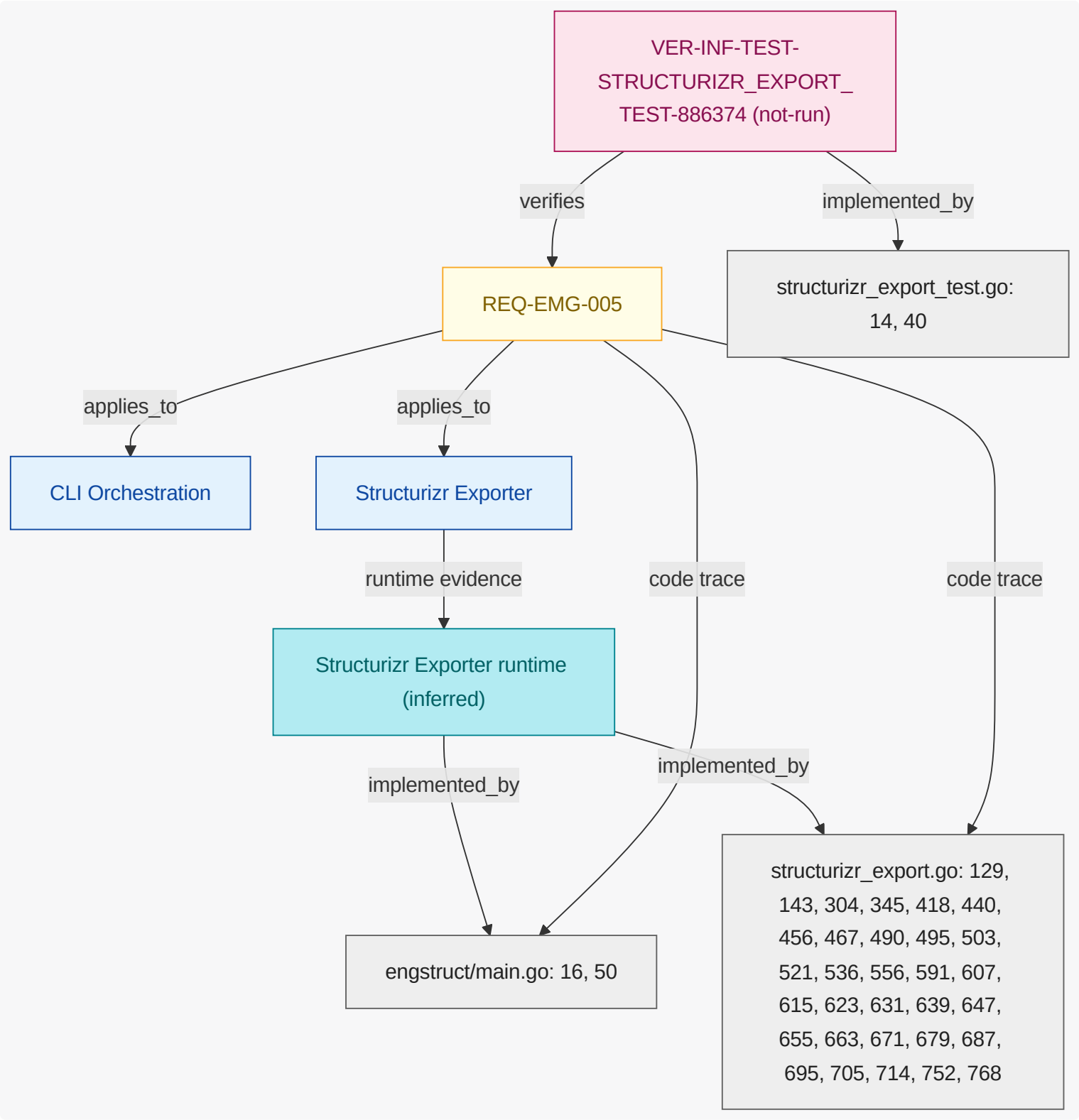
11.1.4. REQ-EMG-005

Where structurizr export is enabled , when generation run is requested , the engineering model go shall export structurizr dsl artifact .

NOTE

Structurizr DSL is exported for C4 visualization; the exported DSL is validated best-effort by the validate-all gauntlet when a Structurizr validator is available.

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



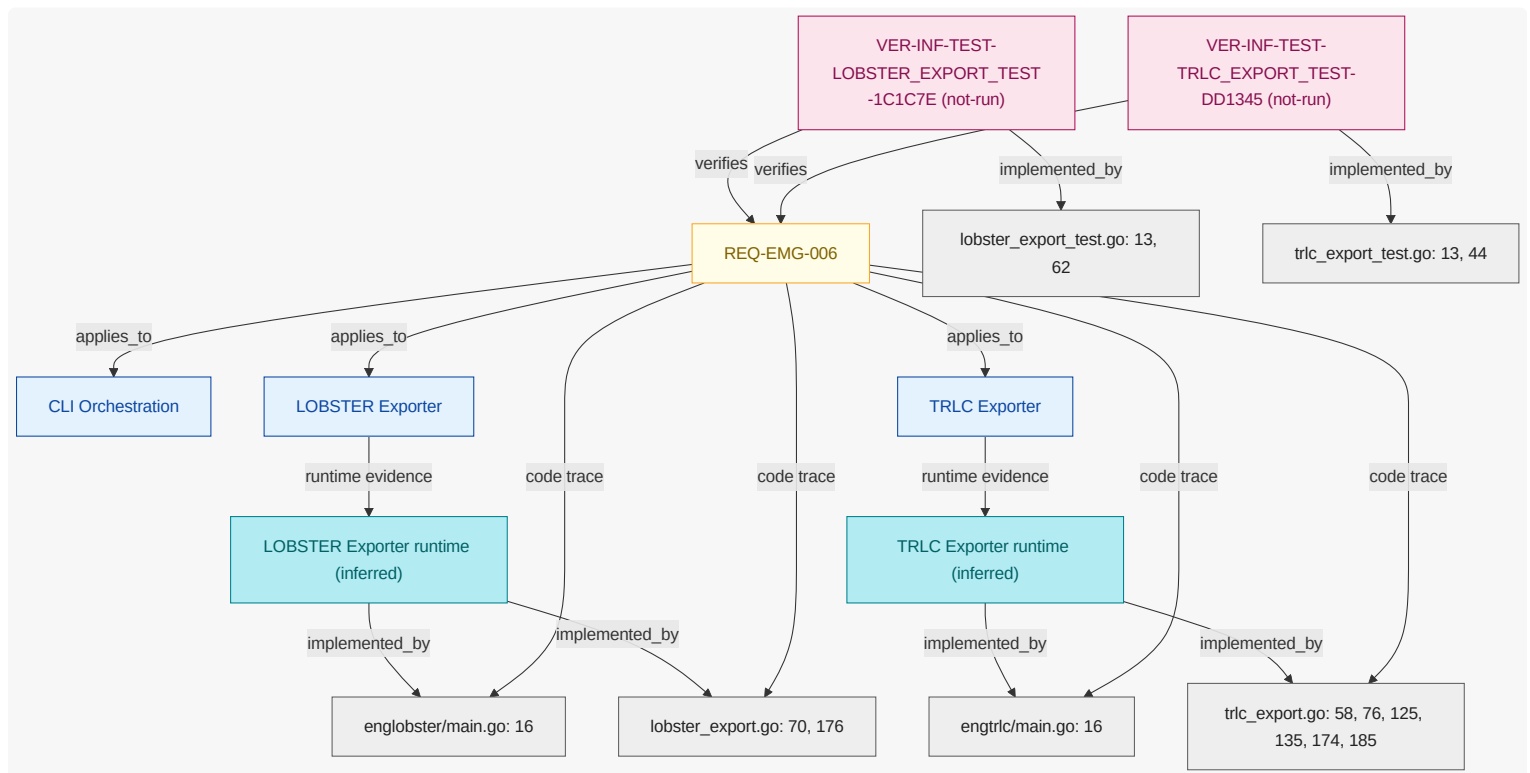
11.1.5. REQ-EMG-006

Where traceability export is enabled, when generation run is requested, the engineering model go shall produce trlc package and lobster trace report artifacts.

NOTE Traceability outputs required for assurance workflows.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



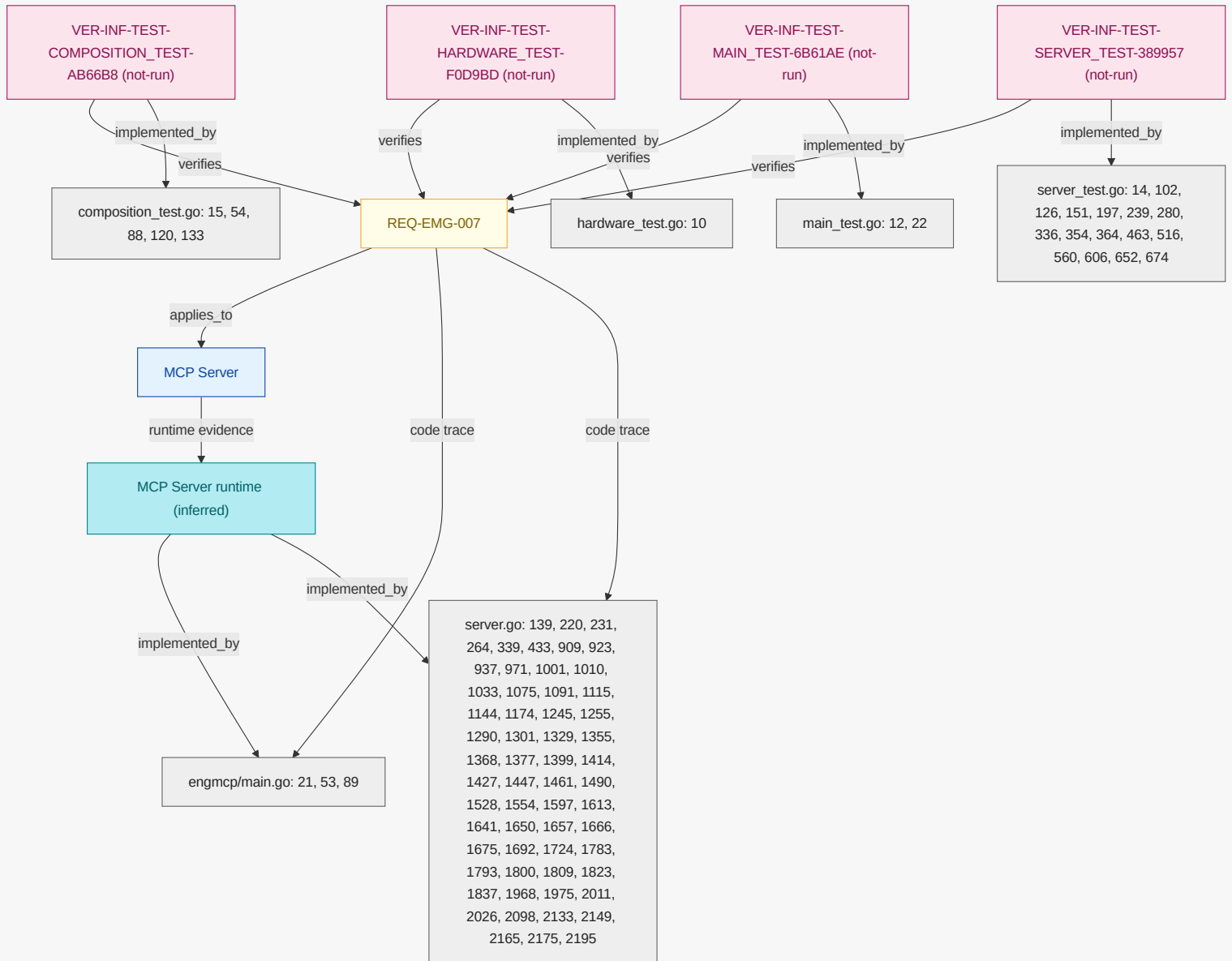
11.1.6. REQ-EMG-007

While mcp server is enabled, when mcp tool call is received and catalog references resolve successfully, the engineering model go shall return mcp tool result with schema version and structured error fields.

NOTE MCP response contract stability requirement.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.7. REQ-EMG-008

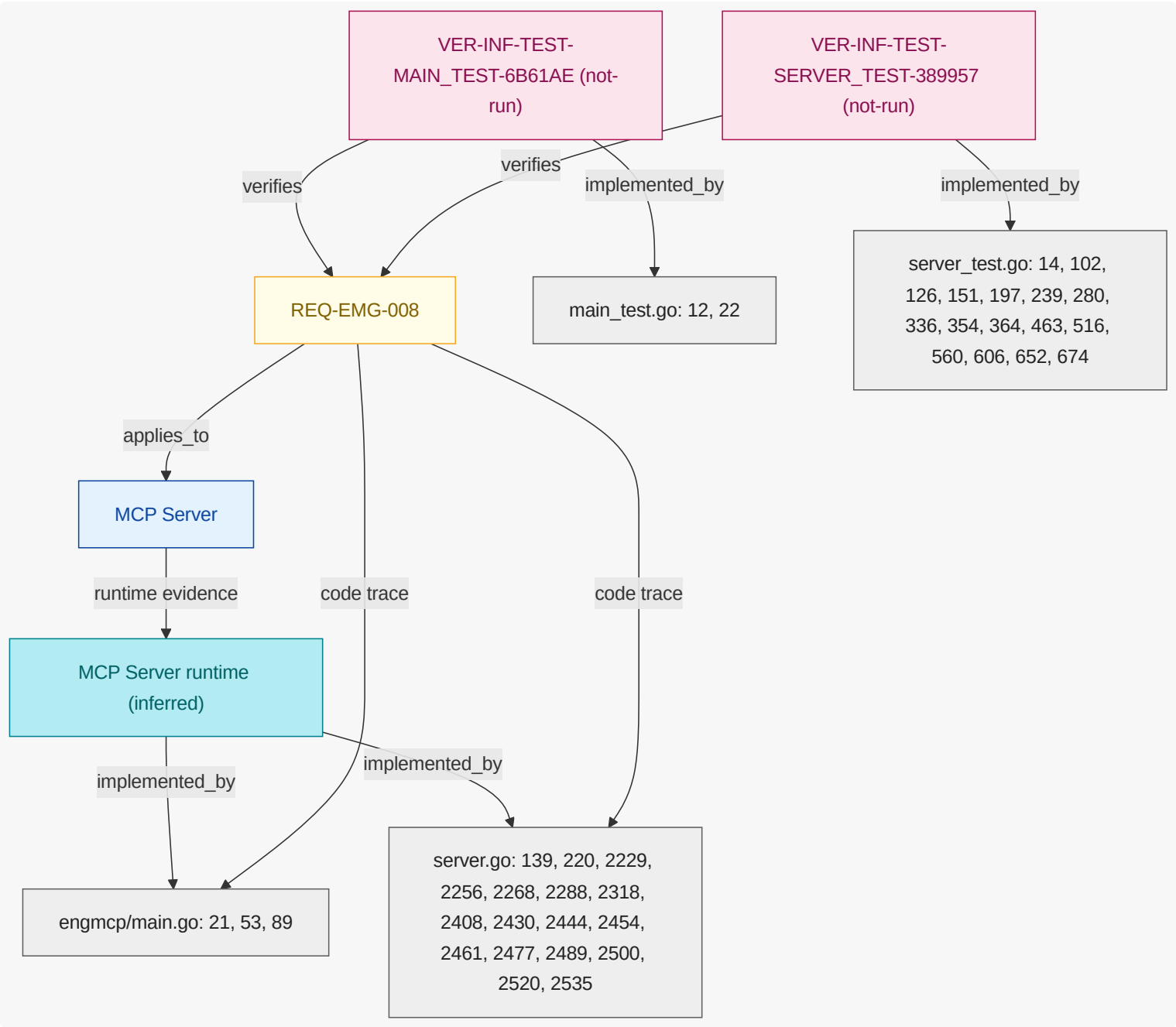
When mcp tool call is received , the engineering model go shall reject path arguments outside repository workspace boundary .

NOTE

Filesystem trust boundary guardrail.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.8. REQ-EMG-009

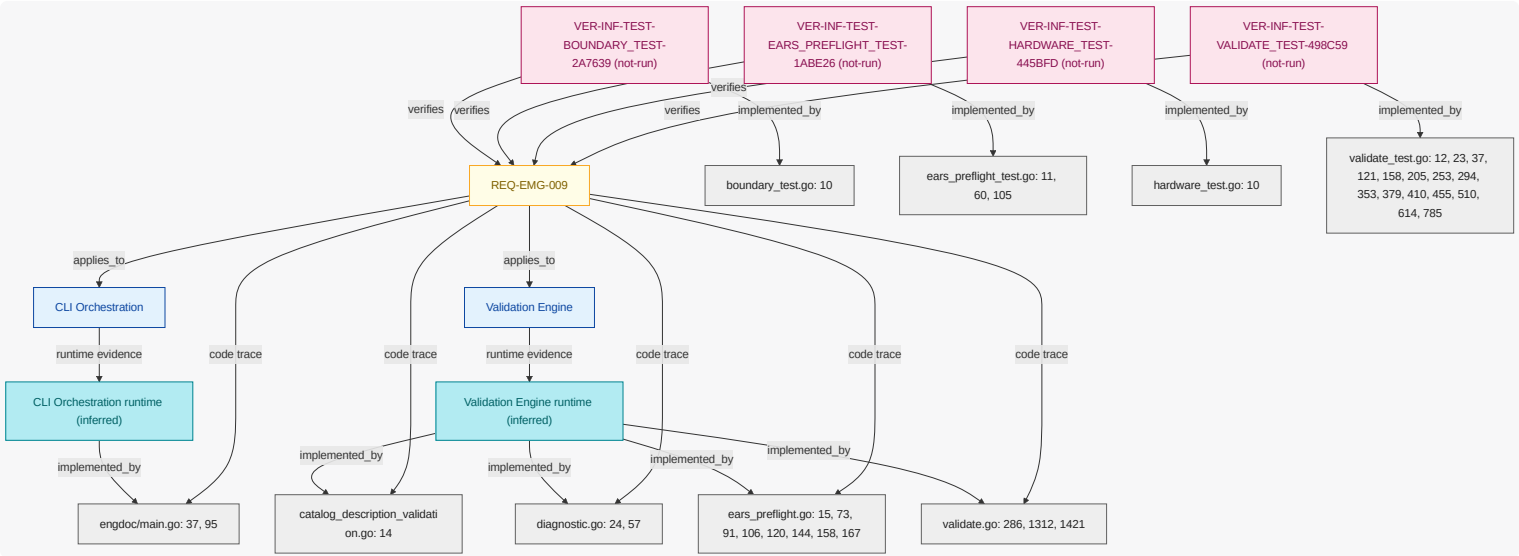
While ci validation mode is active , when generation run is requested , the engineering model go shall execute validation plus tests before artifact generated event is raised .

NOTE

CI installs only the runtimes required by validation , uses mutually compatible exact validator versions and supported command-line semantics for those versions, and configures dependency caches only when their native project metadata exists, so setup and validation do not fail before the validation gauntlet completes.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.9. REQ-EMG-010

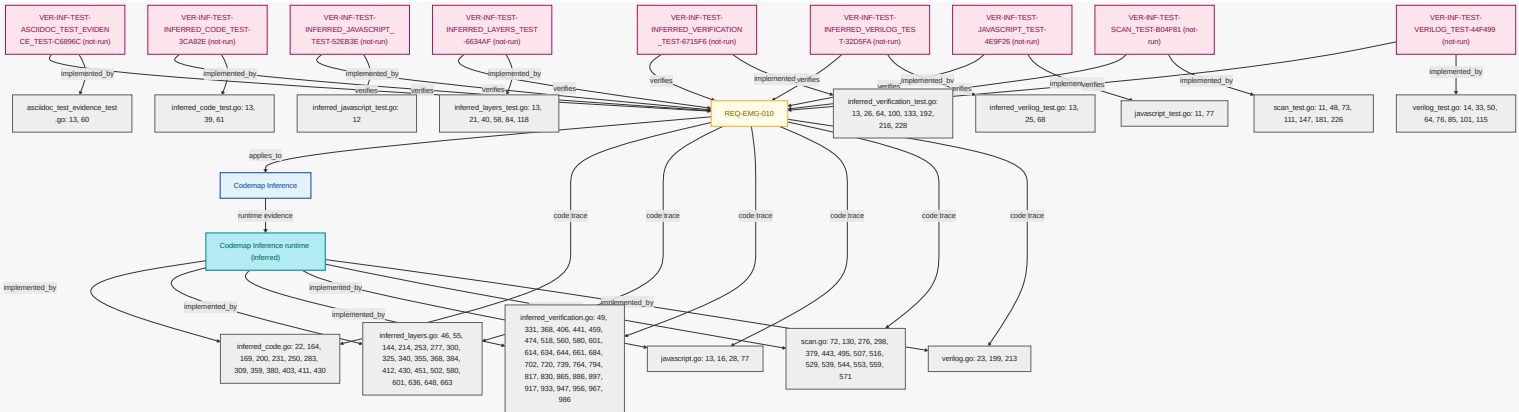
Where requirement trace is complete, when model updated event is received, the engineering model go shall infer traceability coverage from trace markers.

NOTE

Links runtime, code, and test markers to modeled requirement coverage. JavaScript declarations and lexical Verilog modules support explicit source links; Verilog extraction does not elaborate HDL or expand macros. Individual source files and overlapping source directories preserve distinct file identities without double counting. Code and metadata scanning excludes dependency caches (node_modules, vendor) and dot-directories (.git, .engmod, tool caches) so coverage reflects only committed first-party source and stays reproducible.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.10. REQ-EMG-011

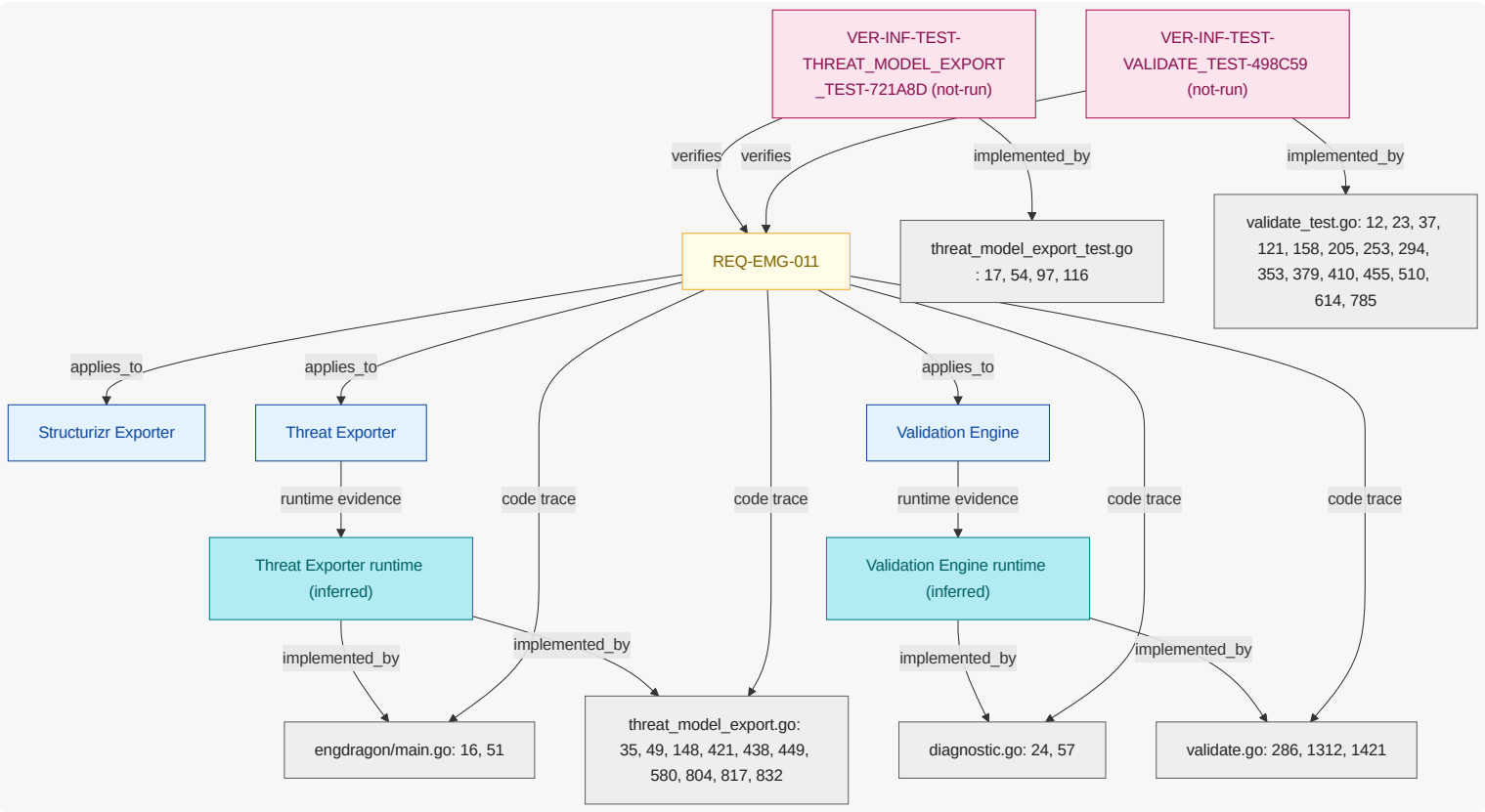
Where external schema dependencies are available, when generation run is requested, the engineering model go shall fail validation if schema checks do not match expected constraints.

NOTE

Protect export integrity when schema trust fails.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.11. REQ-EMG-012

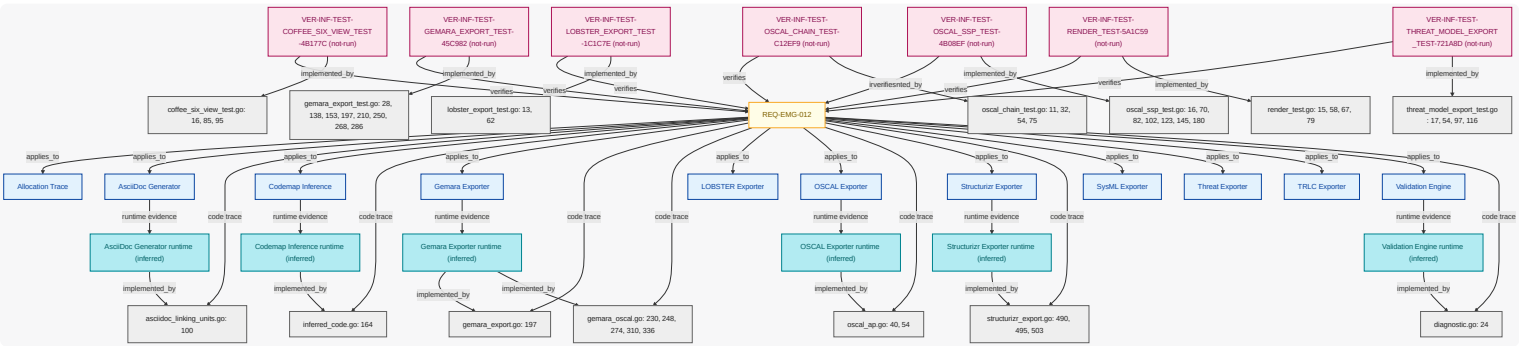
When generation run is requested , the engineering model go shall generate artifacts deterministically so that regeneration is reproducible.

NOTE

Generation uses stable ordering and controlled metadata so maintained artifacts are bit-for-bit reproducible. The validation gauntlet regenerates every maintained example and applies authoritative parsers or schemas where available plus semantic and idiomatic checks for each published format.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.12. REQ-EMG-013

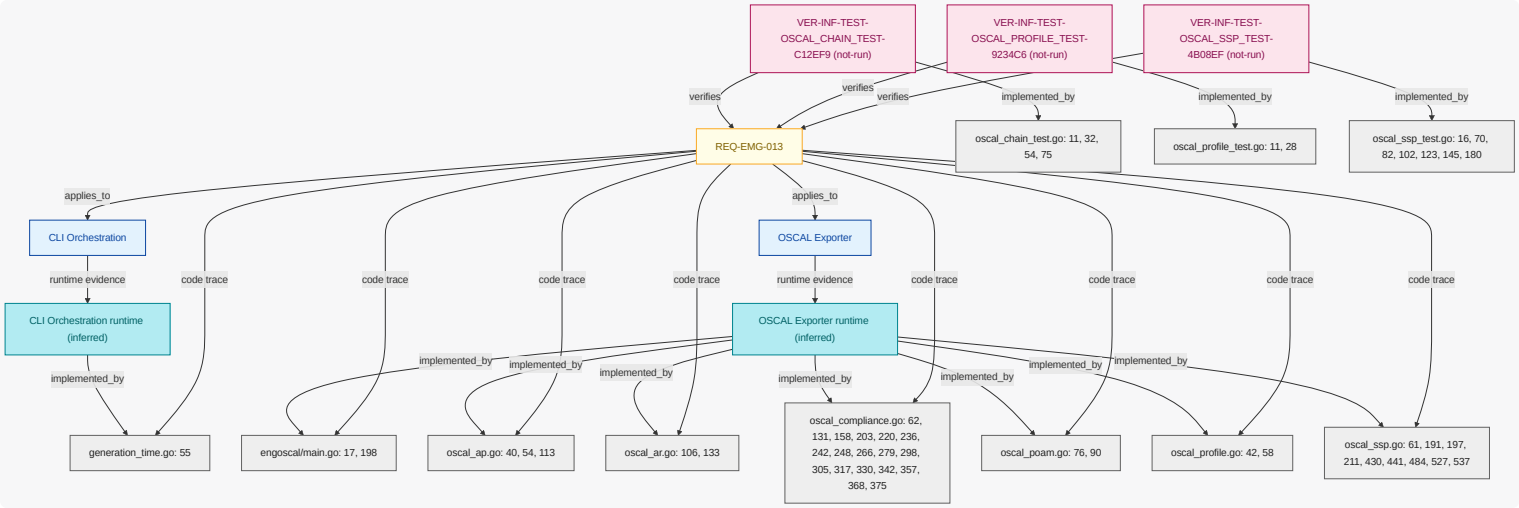
Where traceability export is enabled , when generation run is requested , the engineering model go shall export an OSCAL aggregate profile plus SSP plus assessment plan plus assessment results plus POA&M artifacts when the corresponding compliance and remediation content is applicable.

NOTE

OSCAL documents use version 1.1.2 structures, deterministic metadata, resolvable sibling import links, aggregate every implemented compliance framework, and omit inapplicable documents rather than emitting schema-invalid empty artifacts.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.13. REQ-EMG-014

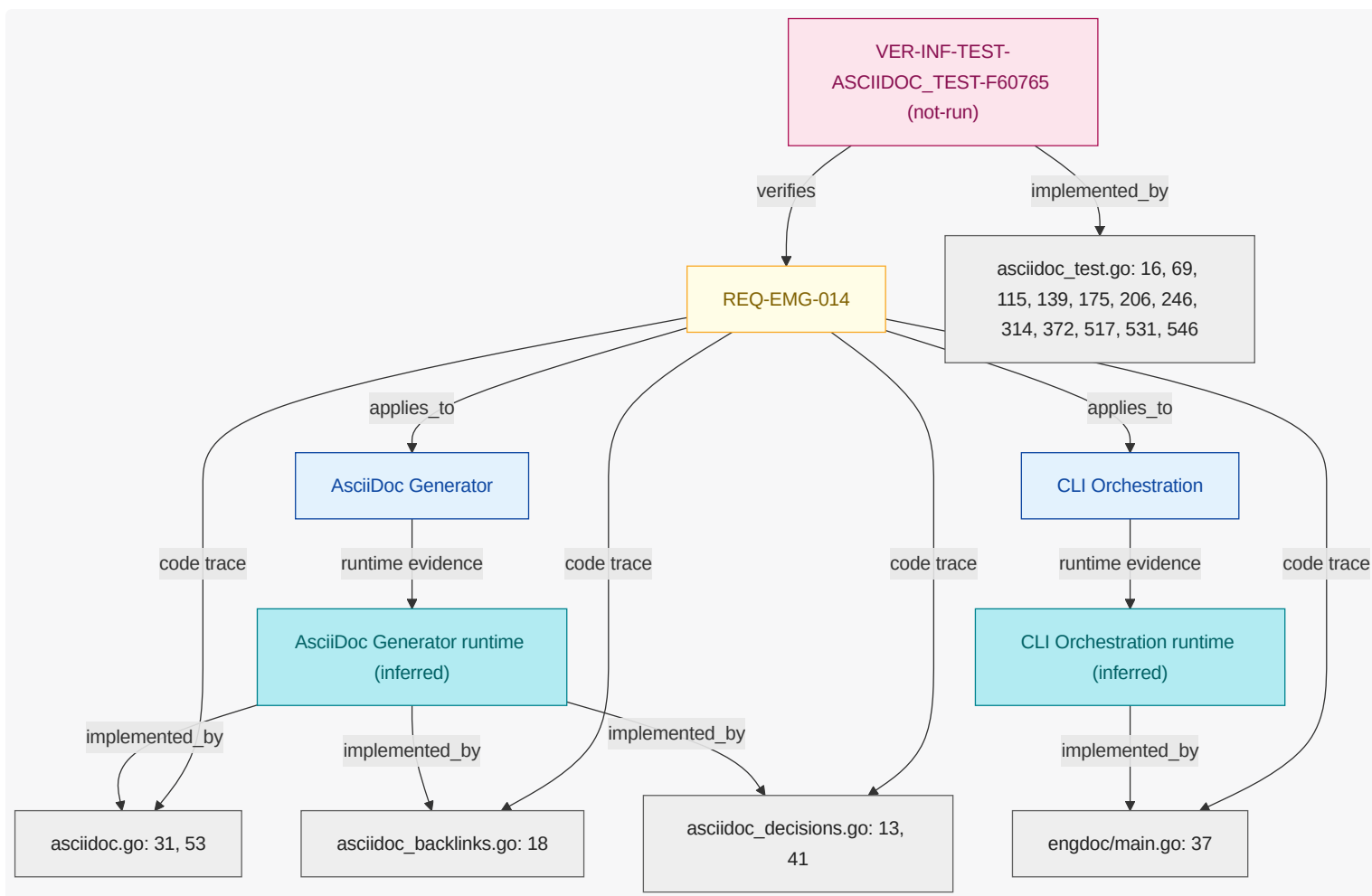
Where architecture decision records are authored , when generation run is requested , the engineering model go shall publish architecture decision records document linked from AsciiDoc architecture publication output.

NOTE

ADRs are first-class model content but are published separately from the main architecture narrative.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.14. REQ-EMG-015

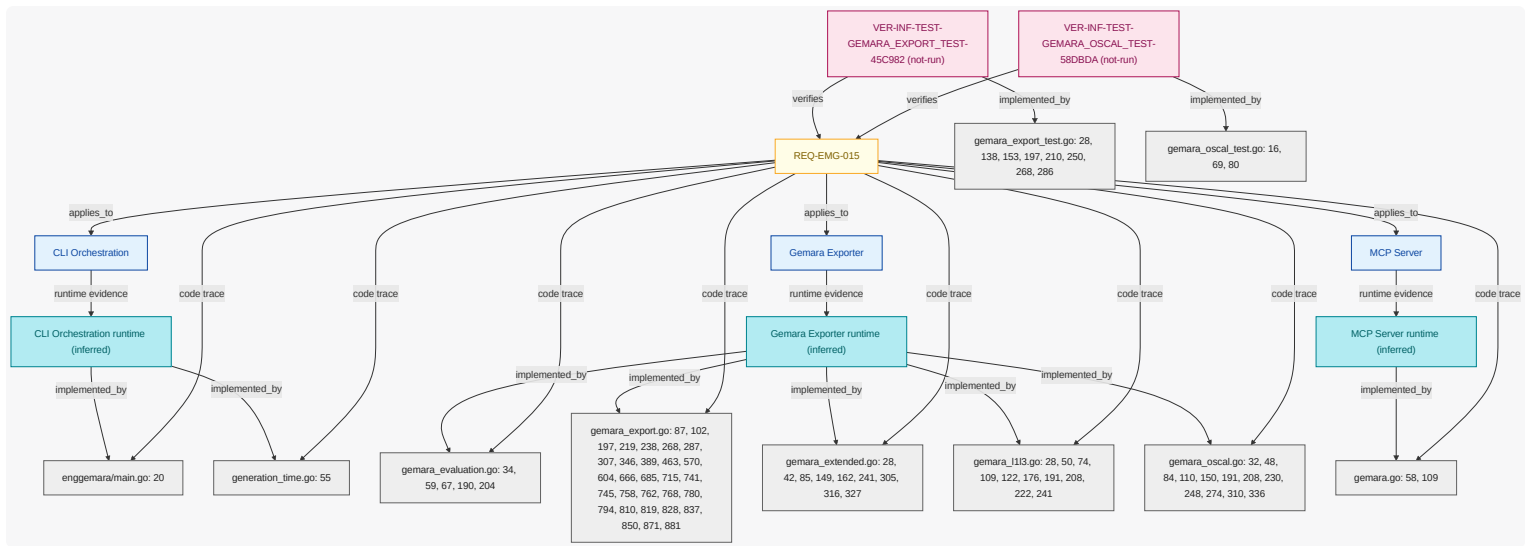
Where `gemara export` is enabled, when `generation run` is requested, the `engineering model go` shall render OpenSSF Gemara `model` artifacts.

NOTE

Gemara is a first-class GRC rendering of the `model`, validated against the published Gemara schemas via the `go-gemara SDK`.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.15. REQ-EMG-016

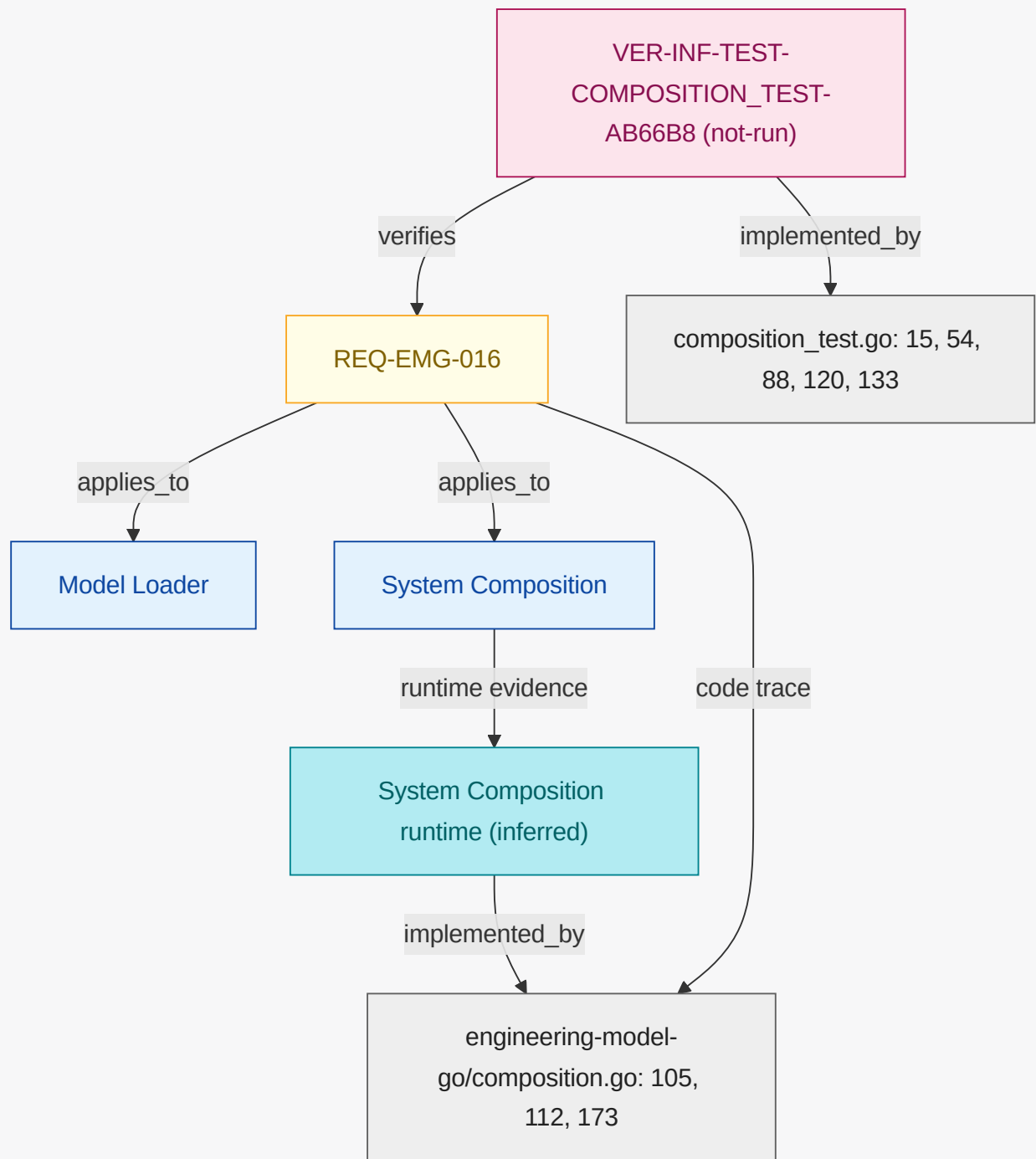
Where system composition is enabled , when generation run is requested , the engineering model go shall resolve the subsystem references declared by the parent system from local subdirectory paths.

NOTE

Recursive system-of-systems composition ; local subdirectory references, with external git repositories cloned into the .engmod cache (REQ-EMG-027 , ADR-EMG-008, ADR-EMG-010).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.16. REQ-EMG-017

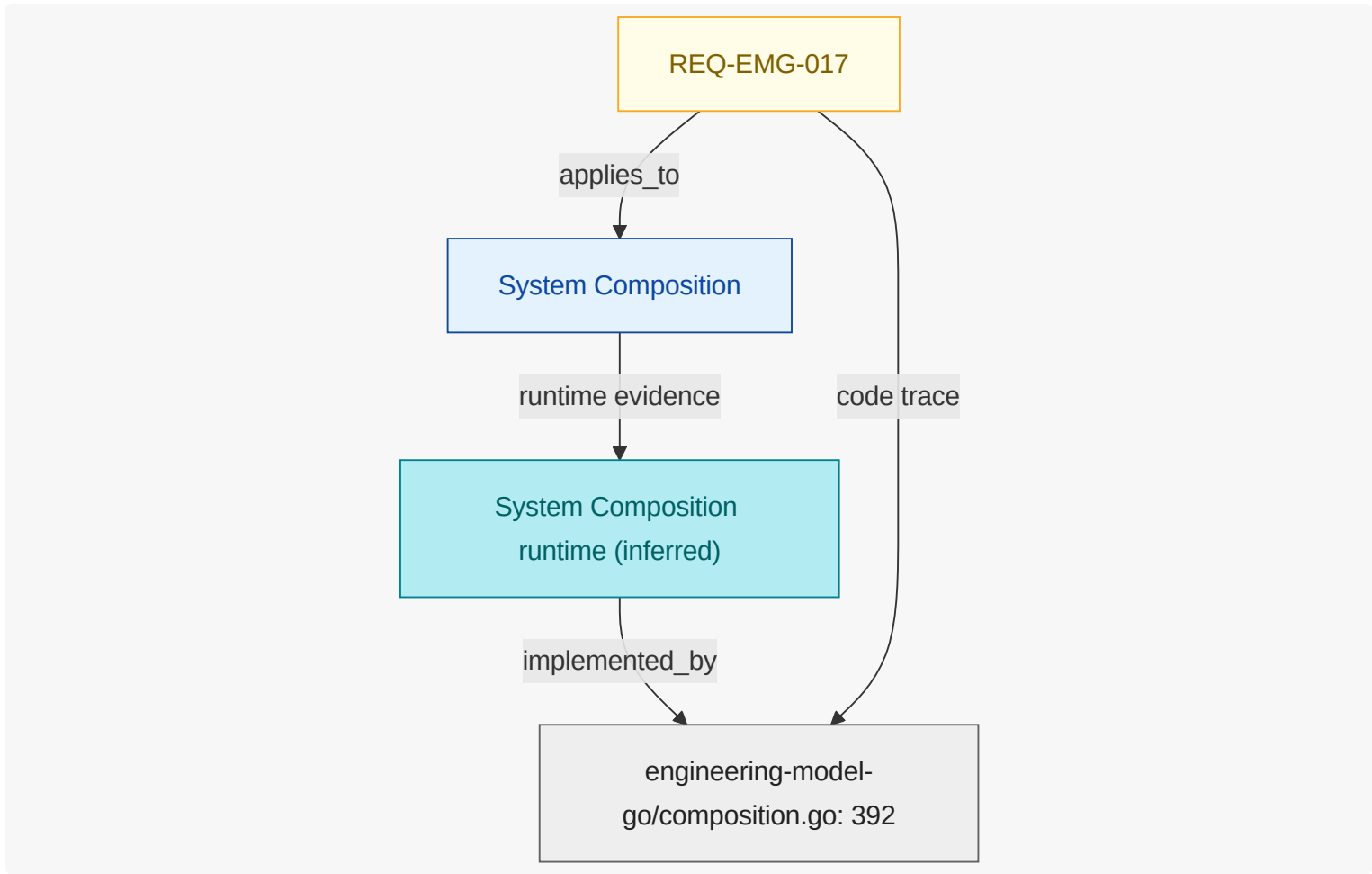
Where `system composition is enabled`, when `generation run is requested`, the `engineering model go` shall reject any subsystem reference that resolves outside the `repository workspace boundary`.

NOTE

Downward-only references remain within the workspace; mirrors the MCP path-boundary guardrail.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.17. REQ-EMG-018

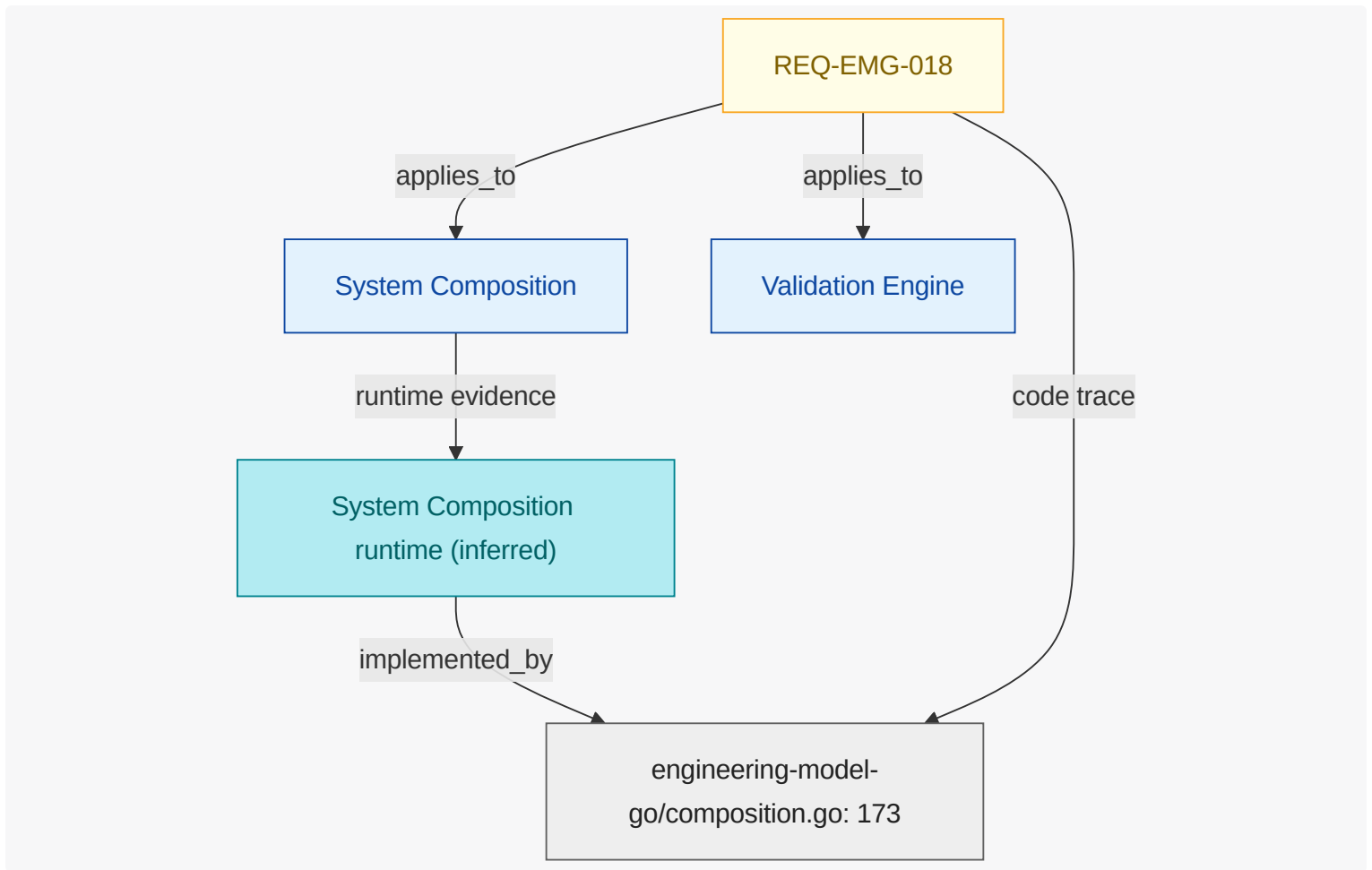
Where system composition is enabled, when generation run is requested, the engineering model go shall reject a subsystem reference graph that contains a cycle.

NOTE

Subsystem references must form a directed acyclic graph (ADR-EMG-009).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.18. REQ-EMG-019

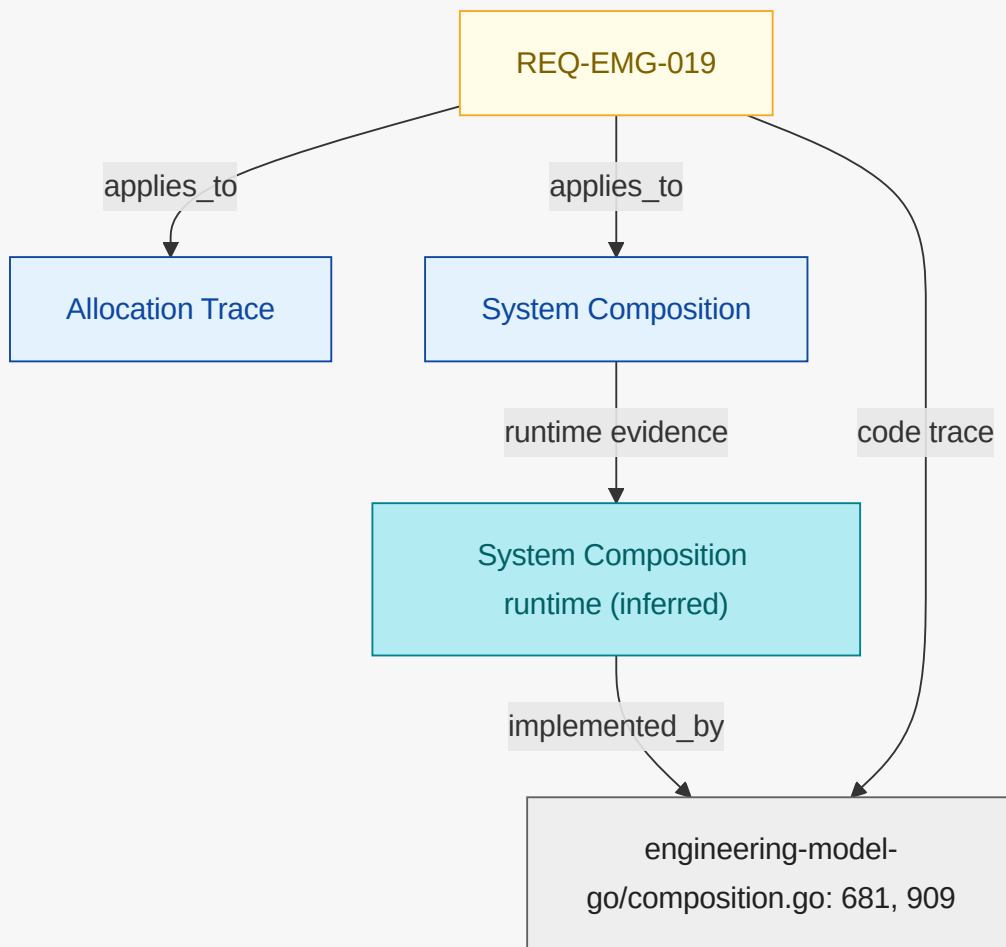
Where `system composition` is enabled, when `generation run` is requested, the `engineering model go` shall resolve cross-system references using subsystem-qualified identifiers.

NOTE

Identifiers are scoped per system; cross-system references are qualified by subsystem to avoid collisions.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.19. REQ-EMG-020

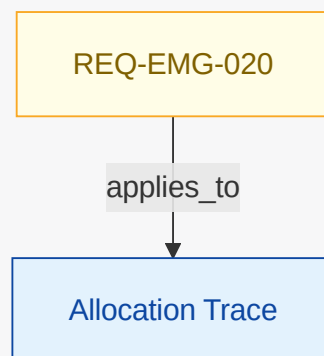
Where **system composition** is enabled, when **generation run** is requested, the **engineering model go** shall materialize parent-to-subsystem **allocation trace** ability in the composed output without modifying any subsystem **model**.

NOTE

A subsystem is parent-agnostic; parent **allocation** is materialized only in composed artifacts, never written back to the child (ADR-EMG-009, ADR-EMG-011).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.20. REQ-EMG-021

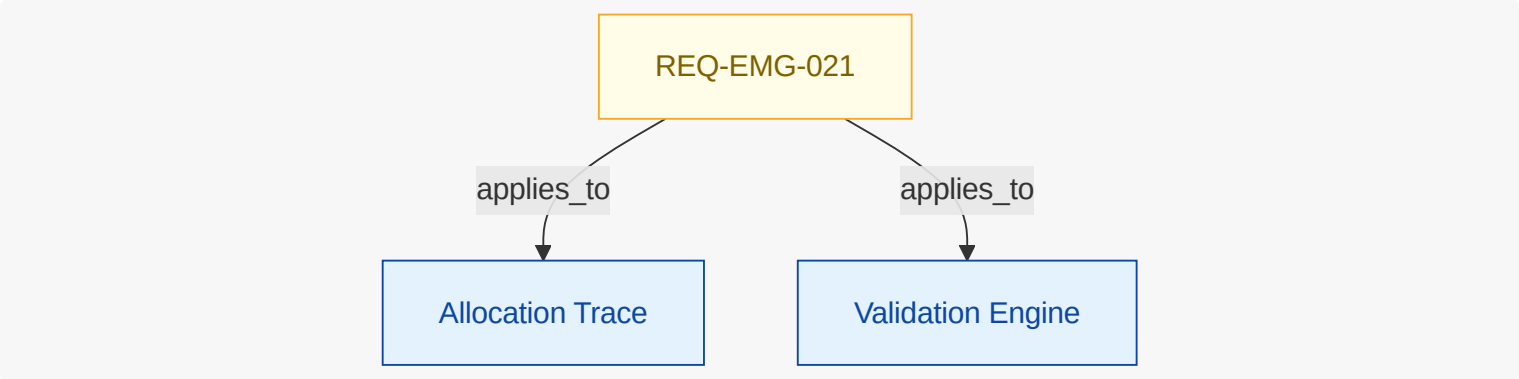
Where system composition is enabled, when generation run is requested, the engineering model go shall report each subsystem required interface that the parent system does not satisfy.

NOTE

Subsystems publish a provides-and-requires contract; the parent must satisfy each required interface.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.21. REQ-EMG-022

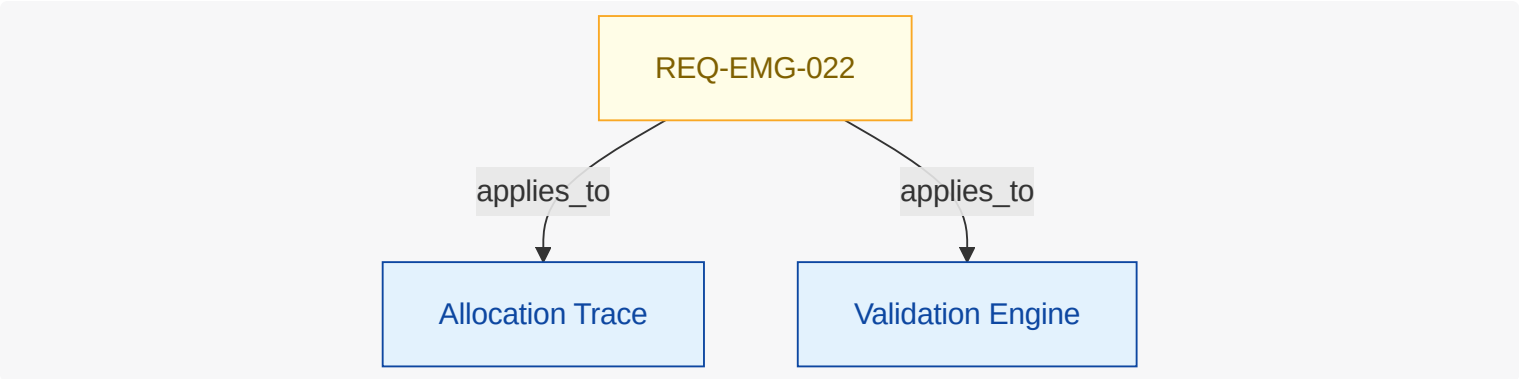
Where system composition is enabled, when generation run is requested, the engineering model go shall reject an allocation that targets a subsystem identifier not published in the subsystem contract.

NOTE

Parents may bind only to a subsystem’s public contract surface, not its internal identifiers.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.22. REQ-EMG-023

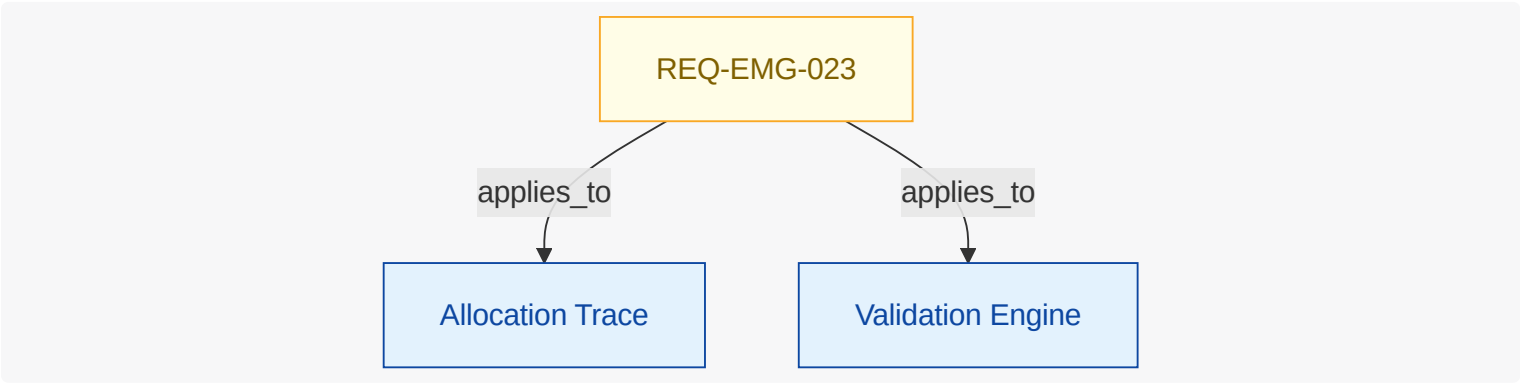
When generation run is requested, the engineering model go shall flag any requirement that links to another requirement in the same document.

NOTE

Requirements carry no tiers. A requirement is either delegated to a subsystem through a downward composition allocation or implemented here, and the delegation link alone tells the two apart. Linking requirements to one another inside a document recreates tier coupling and is confusing across system boundaries, so it is flagged.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.23. REQ-EMG-024

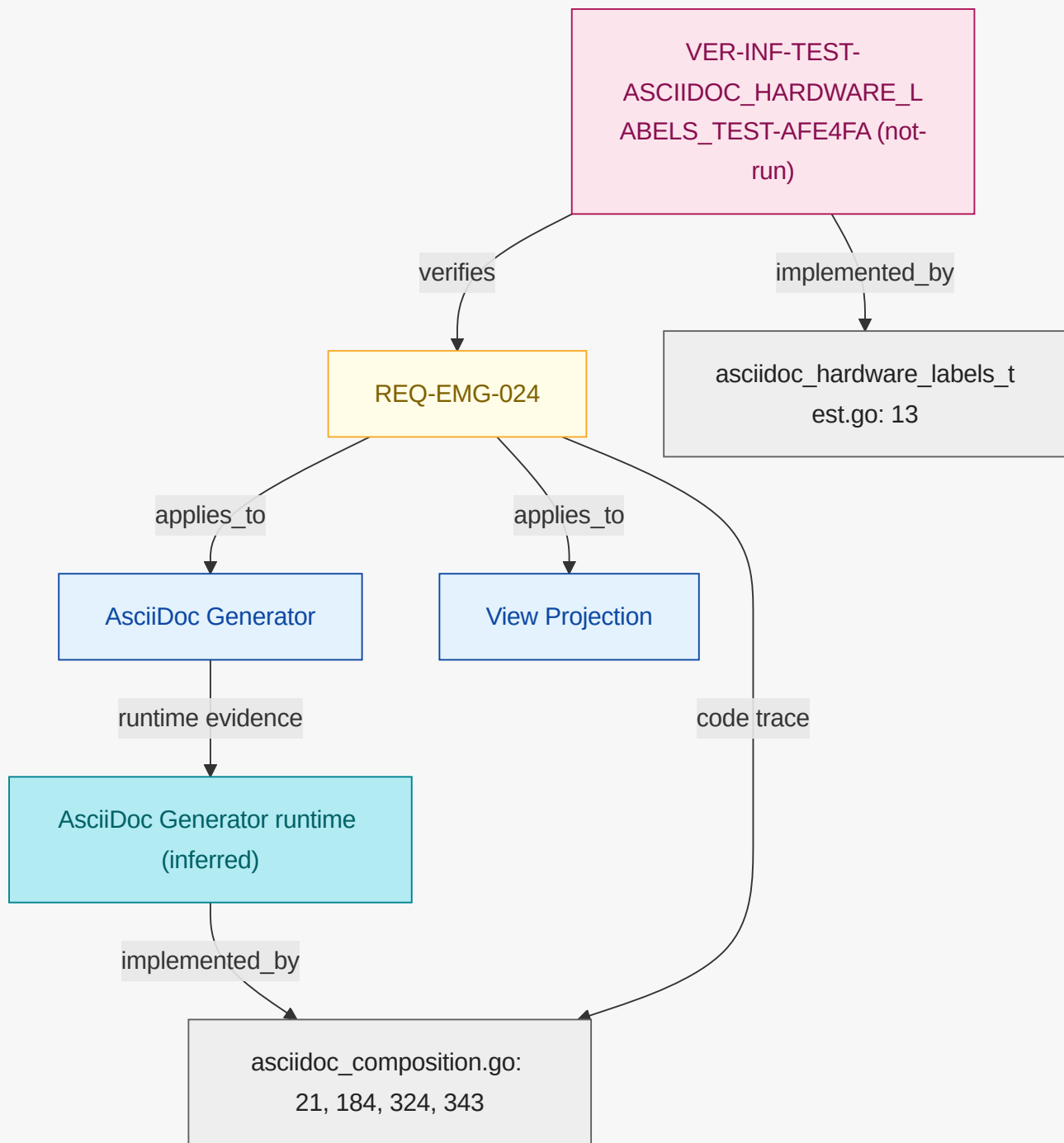
Where hardware interfaces are authored , when generation run is requested , the engineering model go shall render the hardware to software interface allocation view .

NOTE

Hardware items and hardware interfaces capture the HW/SW interface (ICD/IRS) for the interface-control view .

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



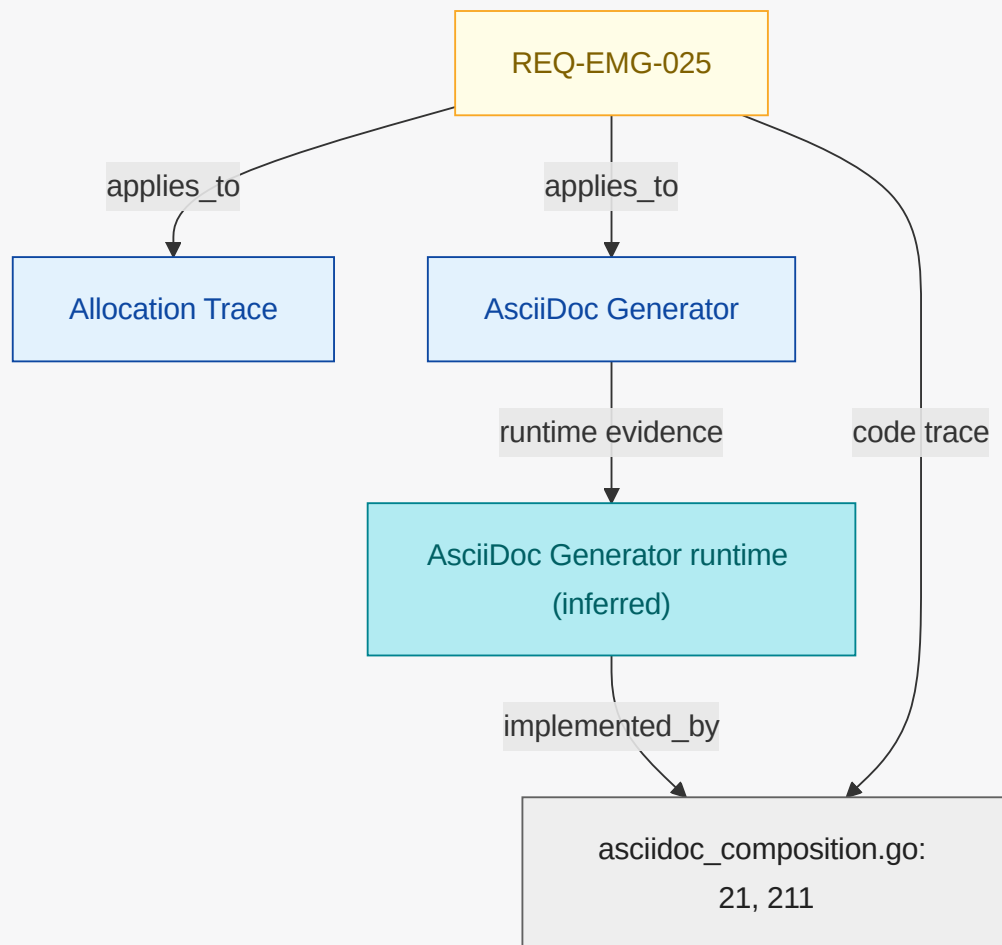
11.1.24. REQ-EMG-025

Where `system composition` is enabled, when `generation run` is requested, the `engineering model go` shall produce a system requirement allocation matrix aggregated across the composed subsystems.

NOTE Aggregate roll-up `generation` mode for the integrating system.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.25. REQ-EMG-026

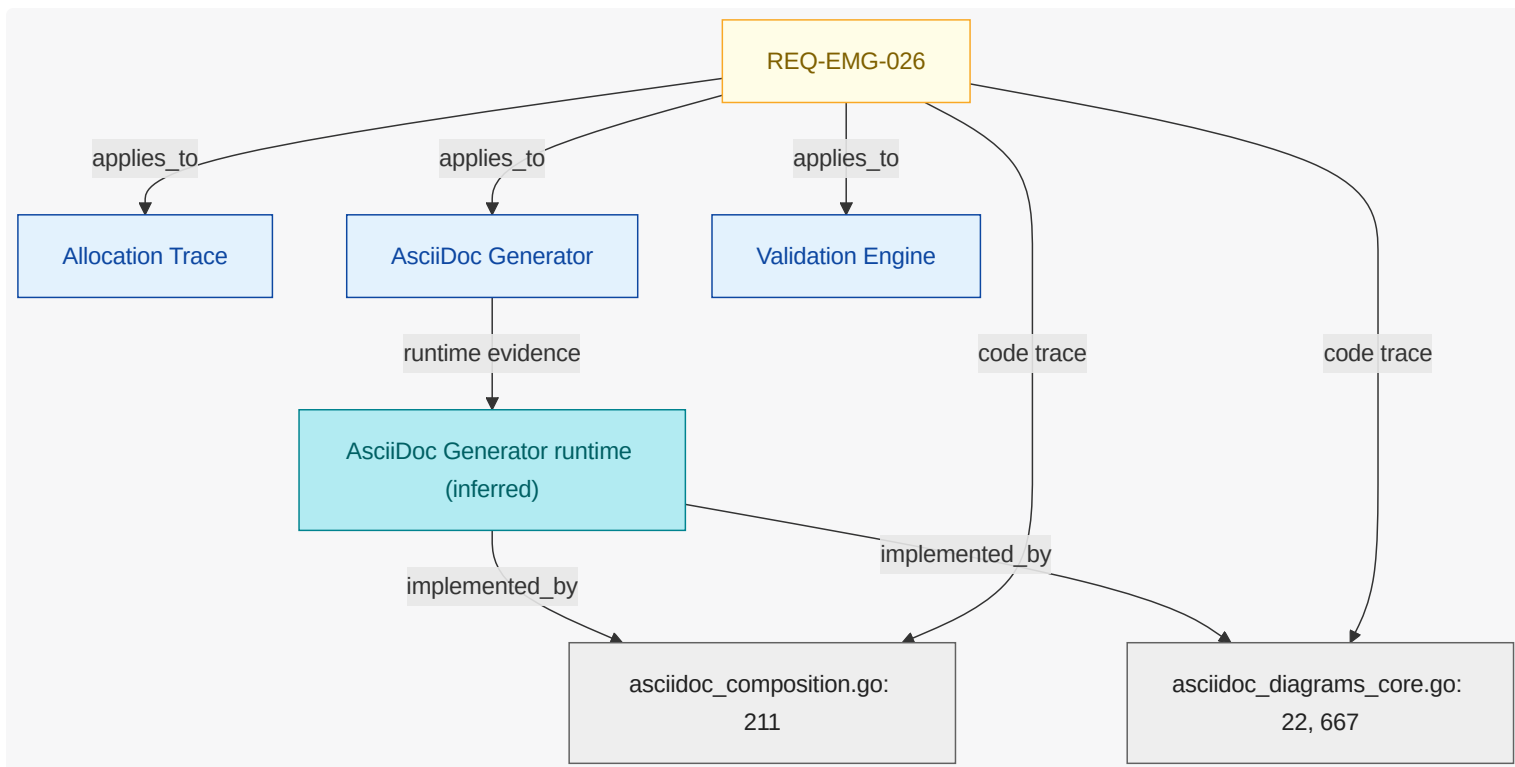
Where system composition is enabled, when generation run is requested, the engineering model go shall render in the requirement view the delegation link from each delegated requirement to the specific subsystem requirement that satisfies it.

NOTE

A provided contract entry names (ref) the subsystem requirement that realizes it, so a parent delegation traces to exactly the requirement that satisfies it. An allocated capability with no such ref is flagged as untraceable, and a ref to a requirement the subsystem does not define is rejected.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.26. REQ-EMG-027

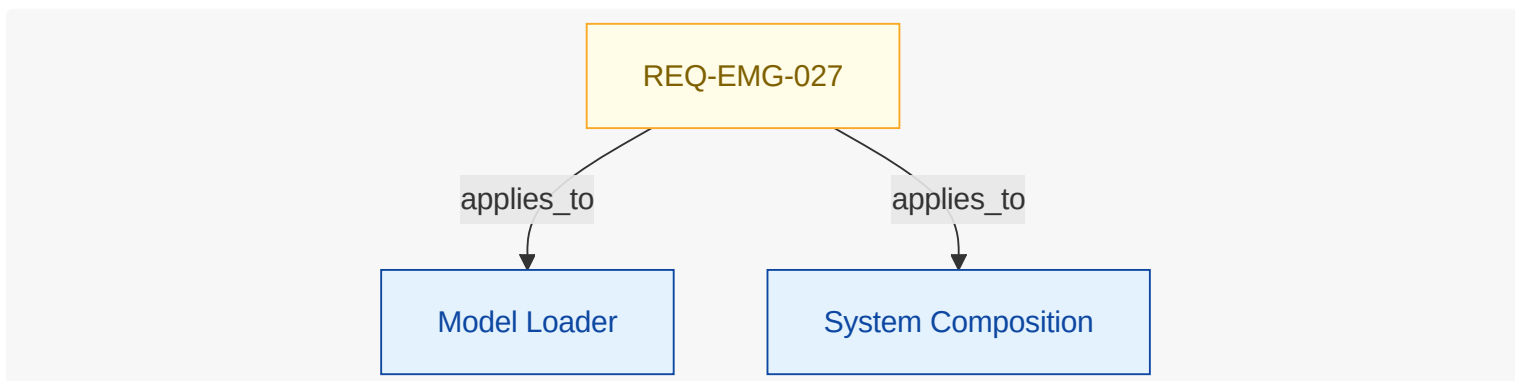
Where **system composition is enabled**, when **generation run is requested**, the **engineering model go** shall resolve an external git subsystem reference by cloning the repository into the workspace **.engmod cache**.

NOTE

A subsystem may reference an external repository with git (and an optional rev and path). The **model** is cloned once into **.engmod/subsystems/<id>** — a gitignored, temporary cache reused while it points at the same repository — and resolved like a local subsystem, giving a full cross-repository composed **view**. References stay downward-only.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.27. REQ-EMG-028

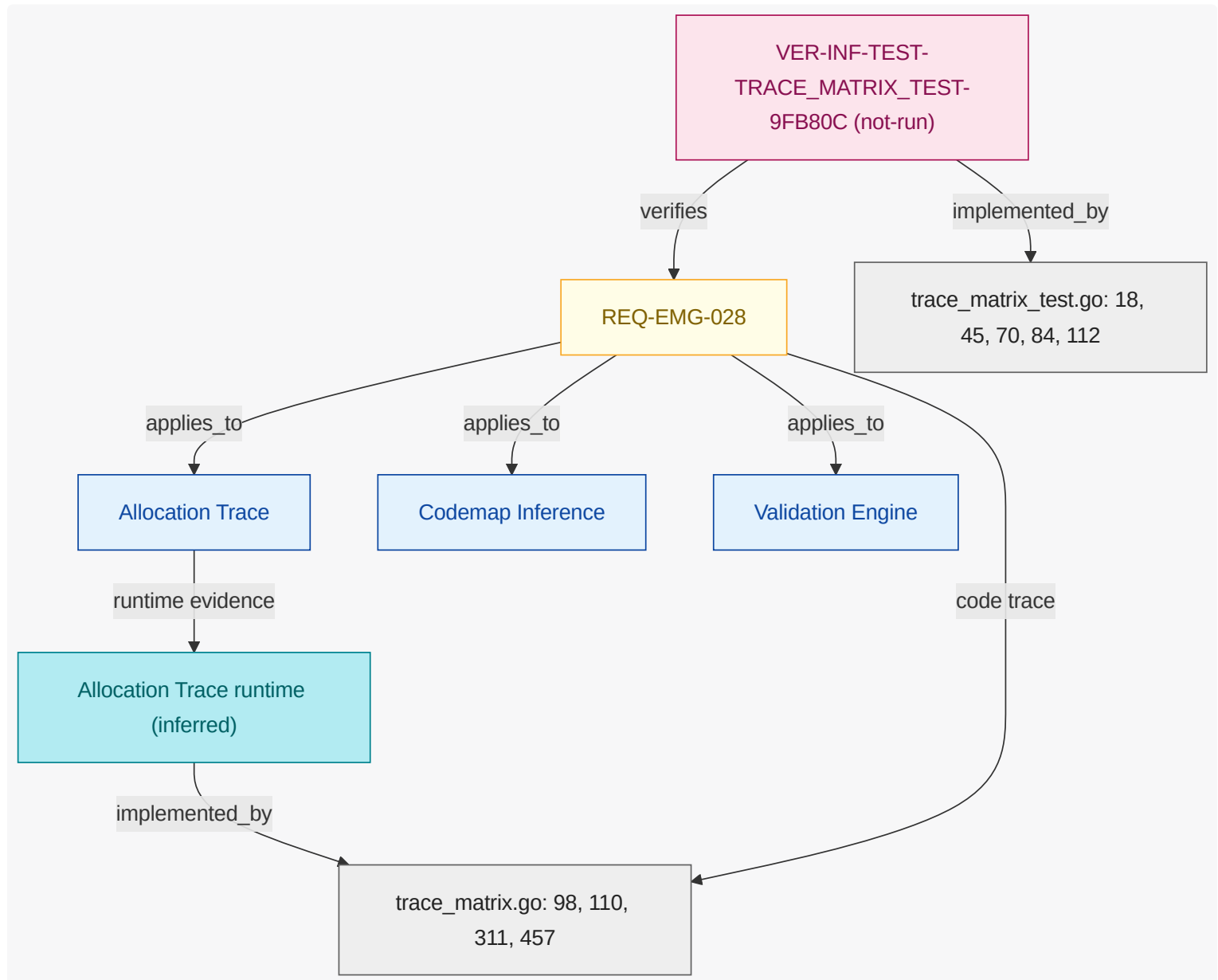
When **generation run is requested**, the **engineering model go** shall reject a code trace link that resolves to no **requirement** or **model element**.

NOTE

Code markers (TRLC-LINKS to a requirement, ENGMODEL-LINKS to a model element) are existence-checked against the model, not only shape-checked, so a link to a deleted or mistyped id is a dangling link and fails the build. A trace link is only checked for the model that owns its code (see REQ-EMG-030 scoping).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.28. REQ-EMG-029

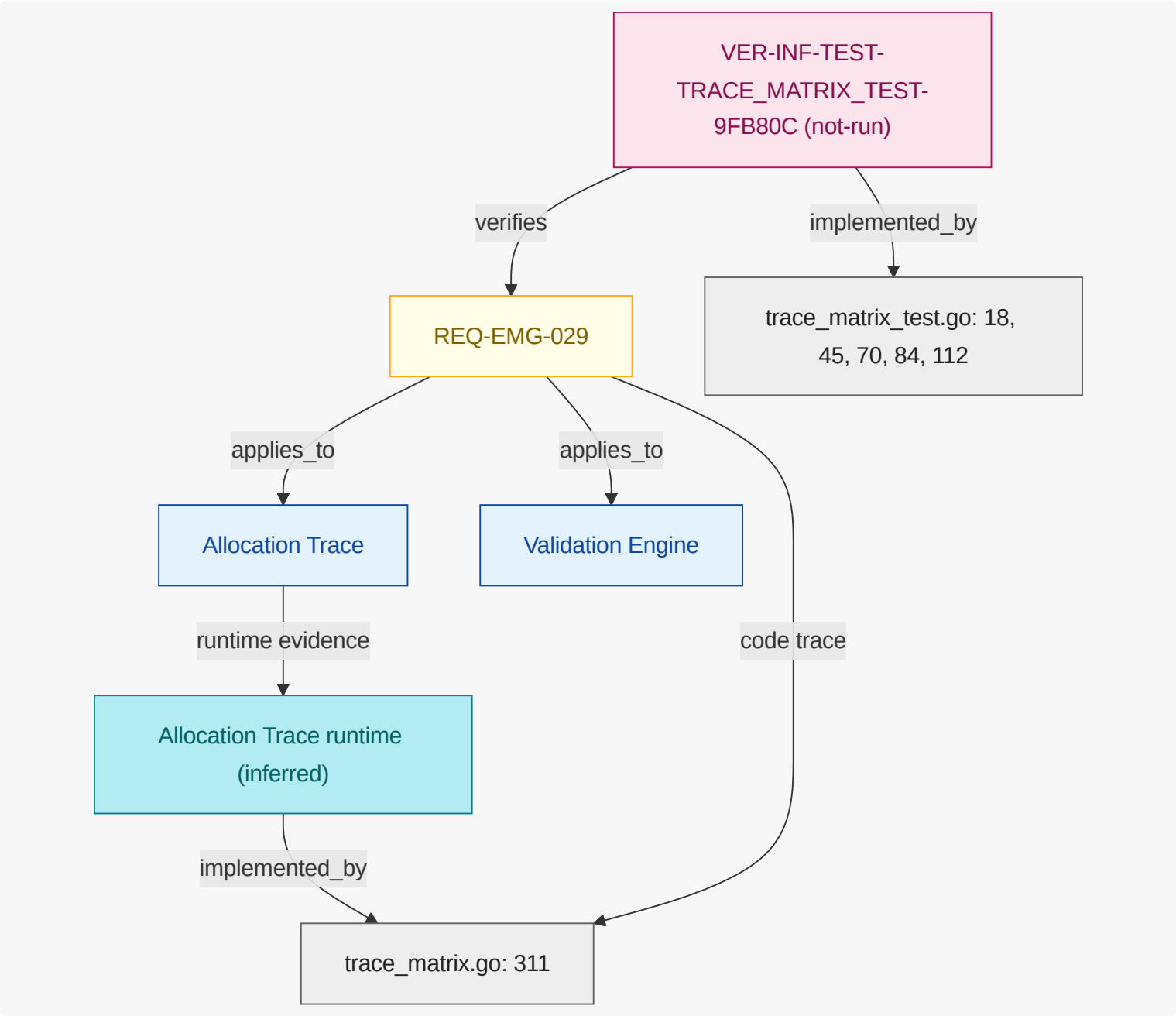
When generation run is requested, the engineering model go shall flag each requirement that is traced by neither code nor verification nor a downward delegation.

NOTE

An orphan requirement has no implementing code symbol, no verification, and is not delegated to a subsystem; it is a coverage gap surfaced as a warning so it is visible without breaking the build.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.29. REQ-EMG-030

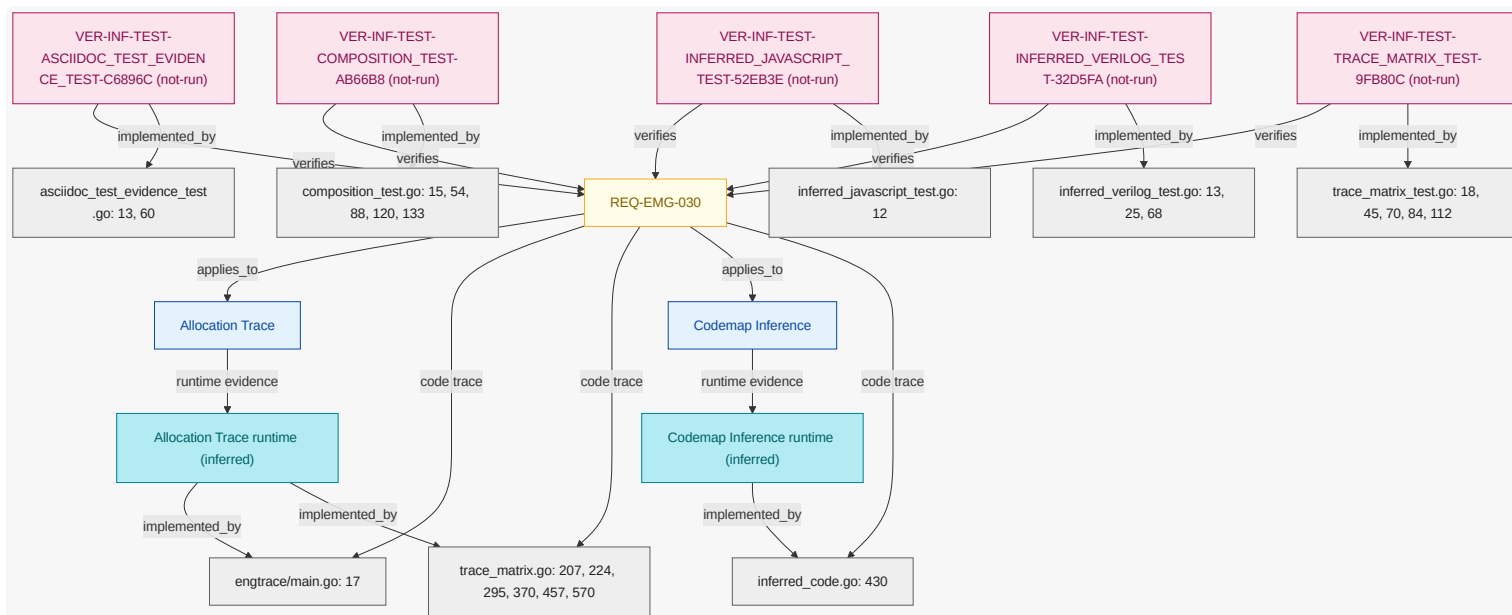
When generation run is requested, the engineering model go shall export a machine-readable traceability matrix that links each requirement to its implementing code and verification and delegated subsystem requirement.

NOTE

One consolidated matrix (JSON) replaces having to read it out of prose and diagrams. Test-only source links count as verification, not production implementation. Code is attributed to the model whose engmod.yml is its nearest enclosing root, so a parent model never validates or counts a child model's trace links (monorepo scoping).

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.30. REQ-EMG-031

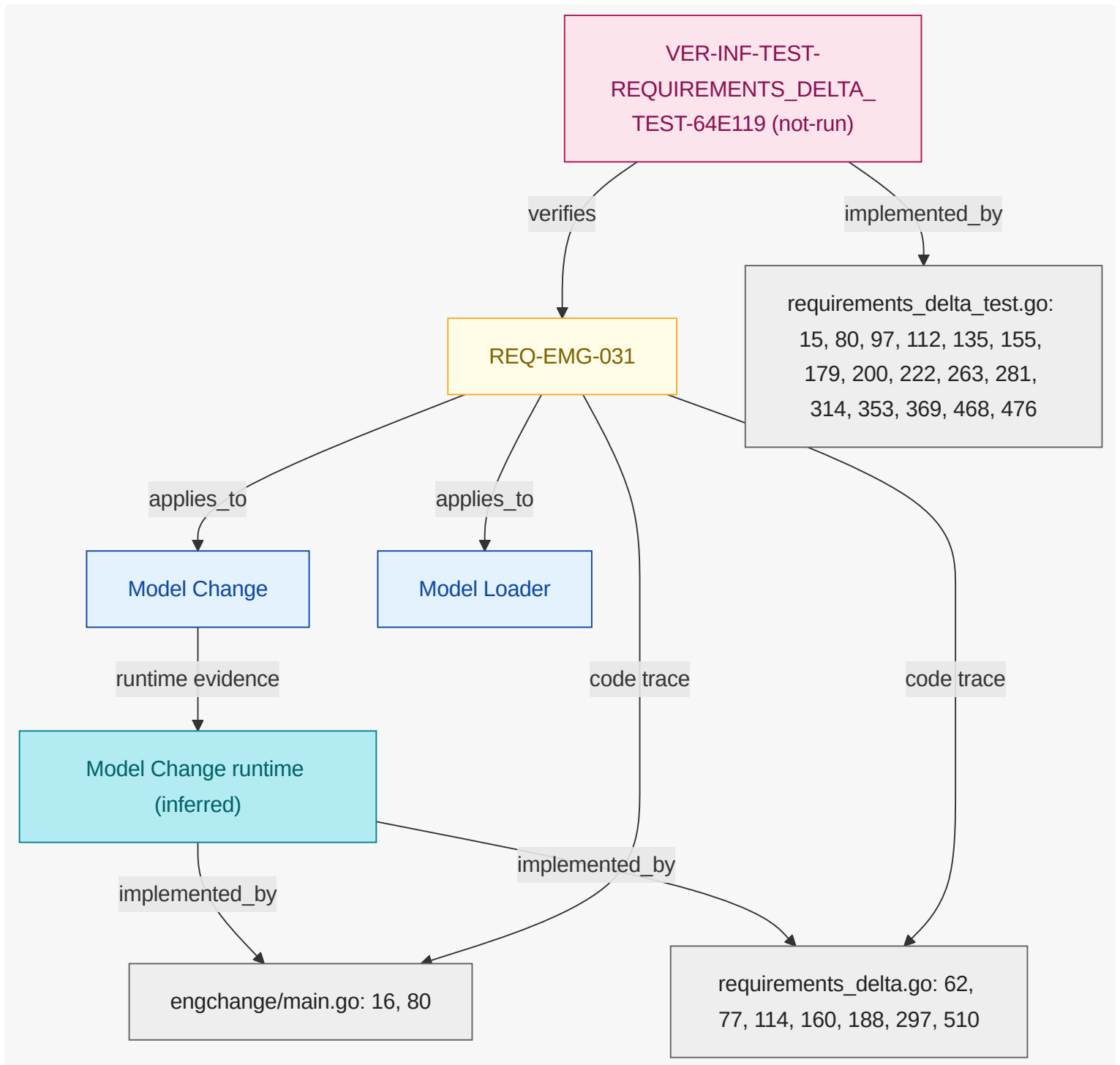
When requirements delta is received, the engineering model go shall strictly validate requirements delta schema plus stable identifier operation uniqueness.

NOTE

Version 1 supports add, whole-entity update, and remove operations. Unknown fields, empty operations, duplicate operations, adds of existing IDs, and updates or removals of missing IDs are rejected.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.31. REQ-EMG-032

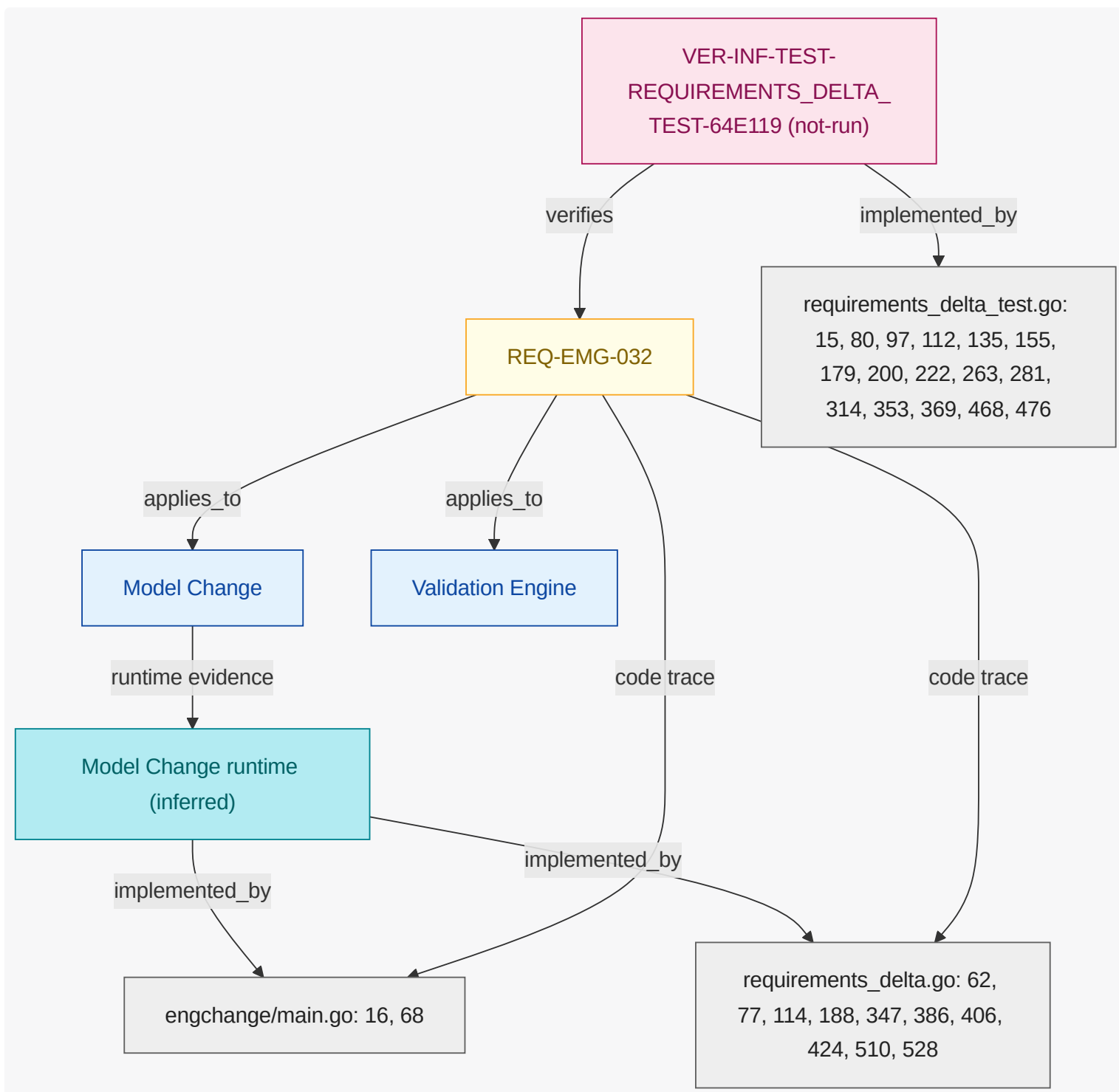
While **dry run mode** is active, when **requirements delta** is received, the **engineering model go** shall validate **candidate architecture model** without modifying **requirements document**.

NOTE

The candidate reuses complete-model structural, **catalog**, **EARS**, and internal-link **validation** before any write is permitted.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.32. REQ-EMG-033

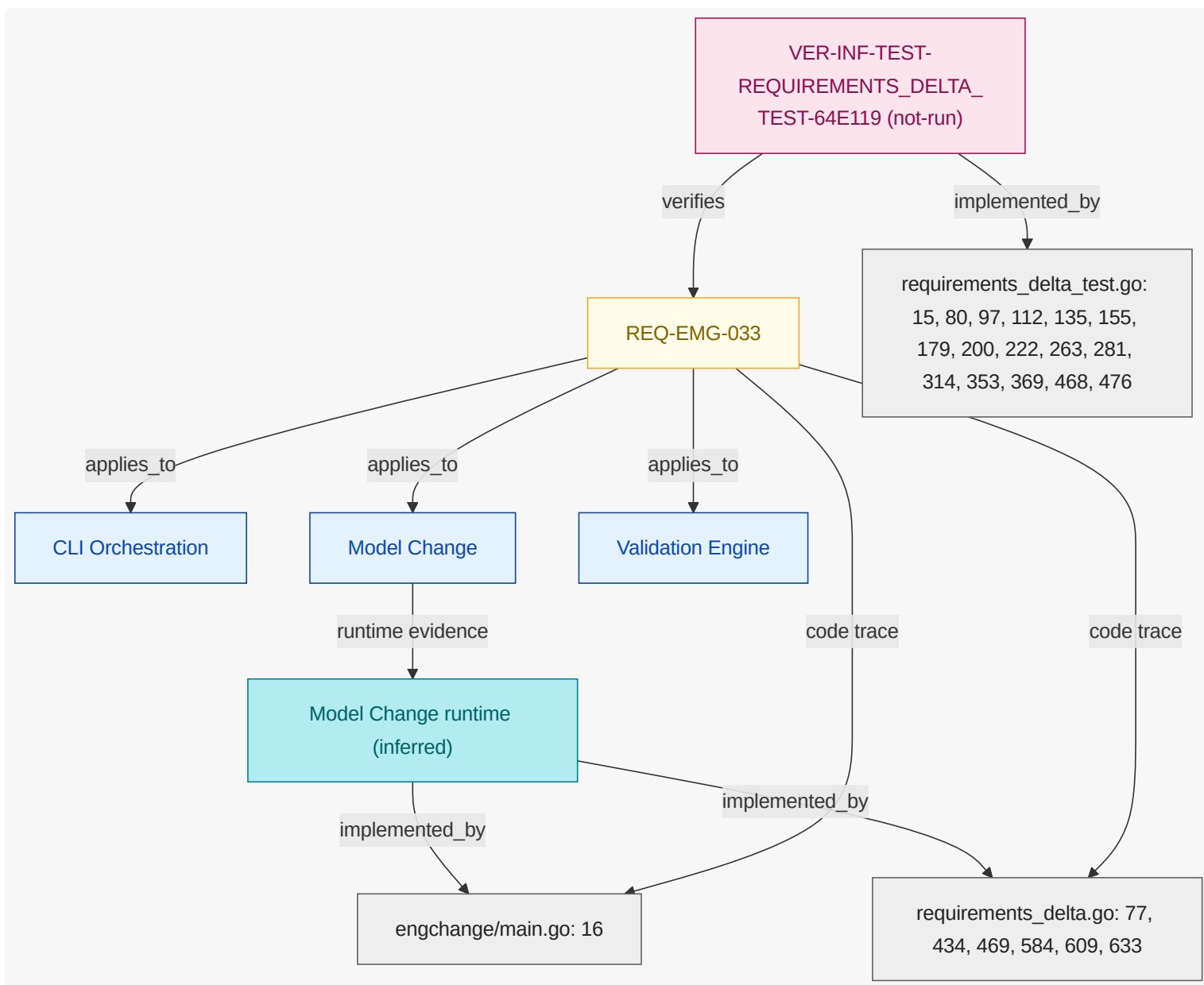
While model is valid, when requirements delta apply is requested, the engineering model go shall atomically replace requirements document.

NOTE

The YAML AST merger preserves unchanged nodes and comments. Atomic replacement preserves file permissions and refuses a candidate with blocking diagnostics.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.33. REQ-EMG-034

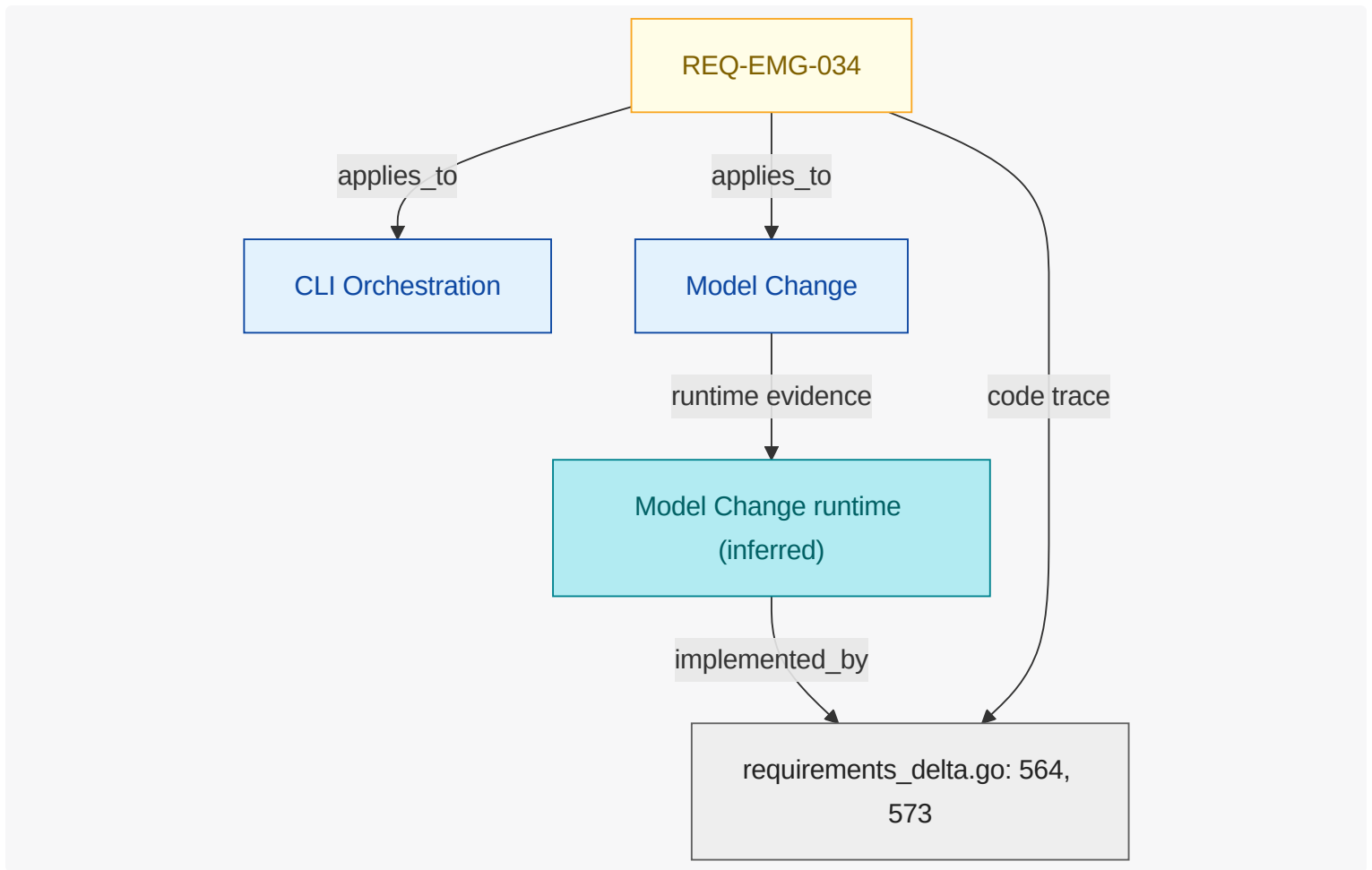
When requirements delta is received, the engineering model go shall produce deterministic requirements change summary.

NOTE

The summary sorts added, updated, and removed stable IDs and reports changed fields for whole-entity updates.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.34. REQ-EMG-035

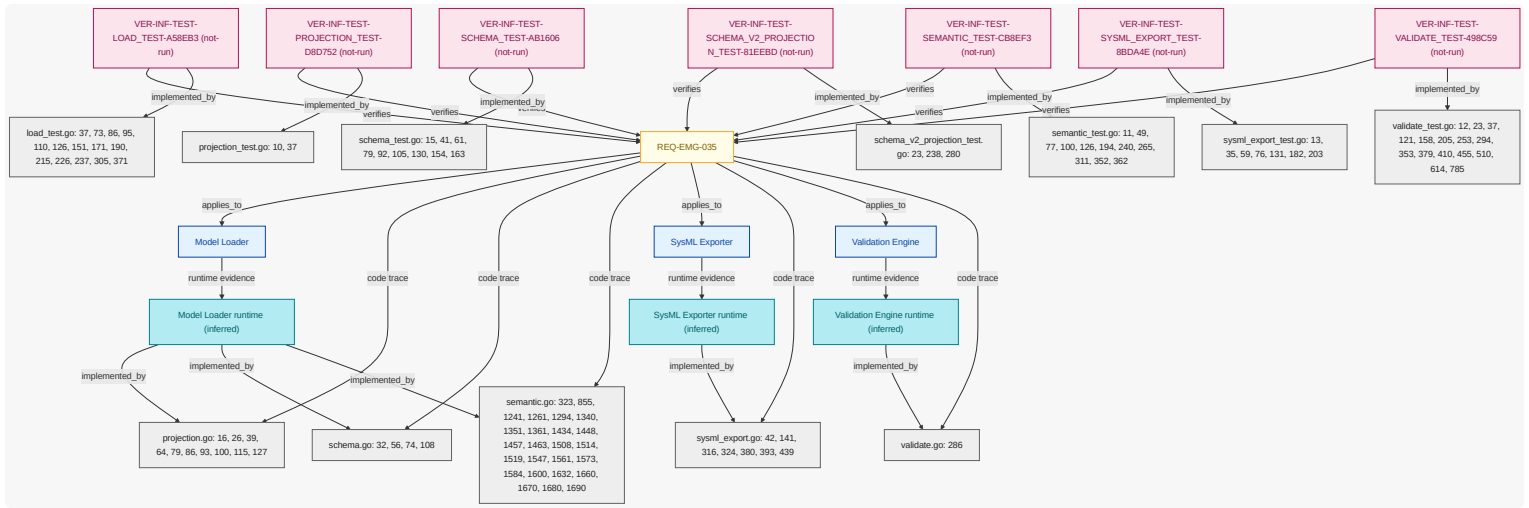
When model updated event is received , the engineering model go shall normalize architecture model plus requirements document into canonical semantic model .

NOTE

The canonical semantic model is SysML v2 and KerML aligned. Existing Engineering Model concepts map into that model once; legacy YAML remains an input adapter during migration and must not create a parallel set of parts, interfaces, flows, requirements , allocations, or views.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.35. REQ-EMG-036

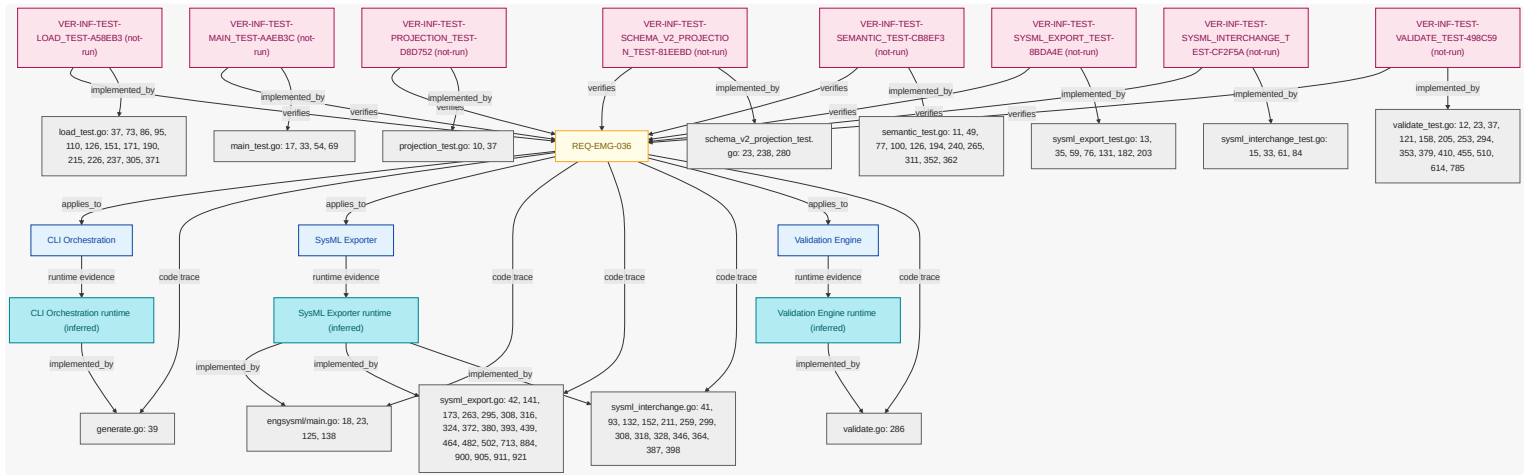
Where sysml export is enabled, when generation run is requested, the engineering model go shall generate sysml v2 model artifact from canonical semantic model.

NOTE

The exporter preserves stable identifiers and maps standard concepts to native SysML v2 or KerML semantics. Engineering-specific concepts may use explicit typed SysML metadata definitions, but canonical data without a native target representation must be omitted rather than encoded in descriptions, opaque payloads, generic property bags, or sidecar files.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.36. REQ-EMG-037

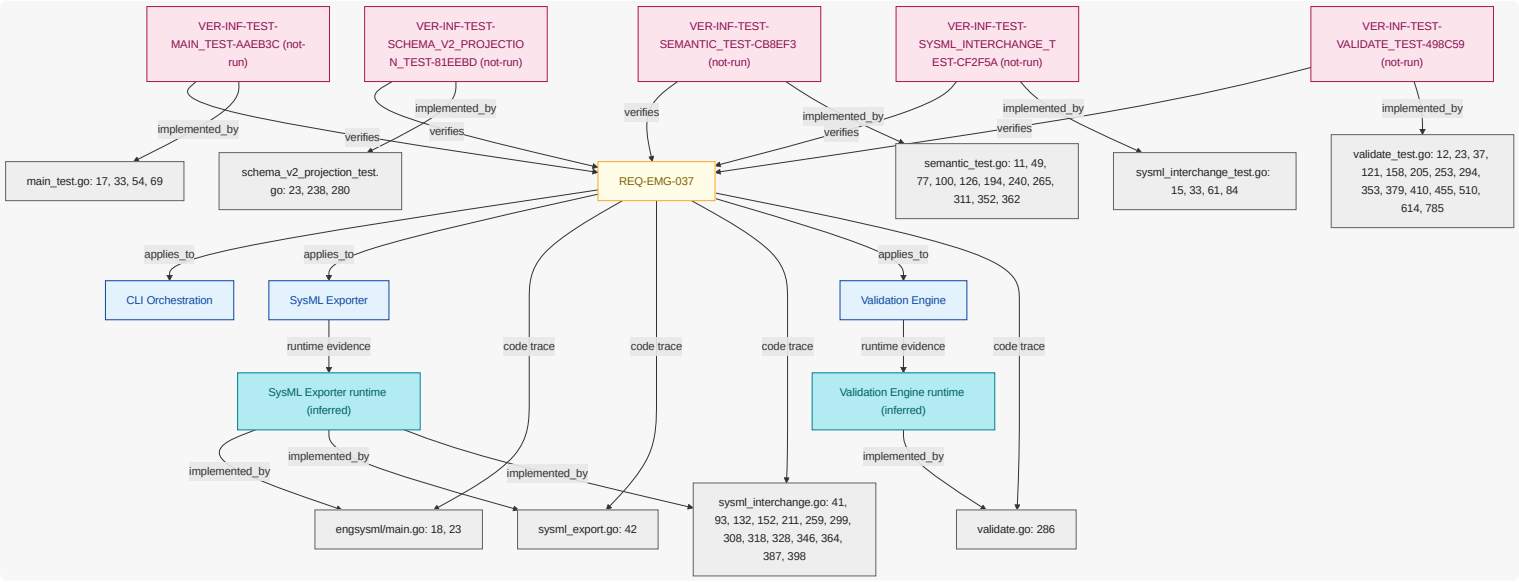
While ci validation mode is active, when generation run is requested, the engineering model go shall validate sysml v2 model artifact before artifact generated event is raised.

NOTE

Validation must report unsupported or lossy mappings and progress toward abstract-syntax, textual-syntax, semantic, and model-interchange conformance. A syntactically valid subset export must not be described as a complete SysML v2 semantic superset.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.37. REQ-EMG-038

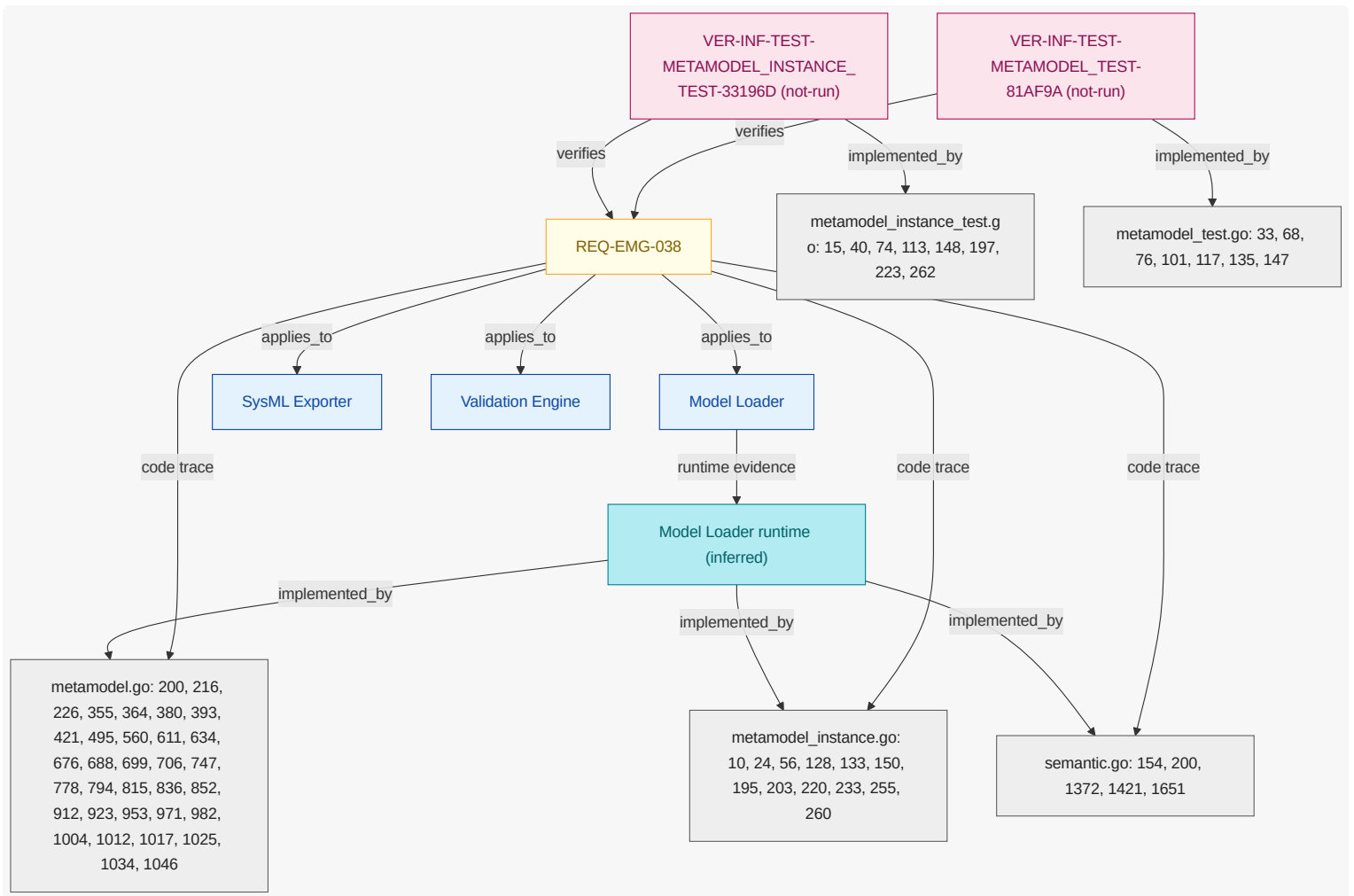
Where sysml export is enabled , when generation run is requested , the engineering model go shall validate canonical semantic model against sysml v2 metamodel .

NOTE

The pinned formal SysML 2.0 and KerML 1.0 metamodel is the source of truth for supported classes, properties, relationships, ownership, and multiplicities. Generated CUE constraints and semantic registries must not be maintained as independent handwritten copies.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.38. REQ-EMG-039

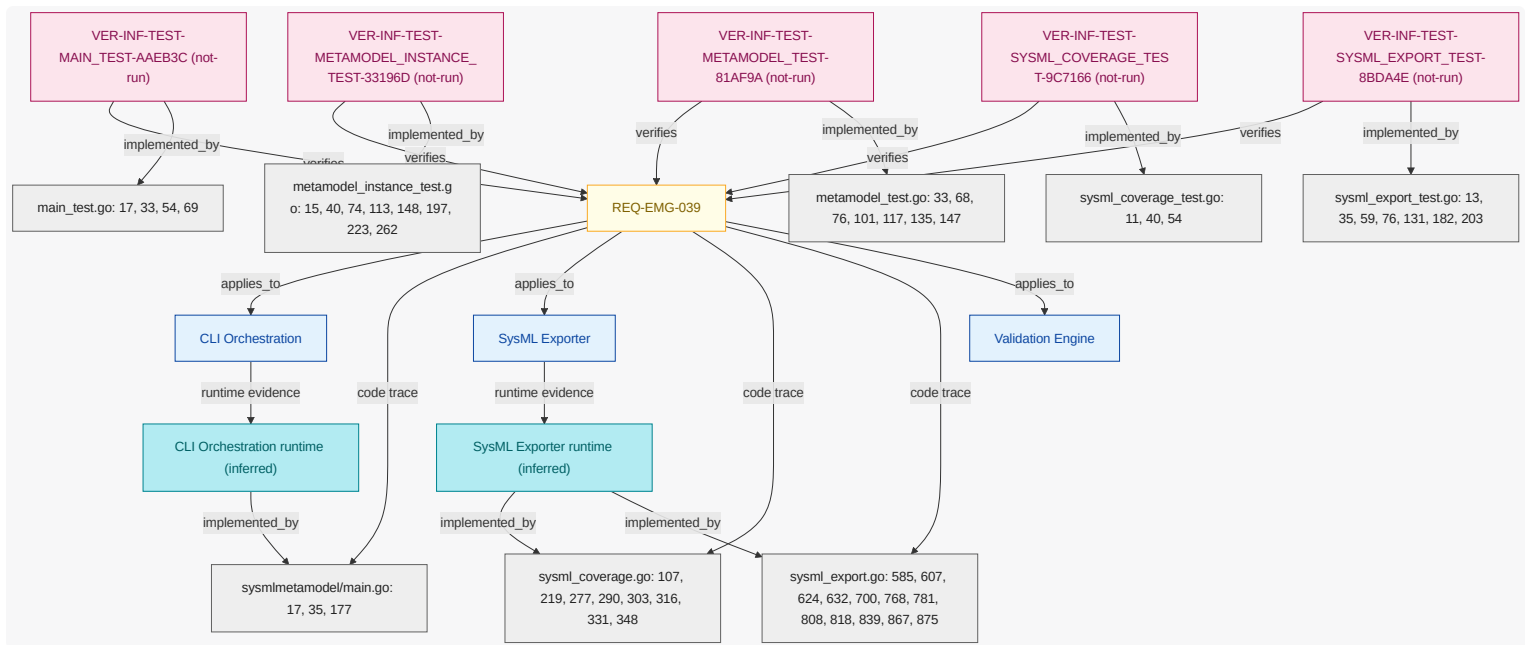
While ci validation mode is active , when generation run is requested , the engineering model go shall reject incomplete sysml v2 metamodel coverage.

NOTE

Coverage is complete only when every normative metaclass and property in the pinned formal release is represented, explicitly excluded as derived, or implemented as a typed Engineering Model extension with executable evidence .

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.39. REQ-EMG-040

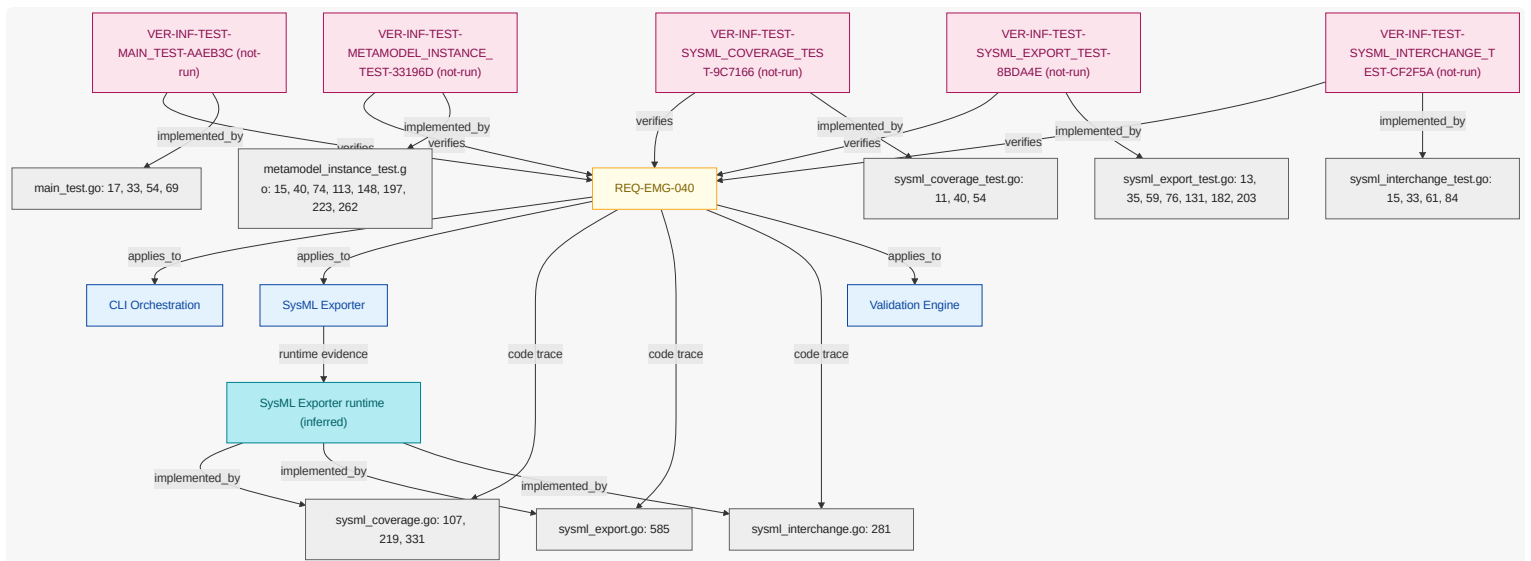
When generation run is requested , the engineering model go shall preserve the deterministic native sysml v2 source through project interchange round trip.

NOTE

Sysand owns normative KPAR packaging and reopening. Verification compares the reopened SysML source byte-for-byte with a fresh projection and validates both sources with the pinned official parser; it does not claim lossless reconstruction of canonical concepts omitted from SysML .

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.40. REQ-EMG-041

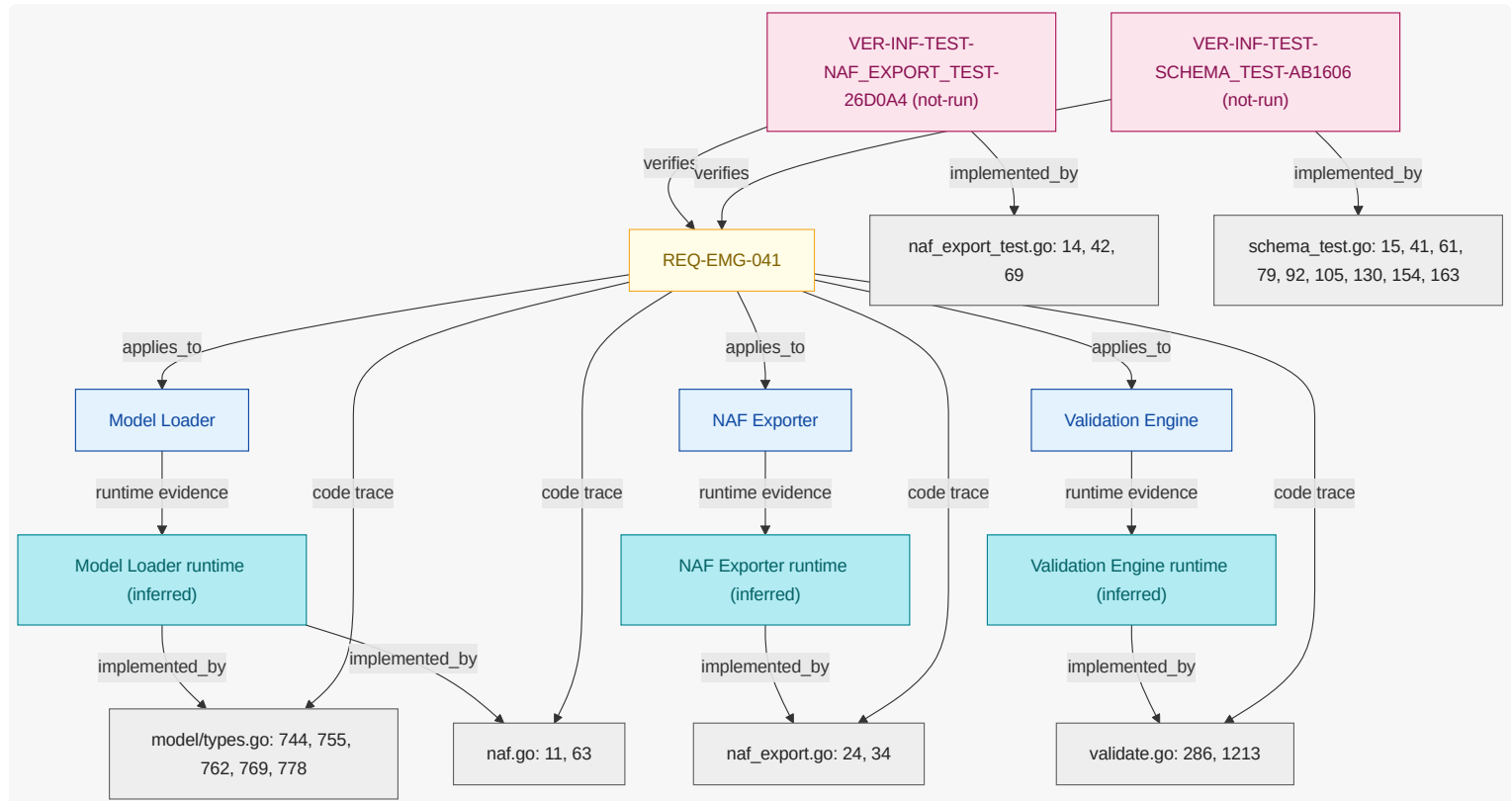
Where naf profile authoring is enabled , when model updated event is received , the engineering model go shall validate naf v4.1 stakeholder plus concern plus viewpoint metadata against canonical model references.

NOTE

NAF-specific YAML contains profile metadata only. Capability, service, logical, resource, behavior, interface, and requirement concepts remain represented once in the canonical SysML and KerML aligned semantic graph.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.41. REQ-EMG-042

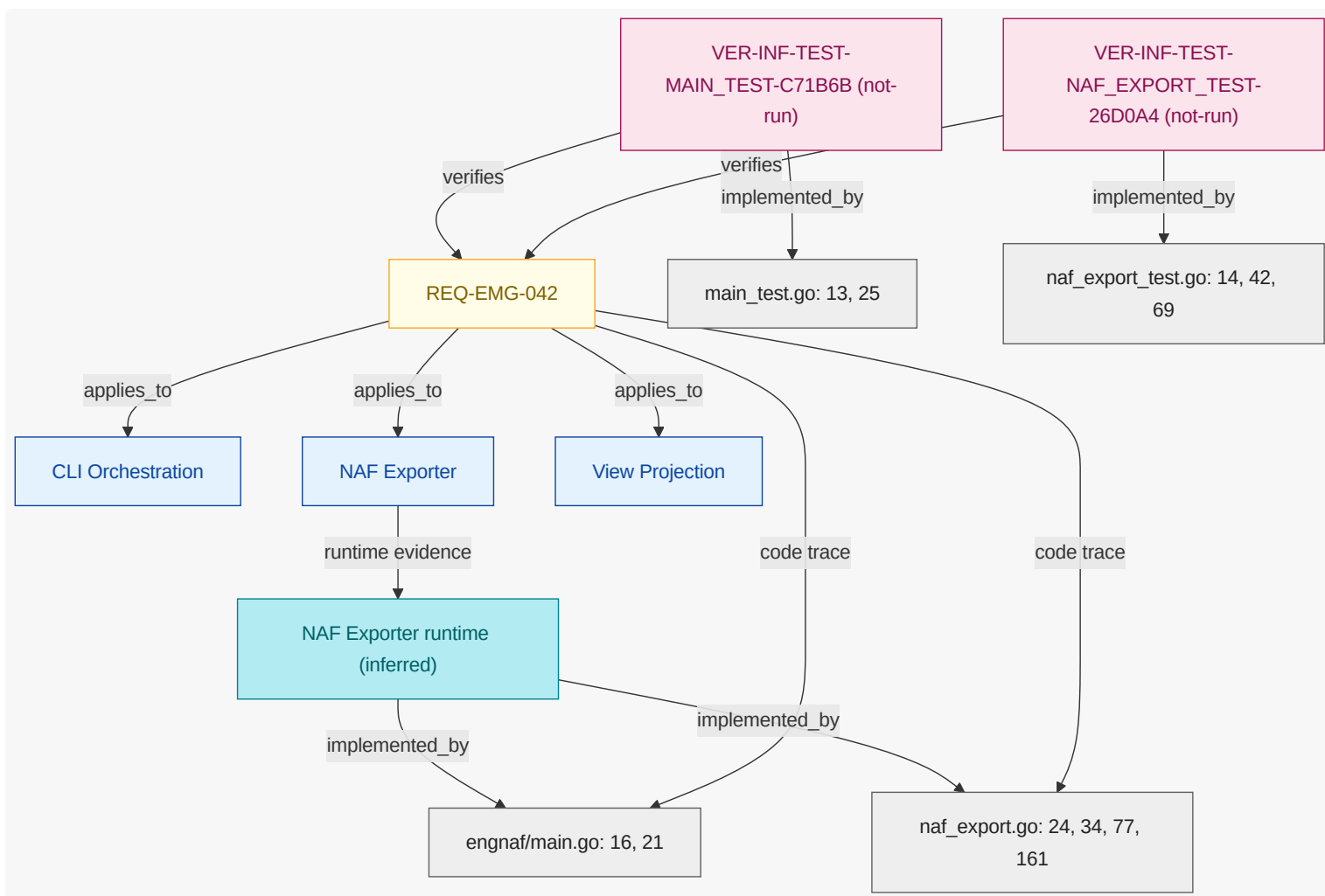
Where naf export is enabled, when generation run is requested, the engineering model go shall generate deterministic naf v4.1 architecture document from referenced canonical model views.

NOTE

Each NAF architecture product uses an official NAF v4.1 viewpoint code and references an existing Engmod view so generated content preserves canonical identities and relationships.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.42. REQ-EMG-043

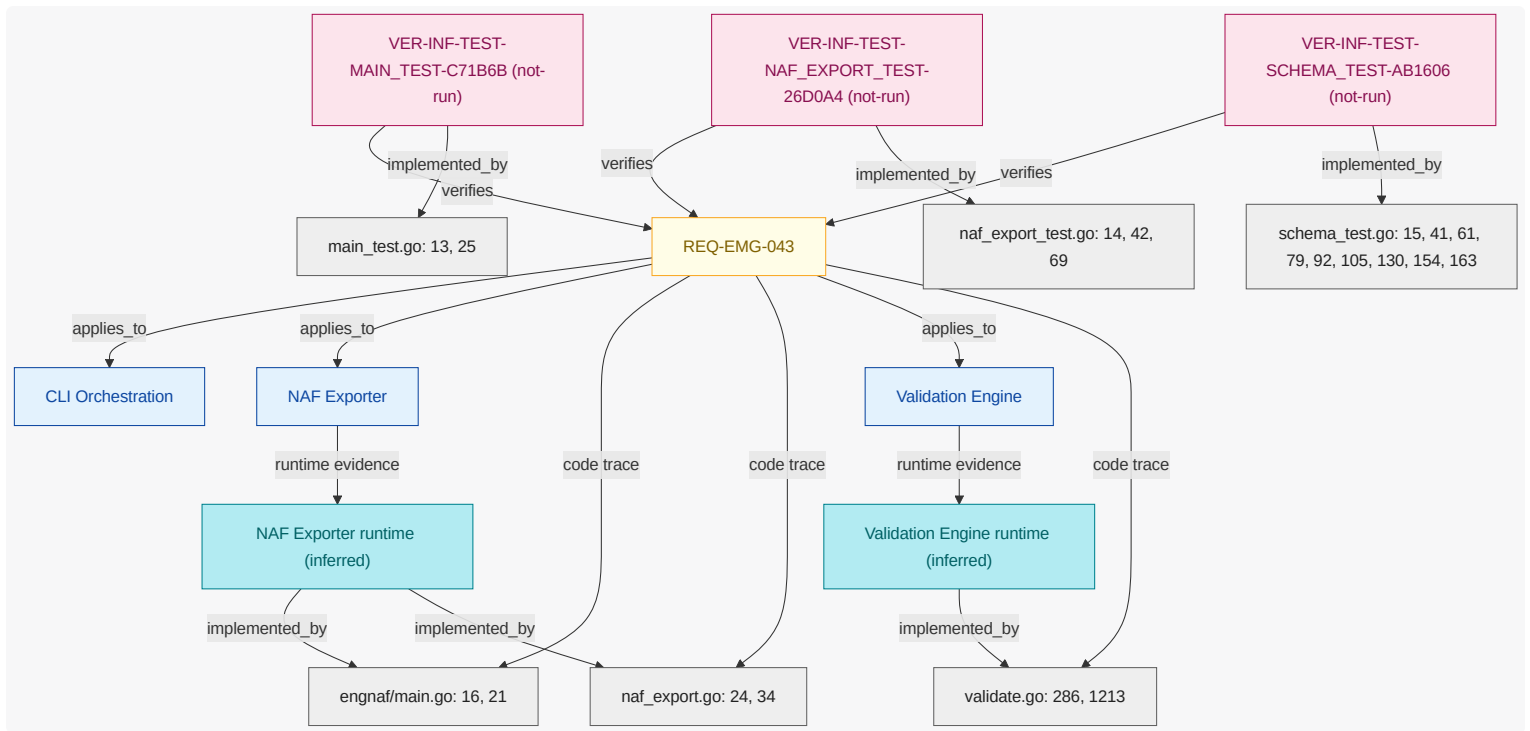
While `ci validation mode` is active, when `generation run` is requested, the `engineering model go` shall reject unsupported `naf version` plus unknown `viewpoint code` plus dangling `profile reference` before `artifact generated event` is raised.

NOTE

Validation covers stakeholder `actor` bindings, concern ownership, architecture-product concern references, and Engmod `view` references.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.43. REQ-EMG-044

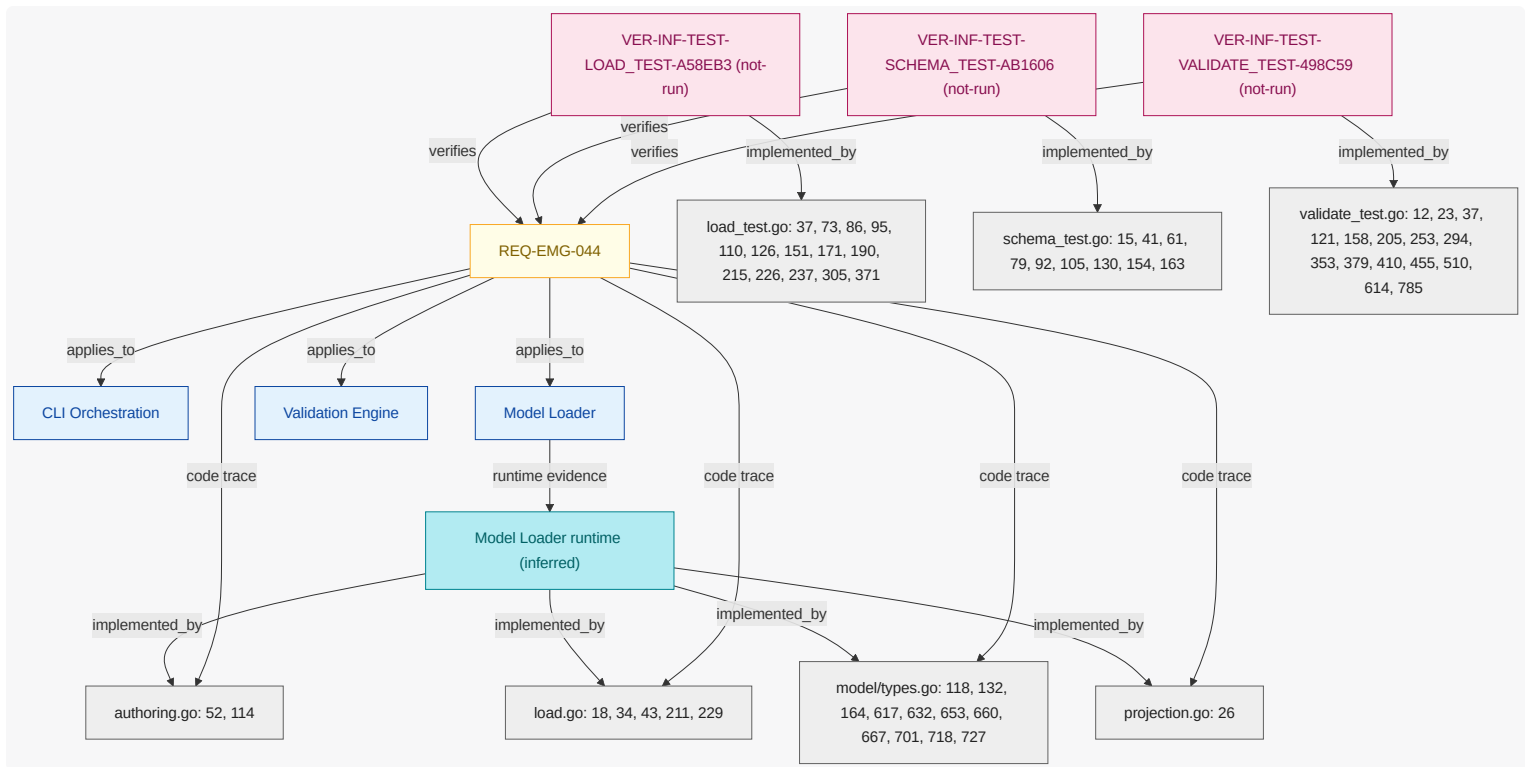
When model updated event is received, the engineering model go shall load schema version 2 engmod manifest plus catalog plus requirements plus architecture plus behavior plus assurance plus compliance plus views plus decisions documents into one canonical semantic model.

NOTE

engmod.yml is the only model entry point and every document path is explicit. YAML is the authoring format and CUE is the schema authority. Requirements may carry format-neutral qualification metadata such as title, category, rationale, source references, verification methods and criteria, priority, criticality, status, derived disposition, and tags. Views own design narratives plus generic document definitions, document-control metadata, content selections, authored sections, and external references; architecture does not contain behavior, assurance, compliance, dependency source, NAF, or projection-specific declarations.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.44. REQ-EMG-045

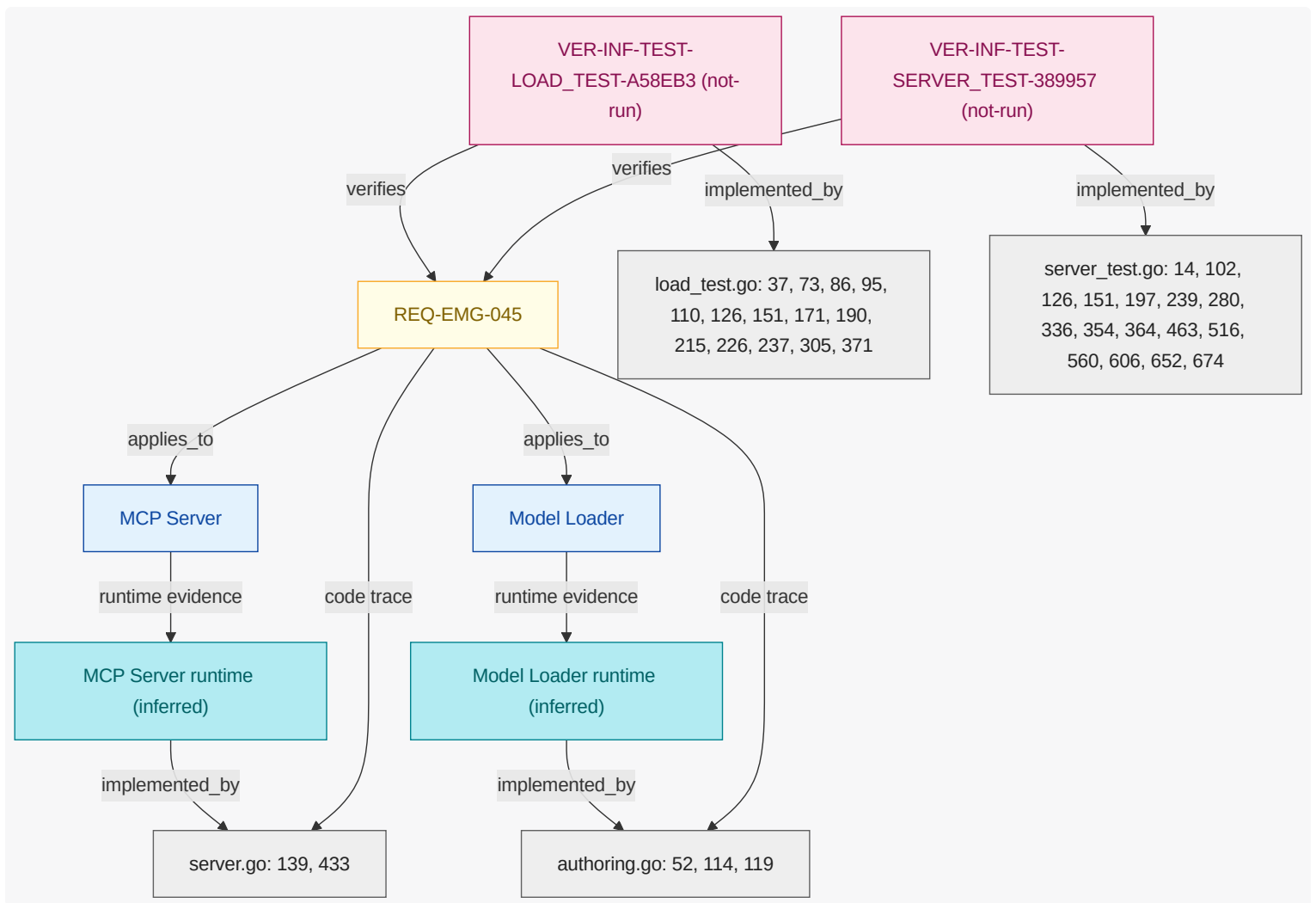
While mcp server is enabled , when mcp tool call is received and catalog references resolve successfully , the engineering model go shall return schema version 2 model authoring contract .

NOTE

The contract contains the manifest document map, authoritative CUE schema for each domain document, stable-ID and typed-relationship conventions, dependency publication aliases, valid references, and recommended task-scoped authoring order.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.45. REQ-EMG-046

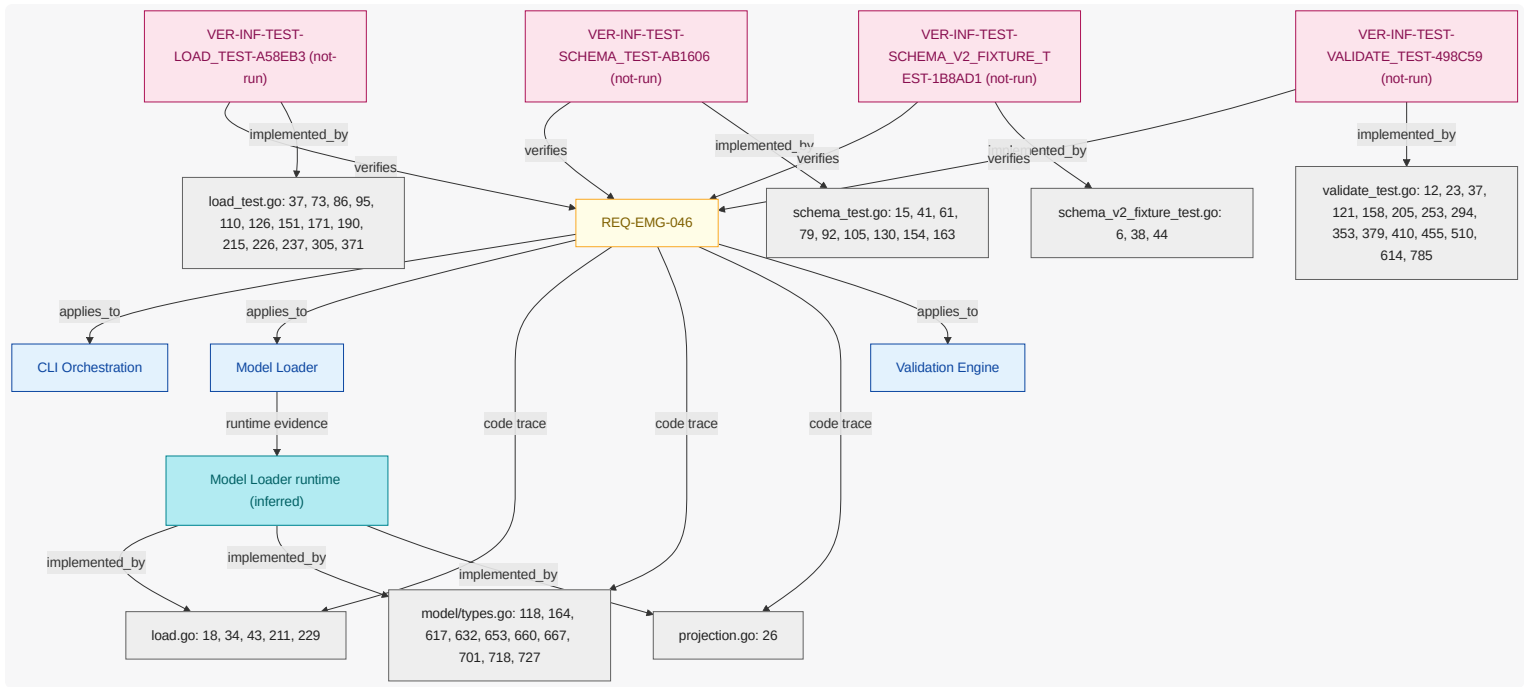
While ci validation mode is active , when generation run is requested , the engineering model go shall reject model input that violates the schema version 2 manifest or any referenced domain document contract before artifact generated event is raised .

NOTE

Missing documents, unsupported versions, unknown fields, duplicate concepts, invalid document-control values, incomplete derived requirement rationale, unresolved document content selections, unresolved local or qualified references, dependency publication violations, and cross-domain ownership violations are blocking.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.46. REQ-EMG-047

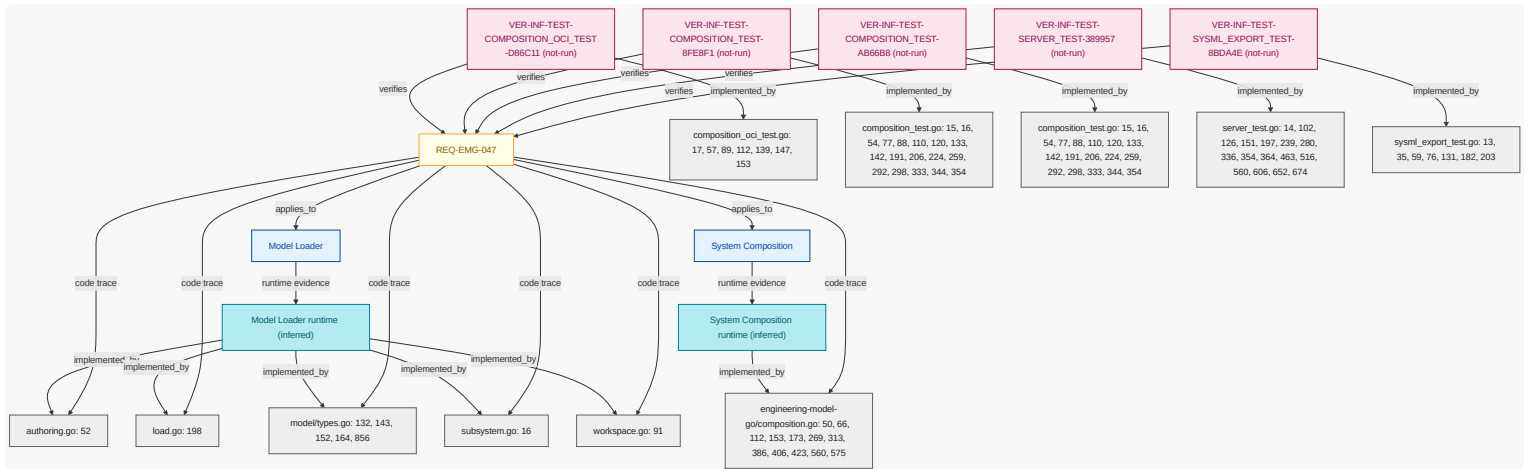
Where system composition is enabled, when generation run is requested, the engineering model go shall resolve each exact CUE module version from the OCI registry configured by CUE_REGISTRY.

NOTE

Dependencies are declared only in engmod.yml with an alias, major-qualified module path, exact semantic version, and selected publications. The resolved module contains engmod.yml, its domain documents, and cue.mod/module.cue.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.47. REQ-EMG-048

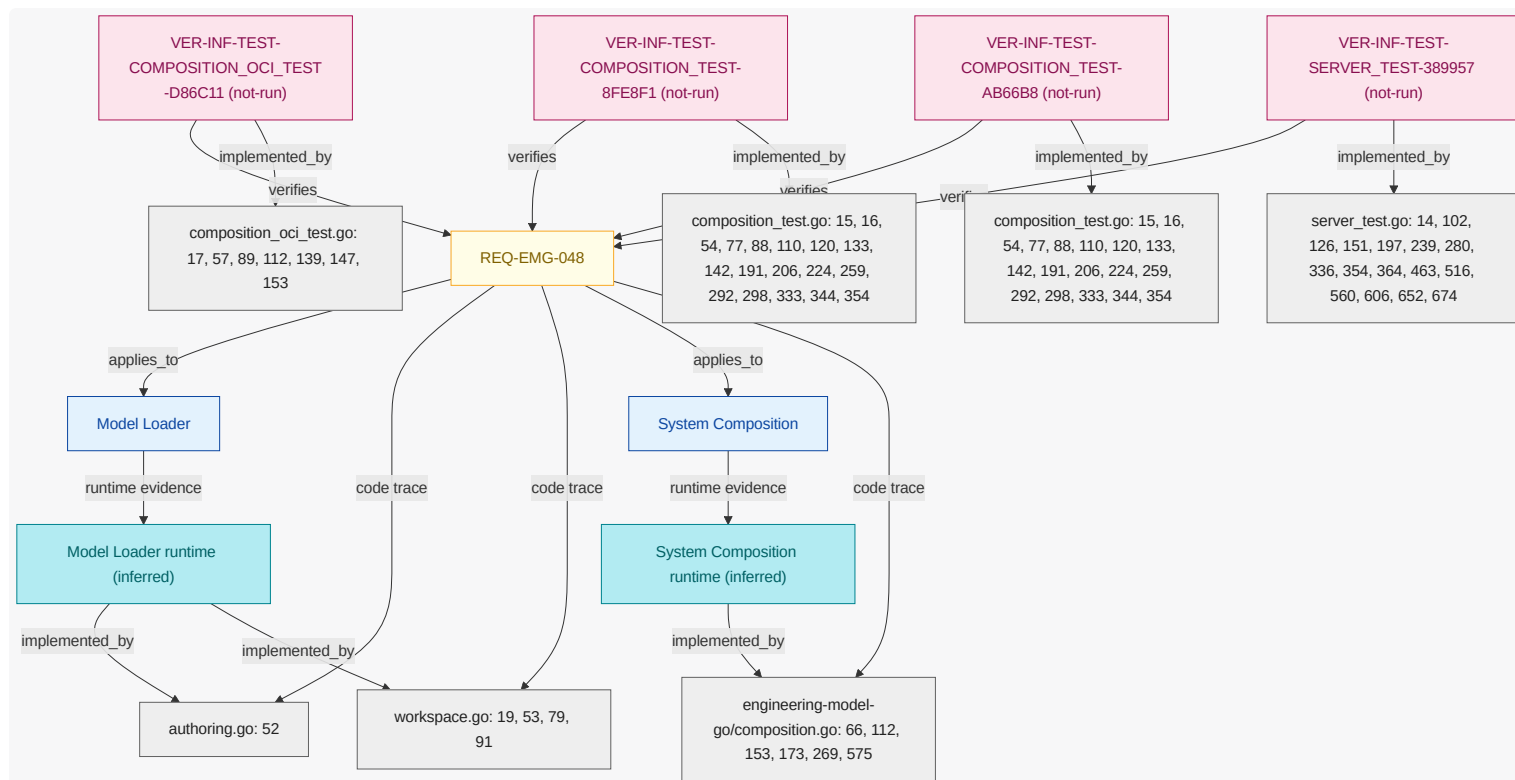
Where a model workspace replacement is declared, when generation run is requested, the engineering model go shall resolve the matching manifest dependency from the replacement repository without changing module identity or selected publications.

NOTE

engmod.work.yml maps module paths to local repositories for offline multi-repository development. A generated engmod.lock.yml records exact module versions and OCI digests but is not an authored semantic document.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.48. REQ-EMG-049

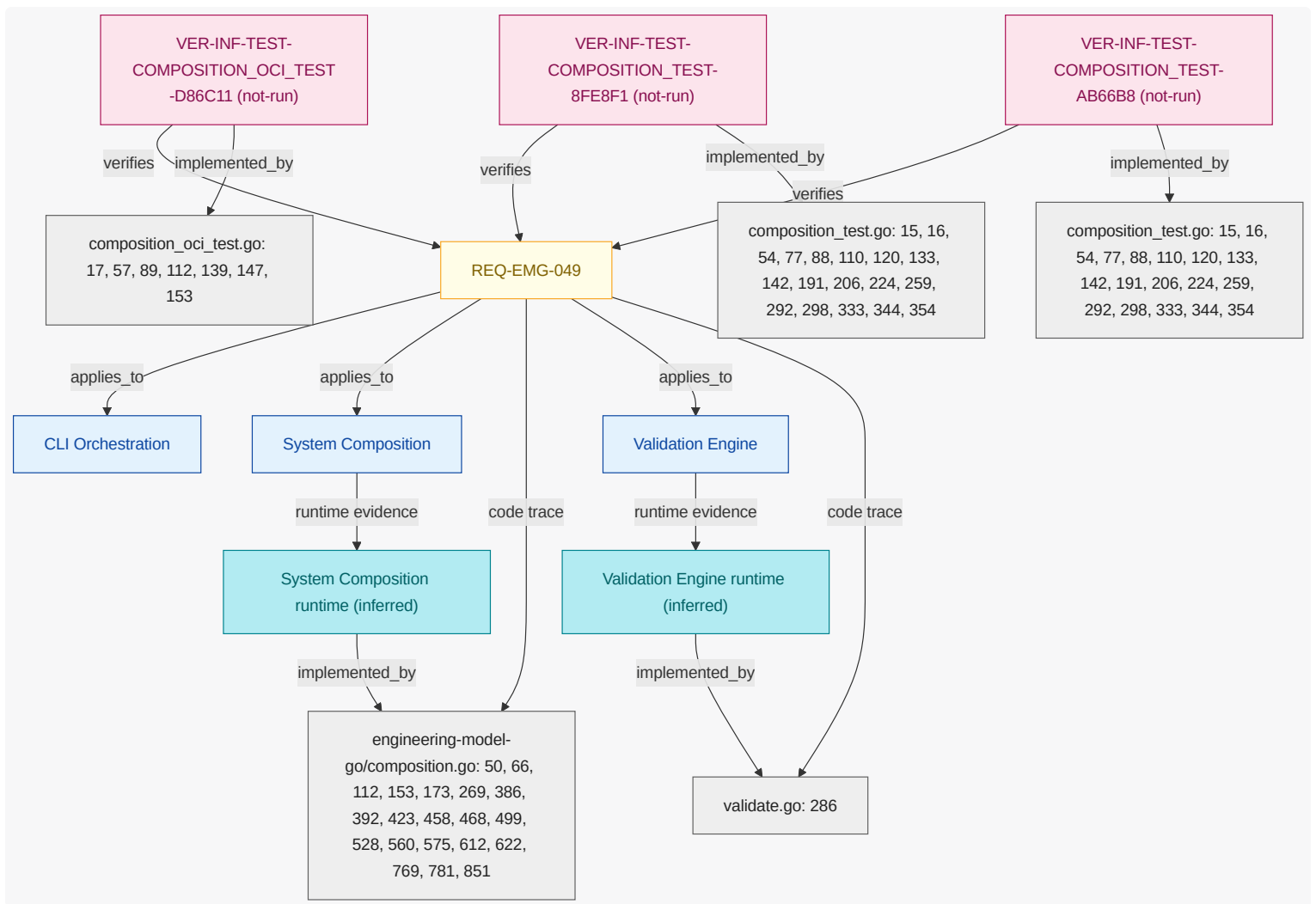
While ci validation mode is active , when generation run is requested , the engineering model go shall reject invalid manifest dependency plus publication plus workspace replacement plus qualified reference declarations.

NOTE

Every dependency requires one alias, one major-qualified module path, one exact version, and at least one exported publication. Aliases are unique, module identities match cue.mod/module.cue, workspace paths remain contained, imported references use alias double-colon stable-id syntax, and transitive entities are unavailable unless re-exported.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.49. REQ-EMG-050

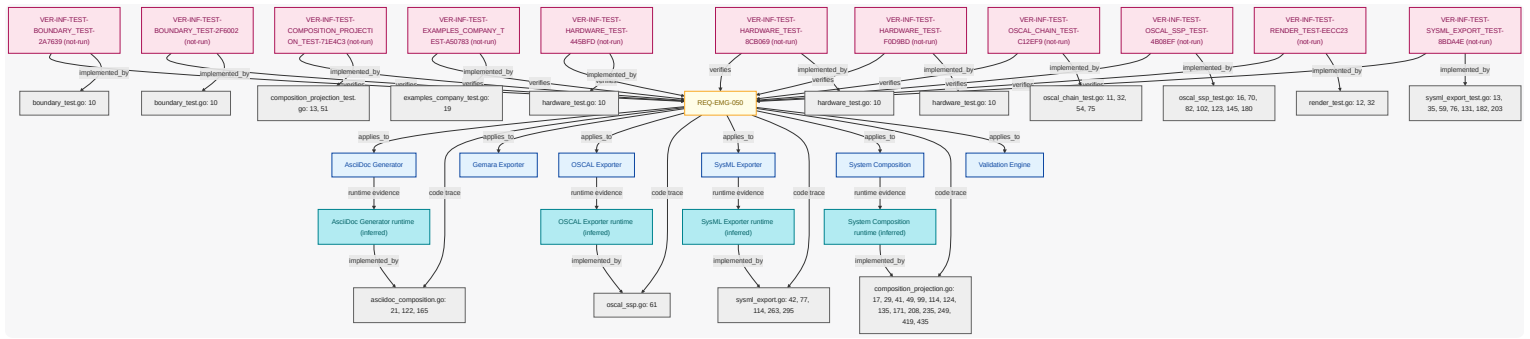
When example validation is requested, the engineering model go shall validate a schema version 2 company portfolio composed from multiple product systems that reuse published hardware plus software plus security plus compliance facets across repository boundaries.

NOTE

The reference workspace demonstrates explicit domain documents, dependency publication selection, qualified imported references, requirement delegation, contract satisfaction, inherited company controls, technical security controls, employee lifecycle compliance, provenance, exact OCI coordinates, local workspace replacements, and complete native format projections.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.50. REQ-EMG-051

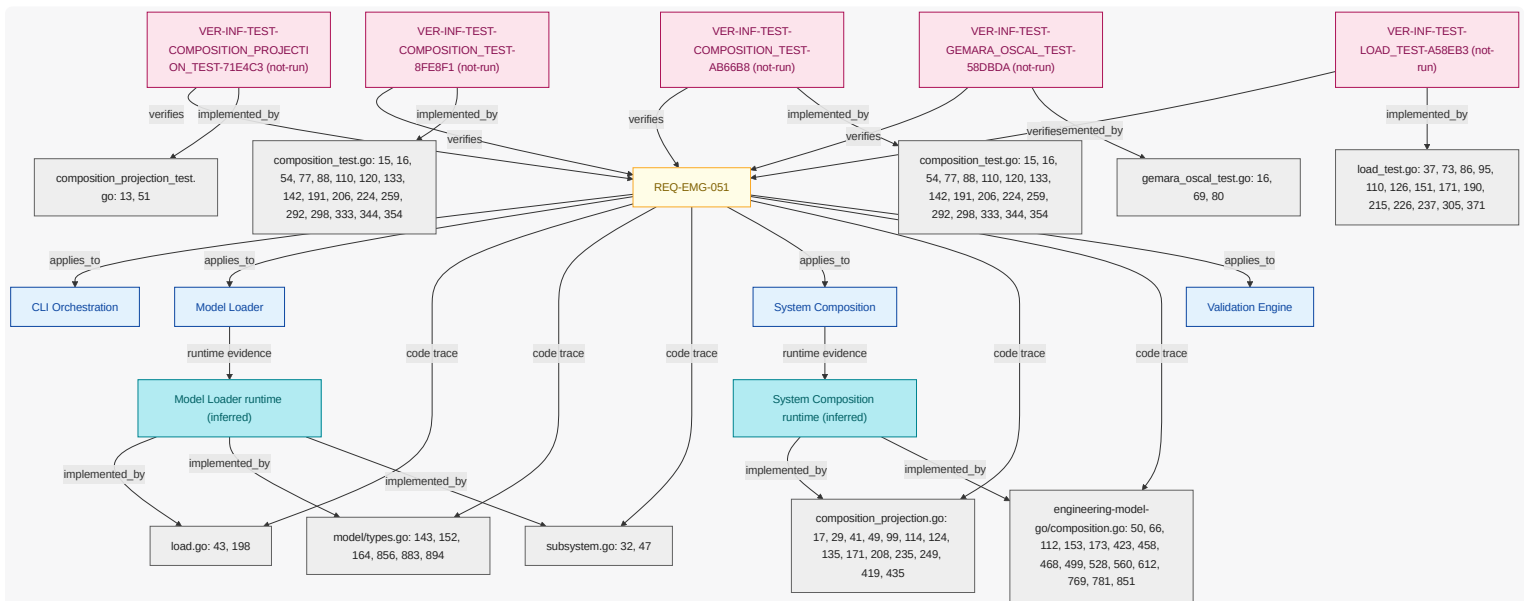
Where system composition is enabled, when generation run is requested, the engineering model go shall import only explicitly selected dependency publications with qualified identity plus source provenance.

NOTE

Each dependency publication exposes typed architecture, behavior, assurance, compliance, requirement, and view references. Imported entities retain module path, exact version, digest, source model, and publication identity; unexported dependency entities remain inaccessible.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.51. REQ-EMG-052

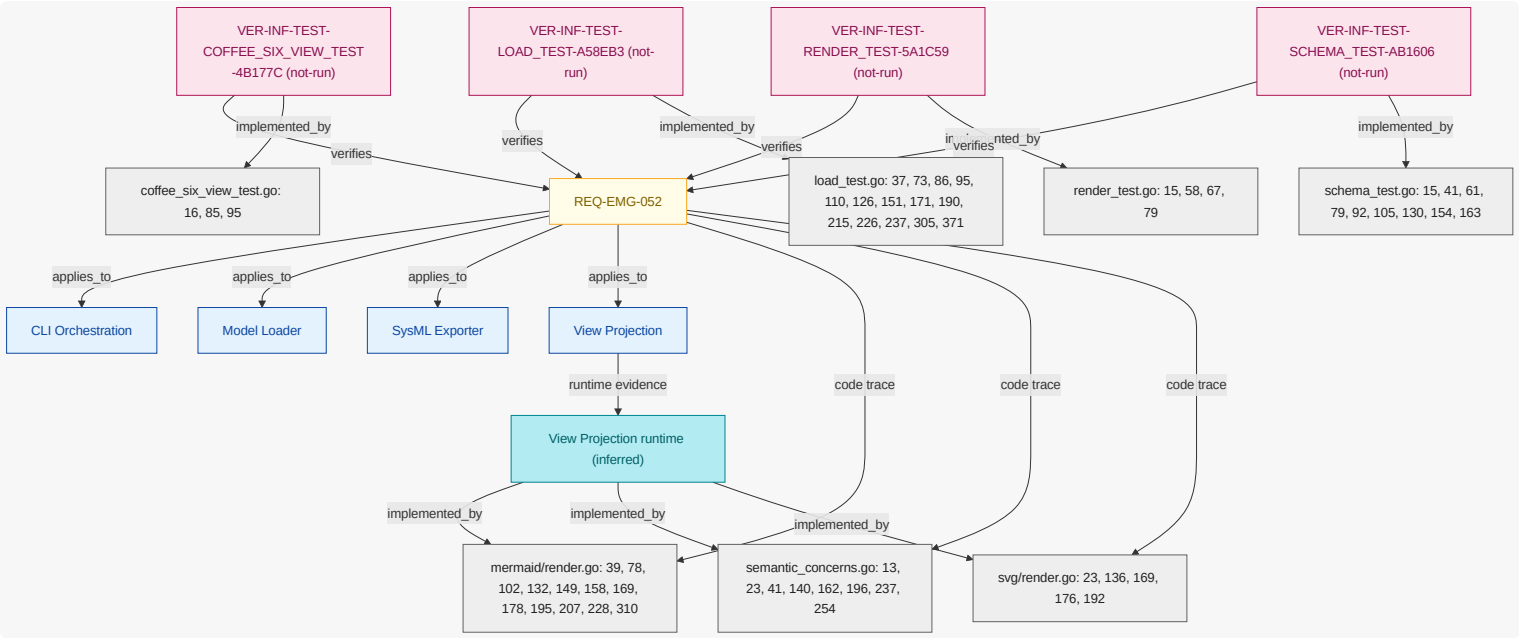
Where six-view example output is enabled, when generation run is requested, the engineering model go shall generate deterministic use-case plus logical plus process plus physical plus implementation plus deployment view artifacts and native SysML from one schema version 2 YAML bundle.

NOTE

The acceptance example preserves viewpoint separation, hierarchical capabilities, ordered typed interactions, nested hardware and software structures, ports, connections, engineering values, and software-to-hardware allocations. Mermaid sources and SVG renderings are generated without depending on authored SysML source files.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



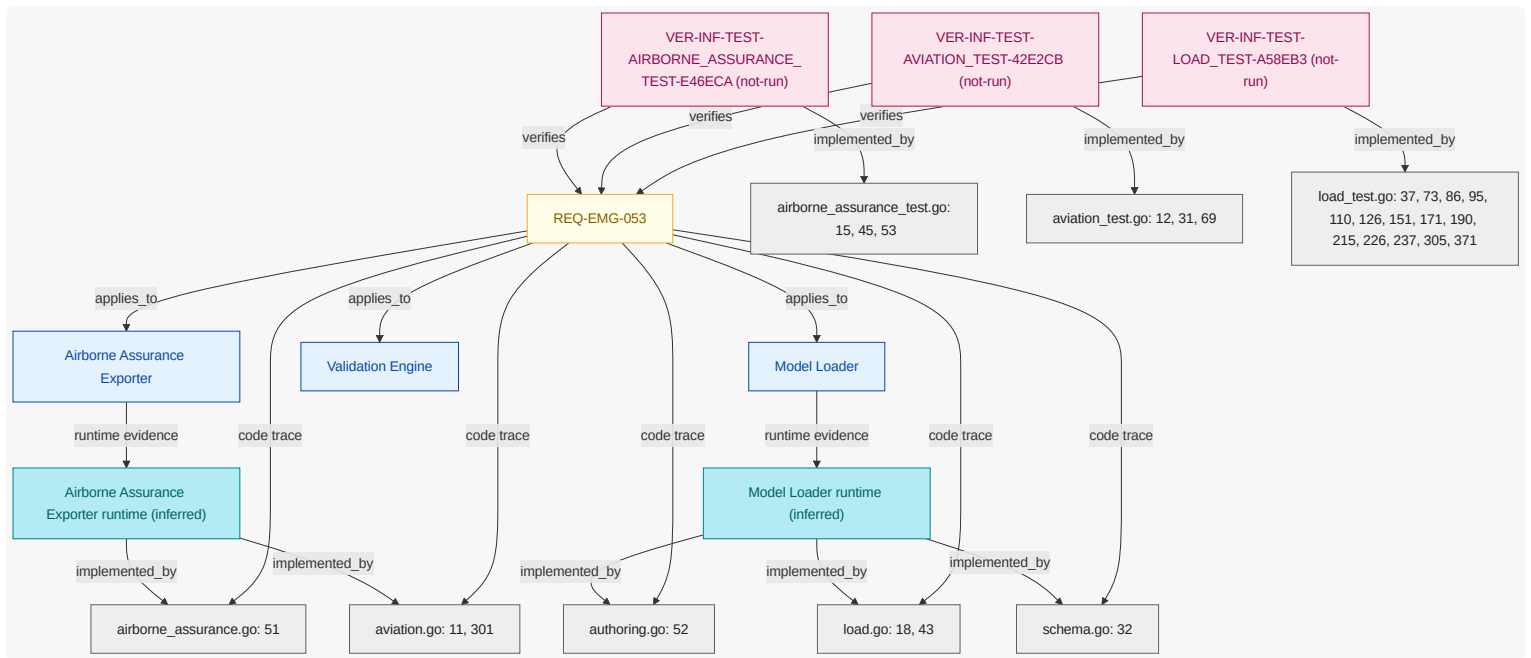
11.1.52. REQ-EMG-053

Where do-178c dal c readiness output is enabled , when model updated event is received , the engineering model go shall load a strict aviation assurance profile plus lifecycle data plus verification evidence plus configuration baselines plus tool assessments plus structural coverage summaries.

NOTE	The model references DO-178C or ED-12C and customer-controlled licensed objective identifiers without reproducing normative objective text. It distinguishes declared, planned, executed, reviewed, accepted, not-applicable, and authority-accepted evidence states .
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Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.53. REQ-EMG-054

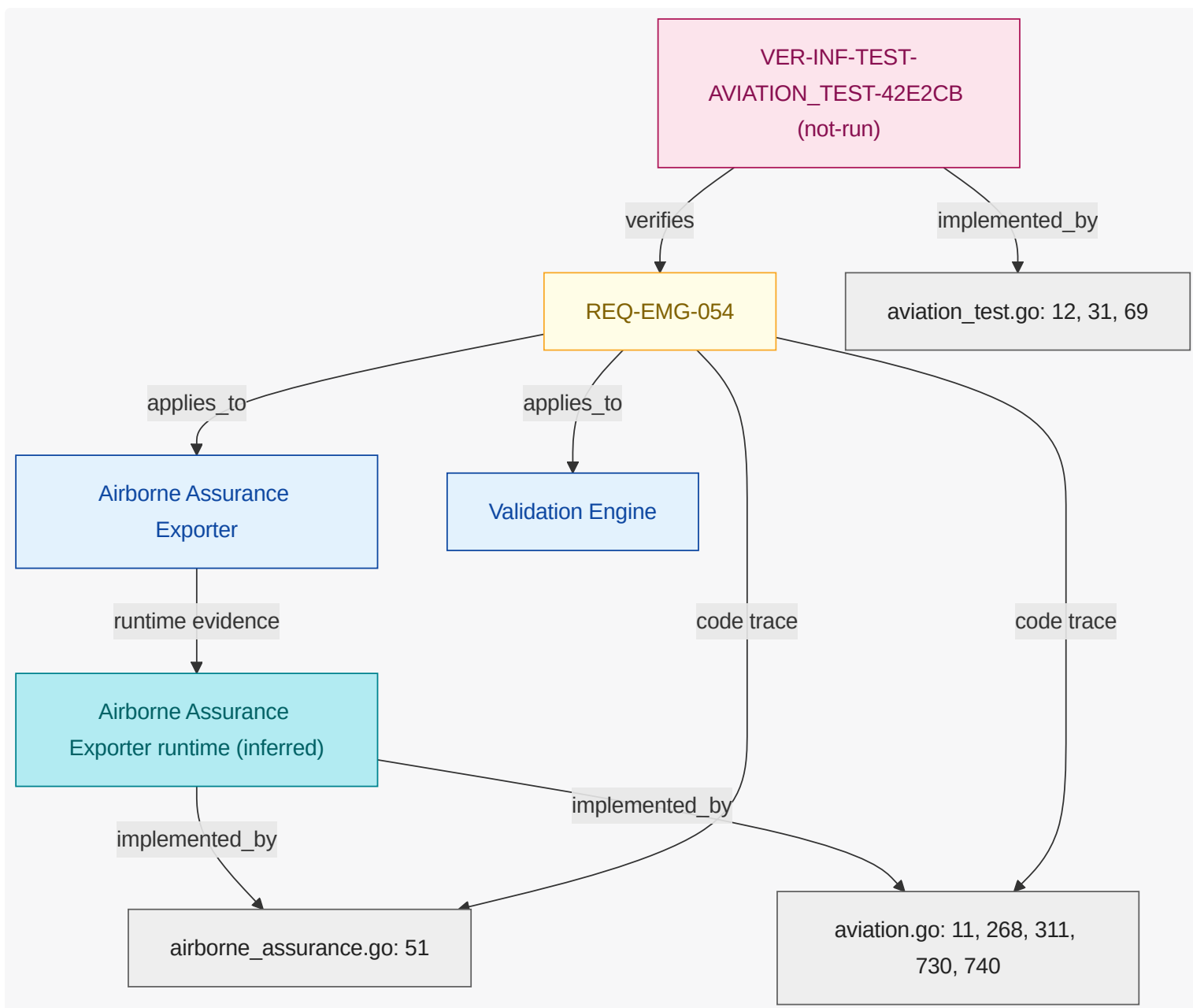
Where do-178c dal c readiness output is enabled , when generation run is requested , the engineering model go shall reject incomplete or inconsistent requirement classification plus derived requirement disposition plus verification chain plus independence plus baseline plus structural coverage plus tool assessment evidence .

NOTE

DAL assignment and objective applicability remain externally authorized inputs. Statement coverage is imported evidence tied to an exact build and environment; the tool does not perform safety assessment, coverage measurement, reviewer qualification, tool qualification, or certification approval.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.54. REQ-EMG-055

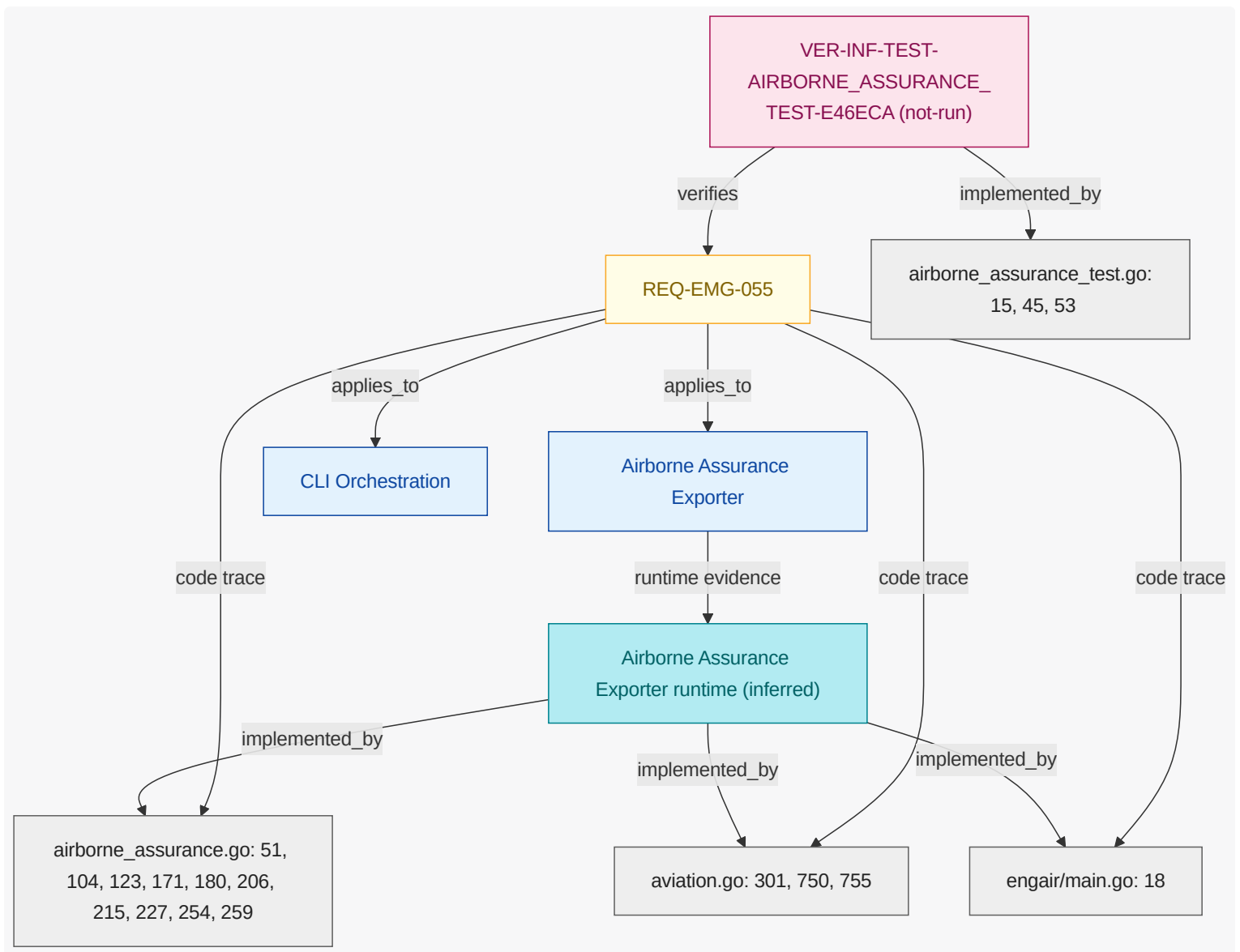
Where do-178c dal c readiness output is enabled , when generation run is requested , the engineering model go shall generate deterministic draft lifecycle data index plus objective evidence matrix plus requirements trace matrix plus verification summary plus structural coverage summary plus tool assessment register plus configuration index plus open gap report plus software accomplishment summary.

NOTE

Every output is visibly marked as draft evidence-readiness material, identifies its source baseline and unresolved gaps, and states that responsible human review plus configuration control plus certification authority acceptance remain required.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.1.55. REQ-EMG-056

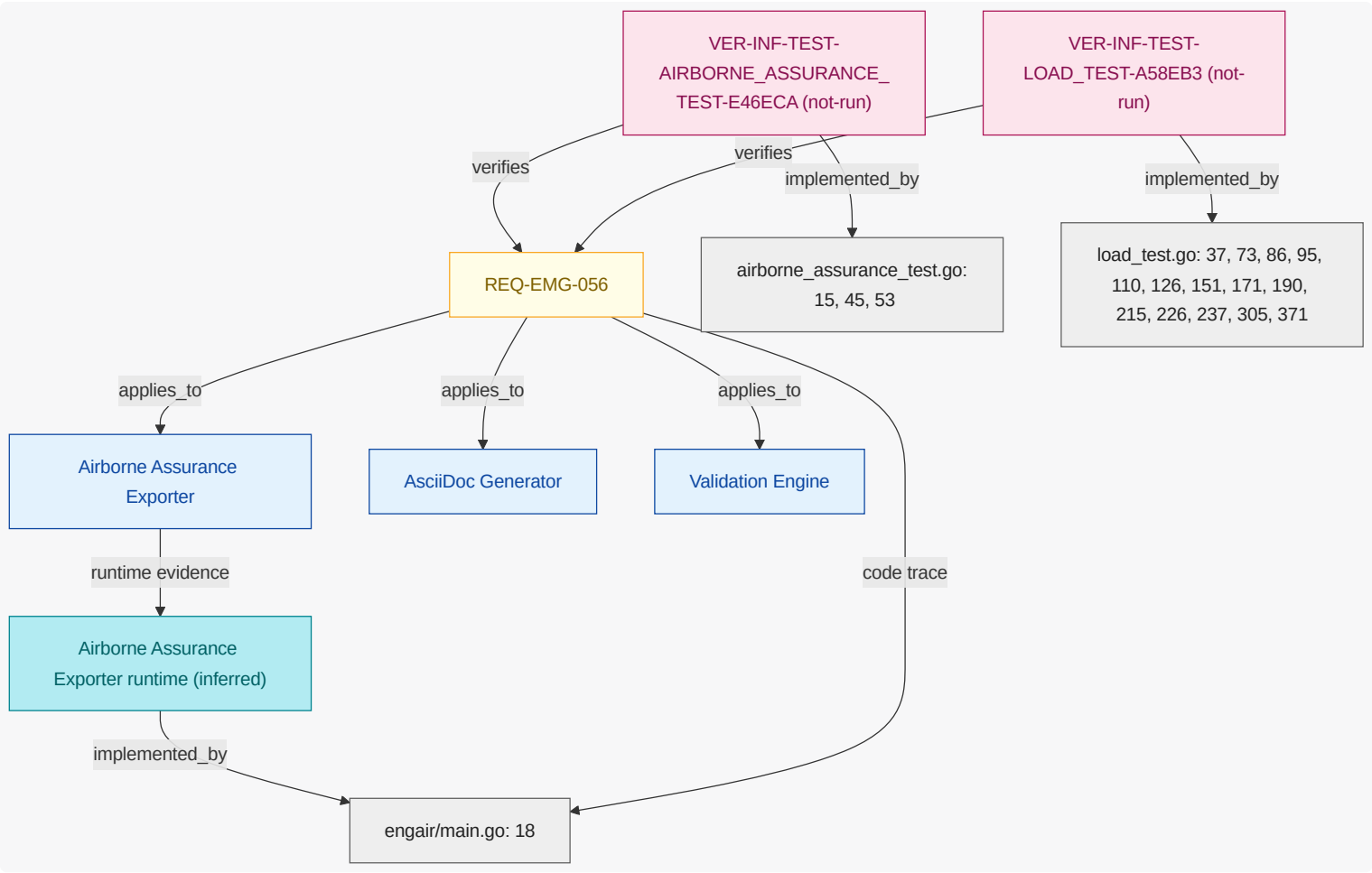
When example validation is requested, the engineering model go shall validate a schema version 2 DAL C flight control software example with placeholder planning plus requirements plus design plus code plus verification plus coverage plus configuration plus quality assurance plus certification liaison evidence.

NOTE

The example demonstrates realistic output shapes and intentionally open findings without claiming DO-178C compliance or certification.

Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



11.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

11.2.1. Authorized Checkout Flow

Field	Value
ID	VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367
Kind	e2e
Status	not-run
Verifies Requirements	REQ-PAY-001
Test Code Elements	examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml
Notes	executes authorized checkout flow from session start through payment authorization success

11.2.2. Ota Rollout Flow

Field	Value
ID	VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC
Kind	e2e
Status	not-run
Verifies Requirements	REQ-COF-003, REQ-COF-005
Test Code Elements	examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml
Notes	runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

11.2.3. Pr Opened Review Flow

Field	Value
ID	VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6
Kind	e2e
Status	not-run
Verifies Requirements	REQ-PRR-001, REQ-PRR-002, REQ-PRR-003, REQ-PRR-005
Test Code Elements	examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml
Derived Functional Owners	none
Result Summary	not-run:4
Evidence	examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml
Notes	exercises end-to-end pull request opened flow from webhook ingest to published review

11.2.4. Audit Record Persistence Test

Field	Value
ID	VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4
Kind	integration
Status	not-run

Field	Value
Verifies Requirements	REQ-PAY-005
Test Code Elements	audit_record_persistence_test.ts:5
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts
Notes	checks audit record persistence includes payment id and computed risk score

11.2.5. Audit Redaction Test

Field	Value
ID	VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4
Kind	integration
Status	not-run
Verifies Requirements	REQ-PRR-007
Test Code Elements	audit_redaction_test.ts:3
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts
Notes	checks audit evidence redaction for secrets and sensitive payload fields

11.2.6. Fraud Gate Test

Field	Value
ID	VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163
Kind	integration
Status	not-run
Verifies Requirements	REQ-PAY-002, REQ-PAY-004
Test Code Elements	fraud_gate_test.rs:6
Derived Functional Owners	none

Field	Value
Result Summary	not-run:2
Evidence	examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs
Notes	validates fraud gate escalation and block decisions before authorization completion

11.2.7. Logging Pipeline Test

Field	Value
ID	VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8
Kind	integration
Status	not-run
Verifies Requirements	REQ-COF-002, REQ-COF-007
Test Code Elements	logging_pipeline_test.ts:4
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts
Notes	validates telemetry logging pipeline persistence and operational reporting signals

11.2.8. Policy Only Fallback Test

Field	Value
ID	VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003
Kind	integration
Status	not-run
Verifies Requirements	REQ-PRR-004, REQ-PRR-006
Test Code Elements	policy_only_fallback_test.rs:5
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/bedrock-pr-review-github-app-sample/tests/integration/policy_only_fallback_test.rs
Notes	verifies policy-only fallback when Bedrock inference is unavailable

11.2.9. Airborne Assurance Test

Field	Value
ID	VER-INF-TEST-AIRBORNE_ASSURANCE_TEST-E46ECA
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-053 , REQ-EMG-055 , REQ-EMG-056
Test Code Elements	airborne_assurance_test.go:15, 45, 53
Derived Functional Owners	FU-AIRBORNE-ASSURANCE-EXPORTER , FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	airborne_assurance_test.go
Notes	Inferred from test source artifact.

11.2.10. AsciiDoc Complete Diagram Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_COMPLETE_DIAGRAM_TEST-8C18F7
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	asciidoc_complete_diagram_test.go:13
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_complete_diagram_test.go
Notes	Inferred from test source artifact.

11.2.11. AsciiDoc Diagrams Core Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003

Field	Value
Test Code Elements	asciidoc_diagrams_core_test.go:14, 41, 135, 161, 197, 223, 254, 273, 293, 327, 355, 383, 411, 431, 447, 459, 472
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_diagrams_core_test.go
Notes	Inferred from test source artifact.

11.2.12. AsciiDoc Diagrams Runtime Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	asciidoc_diagrams_runtime_test.go:12, 50, 72, 95
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_diagrams_runtime_test.go
Notes	Inferred from test source artifact.

11.2.13. AsciiDoc Evidence Table Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_EVIDENCE_TABLE_TEST-963640
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	asciidoc_evidence_table_test.go:11, 40
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1

Field	Value
Evidence	asciidoc_evidence_table_test.go
Notes	Inferred from test source artifact.

11.2.14. AsciiDoc Hardware Labels Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_HARDWARE_LABELS_TEST-AFE4FA
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-024
Test Code Elements	asciidoc_hardware_labels_test.go:13
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_hardware_labels_test.go
Notes	Inferred from test source artifact.

11.2.15. AsciiDoc Linking Units Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_LINKING_UNITS_TEST-22FDE2
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	asciidoc_linking_units_test.go:10, 46, 75, 101, 129, 148, 167, 186
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_linking_units_test.go
Notes	Inferred from test source artifact.

11.2.16. AsciiDoc Publication Scope Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_PUBLICATION_SCOPE_TEST-1BE464
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	asciidoc_publication_scope_test.go:13, 35, 75
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_publication_scope_test.go
Notes	Inferred from test source artifact.

11.2.17. AsciiDoc Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_TEST-F60765
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003 , REQ-EMG-014
Test Code Elements	asciidoc_test.go:16, 69, 115, 139, 175, 206, 246, 314, 372, 517, 531, 546
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:2
Evidence	asciidoc_test.go
Notes	Inferred from test source artifact.

11.2.18. AsciiDoc Test Evidence Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_TEST_EVIDENCE_TEST-C6896C
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003 , REQ-EMG-010 , REQ-EMG-030

Field	Value
Test Code Elements	asciidoc_test_evidence_test.go:13, 60
Derived Functional Owners	FU-ALLOCATION-TRACE , FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-CODEMAP-INFERENCE , FU-VIEW-PROJECTION
Result Summary	not-run:3
Evidence	asciidoc_test_evidence_test.go
Notes	Inferred from test source artifact.

11.2.19. AsciiDoc Views Test

Field	Value
ID	VER-INF-TEST-ASCIIDOC_VIEWS_TEST-BFC7F2
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	asciidoc_views_test.go:11
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	asciidoc_views_test.go
Notes	Inferred from test source artifact.

11.2.20. Aviation Test

Field	Value
ID	VER-INF-TEST-AVIATION_TEST-42E2CB
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-053 , REQ-EMG-054
Test Code Elements	aviation_test.go:12, 31, 69
Derived Functional Owners	FU-AIRBORNE-ASSURANCE-EXPORTER , FU-MODEL-LOADER , FU-VALIDATION-ENGINE
Result Summary	not-run:2

Field	Value
Evidence	model/aviation_test.go
Notes	Inferred from test source artifact.

11.2.21. Boundary Test

Field	Value
ID	VER-INF-TEST-BOUNDARY_TEST-2A7639
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001 , REQ-EMG-009 , REQ-EMG-050
Test Code Elements	boundary_test.go:10
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-SYSML-EXPORTER , FU-SYSTEM-COMPOSITION , FU-THREAT-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	validate/boundary_test.go
Notes	Inferred from test source artifact.

11.2.22. Boundary Test

Field	Value
ID	VER-INF-TEST-BOUNDARY_TEST-2F6002
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003 , REQ-EMG-050
Test Code Elements	boundary_test.go:10
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-OSCAL-EXPORTER , FU-SYSML-EXPORTER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE , FU-VIEW-PROJECTION
Result Summary	not-run:2
Evidence	view /boundary_test.go
Notes	Inferred from test source artifact.

11.2.23. Catalog Systems Test

Field	Value
ID	VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001
Test Code Elements	catalog_systems_test.go:11, 36, 82
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:1
Evidence	catalog_systems_test.go
Notes	Inferred from test source artifact.

11.2.24. Checkout Decline Response Test

Field	Value
ID	VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671
Kind	test
Status	not-run
Verifies Requirements	REQ-PAY-003, REQ-PAY-006
Test Code Elements	checkout_decline_response_test.go:7
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/payments-engineering-sample/tests/contract/checkout_decline_response_test.go
Notes	validates checkout decline response contract includes reason and retry guidance

11.2.25. Coffee Six View Test

Field	Value
ID	VER-INF-TEST-COFFEE_SIX_VIEW_TEST-4B177C
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-012, REQ-EMG-052

Field	Value
Test Code Elements	coffee_six_view_test.go:16, 85, 95
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-CODEMAP-INFERENCE, FU-GEMARA-EXPORTER, FU-LOBSTER-EXPORTER, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-SYSML-EXPORTER, FU-THREAT-EXPORTER, FU-TRL-EXPORTER, FU-VALIDATION-ENGINE, FU-VIEW-PROJECTION
Result Summary	not-run:2
Evidence	coffee_six_view_test.go
Notes	Inferred from test source artifact.

11.2.26. Composition Oci Test

Field	Value
ID	VER-INF-TEST-COMPOSITION_OCI_TEST-D86C11
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-047, REQ-EMG-048, REQ-EMG-049
Test Code Elements	composition_oci_test.go:17, 57, 89, 112, 139, 147, 153
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-SYSTEM-COMPOSITION, FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	composition_oci_test.go
Notes	Inferred from test source artifact.

11.2.27. Composition Projection Test

Field	Value
ID	VER-INF-TEST-COMPOSITION_PROJECTION_TEST-71E4C3
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-050, REQ-EMG-051
Test Code Elements	composition_projection_test.go:13, 51
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-GEMARA-EXPORTER, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-SYSML-EXPORTER, FU-SYSTEM-COMPOSITION, FU-VALIDATION-ENGINE

Field	Value
Result Summary	not-run:2
Evidence	composition_projection_test.go
Notes	Inferred from test source artifact.

11.2.28. Composition Test

Field	Value
ID	VER-INF-TEST-COMPOSITION_TEST-8FE8F1
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-047 , REQ-EMG-048 , REQ-EMG-049 , REQ-EMG-051
Test Code Elements	composition_test.go:16, 77, 110, 142, 191, 206, 224, 259, 292, 298, 333, 344, 354
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE
Result Summary	not-run:4
Evidence	composition_test.go
Notes	Inferred from test source artifact.

11.2.29. Composition Test

Field	Value
ID	VER-INF-TEST-COMPOSITION_TEST-AB66B8
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-007 , REQ-EMG-016 , REQ-EMG-030 , REQ-EMG-047 , REQ-EMG-048 , REQ-EMG-049 , REQ-EMG-051
Test Code Elements	composition_test.go:15, 54, 88, 120, 133
Derived Functional Owners	FU-ALLOCATION-TRACE , FU-CLI-ORCHESTRATION , FU-CODEMAP-INFERENCE , FU-MCP-SERVER , FU-MODEL-LOADER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE
Result Summary	not-run:7
Evidence	mcp/composition_test.go
Notes	Inferred from test source artifact.

11.2.30. E2e Test

Field	Value
ID	VER-INF-TEST-E2E_TEST-BE56F8
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001
Test Code Elements	e2e_test.go:14, 39
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:1
Evidence	e2e_test.go
Notes	Inferred from test source artifact.

11.2.31. Ears Preflight Test

Field	Value
ID	VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001, REQ-EMG-009
Test Code Elements	ears_preflight_test.go:11, 60, 105
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-MODEL-LOADER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-THREAT-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	ears_preflight_test.go
Notes	Inferred from test source artifact.

11.2.32. Examples Company Test

Field	Value
ID	VER-INF-TEST-EXAMPLES_COMPANY_TEST-A50783
Kind	test
Status	not-run

Field	Value
Verifies Requirements	REQ-EMG-050
Test Code Elements	examples_company_test.go:19
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-GEMARA-EXPORTER, FU-OSCAL-EXPORTER, FU-SYSML-EXPORTER, FU-SYSTEM-COMPOSITION, FU-VALIDATION-ENGINE
Result Summary	not-run:1
Evidence	examples_company_test.go
Notes	Inferred from test source artifact.

11.2.33. Gemara Export Test

Field	Value
ID	VER-INF-TEST-GEMARA_EXPORT_TEST-45C982
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-012, REQ-EMG-015
Test Code Elements	gemara_export_test.go:28, 138, 153, 197, 210, 250, 268, 286
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-CODEMAP-INFERENCE, FU-GEMARA-EXPORTER, FU-LOBSTER-EXPORTER, FU-MCP-SERVER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-SYSML-EXPORTER, FU-THREAT-EXPORTER, FU-TRLC-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	gemara_export_test.go
Notes	Inferred from test source artifact.

11.2.34. Gemara Oscal Test

Field	Value
ID	VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-015, REQ-EMG-051
Test Code Elements	gemara_oscal_test.go:16, 69, 80

Field	Value
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-MCP-SERVER , FU-MODEL-LOADER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	gemara_oscal_test.go
Notes	Inferred from test source artifact.

11.2.35. Github Check Run Contract Test

Field	Value
ID	VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F
Kind	test
Status	not-run
Verifies Requirements	REQ-PRR-005
Test Code Elements	github_check_run_contract_test.go:8
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/bedrock-pr-review-github-app-sample/tests/contract/github_check_run_contract_test.go
Notes	validates GitHub check-run contract shape and required review result fields

11.2.36. Hardware Test

Field	Value
ID	VER-INF-TEST-HARDWARE_TEST-445BFD
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001 , REQ-EMG-009 , REQ-EMG-050
Test Code Elements	hardware_test.go:10
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-SYSML-EXPORTER , FU-SYSTEM-COMPOSITION , FU-THREAT-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	validate/hardware_test.go

Field	Value
Notes	Inferred from test source artifact.

11.2.37. Hardware Test

Field	Value
ID	VER-INF-TEST-HARDWARE_TEST-8CB069
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003, REQ-EMG-050
Test Code Elements	hardware_test.go:10
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-GEMARA-EXPORTER, FU-OSCAL-EXPORTER, FU-SYSML-EXPORTER, FU-SYSTEM-COMPOSITION, FU-VALIDATION-ENGINE, FU-VIEW-PROJECTION
Result Summary	not-run:2
Evidence	view /hardware_test.go
Notes	Inferred from test source artifact.

11.2.38. Hardware Test

Field	Value
ID	VER-INF-TEST-HARDWARE_TEST-F0D9BD
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-007, REQ-EMG-050
Test Code Elements	hardware_test.go:10
Derived Functional Owners	FU-ASCIIDOC-GENERATOR, FU-GEMARA-EXPORTER, FU-MCP-SERVER, FU-OSCAL-EXPORTER, FU-SYSML-EXPORTER, FU-SYSTEM-COMPOSITION, FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	mcp/hardware_test.go
Notes	Inferred from test source artifact.

11.2.39. Inferred Code Test

Field	Value
ID	VER-INF-TEST-INFERRED_CODE_TEST-3CA82E

Field	Value
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010
Test Code Elements	inferred_code_test.go:13, 39, 61
Derived Functional Owners	FU-CODEMAP-INFERENCE
Result Summary	not-run:1
Evidence	inferred_code_test.go
Notes	Inferred from test source artifact.

11.2.40. Inferred Javascript Test

Field	Value
ID	VER-INF-TEST-INFERRED_JAVASCRIPT_TEST-52EB3E
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010 , REQ-EMG-030
Test Code Elements	inferred_javascript_test.go:12
Derived Functional Owners	FU-ALLOCATION-TRACE , FU-CODEMAP-INFERENCE
Result Summary	not-run:2
Evidence	inferred_javascript_test.go
Notes	Inferred from test source artifact.

11.2.41. Inferred Layers Test

Field	Value
ID	VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010
Test Code Elements	inferred_layers_test.go:13, 21, 40, 58, 84, 118

Field	Value
Derived Functional Owners	FU-CODEMAP-INFERENCE
Result Summary	not-run:1
Evidence	inferred_layers_test.go
Notes	Inferred from test source artifact.

11.2.42. Inferred Verification Test

Field	Value
ID	VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010
Test Code Elements	inferred_verification_test.go:13, 26, 64, 100, 133, 192, 216, 228
Derived Functional Owners	FU-CODEMAP-INFERENCE
Result Summary	not-run:1
Evidence	inferred_verification_test.go
Notes	Inferred from test source artifact.

11.2.43. Inferred Verilog Test

Field	Value
ID	VER-INF-TEST-INFERRED_VERILOG_TEST-32D5FA
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010 , REQ-EMG-030
Test Code Elements	inferred_verilog_test.go:13, 25, 68
Derived Functional Owners	FU-ALLOCATION-TRACE , FU-CODEMAP-INFERENCE
Result Summary	not-run:2
Evidence	inferred_verilog_test.go

Field	Value
Notes	Inferred from test source artifact.

11.2.44. Javascript Test

Field	Value
ID	VER-INF-TEST-JAVASCRIPT_TEST-4E9F26
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010 , REQ-WEB-001
Test Code Elements	javascript_test.go:11, 77
Derived Functional Owners	FU-CODEMAP-INFERENCE
Result Summary	not-run:2
Evidence	codemap/javascript_test.go
Notes	Inferred from test source artifact.

11.2.45. Load Test

Field	Value
ID	VER-INF-TEST-LOAD_TEST-A58EB3
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-044 , REQ-EMG-045 , REQ-EMG-046 , REQ-EMG-051 , REQ-EMG-052 , REQ-EMG-053 , REQ-EMG-056
Test Code Elements	load_test.go:37, 73, 86, 95, 110, 126, 151, 171, 190, 215, 226, 237, 305, 371
Derived Functional Owners	FU-AIRBORNE-ASSURANCE-EXPORTER , FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-MCP-SERVER , FU-MODEL-LOADER , FU-SYSML-EXPORTER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE , FU-VIEW-PROJECTION
Result Summary	not-run:9
Evidence	model/load_test.go
Notes	Inferred from test source artifact.

11.2.46. Lobster Export Test

Field	Value
ID	VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-006 , REQ-EMG-012
Test Code Elements	lobster_export_test.go:13, 62
Derived Functional Owners	FU-ALLOCATION-TRACE , FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-CODEMAP-INFERENCE , FU-GEMARA-EXPORTER , FU-LOBSTER-EXPORTER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-SYSML-EXPORTER , FU-THREAT-EXPORTER , FU-TRLC-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	lobster_export_test.go
Notes	Inferred from test source artifact.

11.2.47. Main Test

Field	Value
ID	VER-INF-TEST-MAIN_TEST-6B61AE
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-007 , REQ-EMG-008
Test Code Elements	main_test.go:12, 22
Derived Functional Owners	FU-MCP-SERVER
Result Summary	not-run:2
Evidence	cmd/ engmcp /main_test.go
Notes	Inferred from test source artifact.

11.2.48. Main Test

Field	Value
ID	VER-INF-TEST-MAIN_TEST-AAEB3C
Kind	test
Status	not-run

Field	Value
Verifies Requirements	REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-039 , REQ-EMG-040
Test Code Elements	main_test.go:17, 33, 54, 69
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:4
Evidence	cmd/ engsysml /main_test.go
Notes	Inferred from test source artifact.

11.2.49. Main Test

Field	Value
ID	VER-INF-TEST-MAIN_TEST-C71B6B
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-042 , REQ-EMG-043
Test Code Elements	main_test.go:13, 25
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-NAF-EXPORTER , FU-VALIDATION-ENGINE , FU-VIEW-PROJECTION
Result Summary	not-run:2
Evidence	cmd/ engnaf /main_test.go
Notes	Inferred from test source artifact.

11.2.50. Metamodel Instance Test

Field	Value
ID	VER-INF-TEST-METAMODEL_INSTANCE_TEST-33196D
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-038 , REQ-EMG-039 , REQ-EMG-040
Test Code Elements	metamodel_instance_test.go:15, 40, 74, 113, 148, 197, 223, 262
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE

Field	Value
Result Summary	not-run:3
Evidence	model/metamodel_instance_test.go
Notes	Inferred from test source artifact.

11.2.51. Metamodel Test

Field	Value
ID	VER-INF-TEST-METAMODEL_TEST-81AF9A
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-038 , REQ-EMG-039
Test Code Elements	metamodel_test.go:33, 68, 76, 101, 117, 135, 147
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	internal/sysmlmetamodel/metamodel_test.go
Notes	Inferred from test source artifact.

11.2.52. Naf Export Test

Field	Value
ID	VER-INF-TEST-NAF_EXPORT_TEST-26D0A4
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-041 , REQ-EMG-042 , REQ-EMG-043
Test Code Elements	naf_export_test.go:14, 42, 69
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-NAF-EXPORTER , FU-VALIDATION-ENGINE , FU-VIEW-PROJECTION
Result Summary	not-run:3
Evidence	naf_export_test.go
Notes	Inferred from test source artifact.

11.2.53. Oscar Chain Test

Field	Value
ID	VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-012, REQ-EMG-013, REQ-EMG-050
Test Code Elements	oscal_chain_test.go:11, 32, 54, 75
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-CODEMAP-INFERENCE, FU-GEMARA-EXPORTER, FU-LOBSTER-EXPORTER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-SYSML-EXPORTER, FU-SYSTEM-COMPOSITION, FU-THREAT-EXPORTER, FU-TRLIC-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	oscal_chain_test.go
Notes	Inferred from test source artifact.

11.2.54. Oscal Profile Test

Field	Value
ID	VER-INF-TEST-OSCAL_PROFILE_TEST-9234C6
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-013
Test Code Elements	oscal_profile_test.go:11, 28
Derived Functional Owners	FU-CLI-ORCHESTRATION, FU-OSCAL-EXPORTER
Result Summary	not-run:1
Evidence	oscal_profile_test.go
Notes	Inferred from test source artifact.

11.2.55. Oscal Ssp Test

Field	Value
ID	VER-INF-TEST-OSCAL_SSP_TEST-4B08EF
Kind	test
Status	not-run

Field	Value
Verifies Requirements	REQ-EMG-012 , REQ-EMG-013 , REQ-EMG-050
Test Code Elements	oscal_ssp_test.go:16, 70, 82, 102, 123, 145, 180
Derived Functional Owners	FU-ALLOCATION-TRACE , FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-CODEMAP-INFERENCE , FU-GEMARA-EXPORTER , FU-LOBSTER-EXPORTER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-SYSML-EXPORTER , FU-SYSTEM-COMPOSITION , FU-THREAT-EXPORTER , FU-TRLC-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	oscal_ssp_test.go
Notes	Inferred from test source artifact.

11.2.56. Projection Test

Field	Value
ID	VER-INF-TEST-PROJECTION_TEST-D8D752
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-035 , REQ-EMG-036
Test Code Elements	projection_test.go:10, 37
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	model/projection_test.go
Notes	Inferred from test source artifact.

11.2.57. Project Test

Field	Value
ID	VER-INF-TEST-PROJECT_TEST-2ECF72
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003
Test Code Elements	project_test.go:11, 52, 86, 137, 168

Field	Value
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-VIEW-PROJECTION
Result Summary	not-run:1
Evidence	view /project_test.go
Notes	Inferred from test source artifact.

11.2.58. Render Test

Field	Value
ID	VER-INF-TEST-RENDER_TEST-5A1C59
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-012 , REQ-EMG-052
Test Code Elements	render_test.go:15, 58, 67, 79
Derived Functional Owners	FU-ALLOCATION-TRACE , FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-CODEMAP-INFERENCE , FU-GEMARA-EXPORTER , FU-LOBSTER-EXPORTER , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-SYSML-EXPORTER , FU-THREAT-EXPORTER , FU-TRL-EXPORTER , FU-VALIDATION-ENGINE , FU-VIEW-PROJECTION
Result Summary	not-run:2
Evidence	render/svg/render_test.go
Notes	Inferred from test source artifact.

11.2.59. Render Test

Field	Value
ID	VER-INF-TEST-RENDER_TEST-EECC23
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-003 , REQ-EMG-050
Test Code Elements	render_test.go:12, 32
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-OSCAL-EXPORTER , FU-SYSML-EXPORTER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE , FU-VIEW-PROJECTION
Result Summary	not-run:2

Field	Value
Evidence	render/mermaid/render_test.go
Notes	Inferred from test source artifact.

11.2.60. Requirements Delta Test

Field	Value
ID	VER-INF-TEST-REQUIREMENTS_DELTA_TEST-64E119
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-031 , REQ-EMG-032 , REQ-EMG-033
Test Code Elements	requirements_delta_test.go:15, 80, 97, 112, 135, 155, 179, 200, 222, 263, 281, 314, 353, 369, 468, 476
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-CHANGE , FU-MODEL-LOADER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	requirements_delta_test.go
Notes	Inferred from test source artifact.

11.2.61. Scan Test

Field	Value
ID	VER-INF-TEST-SCAN_TEST-B04F81
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010 , REQ-SAMPLE-001, REQ-SAMPLE-002
Test Code Elements	scan_test.go:11, 48, 73, 111, 147, 181, 226
Derived Functional Owners	FU-CODEMAP-INFERENCE
Result Summary	not-run:3
Evidence	codemap/scan_test.go
Notes	Inferred from test source artifact.

11.2.62. Schema Test

Field	Value
ID	VER-INF-TEST-SCHEMA_TEST-AB1606
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-035 , REQ-EMG-041 , REQ-EMG-043 , REQ-EMG-044 , REQ-EMG-046 , REQ-EMG-052
Test Code Elements	schema_test.go:15, 41, 61, 79, 92, 105, 130, 154, 163
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-NAF-EXPORTER , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE , FU-VIEW-PROJECTION
Result Summary	not-run:6
Evidence	model/schema_test.go
Notes	Inferred from test source artifact.

11.2.63. Schema V2 Fixture Test

Field	Value
ID	VER-INF-TEST-SCHEMA_V2_FIXTURE_TEST-1B8AD1
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-046
Test Code Elements	schema_v2_fixture_test.go:6, 38, 44
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-VALIDATION-ENGINE
Result Summary	not-run:1
Evidence	schema_v2_fixture_test.go
Notes	Inferred from test source artifact.

11.2.64. Schema V2 Projection Test

Field	Value
ID	VER-INF-TEST-SCHEMA_V2_PROJECTION_TEST-81EEBD
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-037

Field	Value
Test Code Elements	schema_v2_projection_test.go:23, 238, 280
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	schema_v2_projection_test.go
Notes	Inferred from test source artifact.

11.2.65. Semantic Test

Field	Value
ID	VER-INF-TEST-SEMANTIC_TEST-CB8EF3
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-037
Test Code Elements	semantic_test.go:11, 49, 77, 100, 126, 194, 240, 265, 311, 352, 362
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	model/semantic_test.go
Notes	Inferred from test source artifact.

11.2.66. Server Test

Field	Value
ID	VER-INF-TEST-SERVER_TEST-389957
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-007 , REQ-EMG-008 , REQ-EMG-045 , REQ-EMG-047 , REQ-EMG-048
Test Code Elements	server_test.go:14, 102, 126, 151, 197, 239, 280, 336, 354, 364, 463, 516, 560, 606, 652, 674
Derived Functional Owners	FU-MCP-SERVER , FU-MODEL-LOADER , FU-SYSTEM-COMPOSITION
Result Summary	not-run:5

Field	Value
Evidence	mcp/server_test.go
Notes	Inferred from test source artifact.

11.2.67. Structurizr Export Test

Field	Value
ID	VER-INF-TEST-STRUCTURIZR_EXPORT_TEST-886374
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-005
Test Code Elements	structurizr_export_test.go:14, 40
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-STRUCTURIZR-EXPORTER
Result Summary	not-run:1
Evidence	structurizr_export_test.go
Notes	Inferred from test source artifact.

11.2.68. Sysml Coverage Test

Field	Value
ID	VER-INF-TEST-SYSML_COVERAGE_TEST-9C7166
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-039 , REQ-EMG-040
Test Code Elements	sysml_coverage_test.go:11, 40, 54
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:2
Evidence	sysml_coverage_test.go
Notes	Inferred from test source artifact.

11.2.69. Sysml Export Test

Field	Value
ID	VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-039 , REQ-EMG-040 , REQ-EMG-047 , REQ-EMG-050
Test Code Elements	sysml_export_test.go:13, 35, 59, 76, 131, 182, 203
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-GEMARA-EXPORTER , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-SYSML-EXPORTER , FU-SYSTEM-COMPOSITION , FU-VALIDATION-ENGINE
Result Summary	not-run:6
Evidence	sysml_export_test.go
Notes	Inferred from test source artifact.

11.2.70. Sysml Interchange Test

Field	Value
ID	VER-INF-TEST-SYSML_INTERCHANGE_TEST-CF2F5A
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-040
Test Code Elements	sysml_interchange_test.go:15, 33, 61, 84
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-SYSML-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	sysml_interchange_test.go
Notes	Inferred from test source artifact.

11.2.71. Telemetry Ingest Contract Test

Field	Value
ID	VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2
Kind	test
Status	not-run
Verifies Requirements	REQ-COF-001

Field	Value
Test Code Elements	telemetry_ingest_contract_test.go:7
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/contract/telemetry_ingest_contract_test.go
Notes	validates telemetry ingest contract fields and accepted request semantics

11.2.72. Threat Model Export Test

Field	Value
ID	VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-004, REQ-EMG-011, REQ-EMG-012
Test Code Elements	threat_model_export_test.go:17, 54, 97, 116
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-ASCIIDOC-GENERATOR, FU-CLI-ORCHESTRATION, FU-CODEMAP-INFERENCE, FU-GEMARA-EXPORTER, FU-LOBSTER-EXPORTER, FU-OSCAL-EXPORTER, FU-STRUCTURIZR-EXPORTER, FU-SYSML-EXPORTER, FU-THREAT-EXPORTER, FU-TRLC-EXPORTER, FU-VALIDATION-ENGINE
Result Summary	not-run:3
Evidence	threat_model_export_test.go
Notes	Inferred from test source artifact.

11.2.73. Trace Matrix Test

Field	Value
ID	VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-028, REQ-EMG-029, REQ-EMG-030
Test Code Elements	trace_matrix_test.go:18, 45, 70, 84, 112
Derived Functional Owners	FU-ALLOCATION-TRACE, FU-CODEMAP-INFERENCE, FU-VALIDATION-ENGINE

Field	Value
Result Summary	not-run:3
Evidence	trace_matrix_test.go
Notes	Inferred from test source artifact.

11.2.74. Trlc Export Test

Field	Value
ID	VER-INF-TEST-TRLC_EXPORT_TEST-DD1345
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-006
Test Code Elements	trlc_export_test.go:13, 44
Derived Functional Owners	FU-CLI-ORCHESTRATION , FU-LOBSTER-EXPORTER , FU-TRLC-EXPORTER
Result Summary	not-run:1
Evidence	trlc_export_test.go
Notes	Inferred from test source artifact.

11.2.75. Validate Test

Field	Value
ID	VER-INF-TEST-VALIDATE_TEST-498C59
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-001 , REQ-EMG-009 , REQ-EMG-011 , REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-044 , REQ-EMG-046
Test Code Elements	validate_test.go:12, 23, 37, 121, 158, 205, 253, 294, 353, 379, 410, 455, 510, 614, 785
Derived Functional Owners	FU-ASCIIDOC-GENERATOR , FU-CLI-ORCHESTRATION , FU-MODEL-LOADER , FU-OSCAL-EXPORTER , FU-STRUCTURIZR-EXPORTER , FU-SYSML-EXPORTER , FU-THREAT-EXPORTER , FU-VALIDATION-ENGINE
Result Summary	not-run:8
Evidence	validate/validate_test.go
Notes	Inferred from test source artifact.

11.2.76. Verilog Test

Field	Value
ID	VER-INF-TEST-VERILOG_TEST-44F499
Kind	test
Status	not-run
Verifies Requirements	REQ-EMG-010
Test Code Elements	verilog_test.go:14, 33, 50, 64, 76, 85, 101, 115
Derived Functional Owners	FU-CODEMAP-INFERENCE
Result Summary	not-run:1
Evidence	codemap/verilog_test.go
Notes	Inferred from test source artifact.

11.2.77. Checkout Decline Reason Test

Field	Value
ID	VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD
Kind	unit
Status	not-run
Verifies Requirements	REQ-PAY-003, REQ-PAY-008
Test Code Elements	checkout_decline_reason_test.go:7, 15, 20
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/payments-engineering-sample/tests/unit/checkout_decline_reason_test.go
Notes	validates checkout decline reason normalization and retry policy messaging

11.2.78. Ota Campaign Test

Field	Value
ID	VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB
Kind	unit
Status	not-run

Field	Value
Verifies Requirements	REQ-COF-003, REQ-COF-004
Test Code Elements	ota_campaign_test.go:8
Derived Functional Owners	none
Result Summary	not-run:2
Evidence	examples/coffee-fleet-ota-cloud-sample/tests/unit/ota_campaign_test.go
Notes	verifies OTA campaign signature validation and rollout trigger behavior

11.2.79. Rate Limit Deferral Test

Field	Value
ID	VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE
Kind	unit
Status	not-run
Verifies Requirements	REQ-PRR-008
Test Code Elements	rate_limit_deferral_test.go:8
Derived Functional Owners	none
Result Summary	not-run:1
Evidence	examples/bedrock-pr-review-github-app-sample/tests/unit/rate_limit_deferral_test.go
Notes	verifies rate-limit deferral behavior for pull request review processing

12. Gemara GRC Model

This chapter renders the model as OpenSSF Gemara (<https://gemara.openssf.org>) documents. Each artifact is built with the official `go-gemara` SDK types and validated against the published Gemara CUE schemas. Generate the full YAML set with `enggemara` ; query it via the `gemara.*` MCP tools.

Gemara Layer	Artifact	Entries
L1	Vector Catalog	4
L1	Principle Catalog	5
L1	Guidance Catalog	2
L2	Capability Catalog	20
L2	Control Catalog	5
L2	Threat Catalog	2
L3	Risk Catalog	3
L3	Policy	2
-	Control → Threat Mapping	3
-	Lexicon	83
L7	Audit Log	5
L6	Enforcement Log	2

12.1. Controls (L2)

Control	Objective	Group	Assessment Requirements
CTRL-MCP-PATH-BOUNDARY	Enforce repo-root path constraints and reject traversal paths in MCP tools.	input-validation	1
CTRL-STRICT-MCP-INPUT-SCHEMA	Enforce per-tool input schema and reject unknown arguments.	input-validation	1
CTRL-TRACEABILITY-COVERAGE	Require requirement-linked verification evidence for modeled behavior.	assurance	1
CTRL-TRACE-LINK-INTEGRITY	Reject code trace links (TRLINKS, ENGMODEL-LINKS) that resolve to no requirement or model element, and export a consolidated traceability...	assurance	1

Control	Objective	Group	Assessment Requirements
CTRL-ARTIFACT-FRESHNESS-GATE	Generate artifacts deterministically and fail continuous integration when regenerated documents differ from the committed artifacts.	assurance	1

12.2. Threats (L2)

Threat	Title	Group
TS-MCP-PATH-TRAVERSAL	MCP path traversal accesses non-repo files	stride-tampering
TS-TRACEABILITY-DRIFT	Requirement traceability drifts from implementation	stride-repudiation

12.3. Risks (L3)

Risk	Title	Severity
RISK-MCP-PATH-ESCAPE	MCP path traversal exposes non-repository files	High
RISK-TRACEABILITY-DRIFT	Traceability drift weakens assurance evidence	Medium
RISK-STALE-ARTIFACTS	Stale generated artifacts misrepresent the system	Low

13. Reference Index

13.1. Authored References

CTRL-ARTIFACT-FRESHNESS-GATE

category=assurance; description=Generate artifacts deterministically and fail continuous integration when regenerated documents differ from the committed artifacts.

Key	Value
Kind	Control
Name	Generated Artifact Freshness Gate
Mentioned In	n/a

CTRL-MCP-PATH-BOUNDARY

category=input-validation; description=Enforce repo-root path constraints and reject traversal paths in MCP tools.

Key	Value
Kind	Control
Name	MCP Path Boundary Enforcement
Mentioned In	Security View

CTRL-STRICT-MCP-INPUT-SCHEMA

category=input-validation; description=Enforce per-tool input schema and reject unknown arguments.

Key	Value
Kind	Control
Name	Strict MCP Input Schemas
Mentioned In	Security View

CTRL-TRACE-LINK-INTEGRITY

category=assurance; description=Reject code trace links (TRLC-LINKS, ENGMODEL-LINKS) that resolve to no requirement or model element, and export a consolidated traceability matrix with an implemented, verified, delegated, and orphan rollup.

Key	Value
Kind	Control
Name	Trace Link Integrity
Mentioned In	n/a

CTRL-TRACEABILITY-COVERAGE

category=assurance; description=Require requirement-linked verification evidence for modeled behavior.

Key	Value
Kind	Control
Name	Requirement Traceability Coverage
Mentioned In	Security View

DO-ARCHITECTURE-MODEL

termRef=DATA-ARCHITECTURE-MODEL; schemaRef=model/schema/architecture.cue; sensitivity=internal

Key	Value
Kind	Data Object
Name	Architecture Model
Mentioned In	Security View , REQ-EMG-001 , REQ-EMG-035

DO-CANONICAL-SEMANTIC-MODEL

termRef=DATA-CANONICAL-SEMANTIC-MODEL; schemaRef=model/semantic.go; sensitivity=internal

Key	Value
Kind	Data Object
Name	Canonical Semantic Model
Mentioned In	Architecture Intent View , REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-038 , REQ-EMG-044

DO-MCP-TOOL-RESULT

termRef=DATA-MCP-TOOL-RESULT; schemaRef=mcp.tool-response.v1; sensitivity=internal

Key	Value
Kind	Data Object
Name	MCP Tool Result
Mentioned In	Security View , REQ-EMG-007

DO-MODEL-AUTHORING-CONTRACT

termRef=DATA-MODEL-AUTHORING-CONTRACT; schemaRef=model/authoring.go; sensitivity=internal

Key	Value
Kind	Data Object

Key	Value
Name	Model Authoring Contract
Mentioned In	REQ-EMG-045

DO-NAF-V4-ARCHITECTURE

termRef=DATA-NAF-V4-ARCHITECTURE; schemaRef=generated/ARCHITECTURE.naf.adoc; sensitivity=internal

Key	Value
Kind	Data Object
Name	NAF v4.1 Architecture Document
Mentioned In	REQ-EMG-042

DO-REQUIREMENTS-DELTA

termRef=DATA-REQUIREMENTS-DELTA; schemaRef=requirements_delta.go; sensitivity=internal

Key	Value
Kind	Data Object
Name	Requirements Delta
Mentioned In	Security View , REQ-EMG-031

DO-REQUIREMENTS-DIFF

termRef=DATA-REQUIREMENTS-DIFF; schemaRef=requirements_delta.go; sensitivity=internal

Key	Value
Kind	Data Object
Name	Requirements Change Summary
Mentioned In	Security View , REQ-EMG-034

DO-REQUIREMENTS-DOCUMENT

termRef=DATA-REQUIREMENTS-DOCUMENT; schemaRef=model/schema/requirements.cue; sensitivity=internal

Key	Value
Kind	Data Object
Name	Requirements Document
Mentioned In	Security View , REQ-EMG-032 , REQ-EMG-033 , REQ-EMG-035

DO-STRUCTURIZR-DSL

termRef=DATA-STRUCTURIZR-DSL; schemaRef=generated/STRUCTURIZR.dsl; sensitivity=internal

Key	Value
Kind	Data Object
Name	Structurizr DSL
Mentioned In	Architecture Intent View , Security View , REQ-EMG-005

DO-SYSML-V2-METAMODEL

termRef=DATA-SYSML-V2-METAMODEL; schemaRef=tools/sysml/metamodel; sensitivity=public

Key	Value
Kind	Data Object
Name	SysML v2 Metamodel
Mentioned In	REQ-EMG-038 , REQ-EMG-039

DO-SYSML-V2-MODEL

termRef=DATA-SYSML-V2-MODEL; schemaRef=generated/ARCHITECTURE.sysml; sensitivity=internal

Key	Value
Kind	Data Object
Name	SysML v2 Model
Mentioned In	n/a

DO-THREAT-DRAGON-JSON

termRef=DATA-THREAT-DRAGON-JSON; schemaRef=generated/threat-dragon-v2.json; sensitivity=internal

Key	Value
Kind	Data Object
Name	Threat Dragon JSON
Mentioned In	Security View

DEP-CI-PIPELINE

environment=ci; region=hosted; account=github-actions; cluster=shared-runner; namespace=engineering-model-go; trustZone=automation

Key	Value
Kind	Deployment Target
Name	CI Pipeline Runner
Mentioned In	n/a

DEP-LOCAL-WORKSPACE

environment=dev; region=local; account=workstation; cluster=local-shell; namespace=engineering-model-go; trustZone=developer

Key	Value
Kind	Deployment Target
Name	Local Workspace
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS

entry=edit-model; exits=publish-artifacts; steps=5

Key	Value
Kind	Flow
Name	Model Change to Verified Artifacts Flow
Mentioned In	Security View

FLOW-REQUIREMENTS-DELTA-TO-CANONICAL

entry=load-requirements-delta; exits=replace-requirements; steps=5

Key	Value
Kind	Flow
Name	Requirements Delta to Canonical Requirements Flow
Mentioned In	Security View

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::edit-model

kind=user_action; ref=ACT-IMPLEMENTATION-ENGINEER; action=Update model and requirements artifacts; dataIn=; dataOut=architecture_update; async=false

Key	Value
Kind	Flow Step
Name	Update model and requirements artifacts

Key	Value
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::fail-validation

kind=system_action; ref=FU-VALIDATION-ENGINE; action=Emit validation diagnostics and block publication; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Emit validation diagnostics and block publication
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::generate-artifacts

kind=system_action; ref=FU-CLI-ORCHESTRATION; action=Run generation and export commands; dataIn=; dataOut=generated_artifacts; async=false

Key	Value
Kind	Flow Step
Name	Run generation and export commands
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::publish-artifacts

kind=system_action; ref=FU-ASCIIDOC-GENERATOR; action=Persist generated outputs for downstream use; dataIn=; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Persist generated outputs for downstream use
Mentioned In	n/a

FLOW-MODEL-CHANGE-TO-VERIFIED-ARTIFACTS::validate-model

kind=system_action; ref=FU-VALIDATION-ENGINE; action=Validate authored model references and relationships; dataIn=architecture_update; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Validate authored model references and relationships

Key	Value
Mentioned In	n/a

FLOW-REQUIREMENTS-DELTA-TO-CANONICAL::load-requirements-delta

kind=system_action; ref=FU-MODEL-CHANGE; action=Strictly decode and validate stable-ID operations.; dataIn=DO-REQUIREMENTS-DELTA; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Strictly decode and validate stable-ID operations.
Mentioned In	n/a

FLOW-REQUIREMENTS-DELTA-TO-CANONICAL::merge-requirements-candidate

kind=system_action; ref=FU-MODEL-CHANGE; action=Apply operations to an in-memory YAML candidate while preserving unchanged nodes.; dataIn=DO-REQUIREMENTS-DELTA, DO-REQUIREMENTS-DOCUMENT; dataOut=DO-REQUIREMENTS-DIFF; async=false

Key	Value
Kind	Flow Step
Name	Apply operations to an in-memory YAML candidate while preserving unchanged nodes.
Mentioned In	n/a

FLOW-REQUIREMENTS-DELTA-TO-CANONICAL::reject-requirements-delta

kind=notification; ref=FU-MODEL-CHANGE; action=Return diagnostics without modifying canonical requirements.; dataIn=; dataOut=DO-REQUIREMENTS-DIFF; async=false

Key	Value
Kind	Flow Step
Name	Return diagnostics without modifying canonical requirements.
Mentioned In	n/a

FLOW-REQUIREMENTS-DELTA-TO-CANONICAL::replace-requirements

kind=system_action; ref=FU-MODEL-CHANGE; action=Atomically replace canonical requirements while preserving file permissions.; dataIn=DO-REQUIREMENTS-DOCUMENT; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Atomically replace canonical requirements while preserving file permissions.

Key	Value
Mentioned In	n/a

FLOW-REQUIREMENTS-DELTA-TO-CANONICAL::validate-requirements-candidate

kind=control_action; ref=FU-VALIDATION-ENGINE; action=Run complete-model and EARS validation against the candidate.; dataIn=DO-REQUIREMENTS-DOCUMENT, DO-ARCHITECTURE-MODEL; dataOut=; async=false

Key	Value
Kind	Flow Step
Name	Run complete-model and EARS validation against the candidate.
Mentioned In	n/a

FU-AIRBORNE-ASSURANCE-EXPORTER

Validates aviation assurance evidence-readiness profiles and generates clearly marked draft lifecycle-data indexes, trace matrices, gap reports, and certification-support summaries without claiming compliance or authority acceptance.

Key	Value
Kind	Functional Unit
Name	Airborne Assurance Exporter
Mentioned In	Airborne Assurance Test , Aviation Test , Load Test

IF-CLI-ENGAI

protocol=cli; endpoint=cmd/engair; schemaRef=airborne_assurance.go; owner=FU-AIRBORNE-ASSURANCE-EXPORTER

Key	Value
Kind	Interface
Name	engair CLI
Mentioned In	n/a

IF-CLI-ENGCHANGE

protocol=cli; endpoint=cmd/engchange; schemaRef=requirements_delta.go; owner=FU-MODEL-CHANGE

Key	Value
Kind	Interface
Name	engchange CLI
Mentioned In	n/a

IF-CLI-ENGDOC

protocol=cli; endpoint=cmd/engdoc; schemaRef=docs/asciidoc-architecture-generator-design.md; owner=FU-ASCIIDOC-GENERATOR

Key	Value
Kind	Interface
Name	engdoc CLI
Mentioned In	n/a

IF-CLI-ENGDRAGON

protocol=cli; endpoint=cmd/engdragon; schemaRef=threat_model_export.go; owner=FU-THREAT-EXPORTER

Key	Value
Kind	Interface
Name	engdragon CLI
Mentioned In	n/a

IF-CLI-ENGGEMARA

protocol=cli; endpoint=cmd/enggemara; schemaRef=gemara_export.go; owner=FU-GEMARA-EXPORTER

Key	Value
Kind	Interface
Name	enggemara CLI
Mentioned In	n/a

IF-CLI-ENGLOBSTER

protocol=cli; endpoint=cmd/englobster; schemaRef=lobster_export.go; owner=FU-LOBSTER-EXPORTER

Key	Value
Kind	Interface
Name	englobster CLI
Mentioned In	n/a

IF-CLI-ENGMCP

protocol=stdio-jsonrpc; endpoint=cmd/engmcp; schemaRef=mcp/server.go; owner=FU-MCP-SERVER

Key	Value
Kind	Interface

Key	Value
Name	engmcp CLI
Mentioned In	n/a

IF-CLI-ENGNAF

protocol=cli; endpoint=cmd/engnaf; schemaRef=naf_export.go; owner=FU-NAF-EXPORTER

Key	Value
Kind	Interface
Name	engnaf CLI
Mentioned In	n/a

IF-CLI-ENGOSCAL

protocol=cli; endpoint=cmd/engoscal; schemaRef=oscal_ssp.go; owner=FU-OSCAL-EXPORTER

Key	Value
Kind	Interface
Name	engoscal CLI
Mentioned In	n/a

IF-CLI-ENGSTRUCT

protocol=cli; endpoint=cmd/engstruct; schemaRef=structurizr_export.go; owner=FU-STRUCTURIZR-EXPORTER

Key	Value
Kind	Interface
Name	engstruct CLI
Mentioned In	n/a

IF-CLI-ENGYSML

protocol=cli; endpoint=cmd/engsysml; schemaRef=sysml_export.go; owner=FU-SYSML-EXPORTER

Key	Value
Kind	Interface
Name	engsysml CLI
Mentioned In	n/a

IF-CLI-ENGTRACE

protocol=cli; endpoint=cmd/engtrace; schemaRef=trace_matrix.go; owner=FU-ALLOCATION-TRACE

Key	Value
Kind	Interface
Name	engtrace CLI
Mentioned In	n/a

IF-CLI-ENGTRLC

protocol=cli; endpoint=cmd/engtrlc; schemaRef=trlc_export.go; owner=FU-TRLC-EXPORTER

Key	Value
Kind	Interface
Name	engtrlc CLI
Mentioned In	n/a

IF-CLI-ENGVIEW

protocol=cli; endpoint=cmd/engview; schemaRef=view/project.go; owner=FU-VIEW-PROJECTION

Key	Value
Kind	Interface
Name	engview CLI
Mentioned In	n/a

REQ-EMG-001

While local development mode is active, when model updated event is received, the engineering model go shall validate architecture model before generation run is requested.

Key	Value
Kind	Requirement
Name	REQ-EMG-001
Mentioned In	Architecture Intent View , Interaction Flow View , Boundary Test , Catalog Systems Test , E2e Test , Ears Preflight Test , Hardware Test , Validate Test

REQ-EMG-003

While model is valid, when generation run is requested, the engineering model go shall produce AsciiDoc architecture publication output without whitespace-only lines.

Key	Value
Kind	Requirement
Name	REQ-EMG-003
Mentioned In	Architecture Intent View , Interaction Flow View , AsciiDoc Complete Diagram Test , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Evidence Table Test , AsciiDoc Linking Units Test , AsciiDoc Publication Scope Test , AsciiDoc Test , AsciiDoc Test Evidence Test , AsciiDoc Views Test , Boundary Test , Hardware Test , Project Test , Render Test

REQ-EMG-004

Where threat export is enabled, when generation run is requested, the engineering model go shall export threat dragon json artifact plus open threat model artifact.

Key	Value
Kind	Requirement
Name	REQ-EMG-004
Mentioned In	Architecture Intent View , Interaction Flow View , Threat Model Export Test

REQ-EMG-005

Where structurizr export is enabled, when generation run is requested, the engineering model go shall export structurizr dsl artifact.

Key	Value
Kind	Requirement
Name	REQ-EMG-005
Mentioned In	Architecture Intent View , Interaction Flow View , Structurizr Export Test

REQ-EMG-006

Where traceability export is enabled, when generation run is requested, the engineering model go shall produce trlc package and lobster trace report artifacts.

Key	Value
Kind	Requirement
Name	REQ-EMG-006
Mentioned In	Architecture Intent View , Interaction Flow View , Lobster Export Test , Trlc Export Test

REQ-EMG-007

While mcp server is enabled, when mcp tool call is received and catalog references resolve successfully, the engineering model go shall return mcp tool result with schema version and structured error fields.

Key	Value
Kind	Requirement
Name	REQ-EMG-007
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Test , Hardware Test , Main Test , Server Test

REQ-EMG-008

When mcp tool call is received, the engineering model go shall reject path arguments outside repository workspace boundary.

Key	Value
Kind	Requirement
Name	REQ-EMG-008
Mentioned In	Architecture Intent View , Interaction Flow View , Main Test , Server Test

REQ-EMG-009

While ci validation mode is active, when generation run is requested, the engineering model go shall execute validation plus tests before artifact generated event is raised.

Key	Value
Kind	Requirement
Name	REQ-EMG-009
Mentioned In	Architecture Intent View , Interaction Flow View , Boundary Test , Ears Preflight Test , Hardware Test , Validate Test

REQ-EMG-010

Where requirement trace is complete, when model updated event is received, the engineering model go shall infer traceability coverage from trace markers.

Key	Value
Kind	Requirement
Name	REQ-EMG-010
Mentioned In	Architecture Intent View , Interaction Flow View , AsciiDoc Test Evidence Test , Inferred Code Test , Inferred Javascript Test , Inferred Layers Test , Inferred Verification Test , Inferred Verilog Test , Javascript Test , Scan Test , Verilog Test

REQ-EMG-011

Where external schema dependencies are available, when generation run is requested, the engineering model go shall fail validation if schema checks do not match expected constraints.

Key	Value
Kind	Requirement
Name	REQ-EMG-011
Mentioned In	Architecture Intent View , Interaction Flow View , Threat Model Export Test , Validate Test

REQ-EMG-012

When generation run is requested, the engineering model go shall generate artifacts deterministically so that regeneration is reproducible.

Key	Value
Kind	Requirement
Name	REQ-EMG-012
Mentioned In	Architecture Intent View , Interaction Flow View , Coffee Six View Test , Gemara Export Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Threat Model Export Test

REQ-EMG-013

Where traceability export is enabled, when generation run is requested, the engineering model go shall export an OSCAL aggregate profile plus SSP plus assessment plan plus assessment results plus POA&M artifacts when the corresponding compliance and remediation content is applicable.

Key	Value
Kind	Requirement
Name	REQ-EMG-013
Mentioned In	Architecture Intent View , Interaction Flow View , Oscal Chain Test , Oscal Profile Test , Oscal Ssp Test

REQ-EMG-014

Where architecture decision records are authored, when generation run is requested, the engineering model go shall publish architecture decision records document linked from AsciiDoc architecture publication output.

Key	Value
Kind	Requirement
Name	REQ-EMG-014
Mentioned In	Architecture Intent View , Interaction Flow View , AsciiDoc Test

REQ-EMG-015

Where gemara export is enabled, when generation run is requested, the engineering model go shall render OpenSSF Gemara model artifacts.

Key	Value
Kind	Requirement
Name	REQ-EMG-015
Mentioned In	Architecture Intent View , Interaction Flow View , Gemara Export Test , Gemara Oscal Test

REQ-EMG-016

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve the subsystem references declared by the parent system from local subdirectory paths.

Key	Value
Kind	Requirement
Name	REQ-EMG-016
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Test

REQ-EMG-017

Where system composition is enabled, when generation run is requested, the engineering model go shall reject any subsystem reference that resolves outside the repository workspace boundary.

Key	Value
Kind	Requirement
Name	REQ-EMG-017
Mentioned In	Architecture Intent View

REQ-EMG-018

Where system composition is enabled, when generation run is requested, the engineering model go shall reject a subsystem reference graph that contains a cycle.

Key	Value
Kind	Requirement
Name	REQ-EMG-018
Mentioned In	Architecture Intent View

REQ-EMG-019

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve cross-system references using subsystem-qualified identifiers.

Key	Value
Kind	Requirement
Name	REQ-EMG-019
Mentioned In	Architecture Intent View

REQ-EMG-020

Where system composition is enabled, when generation run is requested, the engineering model go shall materialize parent-to-subsystem allocation traceability in the composed output without modifying any subsystem model.

Key	Value
Kind	Requirement
Name	REQ-EMG-020
Mentioned In	Architecture Intent View

REQ-EMG-021

Where system composition is enabled, when generation run is requested, the engineering model go shall report each subsystem required interface that the parent system does not satisfy.

Key	Value
Kind	Requirement
Name	REQ-EMG-021
Mentioned In	Architecture Intent View

REQ-EMG-022

Where system composition is enabled, when generation run is requested, the engineering model go shall reject an allocation that targets a subsystem identifier not published in the subsystem contract.

Key	Value
Kind	Requirement
Name	REQ-EMG-022
Mentioned In	Architecture Intent View

REQ-EMG-023

When generation run is requested, the engineering model go shall flag any requirement that links to another requirement in the same document.

Key	Value
Kind	Requirement
Name	REQ-EMG-023
Mentioned In	Architecture Intent View

REQ-EMG-024

Where hardware interfaces are authored, when generation run is requested, the engineering model go shall render the hardware to software interface allocation view.

Key	Value
Kind	Requirement
Name	REQ-EMG-024
Mentioned In	Architecture Intent View , Interaction Flow View , AsciiDoc Hardware Labels Test

REQ-EMG-025

Where system composition is enabled, when generation run is requested, the engineering model go shall produce a system requirement allocation matrix aggregated across the composed subsystems.

Key	Value
Kind	Requirement
Name	REQ-EMG-025
Mentioned In	Architecture Intent View

REQ-EMG-026

Where system composition is enabled, when generation run is requested, the engineering model go shall render in the requirement view the delegation link from each delegated requirement to the specific subsystem requirement that satisfies it.

Key	Value
Kind	Requirement
Name	REQ-EMG-026
Mentioned In	Architecture Intent View

REQ-EMG-027

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve an external git subsystem reference by cloning the repository into the workspace .engmod cache.

Key	Value
Kind	Requirement
Name	REQ-EMG-027
Mentioned In	Architecture Intent View , REQ-EMG-016

REQ-EMG-028

When generation run is requested, the engineering model go shall reject a code trace link that resolves to no requirement or model element.

Key	Value
Kind	Requirement
Name	REQ-EMG-028
Mentioned In	Architecture Intent View , Interaction Flow View , Trace Matrix Test

REQ-EMG-029

When generation run is requested, the engineering model go shall flag each requirement that is traced by neither code nor verification nor a downward delegation.

Key	Value
Kind	Requirement
Name	REQ-EMG-029
Mentioned In	Architecture Intent View , Interaction Flow View , Trace Matrix Test

REQ-EMG-030

When generation run is requested, the engineering model go shall export a machine-readable traceability matrix that links each requirement to its implementing code and verification and delegated subsystem requirement.

Key	Value
Kind	Requirement
Name	REQ-EMG-030
Mentioned In	Architecture Intent View , Interaction Flow View , REQ-EMG-028 , AsciiDoc Test Evidence Test , Composition Test , Inferred Javascript Test , Inferred Verilog Test , Trace Matrix Test

REQ-EMG-031

When requirements delta is received, the engineering model go shall strictly validate requirements delta schema plus stable identifier operation uniqueness.

Key	Value
Kind	Requirement
Name	REQ-EMG-031
Mentioned In	Architecture Intent View , Interaction Flow View , Requirements Delta Test

REQ-EMG-032

While dry run mode is active, when requirements delta is received, the engineering model go shall validate candidate architecture model without modifying requirements document.

Key	Value
Kind	Requirement
Name	REQ-EMG-032
Mentioned In	Architecture Intent View , Interaction Flow View , Requirements Delta Test

REQ-EMG-033

While model is valid, when requirements delta apply is requested, the engineering model go shall atomically replace requirements document.

Key	Value
Kind	Requirement
Name	REQ-EMG-033
Mentioned In	Architecture Intent View , Interaction Flow View , Requirements Delta Test

REQ-EMG-034

When requirements delta is received, the engineering model go shall produce deterministic requirements change summary.

Key	Value
Kind	Requirement
Name	REQ-EMG-034
Mentioned In	Architecture Intent View

REQ-EMG-035

When model updated event is received, the engineering model go shall normalize architecture model plus requirements document into canonical semantic model.

Key	Value
Kind	Requirement

Key	Value
Name	REQ-EMG-035
Mentioned In	Architecture Intent View , Interaction Flow View , Load Test , Projection Test , Schema Test , Schema V2 Projection Test , Semantic Test , Sysml Export Test , Validate Test

REQ-EMG-036

Where sysml export is enabled, when generation run is requested, the engineering model go shall generate sysml v2 model artifact from canonical semantic model.

Key	Value
Kind	Requirement
Name	REQ-EMG-036
Mentioned In	Architecture Intent View , Interaction Flow View , Load Test , Main Test , Projection Test , Schema V2 Projection Test , Semantic Test , Sysml Export Test , Sysml Interchange Test , Validate Test

REQ-EMG-037

While ci validation mode is active, when generation run is requested, the engineering model go shall validate sysml v2 model artifact before artifact generated event is raised.

Key	Value
Kind	Requirement
Name	REQ-EMG-037
Mentioned In	Architecture Intent View , Interaction Flow View , Main Test , Schema V2 Projection Test , Semantic Test , Sysml Interchange Test , Validate Test

REQ-EMG-038

Where sysml export is enabled, when generation run is requested, the engineering model go shall validate canonical semantic model against sysml v2 metamodel.

Key	Value
Kind	Requirement
Name	REQ-EMG-038
Mentioned In	Architecture Intent View , Interaction Flow View , Metamodel Instance Test , Metamodel Test

REQ-EMG-039

While ci validation mode is active, when generation run is requested, the engineering model go shall reject incomplete sysml v2 metamodel coverage.

Key	Value
Kind	Requirement
Name	REQ-EMG-039
Mentioned In	Architecture Intent View , Interaction Flow View , Main Test , Metamodel Instance Test , Metamodel Test , Sysml Coverage Test , Sysml Export Test

REQ-EMG-040

When generation run is requested, the engineering model go shall preserve the deterministic native sysml v2 source through project interchange round trip.

Key	Value
Kind	Requirement
Name	REQ-EMG-040
Mentioned In	Architecture Intent View , Interaction Flow View , Main Test , Metamodel Instance Test , Sysml Coverage Test , Sysml Export Test , Sysml Interchange Test

REQ-EMG-041

Where naf profile authoring is enabled, when model updated event is received, the engineering model go shall validate naf v4.1 stakeholder plus concern plus viewpoint metadata against canonical model references.

Key	Value
Kind	Requirement
Name	REQ-EMG-041
Mentioned In	Architecture Intent View , Interaction Flow View , Naf Export Test , Schema Test

REQ-EMG-042

Where naf export is enabled, when generation run is requested, the engineering model go shall generate deterministic naf v4.1 architecture document from referenced canonical model views.

Key	Value
Kind	Requirement
Name	REQ-EMG-042
Mentioned In	Architecture Intent View , Interaction Flow View , Main Test , Naf Export Test

REQ-EMG-043

While ci validation mode is active, when generation run is requested, the engineering model go shall reject unsupported naf version plus unknown viewpoint code plus dangling profile reference before artifact generated event is raised.

Key	Value
Kind	Requirement
Name	REQ-EMG-043
Mentioned In	Architecture Intent View , Interaction Flow View , Main Test , Naf Export Test , Schema Test

REQ-EMG-044

When model updated event is received, the engineering model go shall load schema version 2 engmod manifest plus catalog plus requirements plus architecture plus behavior plus assurance plus compliance plus views plus decisions documents into one canonical semantic model.

Key	Value
Kind	Requirement
Name	REQ-EMG-044
Mentioned In	Architecture Intent View , Interaction Flow View , Load Test , Schema Test , Validate Test

REQ-EMG-045

While mcp server is enabled, when mcp tool call is received and catalog references resolve successfully, the engineering model go shall return schema version 2 model authoring contract.

Key	Value
Kind	Requirement
Name	REQ-EMG-045
Mentioned In	Architecture Intent View , Interaction Flow View , Load Test , Server Test

REQ-EMG-046

While ci validation mode is active, when generation run is requested, the engineering model go shall reject model input that violates the schema version 2 manifest or any referenced domain document contract before artifact generated event is raised.

Key	Value
Kind	Requirement
Name	REQ-EMG-046
Mentioned In	Architecture Intent View , Interaction Flow View , Load Test , Schema Test , Schema V2 Fixture Test , Validate Test

REQ-EMG-047

Where system composition is enabled, when generation run is requested, the engineering model go shall resolve each exact CUE module version from the OCI registry configured by CUE_REGISTRY.

Key	Value
Kind	Requirement
Name	REQ-EMG-047
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Oci Test , Composition Test , Composition Test , Server Test , Sysml Export Test

REQ-EMG-048

Where a model workspace replacement is declared, when generation run is requested, the engineering model go shall resolve the matching manifest dependency from the replacement repository without changing module identity or selected publications.

Key	Value
Kind	Requirement
Name	REQ-EMG-048
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Oci Test , Composition Test , Composition Test , Server Test

REQ-EMG-049

While ci validation mode is active, when generation run is requested, the engineering model go shall reject invalid manifest dependency plus publication plus workspace replacement plus qualified reference declarations.

Key	Value
Kind	Requirement
Name	REQ-EMG-049
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Oci Test , Composition Test , Composition Test

REQ-EMG-050

When example validation is requested, the engineering model go shall validate a schema version 2 company portfolio composed from multiple product systems that reuse published hardware plus software plus security plus compliance facets across repository boundaries.

Key	Value
Kind	Requirement
Name	REQ-EMG-050
Mentioned In	Architecture Intent View , Interaction Flow View , Boundary Test , Boundary Test , Composition Projection Test , Examples Company Test , Hardware Test , Hardware Test , Hardware Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Sysml Export Test

REQ-EMG-051

Where system composition is enabled, when generation run is requested, the engineering model go shall import only explicitly selected dependency publications with qualified identity plus source provenance.

Key	Value
Kind	Requirement
Name	REQ-EMG-051
Mentioned In	Architecture Intent View , Interaction Flow View , Composition Projection Test , Composition Test , Composition Test , Gemara Oscar Test , Load Test

REQ-EMG-052

Where six-view example output is enabled, when generation run is requested, the engineering model go shall generate deterministic use-case plus logical plus process plus physical plus implementation plus deployment view artifacts and native SysML from one schema version 2 YAML bundle.

Key	Value
Kind	Requirement
Name	REQ-EMG-052
Mentioned In	Architecture Intent View , Interaction Flow View , Coffee Six View Test , Load Test , Render Test , Schema Test

REQ-EMG-053

Where do-178c dal c readiness output is enabled, when model updated event is received, the engineering model go shall load a strict aviation assurance profile plus lifecycle data plus verification evidence plus configuration baselines plus tool assessments plus structural coverage summaries.

Key	Value
Kind	Requirement
Name	REQ-EMG-053
Mentioned In	Architecture Intent View , Interaction Flow View , Airborne Assurance Test , Aviation Test , Load Test

REQ-EMG-054

Where do-178c dal c readiness output is enabled, when generation run is requested, the engineering model go shall reject incomplete or inconsistent requirement classification plus derived requirement disposition plus verification chain plus independence plus baseline plus structural coverage plus tool assessment evidence.

Key	Value
Kind	Requirement
Name	REQ-EMG-054

Key	Value
Mentioned In	Architecture Intent View , Interaction Flow View , Aviation Test

REQ-EMG-055

Where do-178c dal c readiness output is enabled, when generation run is requested, the engineering model go shall generate deterministic draft lifecycle data index plus objective evidence matrix plus requirements trace matrix plus verification summary plus structural coverage summary plus tool assessment register plus configuration index plus open gap report plus software accomplishment summary.

Key	Value
Kind	Requirement
Name	REQ-EMG-055
Mentioned In	Architecture Intent View , Interaction Flow View , Airborne Assurance Test

REQ-EMG-056

When example validation is requested, the engineering model go shall validate a schema version 2 DAL C flight control software example with placeholder planning plus requirements plus design plus code plus verification plus coverage plus configuration plus quality assurance plus certification liaison evidence.

Key	Value
Kind	Requirement
Name	REQ-EMG-056
Mentioned In	Architecture Intent View , Interaction Flow View , Airborne Assurance Test , Load Test

TB-EXTERNAL-VALIDATION-TOOLS

Boundary for external validators, schemas, and compliance toolchains.

Key	Value
Kind	Trust Boundary
Name	External Validation Tools Boundary
Mentioned In	n/a

TB-REPO-WORKSPACE

Boundary limiting file operations to repository-root owned paths.

Key	Value
Kind	Trust Boundary
Name	Repository Workspace Boundary

Key	Value
Mentioned In	REQ-EMG-008 , REQ-EMG-017

13.2. Catalog References

ACT-AI-AGENT

Consumes MCP outputs to plan and execute model-guided edits.

Key	Value
Kind	Catalog Actor
Name	ai agent
Mentioned In	n/a

ACT-ARCHITECTURE-AUTHOR

Maintains model structure, mappings, and architecture intent.

Key	Value
Kind	Catalog Actor
Name	architecture author
Mentioned In	Security View

ACT-CI-PIPELINE

Runs automated validation and export checks for pull requests.

Key	Value
Kind	Catalog Actor
Name	ci pipeline
Mentioned In	n/a

ACT-COMPLIANCE-ENGINEER

Reviews controls, risks, and traceability outputs for compliance needs.

Key	Value
Kind	Catalog Actor
Name	compliance engineer
Mentioned In	Security View , REQ-EMG-006 , REQ-EMG-044 , REQ-EMG-051 , REQ-EMG-053 , REQ-EMG-056

ACT-IMPLEMENTATION-ENGINEER

Implements code changes guided by model requirements and support paths.

Key	Value
Kind	Catalog Actor
Name	implementation engineer
Mentioned In	Security View

adr document

Generated document containing authored architecture decision records.

Key	Value
Kind	Catalog Alias
Name	adr document
Mentioned In	n/a

adrs authored

Feature indicating the architecture model includes first-class ADR entries.

Key	Value
Kind	Catalog Alias
Name	adrs authored
Mentioned In	n/a

allocation

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems.

Key	Value
Kind	Catalog Alias
Name	allocation
Mentioned In	Architecture Intent View , REQ-EMG-020 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-024 , REQ-EMG-025

analysis

Validation, inference, and evidence analysis over authored and inferred model content.

Key	Value
Kind	Catalog Alias

Key	Value
Name	analysis
Mentioned In	Security View

architecture.yml

Authored architecture source describing functional intent and mappings.

Key	Value
Kind	Catalog Alias
Name	architecture.yml
Mentioned In	n/a

artifact.generated

Event indicating one or more generated outputs were produced successfully.

Key	Value
Kind	Catalog Alias
Name	artifact.generated
Mentioned In	n/a

assurance

Reviews controls, risks, and traceability outputs for compliance needs.

Key	Value
Kind	Catalog Alias
Name	assurance
Mentioned In	n/a

authoring

Authoring and loading of architecture, catalog, requirements, and design inputs.

Key	Value
Kind	Catalog Alias
Name	authoring
Mentioned In	REQ-EMG-044, REQ-EMG-045

authoring manifest

Machine-readable description of canonical YAML documents, schema versions, explicit references, top-level sections, identifier conventions, and authoring order.

Key	Value
Kind	Catalog Alias
Name	authoring manifest
Mentioned In	n/a

bundle loader

Validates canonical YAML documents against embedded CUE schemas, strictly decodes them into Go runtime representations, and normalizes them for downstream use.

Key	Value
Kind	Catalog Alias
Name	bundle loader
Mentioned In	n/a

candidate architecture model

Authored architecture source describing functional intent and mappings.

Key	Value
Kind	Catalog Alias
Name	candidate architecture model
Mentioned In	REQ-EMG-032

canonical model

SysML v2 and KerML aligned semantic representation used by validation and every output projection.

Key	Value
Kind	Catalog Alias
Name	canonical model
Mentioned In	REQ-EMG-041 , REQ-EMG-042

catalog resolved

Condition requiring all authored references to match known catalog entries.

Key	Value
Kind	Catalog Alias
Name	catalog resolved
Mentioned In	n/a

catalog.yml

Canonical authored vocabulary YAML document used by strict requirement and model validation.

Key	Value
Kind	Catalog Alias
Name	catalog.yml
Mentioned In	n/a

ci

Runs automated validation and export checks for pull requests.

Key	Value
Kind	Catalog Alias
Name	ci
Mentioned In	n/a

ci mode

Automated pipeline mode focused on deterministic checks and artifacts.

Key	Value
Kind	Catalog Alias
Name	ci mode
Mentioned In	n/a

cmd layer

Command entypoints that orchestrate loading, validation, generation, and export flows.

Key	Value
Kind	Catalog Alias
Name	cmd layer

Key	Value
Mentioned In	n/a

coding agent

Consumes MCP outputs to plan and execute model-guided edits.

Key	Value
Kind	Catalog Alias
Name	coding agent
Mentioned In	n/a

compliance

Requirement trace exports and compliance-oriented output generation.

Key	Value
Kind	Catalog Alias
Name	compliance
Mentioned In	Architecture Intent View , REQ-EMG-013 , REQ-EMG-044 , REQ-EMG-050 , REQ-EMG-051 , REQ-EMG-056

composition

Resolves downward subsystem references from local paths, external git repositories, or versioned CUE modules in OCI registries and composes a federated system-of-systems model.

Key	Value
Kind	Catalog Alias
Name	composition
Mentioned In	REQ-EMG-016 , REQ-EMG-023

cue registry

OCI-compliant registry resolved through CUE_REGISTRY and used to distribute immutable versioned Engineering Model modules.

Key	Value
Kind	Catalog Alias
Name	cue registry
Mentioned In	n/a

dal c readiness

Feature enabling evidence-readiness validation and draft lifecycle-data outputs for a DO-178C or ED-12C DAL C project profile.

Key	Value
Kind	Catalog Alias
Name	dal c readiness
Mentioned In	n/a

decisions document

Generated document containing authored architecture decision records.

Key	Value
Kind	Catalog Alias
Name	decisions document
Mentioned In	REQ-EMG-044

decisions.yml

Canonical authored architecture decision YAML document.

Key	Value
Kind	Catalog Alias
Name	decisions.yml
Mentioned In	n/a

delta merger

Plans, validates, summarizes, and atomically applies typed changes to authored model documents.

Key	Value
Kind	Catalog Alias
Name	delta merger
Mentioned In	n/a

design.yml

Canonical authored design-view YAML document.

Key	Value
Kind	Catalog Alias

Key	Value
Name	design.yml
Mentioned In	n/a

dsl

Generated Structurizr workspace DSL document.

Key	Value
Kind	Catalog Alias
Name	dsl
Mentioned In	n/a

dsl export

Feature enabling Structurizr DSL generation and validation.

Key	Value
Kind	Catalog Alias
Name	dsl export
Mentioned In	n/a

engdoc

Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts.

Key	Value
Kind	Catalog Alias
Name	engdoc
Mentioned In	n/a

engdragon

Exports threat model views to Threat Dragon and Open OTM representations.

Key	Value
Kind	Catalog Alias
Name	engdragon
Mentioned In	n/a

enggemara

Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK.

Key	Value
Kind	Catalog Alias
Name	enggemara
Mentioned In	n/a

engineer

Implements code changes guided by model requirements and support paths.

Key	Value
Kind	Catalog Alias
Name	engineer
Mentioned In	n/a

engineering model toolchain

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

Key	Value
Kind	Catalog Alias
Name	engineering model toolchain
Mentioned In	n/a

englobster

Generates LOBSTER traceability inputs and reports from model and verification links.

Key	Value
Kind	Catalog Alias
Name	englobster
Mentioned In	n/a

engmcp

Serves model-backed MCP tools with validated inputs and structured responses.

Key	Value
Kind	Catalog Alias

Key	Value
Name	engmcp
Mentioned In	Security View , Main Test

engmodel

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

Key	Value
Kind	Catalog Alias
Name	engmodel
Mentioned In	n/a

engnaf

Validates NATO Architecture Framework 4.1 profile metadata and generates NAF architecture products from canonical model views.

Key	Value
Kind	Catalog Alias
Name	engnaf
Mentioned In	Main Test

engoscal

Exports the OSCAL system security plan, assessment results, and plan of action and milestones.

Key	Value
Kind	Catalog Alias
Name	engoscal
Mentioned In	n/a

engstruct

Exports Structurizr DSL and supports validation/static-site workflows.

Key	Value
Kind	Catalog Alias
Name	engstruct
Mentioned In	n/a

engsysml

Generates SysML v2 model artifacts from the canonical semantic model and reports conformance coverage.

Key	Value
Kind	Catalog Alias
Name	engsysml
Mentioned In	Main Test

engtrlc

Exports requirements and model structures to TRLC artifacts.

Key	Value
Kind	Catalog Alias
Name	engtrlc
Mentioned In	n/a

example check

Event indicating maintained reference workspaces must be loaded, composed, generated, and checked.

Key	Value
Kind	Catalog Alias
Name	example check
Mentioned In	n/a

gemara generation

Feature enabling OpenSSF Gemara catalog and evaluation log generation.

Key	Value
Kind	Catalog Alias
Name	gemara generation
Mentioned In	n/a

generation

Generation of AsciiDoc and exchange artifacts from model inputs.

Key	Value
Kind	Catalog Alias

Key	Value
Name	generation
Mentioned In	Architecture Intent View , REQ-EMG-001 , REQ-EMG-012 , REQ-EMG-025

generation requested

Event indicating documentation or export generation execution starts.

Key	Value
Kind	Catalog Alias
Name	generation requested
Mentioned In	n/a

go

Go compiler, test runner, and module tooling used by the repository.

Key	Value
Kind	Catalog Alias
Name	go
Mentioned In	n/a

hw interfaces

Feature enabling hardware items and hardware/software interface modeling.

Key	Value
Kind	Catalog Alias
Name	hw interfaces
Mentioned In	n/a

inference

Infers runtime, code, and verification evidence from repository sources and test markers.

Key	Value
Kind	Catalog Alias
Name	inference
Mentioned In	Architecture Intent View , Security View , Policy Only Fallback Test

input contract

Machine-readable description of canonical YAML documents, schema versions, explicit references, top-level sections, identifier conventions, and authoring order.

Key	Value
Kind	Catalog Alias
Name	input contract
Mentioned In	n/a

invalid

State where validation diagnostics include one or more errors.

Key	Value
Kind	Catalog Alias
Name	invalid
Mentioned In	Security View , REQ-EMG-046 , REQ-EMG-049

invalid model

Invalid or ambiguous model content that causes incorrect graph construction.

Key	Value
Kind	Catalog Alias
Name	invalid model
Mentioned In	n/a

kerml

SysML 2.0 and KerML 1.0 with Pilot Implementation 2026-04 artifact 0.59.0, the pinned validator wrapper, and Sysand 0.2.1 for normative KPAR packaging.

Key	Value
Kind	Catalog Alias
Name	kerml
Mentioned In	Architecture Intent View , REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-038 , REQ-EMG-041

kerml metamodel

Pinned formal SysML 2.0 and KerML 1.0 abstract syntax used to derive semantic coverage and schema constraints.

Key	Value
Kind	Catalog Alias
Name	kerml metamodel
Mentioned In	n/a

lobster

LOBSTER core and TRLC integration tools used for traceability reporting.

Key	Value
Kind	Catalog Alias
Name	lobster
Mentioned In	Architecture Intent View

lobster report

Traceability report output generated from TRLC and test links.

Key	Value
Kind	Catalog Alias
Name	lobster report
Mentioned In	n/a

local mode

Local developer mode for iterative model and code updates.

Key	Value
Kind	Catalog Alias
Name	local mode
Mentioned In	n/a

mcp

MCP server interfaces that expose model-aware tooling for AI agents.

Key	Value
Kind	Catalog Alias
Name	mcp

Key	Value
Mentioned In	n/a

mcp enabled

Feature enabling MCP runtime APIs for model-aware tooling.

Key	Value
Kind	Catalog Alias
Name	mcp enabled
Mentioned In	n/a

model author

Maintains model structure, mappings, and architecture intent.

Key	Value
Kind	Catalog Alias
Name	model author
Mentioned In	n/a

model change

Event indicating architecture or requirement model source changed.

Key	Value
Kind	Catalog Alias
Name	model change
Mentioned In	Architecture Intent View , Security View , Traceability View

model dependencies

Resolves downward subsystem references from local paths, external git repositories, or versioned CUE modules in OCI registries and composes a federated system-of-systems model.

Key	Value
Kind	Catalog Alias
Name	model dependencies
Mentioned In	n/a

model registry

OCI-compliant registry resolved through CUE_REGISTRY and used to distribute immutable versioned Engineering Model modules.

Key	Value
Kind	Catalog Alias
Name	model registry
Mentioned In	n/a

naf

Validates NATO Architecture Framework 4.1 profile metadata and generates NAF architecture products from canonical model views.

Key	Value
Kind	Catalog Alias
Name	naf
Mentioned In	n/a

naf architecture

Generated NATO Architecture Framework v4.1 architecture document assembled from profile metadata and canonical model views.

Key	Value
Kind	Catalog Alias
Name	naf architecture
Mentioned In	REQ-EMG-042

naf authoring

Feature enabling NATO Architecture Framework stakeholder, concern, and viewpoint metadata in the architecture model.

Key	Value
Kind	Catalog Alias
Name	naf authoring
Mentioned In	n/a

naf document

Generated NATO Architecture Framework v4.1 architecture document assembled from profile metadata and canonical model views.

Key	Value
Kind	Catalog Alias

Key	Value
Name	naf document
Mentioned In	n/a

naf generation

Feature enabling NATO Architecture Framework architecture document generation.

Key	Value
Kind	Catalog Alias
Name	naf generation
Mentioned In	n/a

naf v4.1

NATO Architecture Framework version 4.1 viewpoint taxonomy and architecture-description guidance.

Key	Value
Kind	Catalog Alias
Name	naf v4.1
Mentioned In	REQ-EMG-041 , REQ-EMG-042

oci registry

OCI-compliant registry resolved through CUE_REGISTRY and used to distribute immutable versioned Engineering Model modules.

Key	Value
Kind	Catalog Alias
Name	oci registry
Mentioned In	Architecture Intent View , REQ-EMG-047

otm schema

Open Threat Model schema used to validate OTM output artifacts.

Key	Value
Kind	Catalog Alias
Name	otm schema
Mentioned In	n/a

path traversal

Host file access attempts through untrusted path arguments in MCP tools.

Key	Value
Kind	Catalog Alias
Name	path traversal
Mentioned In	Security View

preview mode

Change-planning mode that computes and validates a candidate without writing canonical files.

Key	Value
Kind	Catalog Alias
Name	preview mode
Mentioned In	n/a

projection

Builds view-specific projections for architecture communication and traceability outputs.

Key	Value
Kind	Catalog Alias
Name	projection
Mentioned In	Architecture Intent View , Security View , REQ-EMG-040

requirements apply requested

Event indicating a validated requirements change should replace the canonical document.

Key	Value
Kind	Catalog Alias
Name	requirements apply requested
Mentioned In	n/a

requirements change

Versioned add, whole-entity update, and remove operations keyed by requirement stable ID.

Key	Value
Kind	Catalog Alias

Key	Value
Name	requirements change
Mentioned In	n/a

requirements change received

Event indicating a typed requirements change is ready for planning and validation.

Key	Value
Kind	Catalog Alias
Name	requirements change received
Mentioned In	n/a

requirements diff

Deterministic human-readable summary of added, updated, and removed requirements.

Key	Value
Kind	Catalog Alias
Name	requirements diff
Mentioned In	n/a

requirements merger

Plans, validates, summarizes, and atomically applies typed changes to authored model documents.

Key	Value
Kind	Catalog Alias
Name	requirements merger
Mentioned In	n/a

requirements yml

Canonical authored requirements YAML document.

Key	Value
Kind	Catalog Alias
Name	requirements yml
Mentioned In	n/a

schema available

Condition requiring external schema artifacts to be present for validation.

Key	Value
Kind	Catalog Alias
Name	schema available
Mentioned In	n/a

schema tamper

Tampering or drift in external schemas used for export validation.

Key	Value
Kind	Catalog Alias
Name	schema tamper
Mentioned In	n/a

semantic model

SysML v2 and KerML aligned semantic representation used by validation and every output projection.

Key	Value
Kind	Catalog Alias
Name	semantic model
Mentioned In	n/a

six-view output

Feature enabling use-case, logical, process, physical, implementation, and deployment view generation from one model bundle.

Key	Value
Kind	Catalog Alias
Name	six-view output
Mentioned In	n/a

ssp json

Generated OSCAL system security plan JSON output.

Key	Value
Kind	Catalog Alias

Key	Value
Name	ssp json
Mentioned In	n/a

structurizr

Structurizr DSL validation runtime used to verify generated DSL.

Key	Value
Kind	Catalog Alias
Name	structurizr
Mentioned In	n/a

sysml

SysML 2.0 and KerML 1.0 with Pilot Implementation 2026-04 artifact 0.59.0, the pinned validator wrapper, and Sysand 0.2.1 for normative KPAR packaging.

Key	Value
Kind	Catalog Alias
Name	sysml
Mentioned In	Architecture Intent View , REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-038 , REQ-EMG-040 , REQ-EMG-041 , REQ-EMG-052

sysml artifact

Generated SysML v2 textual or project interchange representation of the canonical semantic model.

Key	Value
Kind	Catalog Alias
Name	sysml artifact
Mentioned In	n/a

sysml generation

Feature enabling SysML v2 textual and project interchange generation.

Key	Value
Kind	Catalog Alias
Name	sysml generation

Key	Value
Mentioned In	n/a

sysml metamodel

Pinned formal SysML 2.0 and KerML 1.0 abstract syntax used to derive semantic coverage and schema constraints.

Key	Value
Kind	Catalog Alias
Name	sysml metamodel
Mentioned In	n/a

sysml model

Generated SysML v2 textual or project interchange representation of the canonical semantic model.

Key	Value
Kind	Catalog Alias
Name	sysml model
Mentioned In	n/a

system of systems

Feature enabling recursive composition of subsystem models from local paths, external git repositories, or CUE modules in OCI registries.

Key	Value
Kind	Catalog Alias
Name	system of systems
Mentioned In	n/a

threat dragon

External JSON schemas used to validate Threat Dragon exports.

Key	Value
Kind	Catalog Alias
Name	threat dragon
Mentioned In	Architecture Intent View

threat dragon output

Generated threat model artifact for OWASP Threat Dragon.

Key	Value
Kind	Catalog Alias
Name	threat dragon output
Mentioned In	n/a

threat modeling export

Feature enabling Threat Dragon and Open OTM exports.

Key	Value
Kind	Catalog Alias
Name	threat modeling export
Mentioned In	n/a

tool call

Event indicating the MCP server received a tool invocation request.

Key	Value
Kind	Catalog Alias
Name	tool call
Mentioned In	n/a

tool result

Structured MCP tool response payload carrying model-aware query results.

Key	Value
Kind	Catalog Alias
Name	tool result
Mentioned In	n/a

trace complete

Condition requiring requirements to map to implementation and verification evidence.

Key	Value
Kind	Catalog Alias

Key	Value
Name	trace complete
Mentioned In	n/a

trace drift

Requirement links or tests drifting from implementation over time.

Key	Value
Kind	Catalog Alias
Name	trace drift
Mentioned In	n/a

trace export

Feature enabling TRLC and LOBSTER generation flows.

Key	Value
Kind	Catalog Alias
Name	trace export
Mentioned In	n/a

trlc

TRLC parser and runtime used for requirements language validation.

Key	Value
Kind	Catalog Alias
Name	trlc
Mentioned In	Architecture Intent View

trlc output

Generated TRLC requirements package and model metadata.

Key	Value
Kind	Catalog Alias
Name	trlc output
Mentioned In	n/a

up to date

State where generated outputs reflect the latest authored model inputs.

Key	Value
Kind	Catalog Alias
Name	up to date
Mentioned In	n/a

valid

State where authored model passes validation checks.

Key	Value
Kind	Catalog Alias
Name	valid
Mentioned In	REQ-EMG-037 , REQ-EMG-045

validation failure

Event indicating model or export validation failed.

Key	Value
Kind	Catalog Alias
Name	validation failure
Mentioned In	n/a

validator

Validates IDs, references, and model consistency across authored entities and mappings.

Key	Value
Kind	Catalog Alias
Name	validator
Mentioned In	REQ-EMG-009

workspace replacement

Feature enabling a CUE module identity to resolve from an explicit sibling repository path during local development.

Key	Value
Kind	Catalog Alias

Key	Value
Name	workspace replacement
Mentioned In	REQ-EMG-049 , REQ-EMG-050

AV-MALFORMED-MODEL-INPUT

Invalid or ambiguous model content that causes incorrect graph construction.

Key	Value
Kind	Catalog Attack Vector
Name	malformed model input
Mentioned In	n/a

AV-PATH-TRAVERSAL-IN-MCP

Host file access attempts through untrusted path arguments in MCP tools.

Key	Value
Kind	Catalog Attack Vector
Name	path traversal in mcp calls
Mentioned In	Security View

AV-SCHEMA-SUPPLY-CHAIN-TAMPER

Tampering or drift in external schemas used for export validation.

Key	Value
Kind	Catalog Attack Vector
Name	schema supply chain tamper
Mentioned In	n/a

AV-TRACEABILITY-GAP-DRIFT

Requirement links or tests drifting from implementation over time.

Key	Value
Kind	Catalog Attack Vector
Name	traceability gap drift
Mentioned In	Security View

COND-CATALOG-REFERENCES-RESOLVE

Condition requiring all authored references to match known catalog entries.

Key	Value
Kind	Catalog Condition
Name	catalog references resolve successfully
Mentioned In	REQ-EMG-007 , REQ-EMG-045

COND-EXTERNAL-SCHEMA-AVAILABLE

Condition requiring external schema artifacts to be present for validation.

Key	Value
Kind	Catalog Condition
Name	external schema dependencies are available
Mentioned In	REQ-EMG-011

COND-REQUIREMENT-TRACE-COMPLETE

Condition requiring requirements to map to implementation and verification evidence.

Key	Value
Kind	Catalog Condition
Name	requirement trace is complete
Mentioned In	REQ-EMG-010

DATA-ARCHITECTURE-DECISION-RECORDS

Generated document containing authored architecture decision records.

Key	Value
Kind	Catalog Data Term
Name	architecture decision records document
Mentioned In	REQ-EMG-014

DATA-ARCHITECTURE-MODEL

Authored architecture source describing functional intent and mappings.

Key	Value
Kind	Catalog Data Term

Key	Value
Name	architecture model
Mentioned In	n/a

DATA-CANONICAL-SEMANTIC-MODEL

SysML v2 and KerML aligned semantic representation used by validation and every output projection.

Key	Value
Kind	Catalog Data Term
Name	canonical semantic model
Mentioned In	n/a

DATA-CATALOG-DOCUMENT

Canonical authored vocabulary YAML document used by strict requirement and model validation.

Key	Value
Kind	Catalog Data Term
Name	catalog document
Mentioned In	n/a

DATA-DECISIONS-DOCUMENT

Canonical authored architecture decision YAML document.

Key	Value
Kind	Catalog Data Term
Name	decisions document
Mentioned In	n/a

DATA-DESIGN-DOCUMENT

Canonical authored design-view YAML document.

Key	Value
Kind	Catalog Data Term
Name	design document
Mentioned In	n/a

DATA-LOBSTER-TRACE-REPORT

Traceability report output generated from TRLC and test links.

Key	Value
Kind	Catalog Data Term
Name	lobster trace report
Mentioned In	REQ-EMG-006

DATA-MCP-TOOL-RESULT

Structured MCP tool response payload carrying model-aware query results.

Key	Value
Kind	Catalog Data Term
Name	mcp tool result
Mentioned In	n/a

DATA-MODEL-AUTHORING-CONTRACT

Machine-readable description of canonical YAML documents, schema versions, explicit references, top-level sections, identifier conventions, and authoring order.

Key	Value
Kind	Catalog Data Term
Name	model authoring contract
Mentioned In	n/a

DATA-NAF-V4-ARCHITECTURE

Generated NATO Architecture Framework v4.1 architecture document assembled from profile metadata and canonical model views.

Key	Value
Kind	Catalog Data Term
Name	naf v4.1 architecture document
Mentioned In	n/a

DATA-OSCAL-SSP

Generated OSCAL system security plan JSON output.

Key	Value
Kind	Catalog Data Term
Name	oscal ssp artifact
Mentioned In	n/a

DATA-REQUIREMENTS-DELTA

Versioned add, whole-entity update, and remove operations keyed by requirement stable ID.

Key	Value
Kind	Catalog Data Term
Name	requirements delta
Mentioned In	n/a

DATA-REQUIREMENTS-DIFF

Deterministic human-readable summary of added, updated, and removed requirements.

Key	Value
Kind	Catalog Data Term
Name	requirements change summary
Mentioned In	n/a

DATA-REQUIREMENTS-DOCUMENT

Canonical authored requirements YAML document.

Key	Value
Kind	Catalog Data Term
Name	requirements document
Mentioned In	n/a

DATA-STRUCTURIZR-DSL

Generated Structurizr workspace DSL document.

Key	Value
Kind	Catalog Data Term
Name	structurizr dsl artifact

Key	Value
Mentioned In	REQ-EMG-005

DATA-SYSML-V2-METAMODEL

Pinned formal SysML 2.0 and KerML 1.0 abstract syntax used to derive semantic coverage and schema constraints.

Key	Value
Kind	Catalog Data Term
Name	sysml v2 metamodel
Mentioned In	n/a

DATA-SYSML-V2-MODEL

Generated SysML v2 textual or project interchange representation of the canonical semantic model.

Key	Value
Kind	Catalog Data Term
Name	sysml v2 model artifact
Mentioned In	REQ-EMG-036 , REQ-EMG-037

DATA-THREAT-DRAGON-JSON

Generated threat model artifact for OWASP Threat Dragon.

Key	Value
Kind	Catalog Data Term
Name	threat dragon json artifact
Mentioned In	REQ-EMG-004

DATA-TRLC-PACKAGE

Generated TRLC requirements package and model metadata.

Key	Value
Kind	Catalog Data Term
Name	trlc package
Mentioned In	n/a

EVT-ARTIFACT-GENERATED

Event indicating one or more generated outputs were produced successfully.

Key	Value
Kind	Catalog Event
Name	artifact generated event is raised
Mentioned In	REQ-EMG-009 , REQ-EMG-037 , REQ-EMG-043 , REQ-EMG-046

EVT-EXAMPLE-VALIDATION-REQUESTED

Event indicating maintained reference workspaces must be loaded, composed, generated, and checked.

Key	Value
Kind	Catalog Event
Name	example validation is requested
Mentioned In	REQ-EMG-050 , REQ-EMG-056

EVT-GENERATION-RUN-REQUESTED

Event indicating documentation or export generation execution starts.

Key	Value
Kind	Catalog Event
Name	generation run is requested
Mentioned In	REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-004 , REQ-EMG-005 , REQ-EMG-006 , REQ-EMG-009 , REQ-EMG-011 , REQ-EMG-012 , REQ-EMG-013 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-024 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030 , REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-038 , REQ-EMG-039 , REQ-EMG-040 , REQ-EMG-042 , REQ-EMG-043 , REQ-EMG-046 , REQ-EMG-047 , REQ-EMG-048 , REQ-EMG-049 , REQ-EMG-051 , REQ-EMG-052 , REQ-EMG-054 , REQ-EMG-055

EVT-MCP-TOOL-CALL-RECEIVED

Event indicating the MCP server received a tool invocation request.

Key	Value
Kind	Catalog Event
Name	mcp tool call is received
Mentioned In	REQ-EMG-007 , REQ-EMG-008 , REQ-EMG-045

EVT-MODEL-UPDATED

Event indicating architecture or requirement model source changed.

Key	Value
Kind	Catalog Event
Name	model updated event is received
Mentioned In	REQ-EMG-001 , REQ-EMG-010 , REQ-EMG-035 , REQ-EMG-041 , REQ-EMG-044 , REQ-EMG-053

EVT-REQUIREMENTS-DELTA-APPLY-REQUESTED

Event indicating a validated requirements change should replace the canonical document.

Key	Value
Kind	Catalog Event
Name	requirements delta apply is requested
Mentioned In	REQ-EMG-033

EVT-REQUIREMENTS-DELTA-RECEIVED

Event indicating a typed requirements change is ready for planning and validation.

Key	Value
Kind	Catalog Event
Name	requirements delta is received
Mentioned In	REQ-EMG-031 , REQ-EMG-032 , REQ-EMG-034

EVT-VALIDATION-FAILED

Event indicating model or export validation failed.

Key	Value
Kind	Catalog Event
Name	validation failed event is raised
Mentioned In	n/a

FEAT-ARCHITECTURE-DECISION-RECORDS

Feature indicating the architecture model includes first-class ADR entries.

Key	Value
Kind	Catalog Feature
Name	architecture decision records are authored

Key	Value
Mentioned In	REQ-EMG-014

FEAT-DO178C-DAL-C-READINESS

Feature enabling evidence-readiness validation and draft lifecycle-data outputs for a DO-178C or ED-12C DAL C project profile.

Key	Value
Kind	Catalog Feature
Name	do-178c dal c readiness output is enabled
Mentioned In	REQ-EMG-053 , REQ-EMG-054 , REQ-EMG-055

FEAT-GEMARA-EXPORT

Feature enabling OpenSSF Gemara catalog and evaluation log generation.

Key	Value
Kind	Catalog Feature
Name	gemara export is enabled
Mentioned In	REQ-EMG-015

FEAT-HARDWARE-INTERFACES

Feature enabling hardware items and hardware/software interface modeling.

Key	Value
Kind	Catalog Feature
Name	hardware interfaces are authored
Mentioned In	REQ-EMG-024

FEAT-MCP-SERVER

Feature enabling MCP runtime APIs for model-aware tooling.

Key	Value
Kind	Catalog Feature
Name	mcp server is enabled
Mentioned In	REQ-EMG-007 , REQ-EMG-045

FEAT-MODEL-WORKSPACE-REPLACEMENT

Feature enabling a CUE module identity to resolve from an explicit sibling repository path during local development.

Key	Value
Kind	Catalog Feature
Name	a model workspace replacement is declared
Mentioned In	REQ-EMG-048

FEAT-NAF-EXPORT

Feature enabling NATO Architecture Framework architecture document generation.

Key	Value
Kind	Catalog Feature
Name	naf export is enabled
Mentioned In	REQ-EMG-042

FEAT-NAF-PROFILE

Feature enabling NATO Architecture Framework stakeholder, concern, and viewpoint metadata in the architecture model.

Key	Value
Kind	Catalog Feature
Name	naf profile authoring is enabled
Mentioned In	REQ-EMG-041

FEAT-SIX-VIEW-OUTPUT

Feature enabling use-case, logical, process, physical, implementation, and deployment view generation from one model bundle.

Key	Value
Kind	Catalog Feature
Name	six-view example output is enabled
Mentioned In	REQ-EMG-052

FEAT-STRUCTURIZR-EXPORT

Feature enabling Structurizr DSL generation and validation.

Key	Value
Kind	Catalog Feature
Name	structurizr export is enabled

Key	Value
Mentioned In	REQ-EMG-005

FEAT-SYSML-EXPORT

Feature enabling SysML v2 textual and project interchange generation.

Key	Value
Kind	Catalog Feature
Name	sysml export is enabled
Mentioned In	REQ-EMG-036 , REQ-EMG-038

FEAT-SYSTEM-COMPOSITION

Feature enabling recursive composition of subsystem models from local paths, external git repositories, or CUE modules in OCI registries.

Key	Value
Kind	Catalog Feature
Name	system composition is enabled
Mentioned In	REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027 , REQ-EMG-047 , REQ-EMG-051

FEAT-THREAT-EXPORT

Feature enabling Threat Dragon and Open OTM exports.

Key	Value
Kind	Catalog Feature
Name	threat export is enabled
Mentioned In	REQ-EMG-004

FEAT-TRACEABILITY-EXPORT

Feature enabling TRLC and LOBSTER generation flows.

Key	Value
Kind	Catalog Feature
Name	traceability export is enabled
Mentioned In	REQ-EMG-006 , REQ-EMG-013

FG-ARTIFACT-GENERATION

Generation of AsciiDoc and exchange artifacts from model inputs.

Key	Value
Kind	Catalog Functional Group
Name	artifact generation
Mentioned In	Architecture Intent View

FG-MCP-INTEGRATION

MCP server interfaces that expose model-aware tooling for AI agents.

Key	Value
Kind	Catalog Functional Group
Name	mcp integration
Mentioned In	n/a

FG-MODEL-AUTHORING

Authoring and loading of architecture, catalog, requirements, and design inputs.

Key	Value
Kind	Catalog Functional Group
Name	model authoring
Mentioned In	n/a

FG-TRACEABILITY-COMPLIANCE

Requirement trace exports and compliance-oriented output generation.

Key	Value
Kind	Catalog Functional Group
Name	traceability and compliance
Mentioned In	n/a

FG-VALIDATION-ANALYSIS

Validation, inference, and evidence analysis over authored and inferred model content.

Key	Value
Kind	Catalog Functional Group

Key	Value
Name	validation and analysis
Mentioned In	n/a

FU-ALLOCATION-TRACE

Materializes parent-to-subsystem allocation and bidirectional traceability across composed systems.

Key	Value
Kind	Catalog Functional Unit
Name	allocation trace
Mentioned In	Architecture Intent View , REQ-EMG-020 , AsciiDoc Test Evidence Test , Coffee Six View Test , Composition Test , Gemara Export Test , Inferred Javascript Test , Inferred Verilog Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Threat Model Export Test , Trace Matrix Test

FU-ASCIIDOC-GENERATOR

Renders architecture documentation and view narratives to AsciiDoc and PDF artifacts.

Key	Value
Kind	Catalog Functional Unit
Name	asciidoc generator
Mentioned In	Architecture Intent View , Security View , Traceability View , Airborne Assurance Test , AsciiDoc Complete Diagram Test , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Evidence Table Test , AsciiDoc Hardware Labels Test , AsciiDoc Linking Units Test , AsciiDoc Publication Scope Test , AsciiDoc Test , AsciiDoc Test Evidence Test , AsciiDoc Views Test , Boundary Test , Boundary Test , Catalog Systems Test , Coffee Six View Test , Composition Projection Test , E2e Test , Ears Preflight Test , Examples Company Test , Gemara Export Test , Hardware Test , Hardware Test , Hardware Test , Load Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Project Test , Render Test , Render Test , Sysml Export Test , Threat Model Export Test , Validate Test

FU-CLI-ORCHESTRATION

Command entrypoints that orchestrate loading, validation, generation, and export flows.

Key	Value
Kind	Catalog Functional Unit
Name	cli orchestration

Key	Value
Mentioned In	Architecture Intent View , Security View , Traceability View , Airborne Assurance Test , AsciiDoc Complete Diagram Test , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Evidence Table Test , AsciiDoc Linking Units Test , AsciiDoc Publication Scope Test , AsciiDoc Test , AsciiDoc Test Evidence Test , AsciiDoc Views Test , Boundary Test , Boundary Test , Catalog Systems Test , Coffee Six View Test , Composition Oci Test , Composition Projection Test , Composition Test , Composition Test , E2e Test , Ears Preflight Test , Gemara Export Test , Gemara Oscal Test , Hardware Test , Hardware Test , Load Test , Lobster Export Test , Main Test , Main Test , Metamodel Instance Test , Metamodel Test , Naf Export Test , Oscal Chain Test , Oscal Profile Test , Oscal Ssp Test , Projection Test , Project Test , Render Test , Render Test , Requirements Delta Test , Schema Test , Schema V2 Fixture Test , Schema V2 Projection Test , Semantic Test , Structurizr Export Test , Sysml Coverage Test , Sysml Export Test , Sysml Interchange Test , Threat Model Export Test , Trlc Export Test , Validate Test

FU-CODEMAP-INFERENCE

Infers runtime, code, and verification evidence from repository sources and test markers.

Key	Value
Kind	Catalog Functional Unit
Name	codemap inference
Mentioned In	Architecture Intent View , Security View , Traceability View , AsciiDoc Test Evidence Test , Coffee Six View Test , Composition Test , Gemara Export Test , Inferred Code Test , Inferred Javascript Test , Inferred Layers Test , Inferred Verification Test , Inferred Verilog Test , Javascript Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Scan Test , Threat Model Export Test , Trace Matrix Test , Verilog Test

FU-GEMARA-EXPORTER

Renders the model into OpenSSF Gemara L1-L7 catalogs and logs plus an optional OSCAL bridge, validated by the Gemara SDK.

Key	Value
Kind	Catalog Functional Unit
Name	gemara exporter
Mentioned In	Architecture Intent View , Boundary Test , Boundary Test , Coffee Six View Test , Composition Projection Test , Examples Company Test , Gemara Export Test , Gemara Oscal Test , Hardware Test , Hardware Test , Hardware Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Render Test , Sysml Export Test , Threat Model Export Test

FU-LOBSTER-EXPORTER

Generates LOBSTER traceability inputs and reports from model and verification links.

Key	Value
Kind	Catalog Functional Unit

Key	Value
Name	lobster exporter
Mentioned In	Architecture Intent View , Security View , Traceability View , Coffee Six View Test , Gemara Export Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Threat Model Export Test , Trlc Export Test

FU-MCP-SERVER

Serves model-backed MCP tools with validated inputs and structured responses.

Key	Value
Kind	Catalog Functional Unit
Name	mcp server
Mentioned In	Composition Test , Gemara Export Test , Gemara Oscal Test , Hardware Test , Load Test , Main Test , Server Test

FU-MODEL-CHANGE

Plans, validates, summarizes, and atomically applies typed changes to authored model documents.

Key	Value
Kind	Catalog Functional Unit
Name	model change
Mentioned In	Requirements Delta Test

FU-MODEL-LOADER

Validates canonical YAML documents against embedded CUE schemas, strictly decodes them into Go runtime representations, and normalizes them for downstream use.

Key	Value
Kind	Catalog Functional Unit
Name	model loader
Mentioned In	Architecture Intent View , Security View , Traceability View , Airborne Assurance Test , Aviation Test , Boundary Test , Catalog Systems Test , Coffee Six View Test , Composition Oci Test , Composition Projection Test , Composition Test , Composition Test , E2e Test , Ears Preflight Test , Gemara Oscal Test , Hardware Test , Load Test , Metamodel Instance Test , Metamodel Test , Naf Export Test , Projection Test , Render Test , Requirements Delta Test , Schema Test , Schema V2 Fixture Test , Schema V2 Projection Test , Semantic Test , Server Test , Sysml Export Test , Validate Test

FU-NAF-EXPORTER

Validates NATO Architecture Framework 4.1 profile metadata and generates NAF architecture products from canonical model views.

Key	Value
Kind	Catalog Functional Unit
Name	naf exporter
Mentioned In	Architecture Intent View , Main Test , Naf Export Test , Schema Test

FU-OSCAL-EXPORTER

Exports the OSCAL system security plan, assessment results, and plan of action and milestones.

Key	Value
Kind	Catalog Functional Unit
Name	oscal exporter
Mentioned In	Architecture Intent View , Boundary Test , Boundary Test , Catalog Systems Test , Coffee Six View Test , Composition Projection Test , E2e Test , Ears Preflight Test , Examples Company Test , Gemara Export Test , Hardware Test , Hardware Test , Hardware Test , Lobster Export Test , Oscal Chain Test , Oscal Profile Test , Oscal Ssp Test , Render Test , Render Test , Sysml Export Test , Threat Model Export Test , Validate Test

FU-STRUCTURIZR-EXPORTER

Exports Structurizr DSL and supports validation/static-site workflows.

Key	Value
Kind	Catalog Functional Unit
Name	structurizr exporter
Mentioned In	Architecture Intent View , Security View , Boundary Test , Catalog Systems Test , Coffee Six View Test , E2e Test , Ears Preflight Test , Gemara Export Test , Hardware Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Structurizr Export Test , Threat Model Export Test , Validate Test

FU-SYSML-EXPORTER

Generates SysML v2 model artifacts from the canonical semantic model and reports conformance coverage.

Key	Value
Kind	Catalog Functional Unit
Name	sysml exporter

Key	Value
Mentioned In	Architecture Intent View , Security View , Boundary Test , Boundary Test , Coffee Six View Test , Composition Projection Test , Examples Company Test , Gemara Export Test , Hardware Test , Hardware Test , Hardware Test , Load Test , Lobster Export Test , Main Test , Metamodel Instance Test , Metamodel Test , Oscal Chain Test , Oscal Ssp Test , Projection Test , Render Test , Render Test , Schema Test , Schema V2 Projection Test , Semantic Test , Sysml Coverage Test , Sysml Export Test , Sysml Interchange Test , Threat Model Export Test , Validate Test

FU-SYSTEM-COMPOSITION

Resolves downward subsystem references from local paths, external git repositories, or versioned CUE modules in OCI registries and composes a federated system-of-systems model.

Key	Value
Kind	Catalog Functional Unit
Name	system composition
Mentioned In	Architecture Intent View , Boundary Test , Boundary Test , Composition Oci Test , Composition Projection Test , Composition Test , Composition Test , Examples Company Test , Gemara Oscal Test , Hardware Test , Hardware Test , Hardware Test , Load Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Server Test , Sysml Export Test

FU-THREAT-EXPORTER

Exports threat model views to Threat Dragon and Open OTM representations.

Key	Value
Kind	Catalog Functional Unit
Name	threat exporter
Mentioned In	Architecture Intent View , Security View , Boundary Test , Catalog Systems Test , Coffee Six View Test , E2e Test , Ears Preflight Test , Gemara Export Test , Hardware Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Threat Model Export Test , Validate Test

FU-TRLC-EXPORTER

Exports requirements and model structures to TRLC artifacts.

Key	Value
Kind	Catalog Functional Unit
Name	trlc exporter
Mentioned In	Architecture Intent View , Security View , Traceability View , Coffee Six View Test , Gemara Export Test , Lobster Export Test , Oscal Chain Test , Oscal Ssp Test , Render Test , Threat Model Export Test , Trlc Export Test

FU-VALIDATION-ENGINE

Validates IDs, references, and model consistency across authored entities and mappings.

Key	Value
Kind	Catalog Functional Unit
Name	validation engine
Mentioned In	Architecture Intent View , Security View , Traceability View , Airborne Assurance Test , Aviation Test , Boundary Test , Boundary Test , Catalog Systems Test , Coffee Six View Test , Composition Oci Test , Composition Projection Test , Composition Test , Composition Test , E2e Test , Ears Preflight Test , Examples Company Test , Gemara Export Test , Gemara Oscal Test , Hardware Test , Hardware Test , Hardware Test , Load Test , Lobster Export Test , Main Test , Main Test , Metamodel Instance Test , Metamodel Test , Naf Export Test , Oscal Chain Test , Oscal Ssp Test , Projection Test , Render Test , Render Test , Requirements Delta Test , Schema Test , Schema V2 Fixture Test , Schema V2 Projection Test , Semantic Test , Sysml Coverage Test , Sysml Export Test , Sysml Interchange Test , Threat Model Export Test , Trace Matrix Test , Validate Test

FU-VIEW-PROJECTION

Builds view-specific projections for architecture communication and traceability outputs.

Key	Value
Kind	Catalog Functional Unit
Name	view projection
Mentioned In	Architecture Intent View , AsciiDoc Complete Diagram Test , AsciiDoc Diagrams Core Test , AsciiDoc Diagrams Runtime Test , AsciiDoc Evidence Table Test , AsciiDoc Hardware Labels Test , AsciiDoc Linking Units Test , AsciiDoc Publication Scope Test , AsciiDoc Test , AsciiDoc Test Evidence Test , AsciiDoc Views Test , Boundary Test , Coffee Six View Test , Hardware Test , Load Test , Main Test , Naf Export Test , Project Test , Render Test , Render Test , Schema Test

MODE-CI-VALIDATION

Automated pipeline mode focused on deterministic checks and artifacts.

Key	Value
Kind	Catalog Mode
Name	ci validation mode is active
Mentioned In	REQ-EMG-009 , REQ-EMG-037 , REQ-EMG-039 , REQ-EMG-043 , REQ-EMG-046 , REQ-EMG-049

MODE-DRY-RUN

Change-planning mode that computes and validates a candidate without writing canonical files.

Key	Value
Kind	Catalog Mode
Name	dry run mode is active
Mentioned In	REQ-EMG-032

MODE-LOCAL-DEVELOPMENT

Local developer mode for iterative model and code updates.

Key	Value
Kind	Catalog Mode
Name	local development mode is active
Mentioned In	REQ-EMG-001

REF-CUE-OCI-MODULE-REGISTRY

OCI-compliant registry resolved through CUE_REGISTRY and used to distribute immutable versioned Engineering Model modules.

Key	Value
Kind	Catalog Referenced Element
Name	cue oci module registry
Mentioned In	Architecture Intent View

REF-GO-TOOLCHAIN

Go compiler, test runner, and module tooling used by the repository.

Key	Value
Kind	Catalog Referenced Element
Name	go toolchain
Mentioned In	Architecture Intent View

REF-LOBSTER-TOOLCHAIN

LOBSTER core and TRLC integration tools used for traceability reporting.

Key	Value
Kind	Catalog Referenced Element
Name	lobster toolchain

Key	Value
Mentioned In	Architecture Intent View

REF-NAF-V4-1-SPECIFICATION

NATO Architecture Framework version 4.1 viewpoint taxonomy and architecture-description guidance.

Key	Value
Kind	Catalog Referenced Element
Name	nato architecture framework v4.1
Mentioned In	Architecture Intent View

REF-OPEN-OTM-SCHEMA

Open Threat Model schema used to validate OTM output artifacts.

Key	Value
Kind	Catalog Referenced Element
Name	open otm schema
Mentioned In	Architecture Intent View

REF-STRUCTURIZR-VALIDATOR

Structurizr DSL validation runtime used to verify generated DSL.

Key	Value
Kind	Catalog Referenced Element
Name	structurizr validator
Mentioned In	Architecture Intent View , REQ-EMG-005

REF-SYSML-V2-TOOLCHAIN

SysML 2.0 and KerML 1.0 with Pilot Implementation 2026-04 artifact 0.59.0, the pinned validator wrapper, and Sysand 0.2.1 for normative KPAR packaging.

Key	Value
Kind	Catalog Referenced Element
Name	sysml v2 toolchain
Mentioned In	n/a

REF-THREAT-DRAGON-SCHEMAS

External JSON schemas used to validate Threat Dragon exports.

Key	Value
Kind	Catalog Referenced Element
Name	threat dragon schemas
Mentioned In	Architecture Intent View

REF-TRLC-TOOLCHAIN

TRLC parser and runtime used for requirements language validation.

Key	Value
Kind	Catalog Referenced Element
Name	trlc toolchain
Mentioned In	Architecture Intent View

STATE-ARTIFACTS-FRESH

State where generated outputs reflect the latest authored model inputs.

Key	Value
Kind	Catalog State
Name	generated artifacts are fresh
Mentioned In	n/a

STATE-MODEL-INVALID

State where validation diagnostics include one or more errors.

Key	Value
Kind	Catalog State
Name	model is invalid
Mentioned In	n/a

STATE-MODEL-VALID

State where authored model passes validation checks.

Key	Value
Kind	Catalog State

Key	Value
Name	model is valid
Mentioned In	REQ-EMG-003 , REQ-EMG-033

SYS-ENGINEERING-MODEL-GO

Go-based architecture modeling system that generates docs, analysis views, and compliance artifacts.

Key	Value
Kind	Catalog System
Name	engineering model go
Mentioned In	REQ-EMG-001 , REQ-EMG-003 , REQ-EMG-004 , REQ-EMG-005 , REQ-EMG-006 , REQ-EMG-007 , REQ-EMG-008 , REQ-EMG-009 , REQ-EMG-010 , REQ-EMG-011 , REQ-EMG-012 , REQ-EMG-013 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-016 , REQ-EMG-017 , REQ-EMG-018 , REQ-EMG-019 , REQ-EMG-020 , REQ-EMG-021 , REQ-EMG-022 , REQ-EMG-023 , REQ-EMG-024 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-027 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030 , REQ-EMG-031 , REQ-EMG-032 , REQ-EMG-033 , REQ-EMG-034 , REQ-EMG-035 , REQ-EMG-036 , REQ-EMG-037 , REQ-EMG-038 , REQ-EMG-039 , REQ-EMG-040 , REQ-EMG-041 , REQ-EMG-042 , REQ-EMG-043 , REQ-EMG-044 , REQ-EMG-045 , REQ-EMG-046 , REQ-EMG-047 , REQ-EMG-048 , REQ-EMG-049 , REQ-EMG-050 , REQ-EMG-051 , REQ-EMG-052 , REQ-EMG-053 , REQ-EMG-054 , REQ-EMG-055 , REQ-EMG-056

TERM-ACTOR

A person or operational role that interacts with functional units.

Key	Value
Kind	Engineering Model Term
Name	actor
Mentioned In	REQ-EMG-043

TERM-ASCIIDOC-DIAGRAM

Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc diagram
Mentioned In	n/a

TERM-ASCIIDOC-DOCUMENT

Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc document
Mentioned In	n/a

TERM-ASCIIDOC-SECTION

Structured generated document section or row model used to render authored and inferred architecture content.

Key	Value
Kind	Engineering Model Term
Name	AsciiDoc section
Mentioned In	n/a

TERM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

Key	Value
Kind	Engineering Model Term
Name	attack vector
Mentioned In	n/a

TERM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

Key	Value
Kind	Engineering Model Term
Name	authored mapping
Mentioned In	Security View

TERM-BUNDLE

Loaded architecture, catalog, and companion documents resolved as one working model context.

Key	Value
Kind	Engineering Model Term
Name	model bundle

Key	Value
Mentioned In	n/a

TERM-CATALOG

Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.

Key	Value
Kind	Engineering Model Term
Name	catalog
Mentioned In	Architecture Intent View , REQ-EMG-032 , REQ-EMG-044

TERM-CATALOG-ENTRY

A stable catalog term with a definition and aliases for linting, linking, and generated references.

Key	Value
Kind	Engineering Model Term
Name	catalog entry
Mentioned In	n/a

TERM-CLI-COMMAND

Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.

Key	Value
Kind	Engineering Model Term
Name	CLI command
Mentioned In	n/a

TERM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

Key	Value
Kind	Engineering Model Term
Name	code element
Mentioned In	n/a

TERM-COMPLIANCE-MAPPING

Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.

Key	Value
Kind	Engineering Model Term
Name	compliance mapping
Mentioned In	n/a

TERM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

Key	Value
Kind	Engineering Model Term
Name	control
Mentioned In	Architecture Intent View , Security View , REQ-EMG-003 , REQ-EMG-055 , REQ-EMG-056

TERM-CONTROL-VERIFICATION

Authored control verification record with method, status, threat/risk scope, findings, and evidence.

Key	Value
Kind	Engineering Model Term
Name	control verification
Mentioned In	n/a

TERM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

Key	Value
Kind	Engineering Model Term
Name	data object
Mentioned In	n/a

TERM-DECISION

Authored architecture decision record with status, context, decision text, and consequences.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	architecture decision
Mentioned In	n/a

TERM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

Key	Value
Kind	Engineering Model Term
Name	deployment target
Mentioned In	n/a

TERM-DESIGN

Authored design narrative organized by model entities and architecture views.

Key	Value
Kind	Engineering Model Term
Name	design document
Mentioned In	n/a

TERM-DESIGN-VIEW

View-scoped design narrative attached to an authored functional group or functional unit.

Key	Value
Kind	Engineering Model Term
Name	design view
Mentioned In	n/a

TERM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

Key	Value
Kind	Engineering Model Term
Name	event
Mentioned In	Security View , State Lifecycle View

TERM-EVIDENCE

A file, artifact, or observation used to support control, risk, threat, or verification claims.

Key	Value
Kind	Engineering Model Term
Name	evidence
Mentioned In	Architecture Intent View , Security View , Traceability View , State Lifecycle View , Interaction Flow View , REQ-EMG-039 , REQ-EMG-053 , REQ-EMG-054 , REQ-EMG-055 , REQ-EMG-056 , Audit Redaction Test

TERM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

Key	Value
Kind	Engineering Model Term
Name	flow
Mentioned In	Architecture Intent View , Interaction Flow View , Pr Opened Review Flow

TERM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

Key	Value
Kind	Engineering Model Term
Name	flow step
Mentioned In	n/a

TERM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

Key	Value
Kind	Engineering Model Term
Name	functional group
Mentioned In	n/a

TERM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

Key	Value
Kind	Engineering Model Term
Name	functional unit
Mentioned In	State Lifecycle View

TERM-GENERATION-WORKFLOW

Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.

Key	Value
Kind	Engineering Model Term
Name	generation workflow
Mentioned In	n/a

TERM-INFERENCE-HINT

Authored source and ownership configuration used to discover runtime, code, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	inference hint
Mentioned In	n/a

TERM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

Key	Value
Kind	Engineering Model Term
Name	inferred mapping
Mentioned In	n/a

TERM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

Key	Value
Kind	Engineering Model Term
Name	interface

Key	Value
Mentioned In	REQ-EMG-021 , REQ-EMG-024 , REQ-EMG-041

TERM-LINT-RUN

Requirement lint configuration that controls parsing and quality checks for requirement text.

Key	Value
Kind	Engineering Model Term
Name	lint run
Mentioned In	n/a

TERM-LOBSTER-ACTIVITY-TRACE

Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.

Key	Value
Kind	Engineering Model Term
Name	LOBSTER activity trace
Mentioned In	n/a

TERM-MCP-SERVER

Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.

Key	Value
Kind	Engineering Model Term
Name	MCP server
Mentioned In	Architecture Intent View , Security View , Traceability View

TERM-MCP-STDIO-TRANSPORT

Content-Length framed standard input/output transport used by the MCP command-line server.

Key	Value
Kind	Engineering Model Term
Name	MCP stdio transport
Mentioned In	n/a

TERM-MCP-TOOL

Named MCP operation with a constrained input schema and structured response payload.

Key	Value
Kind	Engineering Model Term
Name	MCP tool
Mentioned In	n/a

TERM-MCP-TOOL-RESPONSE

Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.

Key	Value
Kind	Engineering Model Term
Name	MCP tool response
Mentioned In	n/a

TERM-MODEL

The authored architecture model root that composes functional structure, relationships, inference hints, and views.

Key	Value
Kind	Engineering Model Term
Name	model
Mentioned In	Architecture Intent View , Security View , REQ-EMG-004 , REQ-EMG-014 , REQ-EMG-015 , REQ-EMG-020 , REQ-EMG-027 , REQ-EMG-028 , REQ-EMG-030 , REQ-EMG-035 , REQ-EMG-039 , REQ-EMG-044 , REQ-EMG-046 , REQ-EMG-051 , REQ-EMG-053 , Aviation Test , Load Test , Metamodel Instance Test , Projection Test , Schema Test , Semantic Test

TERM-OPEN-OTM-DOCUMENT

Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.

Key	Value
Kind	Engineering Model Term
Name	Open OTM document
Mentioned In	n/a

TERM-OSCAL-ASSESSMENT-RESULTS

Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	OSCAL assessment results
Mentioned In	n/a

TERM-OSCAL-POAM

Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL POA&M
Mentioned In	n/a

TERM-OSCAL-SSP

Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.

Key	Value
Kind	Engineering Model Term
Name	OSCAL SSP
Mentioned In	n/a

TERM-POAM-ITEM

Plan of action and milestones item linked to a modeled risk and supporting artifacts.

Key	Value
Kind	Engineering Model Term
Name	POA&M item
Mentioned In	n/a

TERM-REFERENCE-INDEX

Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.

Key	Value
Kind	Engineering Model Term
Name	reference index
Mentioned In	n/a

TERM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

Key	Value
Kind	Engineering Model Term
Name	referenced element
Mentioned In	n/a

TERM-REPO-INDEX

Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.

Key	Value
Kind	Engineering Model Term
Name	repository index
Mentioned In	n/a

TERM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

Key	Value
Kind	Engineering Model Term
Name	requirement
Mentioned In	Architecture Intent View , Security View , Traceability View , Interaction Flow View , REQ-EMG-007 , REQ-EMG-010 , REQ-EMG-023 , REQ-EMG-025 , REQ-EMG-026 , REQ-EMG-028 , REQ-EMG-029 , REQ-EMG-030 , REQ-EMG-035 , REQ-EMG-041 , REQ-EMG-044 , REQ-EMG-046 , REQ-EMG-050 , REQ-EMG-051 , REQ-EMG-054 , REQ-EMG-055 , REQ-EMG-056

TERM-RISK

Authored risk record with likelihood, impact, response, scope, controls, and evidence.

Key	Value
Kind	Engineering Model Term
Name	risk
Mentioned In	Architecture Intent View , Audit Record Persistence Test

TERM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

Key	Value
Kind	Engineering Model Term
Name	runtime element
Mentioned In	n/a

TERM-SECURITY-CONTEXT

Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.

Key	Value
Kind	Engineering Model Term
Name	security context
Mentioned In	n/a

TERM-SOURCE-BLOCK

Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.

Key	Value
Kind	Engineering Model Term
Name	source block
Mentioned In	n/a

TERM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

Key	Value
Kind	Engineering Model Term
Name	state
Mentioned In	Security View , State Lifecycle View , REQ-EMG-053 , REQ-EMG-055

TERM-STRUCTURIZR-DSL

Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.

Key	Value
Kind	Engineering Model Term
Name	Structurizr DSL

Key	Value
Mentioned In	n/a

TERM-THREAT-ASSUMPTION

Authored security assumption that records scope, rationale, owner, status, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat assumption
Mentioned In	n/a

TERM-THREAT-DRAGON-DOCUMENT

Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.

Key	Value
Kind	Engineering Model Term
Name	Threat Dragon document
Mentioned In	n/a

TERM-THREAT-MITIGATION

Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat mitigation
Mentioned In	n/a

TERM-THREAT-MODEL-EXPORT

Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.

Key	Value
Kind	Engineering Model Term
Name	threat model export
Mentioned In	n/a

TERM-THREAT-OUT-OF-SCOPE

Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.

Key	Value
Kind	Engineering Model Term
Name	threat out-of-scope decision
Mentioned In	n/a

TERM-THREAT-SCENARIO

Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.

Key	Value
Kind	Engineering Model Term
Name	threat scenario
Mentioned In	n/a

TERM-TRACE-MARKER

Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.

Key	Value
Kind	Engineering Model Term
Name	trace marker
Mentioned In	REQ-EMG-010

TERM-TRL-PACKAGE

Generated TRLC model and requirements package used for formal requirement traceability processing.

Key	Value
Kind	Engineering Model Term
Name	TRL package
Mentioned In	REQ-EMG-006

TERM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

Key	Value
Kind	Engineering Model Term

Key	Value
Name	trust boundary
Mentioned In	Security View , REQ-EMG-008

TERM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Key	Value
Kind	Engineering Model Term
Name	upward linking
Mentioned In	n/a

TERM-VALIDATION

Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.

Key	Value
Kind	Engineering Model Term
Name	validation
Mentioned In	Architecture Intent View , REQ-EMG-001 , REQ-EMG-009 , REQ-EMG-011 , REQ-EMG-012 , REQ-EMG-032 , REQ-EMG-037 , REQ-EMG-043 , Ota Campaign Test

TERM-VALIDATION-DIAGNOSTIC

Structured validation finding with code, severity, message, and optional source path.

Key	Value
Kind	Engineering Model Term
Name	validation diagnostic
Mentioned In	n/a

TERM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

Key	Value
Kind	Engineering Model Term
Name	verification check
Mentioned In	n/a

TERM-VIEW

Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

Key	Value
Kind	Engineering Model Term
Name	view
Mentioned In	Architecture Intent View , Security View , Traceability View , State Lifecycle View , Interaction Flow View , REQ-EMG-024 , REQ-EMG-026 , REQ-EMG-027 , REQ-EMG-042 , REQ-EMG-043 , REQ-EMG-051 , REQ-EMG-052 , Boundary Test , Hardware Test , Project Test

13.3. Inferred Runtime References

13.4. Inferred Code References

[aws-sdk-go-v2/secretsmanager \(secrets_configuration.ts\)](#)

Inferred library_external dependency owned by FU-SECRETS-CONFIGURATION from src/secrets_configuration.ts.

Key	Value
Kind	Code library_external
Owner	FU-SECRETS-CONFIGURATION
Source	src/secrets_configuration.ts
Mentioned In	n/a

[cuelang.org/go/cue \(schema.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/schema.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/schema.go
Mentioned In	n/a

[cuelang.org/go/cue \(schema_test.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/schema_test.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER

Key	Value
Source	model/schema_test.go
Mentioned In	n/a

[cuelang.org/go/cue/build \(schema.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/schema.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/schema.go
Mentioned In	n/a

[cuelang.org/go/cue/cuecontext \(schema.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/schema.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/schema.go
Mentioned In	n/a

[cuelang.org/go/cue/errors \(schema.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/schema.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/schema.go
Mentioned In	n/a

[cuelang.org/go/encoding/yaml \(schema.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/schema.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/schema.go
Mentioned In	n/a

[cuelang.org/go/mod/modconfig \(composition.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

[cuelang.org/go/mod/modfile \(workspace.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/workspace.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/workspace.go
Mentioned In	n/a

[cuelang.org/go/mod/modregistry \(composition.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

[cuelang.org/go/mod/modregistrytest \(composition_oci_test.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from composition_oci_test.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	composition_oci_test.go
Mentioned In	n/a

[cuelang.org/go/mod/module \(composition.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

[cuelang.org/go/mod/module \(examples_company_test.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from examples_company_test.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	examples_company_test.go
Mentioned In	n/a

[cuelang.org/go/mod/module \(subsystem.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/subsystem.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/subsystem.go
Mentioned In	n/a

[cuelang.org/go/mod/module \(workspace.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/workspace.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/workspace.go
Mentioned In	n/a

[cuelang.org/go/mod/modzip \(composition.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

[cuelang.org/go/mod/modzip \(examples_company_test.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from examples_company_test.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	examples_company_test.go
Mentioned In	n/a

[github-rest-sdk \(pr_context_assembly.ts\)](#)

Inferred library_external dependency owned by FU-PR-CONTEXT-ASSEMBLY from src/pr_context_assembly.ts.

Key	Value
Kind	Code library_external
Owner	FU-PR-CONTEXT-ASSEMBLY
Source	src/pr_context_assembly.ts
Mentioned In	n/a

[github-rest-sdk/publication-client \(review_publication.ts\)](#)

Inferred library_external dependency owned by FU-REVIEW-PUBLICATION from src/review_publication.ts.

Key	Value
Kind	Code library_external
Owner	FU-REVIEW-PUBLICATION
Source	src/review_publication.ts
Mentioned In	n/a

github.com/defenseunicorns/go-oscal/src/types/oscal-1-1-3 (gemara_oscal.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_oscal.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara.go)

Inferred library_external dependency owned by FU-MCP-SERVER from mcp/gemara.go.

Key	Value
Kind	Code library_external
Owner	FU-MCP-SERVER
Source	mcp/gemara.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara_evaluation.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_evaluation.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_evaluation.go
Mentioned In	n/a

github.com/gemaraproj/go-gemara (gemara_export.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export.go
Mentioned In	n/a

[github.com/gemaraproj/go-gemara \(gemara_export_test.go\)](#)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export_test.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export_test.go
Mentioned In	n/a

[github.com/gemaraproj/go-gemara \(gemara_extended.go\)](#)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_extended.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_extended.go
Mentioned In	n/a

[github.com/gemaraproj/go-gemara \(gemara_l1l3.go\)](#)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_l1l3.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_l1l3.go
Mentioned In	n/a

[github.com/gemaraproj/go-gemara/fetcher \(gemara_export_test.go\)](#)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export_test.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export_test.go
Mentioned In	n/a

[github.com/gemaraproj/go-gemara/gemaraconv](#) (gemara_oscal.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_oscal.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal.go
Mentioned In	n/a

[github.com/goccy/go-yaml](#) (gemara_export.go)

Inferred library_external dependency owned by FU-GEMARA-EXPORTER from gemara_export.go.

Key	Value
Kind	Code library_external
Owner	FU-GEMARA-EXPORTER
Source	gemara_export.go
Mentioned In	n/a

[github.com/gofrs/flock](#) (composition.go)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

[github.com/gofrs/flock](#) (requirements_delta.go)

Inferred library_external dependency owned by FU-MODEL-CHANGE from requirements_delta.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-CHANGE
Source	requirements_delta.go
Mentioned In	n/a

[github.com/hashicorp/hcl/v2](#) (inferred_layers.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

[github.com/hashicorp/hcl/v2/hclparse](#) (inferred_layers.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

[github.com/labeth/ears-lint-go](#) (ears_preflight.go)

Inferred library_external dependency owned by FU-VALIDATION-ENGINE from ears_preflight.go.

Key	Value
Kind	Code library_external
Owner	FU-VALIDATION-ENGINE
Source	ears_preflight.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go](#) (coffee_six_view_test.go)

Inferred library_external dependency owned by FU-VIEW-PROJECTION from coffee_six_view_test.go.

Key	Value
Kind	Code library_external
Owner	FU-VIEW-PROJECTION
Source	coffee_six_view_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(composition.go\)](#)

Inferred library_external dependency owned by FU-MCP-SERVER from mcp/composition.go.

Key	Value
Kind	Code library_external
Owner	FU-MCP-SERVER
Source	mcp/composition.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(gemara.go\)](#)

Inferred library_external dependency owned by FU-MCP-SERVER from mcp/gemara.go.

Key	Value
Kind	Code library_external
Owner	FU-MCP-SERVER
Source	mcp/gemara.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-AIRBORNE-ASSURANCE-EXPORTER from cmd/engair/main.go.

Key	Value
Kind	Code library_external
Owner	FU-AIRBORNE-ASSURANCE-EXPORTER
Source	cmd/engair/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-MODEL-CHANGE from cmd/engchange/main.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-CHANGE
Source	cmd/engchange/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-CLI-ORCHESTRATION from cmd/engdoc/main.go.

Key	Value
Kind	Code library_external
Owner	FU-CLI-ORCHESTRATION
Source	cmd/engdoc/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-THREAT-EXPORTER from cmd/engdragon/main.go.

Key	Value
Kind	Code library_external
Owner	FU-THREAT-EXPORTER
Source	cmd/engdragon/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-CLI-ORCHESTRATION from cmd/enggemara/main.go.

Key	Value
Kind	Code library_external
Owner	FU-CLI-ORCHESTRATION
Source	cmd/enggemara/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-LOBSTER-EXPORTER from cmd/englobster/main.go.

Key	Value
Kind	Code library_external
Owner	FU-LOBSTER-EXPORTER
Source	cmd/englobster/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-NAF-EXPORTER from cmd/engnaf/main.go.

Key	Value
Kind	Code library_external
Owner	FU-NAF-EXPORTER
Source	cmd/engnaf/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-OSCAL-EXPORTER from cmd/engoscal/main.go.

Key	Value
Kind	Code library_external
Owner	FU-OSCAL-EXPORTER
Source	cmd/engoscal/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-STRUCTURIZR-EXPORTER from cmd/engstruct/main.go.

Key	Value
Kind	Code library_external
Owner	FU-STRUCTURIZR-EXPORTER
Source	cmd/engstruct/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-SYSML-EXPORTER from cmd/engsysml/main.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSML-EXPORTER
Source	cmd/engsysml/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-ALLOCATION-TRACE from cmd/engtrace/main.go.

Key	Value
Kind	Code library_external
Owner	FU-ALLOCATION-TRACE
Source	cmd/engtrace/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-TRLC-EXPORTER from cmd/engtrlc/main.go.

Key	Value
Kind	Code library_external
Owner	FU-TRLC-EXPORTER
Source	cmd/engtrlc/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main.go\)](#)

Inferred library_external dependency owned by FU-CLI-ORCHESTRATION from cmd/engview/main.go.

Key	Value
Kind	Code library_external
Owner	FU-CLI-ORCHESTRATION
Source	cmd/engview/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(main_test.go\)](#)

Inferred library_external dependency owned by FU-SYSML-EXPORTER from cmd/engsysml/main_test.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSML-EXPORTER
Source	cmd/engsysml/main_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go \(schema_v2_projection_test.go\)](#)

Inferred library_external dependency owned by FU-CLI-ORCHESTRATION from schema_v2_projection_test.go.

Key	Value
Kind	Code library_external
Owner	FU-CLI-ORCHESTRATION
Source	schema_v2_projection_test.go
Mentioned In	n/a

[github.com/santhosh-tekuri/jsonschema/v5 \(threat_model_export_test.go\)](#)

Inferred library_external dependency owned by FU-THREAT-EXPORTER from threat_model_export_test.go.

Key	Value
Kind	Code library_external
Owner	FU-THREAT-EXPORTER
Source	threat_model_export_test.go
Mentioned In	n/a

[github.com/tree-sitter/go-tree-sitter \(javascript.go\)](#)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/javascript.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/javascript.go
Mentioned In	n/a

[github.com/tree-sitter/go-tree-sitter \(scan.go\)](#)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

[github.com/tree-sitter/tree-sitter-go/bindings/go](#) (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

[github.com/tree-sitter/tree-sitter-javascript/bindings/go](#) (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

[github.com/tree-sitter/tree-sitter-rust/bindings/go](#) (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

[github.com/tree-sitter/tree-sitter-typescript/bindings/go](#) (scan.go)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from codemap/scan.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	codemap/scan.go
Mentioned In	n/a

[golang.org/x/mod/semver \(subsystem.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/subsystem.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/subsystem.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(checkout_api.go\)](#)

Inferred library_external dependency owned by FU-CHECKOUT from src/checkout_api.go.

Key	Value
Kind	Code library_external
Owner	FU-CHECKOUT
Source	src/checkout_api.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(composition.go\)](#)

Inferred library_external dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_external
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(inferred_layers.go\)](#)

Inferred library_external dependency owned by FU-CODEMAP-INFERENCE from inferred_layers.go.

Key	Value
Kind	Code library_external
Owner	FU-CODEMAP-INFERENCE
Source	inferred_layers.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(load.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/load.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/load.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(load_test.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/load_test.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/load_test.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(metamodel_instance_test.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/metamodel_instance_test.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/metamodel_instance_test.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(requirements_delta.go\)](#)

Inferred library_external dependency owned by FU-MODEL-CHANGE from requirements_delta.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-CHANGE
Source	requirements_delta.go
Mentioned In	n/a

[gopkg.in/yaml.v3 \(semantic.go\)](#)

Inferred library_external dependency owned by FU-MODEL-LOADER from model/semantic.go.

Key	Value
Kind	Code library_external
Owner	FU-MODEL-LOADER
Source	model/semantic.go
Mentioned In	n/a

[serde_json::json \(payment_engine.rs\)](#)

Inferred library_external dependency owned by FU-PAYMENT-AUTHORIZATION from src/payment_engine.rs.

Key	Value
Kind	Code library_external
Owner	FU-PAYMENT-AUTHORIZATION
Source	src/payment_engine.rs
Mentioned In	n/a

[serde_json::json \(review_orchestrator.rs\)](#)

Inferred library_external dependency owned by FU-REVIEW-ORCHESTRATION from src/review_orchestrator.rs.

Key	Value
Kind	Code library_external
Owner	FU-REVIEW-ORCHESTRATION
Source	src/review_orchestrator.rs
Mentioned In	n/a

[zod \(risk_scorer.ts\)](#)

Inferred library_external dependency owned by FU-RISK-SCORING from src/risk_scorer.ts.

Key	Value
Kind	Code library_external
Owner	FU-RISK-SCORING
Source	src/risk_scorer.ts
Mentioned In	n/a

[./support/audit_envelope \(risk_scorer.ts\)](#)

Inferred library_first_party dependency owned by FU-RISK-SCORING from src/risk_scorer.ts.

Key	Value
Kind	Code library_first_party
Owner	FU-RISK-SCORING
Source	src/risk_scorer.ts
Mentioned In	n/a

[crate::domain::events::PaymentEvent \(payment_engine.rs\)](#)

Inferred library_first_party dependency owned by FU-PAYMENT-AUTHORIZATION from src/payment_engine.rs.

Key	Value
Kind	Code library_first_party
Owner	FU-PAYMENT-AUTHORIZATION
Source	src/payment_engine.rs
Mentioned In	n/a

[github.com/labeth/engineering-model-go/codemap \(inferred_code.go\)](#)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_code.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_code.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/codemap \(inferred_verification.go\)](#)

Inferred library_first_party dependency owned by FU-CODEMAP-INFERENCE from inferred_verification.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CODEMAP-INFERENCE
Source	inferred_verification.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/codemap](#) (server.go)

Inferred library_first_party dependency owned by FU-MCP-SERVER from mcp/server.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	mcp/server.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/internal/sysmlmetamodel](#) (main.go)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from cmd/sysmlmetamodel/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	cmd/sysmlmetamodel/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/internal/sysmlmetamodel](#) (sysml_coverage.go)

Inferred library_first_party dependency owned by FU-SYSML-EXPORTER from sysml_coverage.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSML-EXPORTER
Source	sysml_coverage.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/mcp](#) (main.go)

Inferred library_first_party dependency owned by FU-MCP-SERVER from cmd/engmcp/main.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	cmd/engmcp/main.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/mcp \(schema_v2_projection_test.go\)](#)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from schema_v2_projection_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	schema_v2_projection_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(airborne_assurance.go\)](#)

Inferred library_first_party dependency owned by FU-AIRBORNE-ASSURANCE-EXPORTER from airborne_assurance.go.

Key	Value
Kind	Code library_first_party
Owner	FU-AIRBORNE-ASSURANCE-EXPORTER
Source	airborne_assurance.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_complete_diagram_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_complete_diagram_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_complete_diagram_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_composition.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_composition.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_composition.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_decisions.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_decisions.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_decisions.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_design_refs.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_design_refs.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_design_refs.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_diagrams_core.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_core.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_core.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_diagrams_core_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_core_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_core_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_diagrams_runtime.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_runtime.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_runtime.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_diagrams_runtime_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_diagrams_runtime_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_diagrams_runtime_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_gemara.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_gemara.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_gemara.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_hardware_labels_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_hardware_labels_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_hardware_labels_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_linking_units.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_linking_units.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_linking_units.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_publication_scope_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_publication_scope_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_publication_scope_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_test_evidence_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_test_evidence_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_test_evidence_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_views.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_views.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_views.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(asciidoc_views_test.go\)](#)

Inferred library_first_party dependency owned by FU-ASCIIDOC-GENERATOR from asciidoc_views_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-ASCIIDOC-GENERATOR
Source	asciidoc_views_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(boundary_test.go\)](#)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from validate/boundary_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	validate/boundary_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model\(boundary_test.go\)](https://github.com/labeth/engineering-model-go/model(boundary_test.go))

Inferred library_first_party dependency owned by FU-VIEW-PROJECTION from view/boundary_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VIEW-PROJECTION
Source	view/boundary_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model\(catalog_description_validation.go\)](https://github.com/labeth/engineering-model-go/model(catalog_description_validation.go))

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from catalog_description_validation.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	catalog_description_validation.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model\(catalog_systems_test.go\)](https://github.com/labeth/engineering-model-go/model(catalog_systems_test.go))

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from catalog_systems_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	catalog_systems_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model\(composition.go\)](https://github.com/labeth/engineering-model-go/model(composition.go))

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(composition_oci_test.go\)](#)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition_oci_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition_oci_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(composition_projection.go\)](#)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition_projection.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition_projection.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(composition_projection_test.go\)](#)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition_projection_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition_projection_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(composition_test.go\)](#)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from composition_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	composition_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(ears_preflight.go\)](#)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from ears_preflight.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	ears_preflight.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(ears_preflight_test.go\)](#)

Inferred library_first_party dependency owned by FU-VALIDATION-ENGINE from ears_preflight_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-VALIDATION-ENGINE
Source	ears_preflight_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(examples_company_test.go\)](#)

Inferred library_first_party dependency owned by FU-SYSTEM-COMPOSITION from examples_company_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-SYSTEM-COMPOSITION
Source	examples_company_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara.go\)](#)

Inferred library_first_party dependency owned by FU-MCP-SERVER from mcp/gemara.go.

Key	Value
Kind	Code library_first_party
Owner	FU-MCP-SERVER
Source	mcp/gemara.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara_evaluation.go\)](#)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_evaluation.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_evaluation.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara_export.go\)](#)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_export.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_export.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara_export_test.go\)](#)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_export_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_export_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara_extended.go\)](#)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_extended.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_extended.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara_l1l3.go\)](#)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_l1l3.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_l1l3.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara_oscal.go\)](#)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_oscal.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(gemara_oscal_test.go\)](#)

Inferred library_first_party dependency owned by FU-GEMARA-EXPORTER from gemara_oscal_test.go.

Key	Value
Kind	Code library_first_party
Owner	FU-GEMARA-EXPORTER
Source	gemara_oscal_test.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model \(generate.go\)](#)

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from generate.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	generate.go
Mentioned In	n/a

[github.com/labeth/engineering-model-go/model\(generation_time.go\)](https://github.com/labeth/engineering-model-go/model(generation_time.go))

Inferred library_first_party dependency owned by FU-CLI-ORCHESTRATION from generation_time.go.

Key	Value
Kind	Code library_first_party
Owner	FU-CLI-ORCHESTRATION
Source	generation_time.go
Mentioned In	n/a

13.5. Verification References

[VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-1B5367](#)

executes authorized checkout flow from session start through payment authorization success

Key	Value
Kind	e2e
Status	not-run
Source	examples/payments-engineering-sample/tests/e2e/authorized_checkout_flow.yaml
Name	Authorized Checkout Flow
Mentioned In	Interaction Flow View , Authorized Checkout Flow

[VER-INF-E2E-OTA_ROLLOUT_FLOW-45BCCC](#)

runs fleet OTA rollout scenario including scheduling, deployment, and telemetry confirmation

Key	Value
Kind	e2e
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/e2e/ota_rollout_flow.yaml
Name	Ota Rollout Flow

Key	Value
Mentioned In	Interaction Flow View , Ota Rollout Flow

VER-INF-E2E-PR_OPENED_REVIEW_FLOW-7E76F6

exercises end-to-end pull request opened flow from webhook ingest to published review

Key	Value
Kind	e2e
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/e2e/pr_opened_review_flow.yaml
Name	Pr Opened Review Flow
Mentioned In	Interaction Flow View , Pr Opened Review Flow

VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-7F02A4

checks audit record persistence includes payment id and computed risk score

Key	Value
Kind	integration
Status	not-run
Source	examples/payments-engineering-sample/tests/integration/audit_record_persistence_test.ts
Name	Audit Record Persistence Test
Mentioned In	Interaction Flow View , Audit Record Persistence Test

VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-75E1E4

checks audit evidence redaction for secrets and sensitive payload fields

Key	Value
Kind	integration
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/integration/audit_redaction_test.ts
Name	Audit Redaction Test
Mentioned In	Interaction Flow View , Audit Redaction Test

VER-INF-INTEGRATION-FRAUD_GATE_TEST-C0B163

validates fraud gate escalation and block decisions before authorization completion

Key	Value
Kind	integration
Status	not-run
Source	examples/payments-engineering-sample/tests/integration/fraud_gate_test.rs
Name	Fraud Gate Test
Mentioned In	Interaction Flow View , Fraud Gate Test

VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-6E4BE8

validates telemetry logging pipeline persistence and operational reporting signals

Key	Value
Kind	integration
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/integration/logging_pipeline_test.ts
Name	Logging Pipeline Test
Mentioned In	Interaction Flow View , Logging Pipeline Test

VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-99C003

verifies policy-only fallback when Bedrock inference is unavailable

Key	Value
Kind	integration
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/integration/policy_only_fallback_test.rs
Name	Policy Only Fallback Test
Mentioned In	Interaction Flow View , Policy Only Fallback Test

VER-INF-TEST-AIRBORNE_ASSURANCE_TEST-E46ECA

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	airborne_assurance_test.go

Key	Value
Name	Airborne Assurance Test
Mentioned In	Interaction Flow View , Airborne Assurance Test

VER-INF-TEST-ASCIIDOC_COMPLETE_DIAGRAM_TEST-8C18F7

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_complete_diagram_test.go
Name	Asciidoc Complete Diagram Test
Mentioned In	Interaction Flow View , Asciidoc Complete Diagram Test

VER-INF-TEST-ASCIIDOC_DIAGRAMS_CORE_TEST-3FAC84

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_diagrams_core_test.go
Name	Asciidoc Diagrams Core Test
Mentioned In	Interaction Flow View , Asciidoc Diagrams Core Test

VER-INF-TEST-ASCIIDOC_DIAGRAMS_RUNTIME_TEST-A8483D

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_diagrams_runtime_test.go
Name	Asciidoc Diagrams Runtime Test
Mentioned In	Interaction Flow View , Asciidoc Diagrams Runtime Test

VER-INF-TEST-ASCIIDOC_EVIDENCE_TABLE_TEST-963640

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_evidence_table_test.go
Name	Asciidoc Evidence Table Test
Mentioned In	Interaction Flow View , Asciidoc Evidence Table Test

VER-INF-TEST-ASCIIDOC_HARDWARE_LABELS_TEST-AFE4FA

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_hardware_labels_test.go
Name	Asciidoc Hardware Labels Test
Mentioned In	Interaction Flow View , Asciidoc Hardware Labels Test

VER-INF-TEST-ASCIIDOC_LINKING_UNITS_TEST-22FDE2

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_linking_units_test.go
Name	Asciidoc Linking Units Test
Mentioned In	Interaction Flow View , Asciidoc Linking Units Test

VER-INF-TEST-ASCIIDOC_PUBLICATION_SCOPE_TEST-1BE464

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run

Key	Value
Source	asciidoc_publication_scope_test.go
Name	Asciidoc Publication Scope Test
Mentioned In	Interaction Flow View , Asciidoc Publication Scope Test

VER-INF-TEST-ASCIIDOC_TEST-F60765

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_test.go
Name	Asciidoc Test
Mentioned In	Interaction Flow View , Asciidoc Test

VER-INF-TEST-ASCIIDOC_TEST_EVIDENCE_TEST-C6896C

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_test_evidence_test.go
Name	Asciidoc Test Evidence Test
Mentioned In	Interaction Flow View , Asciidoc Test Evidence Test

VER-INF-TEST-ASCIIDOC_VIEWS_TEST-BFC7F2

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	asciidoc_views_test.go
Name	Asciidoc Views Test
Mentioned In	Interaction Flow View , Asciidoc Views Test

VER-INF-TEST-AVIATION_TEST-42E2CB

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	model/aviation_test.go
Name	Aviation Test
Mentioned In	Interaction Flow View , Aviation Test

VER-INF-TEST-BOUNDARY_TEST-2A7639

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	validate/boundary_test.go
Name	Boundary Test
Mentioned In	Interaction Flow View , Boundary Test

VER-INF-TEST-BOUNDARY_TEST-2F6002

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	view/boundary_test.go
Name	Boundary Test
Mentioned In	Interaction Flow View , Boundary Test

VER-INF-TEST-CATALOG_SYSTEMS_TEST-6807D7

Inferred from test source artifact.

Key	Value
Kind	test

Key	Value
Status	not-run
Source	catalog_systems_test.go
Name	Catalog Systems Test
Mentioned In	Interaction Flow View , Catalog Systems Test

VER-INF-TEST-CHECKOUT_DECLINE_RESPONSE_TEST-A14671

validates checkout decline response contract includes reason and retry guidance

Key	Value
Kind	test
Status	not-run
Source	examples/payments-engineering-sample/tests/contract/checkout_decline_response_test.go
Name	Checkout Decline Response Test
Mentioned In	Interaction Flow View , Checkout Decline Response Test

VER-INF-TEST-COFFEE_SIX_VIEW_TEST-4B177C

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	coffee_six_view_test.go
Name	Coffee Six View Test
Mentioned In	Interaction Flow View , Coffee Six View Test

VER-INF-TEST-COMPOSITION_OCI_TEST-D86C11

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	composition_oci_test.go
Name	Composition Oci Test

Key	Value
Mentioned In	Interaction Flow View , Composition Oci Test

VER-INF-TEST-COMPOSITION_PROJECTION_TEST-71E4C3

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	composition_projection_test.go
Name	Composition Projection Test
Mentioned In	Interaction Flow View , Composition Projection Test

VER-INF-TEST-COMPOSITION_TEST-8FE8F1

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	composition_test.go
Name	Composition Test
Mentioned In	Interaction Flow View , Composition Test

VER-INF-TEST-COMPOSITION_TEST-AB66B8

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	mcp/composition_test.go
Name	Composition Test
Mentioned In	Interaction Flow View , Composition Test

VER-INF-TEST-E2E_TEST-BE56F8

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	e2e_test.go
Name	E2e Test
Mentioned In	Interaction Flow View , E2e Test

VER-INF-TEST-EARS_PREFLIGHT_TEST-1ABE26

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	ears_preflight_test.go
Name	Ears Preflight Test
Mentioned In	Interaction Flow View , Ears Preflight Test

VER-INF-TEST-EXAMPLES_COMPANY_TEST-A50783

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	examples_company_test.go
Name	Examples Company Test
Mentioned In	Interaction Flow View , Examples Company Test

VER-INF-TEST-GEMARA_EXPORT_TEST-45C982

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	gemara_export_test.go

Key	Value
Name	Gemara Export Test
Mentioned In	Interaction Flow View , Gemara Export Test

VER-INF-TEST-GEMARA_OSCAL_TEST-58DBDA

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	gemara_oscal_test.go
Name	Gemara Oscal Test
Mentioned In	Interaction Flow View , Gemara Oscal Test

VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-3F4D3F

validates GitHub check-run contract shape and required review result fields

Key	Value
Kind	test
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/contract/github_check_run_contract_test.go
Name	Github Check Run Contract Test
Mentioned In	Interaction Flow View , Github Check Run Contract Test

VER-INF-TEST-HARDWARE_TEST-445BFD

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	validate/hardware_test.go
Name	Hardware Test
Mentioned In	Interaction Flow View , Hardware Test

VER-INF-TEST-HARDWARE_TEST-8CB069

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	view/hardware_test.go
Name	Hardware Test
Mentioned In	Interaction Flow View , Hardware Test

VER-INF-TEST-HARDWARE_TEST-F0D9BD

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	mcp/hardware_test.go
Name	Hardware Test
Mentioned In	Interaction Flow View , Hardware Test

VER-INF-TEST-INFERRED_CODE_TEST-3CA82E

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	inferred_code_test.go
Name	Inferred Code Test
Mentioned In	Interaction Flow View , Inferred Code Test

VER-INF-TEST-INFERRED_JAVASCRIPT_TEST-52EB3E

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run

Key	Value
Source	inferred_javascript_test.go
Name	Inferred Javascript Test
Mentioned In	Interaction Flow View , Inferred Javascript Test

VER-INF-TEST-INFERRED_LAYERS_TEST-6634AF

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	inferred_layers_test.go
Name	Inferred Layers Test
Mentioned In	Interaction Flow View , Inferred Layers Test

VER-INF-TEST-INFERRED_VERIFICATION_TEST-6715F6

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	inferred_verification_test.go
Name	Inferred Verification Test
Mentioned In	Interaction Flow View , Inferred Verification Test

VER-INF-TEST-INFERRED_VERILOG_TEST-32D5FA

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	inferred_verilog_test.go
Name	Inferred Verilog Test
Mentioned In	Interaction Flow View , Inferred Verilog Test

VER-INF-TEST-JAVASCRIPT_TEST-4E9F26

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	codemap/javascript_test.go
Name	Javascript Test
Mentioned In	Interaction Flow View , Javascript Test

VER-INF-TEST-LOAD_TEST-A58EB3

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	model/load_test.go
Name	Load Test
Mentioned In	Interaction Flow View , Load Test

VER-INF-TEST-LOBSTER_EXPORT_TEST-1C1C7E

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	lobster_export_test.go
Name	Lobster Export Test
Mentioned In	Interaction Flow View , Lobster Export Test

VER-INF-TEST-MAIN_TEST-6B61AE

Inferred from test source artifact.

Key	Value
Kind	test

Key	Value
Status	not-run
Source	cmd/engmcp/main_test.go
Name	Main Test
Mentioned In	Interaction Flow View , Main Test

VER-INF-TEST-MAIN_TEST-AAEB3C

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	cmd/engsysml/main_test.go
Name	Main Test
Mentioned In	Interaction Flow View , Main Test

VER-INF-TEST-MAIN_TEST-C71B6B

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	cmd/engnaf/main_test.go
Name	Main Test
Mentioned In	Interaction Flow View , Main Test

VER-INF-TEST-METAMODEL_INSTANCE_TEST-33196D

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	model/metamodel_instance_test.go
Name	Metamodel Instance Test

Key	Value
Mentioned In	Interaction Flow View , Metamodel Instance Test

VER-INF-TEST-METAMODEL_TEST-81AF9A

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	internal/sysmlmetamodel/metamodel_test.go
Name	Metamodel Test
Mentioned In	Interaction Flow View , Metamodel Test

VER-INF-TEST-NAF_EXPORT_TEST-26D0A4

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	naf_export_test.go
Name	Naf Export Test
Mentioned In	Interaction Flow View , Naf Export Test

VER-INF-TEST-OSCAL_CHAIN_TEST-C12EF9

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	oscal_chain_test.go
Name	Oscal Chain Test
Mentioned In	Interaction Flow View , Oscal Chain Test

VER-INF-TEST-OSCAL_PROFILE_TEST-9234C6

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	oscal_profile_test.go
Name	Oscal Profile Test
Mentioned In	Interaction Flow View , Oscal Profile Test

VER-INF-TEST-OSCAL_SSP_TEST-4B08EF

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	oscal_ssp_test.go
Name	Oscal Ssp Test
Mentioned In	Interaction Flow View , Oscal Ssp Test

VER-INF-TEST-PROJECTION_TEST-D8D752

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	model/projection_test.go
Name	Projection Test
Mentioned In	Interaction Flow View , Projection Test

VER-INF-TEST-PROJECT_TEST-2ECF72

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	view/project_test.go

Key	Value
Name	Project Test
Mentioned In	Interaction Flow View , Project Test

VER-INF-TEST-RENDER_TEST-5A1C59

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	render/svg/render_test.go
Name	Render Test
Mentioned In	Interaction Flow View , Render Test

VER-INF-TEST-RENDER_TEST-EECC23

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	render/mermaid/render_test.go
Name	Render Test
Mentioned In	Interaction Flow View , Render Test

VER-INF-TEST-REQUIREMENTS_DELTA_TEST-64E119

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	requirements_delta_test.go
Name	Requirements Delta Test
Mentioned In	Interaction Flow View , Requirements Delta Test

VER-INF-TEST-SCAN_TEST-B04F81

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	codemap/scan_test.go
Name	Scan Test
Mentioned In	Interaction Flow View , Scan Test

VER-INF-TEST-SCHEMA_TEST-AB1606

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	model/schema_test.go
Name	Schema Test
Mentioned In	Interaction Flow View , Schema Test

VER-INF-TEST-SCHEMA_V2_FIXTURE_TEST-1B8AD1

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	schema_v2_fixture_test.go
Name	Schema V2 Fixture Test
Mentioned In	Interaction Flow View , Schema V2 Fixture Test

VER-INF-TEST-SCHEMA_V2_PROJECTION_TEST-81EEBD

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run

Key	Value
Source	schema_v2_projection_test.go
Name	Schema V2 Projection Test
Mentioned In	Interaction Flow View , Schema V2 Projection Test

VER-INF-TEST-SEMANTIC_TEST-CB8EF3

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	model/semantic_test.go
Name	Semantic Test
Mentioned In	Interaction Flow View , Semantic Test

VER-INF-TEST-SERVER_TEST-389957

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	mcp/server_test.go
Name	Server Test
Mentioned In	Interaction Flow View , Server Test

VER-INF-TEST-STRUCTURIZR_EXPORT_TEST-886374

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	structurizr_export_test.go
Name	Structurizr Export Test
Mentioned In	Interaction Flow View , Structurizr Export Test

VER-INF-TEST-SYSML_COVERAGE_TEST-9C7166

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	sysml_coverage_test.go
Name	Sysml Coverage Test
Mentioned In	Interaction Flow View , Sysml Coverage Test

VER-INF-TEST-SYSML_EXPORT_TEST-8BDA4E

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	sysml_export_test.go
Name	Sysml Export Test
Mentioned In	Interaction Flow View , Sysml Export Test

VER-INF-TEST-SYSML_INTERCHANGE_TEST-CF2F5A

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	sysml_interchange_test.go
Name	Sysml Interchange Test
Mentioned In	Interaction Flow View , Sysml Interchange Test

VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-5811E2

validates telemetry ingest contract fields and accepted request semantics

Key	Value
Kind	test

Key	Value
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/contract/telemetry_ingest_contract_test.go
Name	Telemetry Ingest Contract Test
Mentioned In	Interaction Flow View , Telemetry Ingest Contract Test

VER-INF-TEST-THREAT_MODEL_EXPORT_TEST-721A8D

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	threat_model_export_test.go
Name	Threat Model Export Test
Mentioned In	Interaction Flow View , Threat Model Export Test

VER-INF-TEST-TRACE_MATRIX_TEST-9FB80C

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	trace_matrix_test.go
Name	Trace Matrix Test
Mentioned In	Interaction Flow View , Trace Matrix Test

VER-INF-TEST-TRLC_EXPORT_TEST-DD1345

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	trlc_export_test.go
Name	Trlc Export Test

Key	Value
Mentioned In	Interaction Flow View , Trlc Export Test

VER-INF-TEST-VALIDATE_TEST-498C59

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	validate/validate_test.go
Name	Validate Test
Mentioned In	Interaction Flow View , Validate Test

VER-INF-TEST-VERILOG_TEST-44F499

Inferred from test source artifact.

Key	Value
Kind	test
Status	not-run
Source	codemap/verilog_test.go
Name	Verilog Test
Mentioned In	Interaction Flow View , Verilog Test

VER-INF-UNIT-CHECKOUT_DECLINE_REASON_TEST-B8BCCD

validates checkout decline reason normalization and retry policy messaging

Key	Value
Kind	unit
Status	not-run
Source	examples/payments-engineering-sample/tests/unit/checkout_decline_reason_test.go
Name	Checkout Decline Reason Test
Mentioned In	Interaction Flow View , Checkout Decline Reason Test

VER-INF-UNIT-OTA_CAMPAIGN_TEST-9000FB

verifies OTA campaign signature validation and rollout trigger behavior

Key	Value
Kind	unit
Status	not-run
Source	examples/coffee-fleet-ota-cloud-sample/tests/unit/ota_campaign_test.go
Name	Ota Campaign Test
Mentioned In	Interaction Flow View , Ota Campaign Test

VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-4C4AEE

verifies rate-limit deferral behavior for pull request review processing

Key	Value
Kind	unit
Status	not-run
Source	examples/bedrock-pr-review-github-app-sample/tests/unit/rate_limit_deferral_test.go
Name	Rate Limit Deferral Test
Mentioned In	Interaction Flow View , Rate Limit Deferral Test